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Abstract

Meeting the climate targets set by the Paris Agreement requires leaving a large share of proven oil reserves underground. Yet the oil sector comprises tens of thousands of individual deposits that differ substantially in their extraction costs, lifecycle carbon intensities, fiscal regimes, and ownership structures. This heterogeneity, largely overlooked by aggregate analyses, is central to understanding how climate policy affects the oil sector. This thesis uses detailed deposit-level data, complemented by satellite-based gas flaring observations and original estimates of carbon intensity, to study the interaction between oil sector heterogeneity and climate policy.

The first chapter studies the consequences of delaying a carbon tax aligned with the 1.5°C objective from 2016 to 2030. It shows that policy delay results in a 33% overshoot of the oil carbon budget for a constant tax (19% for an increasing tax), driven by both excess production during the delay and long-term infrastructure lock-in from new investments. It introduces the concept of the carbon tax lock-in premium: the additional tax required after the delay to offset committed emissions from investments that would not have occurred under timely policy but become virtually irreversible once facilities are built. While this premium remains modest relative to the carbon tax level, ignoring it leads to a residual overshoot of 17-19 GtCO2eq (over 10% of the remaining oil budget in 2030). The high concentration of lock-in effects in a small number of capital-intensive offshore projects highlights the potential of targeted supply-side policies to mitigate the long-term costs of delayed action.

The second chapter, joint with Renaud Coulomb and Fanny Henriet, examines how carbon taxation reshapes the economic valuation of heterogeneous oil deposits, affecting corporate profits and government fiscal revenues. Using a global simulation model calibrated on over 21,000 deposits, it documents highly heterogeneous effects characterised by a surprisingly weak correlation between a deposit's lifecycle carbon intensity and its

change in value. A theoretical model of intertemporal extraction with heterogeneous deposits explains these counterintuitive outcomes, showing that high-carbon deposits can appreciate while cleaner ones decline. Integrating detailed fiscal data at the asset level, the chapter decomposes value changes into after-tax corporate profits and States revenues, showing that while corporate profits generally fall, moderate carbon taxes can fully offset fiscal revenue losses for a majority of producer countries. The chapter also evaluates second-best instruments, including uniform per-barrel taxes and extraction bans, finding that they impose substantial welfare losses relative to optimal carbon taxation and yield markedly different distributional consequences.

The last chapter, joint with Sanjana Ghosh and Pritam Saha, studies how ownership structure conditions the effectiveness of voluntary environmental commitments in the oil sector, using the World Bank's Zero Routine Flaring initiative as a natural experiment. Combining asset-level ownership data with satellite flaring measurements, it shows that committed firms do reduce flaring on assets they co-own, including when they are not operators, through board participation. However, these gains crucially depend on the presence of at least one committed owner. When the last committed owner exits and sells to an uncommitted buyer, flaring increases sharply. This "last good owner" effect reveals a tension between voice and exit: divestment by committed owners can improve their portfolio-level metrics yet worsen asset-level emissions. These findings highlight the importance of monitoring ownership transactions to limit emissions leakage.

Résumé

Atteindre les objectifs climatiques fixés par l'Accord de Paris suppose de laisser une part substantielle des réserves de pétrole sous terre. Le secteur pétrolier comprend néanmoins des dizaines de milliers de gisements individuels qui diffèrent par leurs coûts d'extraction, leur intensité carbone, leurs régimes fiscaux et leurs structures d'actionnariat. Cette hétérogénéité, largement ignorée par les analyses agrégées, est centrale pour comprendre comment les politiques climatiques reconfigurent le secteur. Cette thèse exploite des données détaillées au niveau des gisements, enrichies d'observations satellitaires du torchage de gaz et d'estimations d'intensité carbone, pour analyser ces interactions.

Le premier chapitre étudie les conséquences du report de 2016 à 2030 d'une taxe carbone alignée sur l'objectif de 1,5°C. Le retard entraîne un dépassement de 33% du budget carbone pétrolier pour une taxe constante (19% pour une taxe croissante), dû à la fois à la surproduction pendant le délai et à des effets de *lock-in* résultant de nouveaux investissements. Le chapitre introduit le concept de prime de *lock-in* sur la taxe carbone : la surtaxe requise après le délai pour compenser les émissions engagées par des investissements dus à l'absence de taxe, qui deviennent irréversibles une fois les infrastructures construites. Bien que modeste en absolu, l'ignorer conduit à un dépassement résiduel de 17-19 GtCO2eq, plus de 10% du budget en 2030. La concentration du *lock-in* dans un nombre réduit de projets offshore intenses en capital souligne le potentiel de politiques d'offre ciblées pour atténuer le coût à long terme de l'inaction climatique.

Le deuxième chapitre, co-écrit avec Renaud Coulomb et Fanny Henriet, examine comment la taxation carbone affecte la valorisation économique de gisements hétérogènes, et ainsi les profits des entreprises et les recettes des États. À partir d'un modèle de simulation mondiale calibré sur plus de 21 000 gisements, il documente des effets très hétérogènes, caractérisés par une corrélation étonnamment faible entre intensité carbone et variation de valeur. Un modèle théorique d'extraction intertemporelle explique ce résul-

tat contre-intuitif, montrant que des gisements à forte intensité carbone peuvent voir leur valeur augmenter tandis que des gisements plus propres déclinent. À l'aide de données fiscales détaillées, le chapitre décompose les variations de valeur entre profits nets des entreprises et revenus des États : si les profits diminuent généralement, des niveaux modérés de taxe carbone peuvent entièrement compenser les pertes de recettes budgétaires pour plusieurs pays producteurs. Il évalue ensuite des instruments de second rang (taxe uniforme et moratoire sur la production), et montre qu'ils imposent des coûts majeurs par rapport à la taxation carbone optimale et produisent des effets distributifs très différents.

Le dernier chapitre, co-écrit avec Sanjana Ghosh et Pritam Saha, étudie la manière dont la structure de propriété conditionne l'efficacité des engagements environnementaux volontaires, à partir de l'initiative « *Zero Routine Flaring* » de la Banque mondiale. Combinant données d'actionnariat au niveau des actifs et mesures satellitaires du torchage de gaz, il montre que les firmes engagées réduisent effectivement le torchage dans les actifs qu'elles co-détiennent, y compris lorsqu'elles ne sont pas opératrices. Cependant, ces gains ne persistent qu'en leur présence : si le dernier propriétaire engagé vend ses parts à une firme sans engagement environnemental, le torchage augmente fortement. Cet effet du « dernier bon propriétaire » met en lumière une tension entre « voix » et « désinvestissement » : les actionnaires engagés peuvent améliorer leurs indicateurs environnementaux en cédant leurs parts dans un actif polluant et ainsi entraîner une hausse des émissions de ce même actif. Ces résultats soulignent l'importance d'encadrer les transactions afin de limiter les fuites d'émissions.

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General introduction

The stone age came to an end not for a lack of stones,
and the oil age will end, but not for a lack of oil.

Sheikh Ahmed Zaki Yamani, former Saudi Arabian Oil Minister

Oil is an inexhaustible resource
that is going to become increasingly rare.

Dominique de Villepin, former French Prime Minister

Oil remains the world's single largest source of primary energy, accounting for approximately 30% of global energy supply. Daily production exceeds 100 million barrels, feeding an industrial system that spans transport, petrochemicals, heating, and electricity generation. Despite growing investment in renewable energy and the acceleration of the energy transition, global oil demand has continued to rise, driven in particular by growth in emerging economies. The centrality of oil in the world economy means that any policy aimed at reducing its production or consumption carries consequences that extend far beyond the energy sector, affecting trade balances, fiscal revenues, and geopolitical dynamics.

Yet oil is not a homogeneous commodity extracted from a representative deposit. Global production originates from tens of thousands of individual assets (fields or portions of fields) spread across more than 80 countries and operated by a wide range of companies, from state-owned national oil companies to small independent firms. Figure 1 displays the geographical distribution of oil producing assets recorded in the Rystad UCube database, which covers more than 100,000 oil and gas assets worldwide, of which around 18,000 were actively producing oil as of 2025. These assets differ profoundly in their geological characteristics, extraction technologies, production costs, carbon intensity, ownership structures, and the fiscal regimes under which they operate. This granularity reveals a degree of heterogeneity hidden in aggregate statistics, that turns out to be central to understanding the interaction between the oil sector and climate policy.

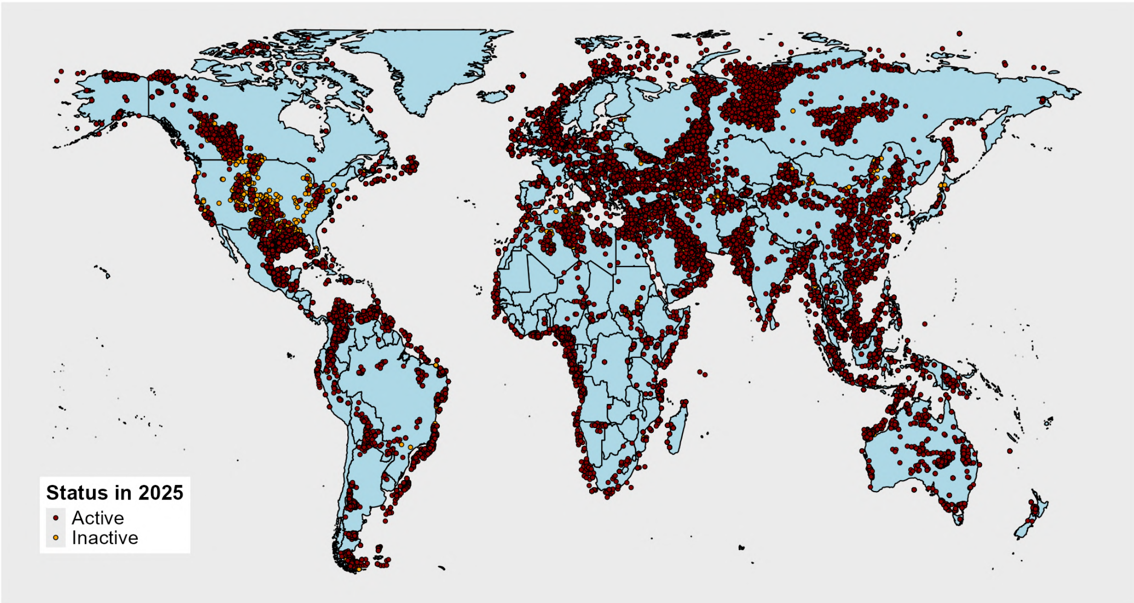

Figure 1: Oil producing assets worldwide (2025).
Source: Rystad Energy UCube database.

The importance of this interaction is well established. Oil extraction, refining, and combustion together represent a major share of global greenhouse gas emissions. The Paris Agreement, signed in 2015, set the objective of keeping the rise in global temperatures to well below 2°C and pursuing efforts to limit it to 1.5°C. Meeting these targets implies operating under a finite carbon budget, a cumulative ceiling on the quantity of CO2 that can still be emitted. Previous studies have translated this constraint into stark implications for the oil sector: McGlade and Ekins (2015) estimate that a third of oil reserves must remain unextracted to stay within a 2°C budget, while Welsby et al. (2021) show that the constraint is even tighter under a 1.5°C target, with more than half of proven reserves needing to stay in the ground. Extracting, refining, and burning all currently proven oil reserves would alone generate approximately 670 GtCO2eq, roughly 1.5 times the total remaining carbon budget for all sources combined under the 1.5°C objective. Oil reserves are, in other words, far too abundant relative to what the climate can accommodate. This constraint entails identifying which reserves should remain underground, but its implications extend much further. Because oil deposits are so heterogeneous, the design and timing of climate policy reshape the entire landscape of the sector: they alter the dynamics of investment in new production infrastructure, redistribute revenues across firms and governments, change the relative value of deposits, and interact with the governance decisions of the companies that own and operate each asset. Understanding these interactions, and the role that deposit-level heterogeneity plays in shaping them, is the purpose of this thesis.

Climate policy for the oil sector

The case for carbon pricing

The economic literature widely recognises carbon taxation as the most effective instrument for curbing fossil fuel use (Nordhaus, 2007; Stern, 2007; Metcalf, 2019). A global carbon tax, applied uniformly across producers, would internalise the environmental externality associated with oil production and combustion by ordering extraction according to the full social cost of each barrel, defined as the sum of private extraction costs and the environmental damage from associated emissions. Under a binding carbon budget, the dynamically efficient carbon tax increases at the social discount rate (Lemoine and Rudik, 2017; Dietz and Venmans, 2019), reflecting the rising scarcity of the remaining carbon budget. In principle, such a tax would allocate the burden of emission reductions where abatement is cheapest, minimising the aggregate cost of meeting climate objectives. The idea of a global carbon price on fossil fuels is not purely theoretical: the International Maritime Organisation agreed in principle in 2025 over a carbon pricing mechanism on international shipping, demonstrating that coordinated instruments covering globally traded commodities are conceivable.

Existing policies and implicit subsidies

Yet, as of 2025, the gap between this theoretical ideal and observed climate policies remains vast. Some jurisdictions have implemented carbon pricing mechanisms, as shown in Figure 2. However, only a handful cover oil production emissions, and the levels of existing taxes fall short of what is required to align with the Paris Agreement objectives (Metcalf, 2019). Among the ten largest oil producing countries, only the United States, China, and Canada have introduced carbon pricing mechanisms of any kind, and none of them comprehensively covers upstream extraction emissions[1]. Most climate regulation targets downstream combustion, through fuel taxes, vehicle emission standards, or subsidies to alternative energy, leaving the heterogeneity in extraction and refining emis-

[1] See table B.1 in appendix.

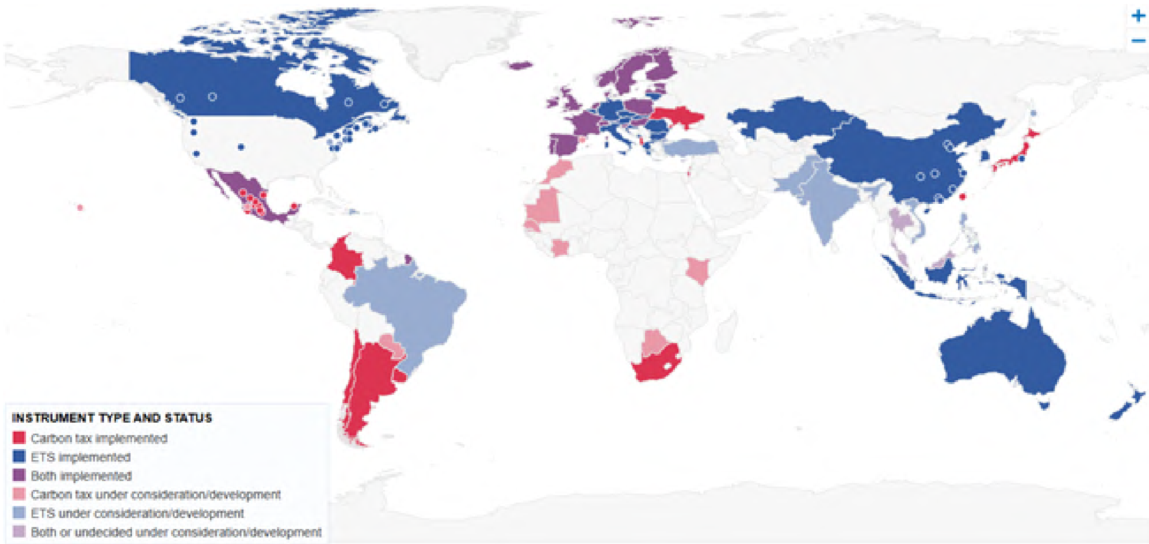

Figure 2: Carbon pricing mechanisms implemented or under consideration (2025).
Source: World Bank Carbon Pricing Dashboard (2025).

sions entirely unpriced. At the same time, fossil fuels continue to benefit from substantial implicit subsidies worldwide. As documented by Parry et al. (2021), global fossil fuel subsidies, including both explicit transfers and the underpricing of environmental externalities, amounted to approximately $7 trillion in 2022, or 7.1% of global GDP. Implicit subsidies directed at oil and gas producers specifically, while much lower in magnitude, can have substantial impacts on production: Erickson et al. (2017) show that in a low oil price environment, nearly half US oil and gas projects are dependent on it. These subsidies actively counteract the incentives that a carbon tax would create, and disproportionately benefit the most carbon-intensive forms of production.

Political constraints

The persistence of weak climate policies in the oil sector reflects deep political economy constraints, rooted in the economic dependence of major producing countries on oil revenues. Opposition to stringent climate policies on fossil fuels is led by two broad coalitions. The first is constituted of OPEC+ countries, for whom oil revenues represent a major share of government budgets and GDP. The International Monetary Fund reports that the fiscal breakeven oil price, the price at which oil-exporting countries balance their budget, exceeds $80/bbl for Saudi Arabia and is even higher for several other OPEC members (International Monetary Fund, 2024). For these countries, a global carbon tax that significantly reduces oil demand or prices would directly threaten fiscal stability. OPEC countries have in turn been the leading opposition to any international moratorium on fossil fuel extraction and new field development (Lujala et al., 2022). The second source of resistance comes from the United States, which has become the world's largest oil producer over the past decade thanks to the rapid expansion of shale oil. At the international level, as Van Asselt (2021) notes, the United Nations Framework Convention on Climate Change emphasises territorial accounting for greenhouse gas emissions, which focuses on fossil fuel consumption rather than supply, further incentivising countries to focus on demand-side policies to fulfil their international commitments while leaving production largely unconstrained.

The stranded assets literature

This political economy context has motivated a large literature studying the consequences of insufficiently ambitious climate policies, and in particular the question of which fossil fuel reserves would become "stranded" under various carbon budget scenarios. McGlade and Ekins (2015) provide a seminal geographical decomposition of unburnable reserves under a 2°C target, while Welsby et al. (2021) update and tighten these estimates for the 1.5°C objective. Other contributions have studied the financial implications of stranded asset risk for investors and firms (Mercure et al., 2018; Van der Ploeg and Rezai, 2020). However, this literature predominantly relies on either theoretical models with representative deposits, or on aggregate supply curves constructed at the country or basin level (Heal and Schlenker, 2019). While these approaches are valuable for establishing broad magnitudes, they inevitably overlook the substantial deposit-level heterogeneity that, as this thesis demonstrates, is central to understanding the actual consequences of climate policy. In practice, climate policy does not operate on aggregate supply curves: it materialises through the specific investment, production, and governance decisions made for each individual asset.

The case for supply-side policies

The persistent difficulty of implementing comprehensive demand-side instruments has led a growing body of literature to advocate for supply-side climate policies, which directly constrain the production of fossil fuels rather than penalising their consumption alone. Erickson et al. (2018) argue that addressing fossil fuel supply should be viewed as complementary to existing demand-side efforts rather than a substitute for them. As explained by Green and Denniss (2018), several political economy arguments justify the use of supply-side approaches: (i) supply-side measures have more direct and predictable impacts on emissions, since preventing a specific project from going forward provides certainty about the quantity of abatement obtained, (ii) they face lower administrative costs, as regulating a small number of producers is simpler than monitoring millions of consumers, and (iii) they tend to have higher public support, because the distributional burden is perceived as fairer and the benefits more tangible than those of consumption-based taxes. The potential for public mobilisation around specific fossil fuel projects has been documented by Piggot (2018) and Carter and McKenzie (2020), who show that most production bans issued thus far have originated from citizen-led campaigns targeting identifiable developments. More recently, Green et al. (2024) have called for an international norm of "no new fossil fuel projects", arguing that such a norm is both feasible and effective. Yet, as Lazarus and van Asselt (2018) observe, the academic literature on supply-side policies remains underdeveloped as compared to the literature on demand-side policies. This thesis contributes to filling this gap, by quantifying carbon lock-in at the asset level to identify which supply-side interventions would be most effective (Chapter 1), by documenting distributional consequences of carbon pricing (Chapter 2), and by examining the effectiveness of a voluntary commitment framework, the World Bank's Zero Routine Flaring initiative, as a softer form of supply-side policy (Chapter 3).

Heterogeneous individual oil deposits

New data sources

A central challenge in studying the interaction between climate policy and the oil sector at the deposit level is the scarcity of comprehensive, globally consistent data. This thesis relies on a combination of proprietary industry databases, satellite observations, and engineering-based models to overcome this challenge. The primary data source, shared by all three chapters, is Rystad Energy's UCube database, one of the most comprehensive datasets on upstream oil and gas worldwide. It covers over 100,000 oil and gas assets, among which more than 20,000 are actively producing as of 2025. For each asset, the database provides a detailed breakdown of annual production by oil type, costs by category (operating expenses, capital expenses, exploration, government take), exploitable reserves, development timelines, as well as administrative, geographic, and geological characteristics. Crucially for Chapter 3, the database also records the ownership structure and operatorship of each asset and their evolution through time.

In addition to this industry database, the thesis makes use of satellite observations of gas flaring from the Visible Infrared Imaging Radiometer Suite (VIIRS), collected by the

Earth Observation Group at the Colorado School of Mines in collaboration with NOAA (Elvidge et al., 2015). Gas flares detected at night by the instrument are converted into estimates of flared volumes using the VIIRS Nightfire algorithm (Zhizhin et al., 2021). As nearly 95% of oil and gas operators do not publicly report their flared volumes, satellite observations constitute the most reliable source of information on this environmentally important practice. I develop in this thesis a multi-stage geographic matching procedure that assigns over 134,000 satellite flaring observations recorded between 2012 and 2024 to individual Rystad assets, using data on well locations and assets' geographic boundaries. This matched dataset is used in Chapter 1 to estimate field-level flaring-to-oil ratios, a key input for upstream carbon intensity estimation, and in Chapter 3 as the primary measure of environmental outcomes.

Finally, the thesis develops novel estimates of lifecycle carbon intensities for each asset by combining three engineering-based models from the Oil-Climate Index: the Oil Production Greenhouse Gas Emissions Estimator (OPGEE) for upstream emissions, the Petroleum Refinery Lifecycle Inventory Model (PRELIM) for midstream emissions, and the Oil Products Emissions Module (OPEM) for downstream emissions. Direct model computations are obtained for a sample of several hundred oil fields, and predictive regression models are fitted to extend these estimates to the full universe of Rystad assets. This combination of direct engineering estimates, satellite-derived flaring data, and predictive modelling provides lifecycle carbon intensity estimates for all producing and not-yet-developed assets worldwide. These estimates are used in Chapters 1 and 2, and the methods used to obtain them are described at length in appendix **??**.

Heterogeneity in extraction costs

Extraction costs vary considerably across oil deposits. Figure 3 reports reserve-weighted average breakeven prices for the twenty largest producing countries. Middle Eastern conventional assets, concentrated in Kuwait, Iraq, Qatar, Saudi Arabia, Iran, and the UAE, display average breakeven prices below $12/bbl. At the other end of the distribution, deposits in Canada and China have average breakeven prices exceeding $33/bbl, approximately three times higher. However, focusing on cross-country averages alone is misleading. As the error bars in the figure indicate, within-country dispersion is substantial: in Canada, the top-10% most expensive assets have breakeven prices exceeding $55/bbl, while the cheapest 10% are below $12/bbl. Similar patterns exist in all major producing countries. This within-country heterogeneity is overlooked by aggregate supply curves constructed at the country or basin level, yet it is central to the analysis conducted in this thesis.

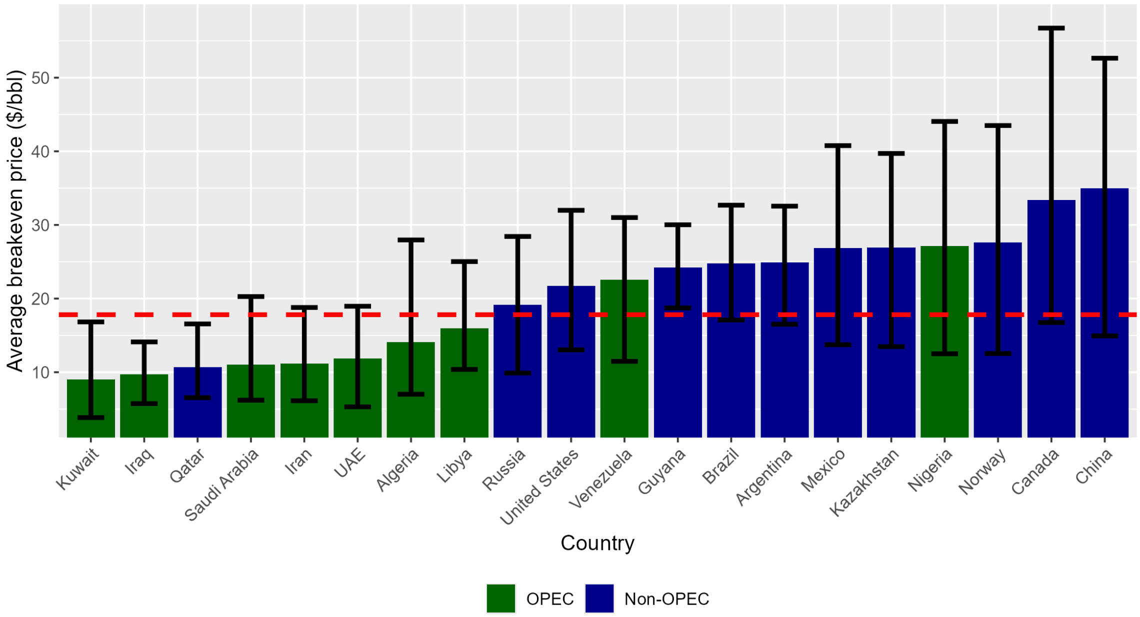

Figure 3: Breakeven prices by country ($/bbl): 2025 reserve-weighted average, top-10% and bottom-10% quantiles.

Notes: Fiscal expenses are ignored in the computations, only capital and operating expenses are taken into account. Source: author's computations from Rystad Energy UCube database.

Besides this substantial heterogeneity across deposits, the dual structure of costs typically associated with oil projects plays an important role in shaping climate policy outcomes. Oil projects require two distinct types of expenses: development costs, which correspond to the upfront capital invested in building production facilities, drilling initial wells, and installing infrastructure before or in the early years of production, and production costs, which are the per-barrel operating expenses incurred once the asset is active. The relative weight of these two components varies substantially across types of deposits. Deepwater offshore projects, for instance, require very large upfront investments (often several billion dollars) but are relatively cheap to operate once the infrastructure is in place. Conversely, shale oil wells involve comparatively modest individual drilling costs but face high per-barrel operating expenses and their production declines rapidly. This dual structure will prove central to understanding the consequences of climate policy delay, as the gap between the full project breakeven price and the production breakeven price determines the extent to which an asset, once developed, becomes resilient to carbon taxation.

Heterogeneity in carbon intensity

Not all barrels of oil carry the same environmental cost. The lifecycle greenhouse gas emissions associated with a barrel of oil, from extraction to final combustion, can vary by a factor of two or more across deposits. Figure 4 illustrates this variation across the twenty largest producing countries. While Qatar, Kazakhstan, the UAE, and Saudi Arabia display average lifecycle carbon intensities around 400 to 440 kgCO2eq/bbl, Canadian and Venezuelan oil exceeds 550 kgCO2eq/bbl on average. As with extraction costs, within-country dispersion can be substantial, reflecting the diversity of oil types, extraction methods, and flaring practices within each country. OPEC members are not a homogeneous group: Middle Eastern producers tend to have much cleaner oil than heavy oil producing countries such as Venezuela or high-flaring countries such as Algeria and Nigeria.

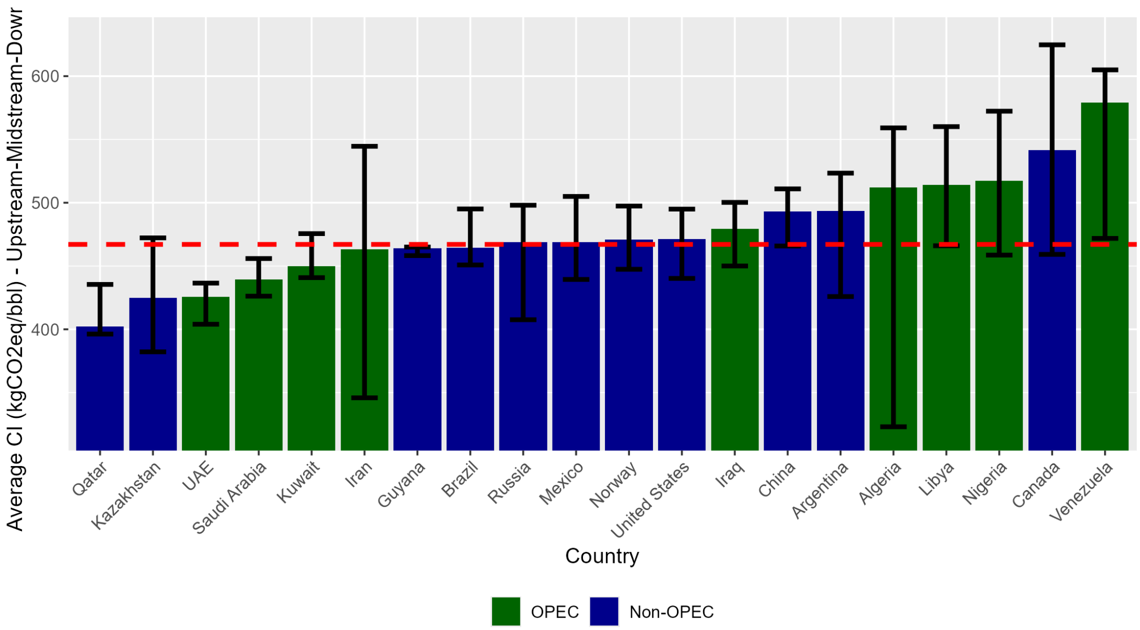

Figure 4: Lifecycle carbon intensity by country (kgCO2eq/bbl): 2025 reserve-weighted average, top-10% and bottom-10% quantiles.

Note: values presented are full lifecycle carbon intensities, including upstream, midstream, and downstream emissions. Source: author's computations from Rystad Energy UCube database.

Lifecycle emissions of oil can be split into three main stages: (i) upstream emissions, associated with the extraction of crude oil, (ii) midstream emissions, originating from its refining, and (iii) downstream emissions, which correspond to the transport and final use of refined oil products by end-users. Although downstream emissions account for approximately 80% of total lifecycle emissions, all three stages contribute to the heterogeneity observed across deposits, with the upstream playing a major role. Understanding the drivers of variation at each stage is essential, as carbon taxation applied to the full lifecycle would affect deposits very differently depending on their characteristics.

Upstream carbon intensity is shaped primarily by two factors. The first is the nature of the oil itself and the extraction methods it requires. Heavier and more viscous oils, such as bitumen, often necessitate steam injection to reduce their viscosity and allow extraction, a process that is highly energy-intensive and translates into elevated upstream emissions. Oil sands operations are a well-known example: when on-site upgrading into synthetic crude is required, upstream emissions can be several times higher than those of conventional light crude (Brandt and Farrell, 2007). The second, and arguably most consequential, driver of upstream variation is the treatment of co-produced gas. When oil is extracted, natural gas is typically brought to the surface as a co-product. This associated gas can be sold on the market if the asset is connected to gas infrastructure, reinjected into the reservoir to maintain pressure, or used to power on-site equipment. When none of these valorisation options are economically viable or technically feasible, the gas is either flared (burned on site) or vented (released into the atmosphere). Venting carries the worst environmental impact, as it releases methane directly, and is banned or severely restricted in most jurisdictions (World Bank, 2022). Flaring remains the default disposal method in the absence of gas valorisation, but it is far from benign: gas flaring represented more than 150 billion cubic metres worldwide in 2024 (World Bank, 2025), accounting for nearly 1% of total global energy-related greenhouse gas emissions[2]. Flaring-intensive assets can therefore have upstream carbon intensities several times higher than the global average. The quantity of co-produced gas, the presence or absence of gas infrastructure, and the operational choices made by the operator regarding the use of this gas thus emerge as critical determinants of an asset's environmental footprint.

Midstream and downstream emissions, while individually less variable than upstream ones, also contribute to the heterogeneity in lifecycle carbon intensity across deposits. Midstream emissions depend on the refinery configuration needed to process a given type of crude: heavy, sour crudes require more intensive treatment (e.g. deep conversion), while light, sweet crudes can be processed through simpler configurations, resulting in substantially lower refining emissions. Downstream emissions are primarily determined by the mix of oil products yielded by refining, which itself depends on both the characteristics of the crude and the refinery configuration. Although downstream emissions are more homogeneous across crude types than upstream ones, they are far from uniform, and the combined midstream and downstream variation reinforces the cross-deposit heterogeneity in total lifecycle emissions documented in Figure 4.

[2]Besides its environmental consequences, this volume of gas would allow to cover more than three times the total Sub-Saharan consumption.

A critical empirical finding, documented in Chapter 2 using deposit-level data, is that lifecycle carbon intensity and private extraction cost are only weakly correlated across deposits. Comparing Figures 3 and 4 provides a first indication: the ranking of countries by extraction cost does not mirror the ranking by carbon intensity. The reason is that flaring and land use, which together account for a large share of upstream emissions, are not positively related to private production costs (Coulomb et al., 2026). This has a profound implication: the cost-efficient extraction order, in which the cheapest barrels are produced first, and the climate-efficient order, in which the cleanest barrels are prioritised, are substantially different. A policy that ignores carbon intensity heterogeneity and relies solely on extraction costs to determine which deposits to develop will generate more emissions than necessary to meet a given carbon budget.

Institutional heterogeneity

Beyond their physical and environmental characteristics, oil deposits also differ in the institutional framework under which they are exploited. Two dimensions of this institutional heterogeneity are particularly relevant for this thesis: fiscal regimes and ownership structure.

Each deposit operates under a specific fiscal regime that dictates the distribution of revenues between private firms and host-country governments. These regimes take various forms, including royalties levied on gross revenues, production-sharing contracts in which the government receives a share of profit oil, income taxes on net profits, and various production-related taxes reported as operating expenses. In practice, fiscal regimes are often tailored to a specific project or extraction basin, with rates and structures sometimes negotiated on a case-by-case basis between governments and oil companies. This project-level specificity means that the effective tax burden can vary substantially even across deposits within the same country. Fiscal regimes shape the incentives of both producers and governments: they affect the profitability of investment decisions, the willingness of governments to renegotiate terms in response to changing market conditions, and the distribution of revenues under carbon taxation. Chapter 1 abstracts from government takes, on the grounds that in the presence of a worldwide carbon tax, producing governments would renegotiate fiscal terms to keep viable assets active and continue extracting a share of the rent. Chapter 2 takes a different approach by integrating fiscal data at the deposit level, covering royalties, taxes, and government profit shares, to decompose changes in oil value into after-tax corporate profits and States revenues.

The second dimension of institutional heterogeneity is the ownership and governance structure of oil assets. Oil production typically requires substantial capital investment, geological risk, and technical expertise, often leading to the use of joint venture structures in which multiple firms share ownership of a given asset. One partner is designated as the operator, responsible for managing field operations and making operational decisions, including those related to drilling, production levels, and environmental practices such as flaring. The remaining partners are non-operating equity owners who share in production revenues and costs in proportion to their stake but do not directly control operations. Major capital investments, such as gas capture infrastructure or flaring reduction equipment, typically require approval from joint operating committees in which non-operating owners have representation and voting rights proportional to their equity share.

The composition of assets' ownership varies considerably, and plays a major role in shaping the response by operators to climate policies. Publicly-listed oil companies (and a fortiori the "majors") face strong environmental, social, and governance (ESG) pressure and disclosure requirements from investors and regulators. Conversely, national oil companies, which produce approximately half of global oil, face substantially less investor scrutiny (Cahill and Swanson, 2023). Non-publicly traded private firms are subject to minimal disclosure obligations and operate largely beyond the oversight of financial and environmental regulators (Matikainen, 2022). When an asset is jointly owned by companies with heterogeneous environmental priorities, the effectiveness of any individual firm's commitment depends on the governance dynamics within the asset's board. Whether the committed firm is the operator, the share of equity it controls, and the environmental posture of the remaining partners all shape the extent to which voluntary pledges translate into concrete operational changes. This governance heterogeneity, and its consequences for the effectiveness of environmental initiatives, is the central object of Chapter 3.

The process of oil extraction: investment dynamics and production rigidity

The dimensions of heterogeneity described above do not simply exist independently to each other: they are jointly assessed by investors when deciding whether and how to develop a new oil deposit. This process of investment and production is what ultimately translates deposit characteristics into actual production, revenues, and environmental outcomes.

When an investor considers developing a new field, it evaluates the expected net present value of the project, which depends on anticipated future oil prices, the deposit's cost structure (both development and production costs), the applicable fiscal regime, the carbon intensity of the deposit and its potential exposure to future environmental regulation, and the partnership structure under which the project will be operated. On the basis of this assessment, the investor determines the optimal level of infrastructure to build, which in turn sets the production capacity and the methods of extraction to be employed (Venables, 2014; Okullo and Reynès, 2016). This initial investment decision is largely irreversible. Once development costs are sunk, the asset enters production and will typically continue operating as long as revenues exceed the much lower production costs, regardless of whether the project as a whole remains economically justified in light of subsequent policy changes.

At the well level, production rigidity is well documented: geophysical and technological constraints dictate the production path to the point that no meaningful adaptation to changes in oil price can be observed (Anderson et al., 2018). At the deposit level, however, some adjustment remains possible. For instance, operators can choose to drill additional producer wells, drill injection wells to increase output, or invest in enhanced recovery to extend the asset's lifetime. A notable exception to the general pattern of production rigidity is provided by OPEC-core countries (Saudi Arabia, Kuwait and UAE), which deliberately maintain spare production capacity to retain the ability to adjust output rapidly in response to shocks (Fattouh, 2007; Kellogg, 2024).

Despite the relative rigidity of production once the initial development decision is made, important operational choices remain throughout the productive life of the asset. Beside deciding when and where to drill new wells or whether to invest in enhanced oil recovery, a critical decision for operators is how to handle co-produced gas: whether to invest in gas gathering and processing infrastructure, connect the asset to a gas market, or continue flaring. These ongoing decisions are shaped not only by economic considerations but also by the governance of the asset, and in particular by whether the companies on the management board have made environmental commitments. The extent to which committed non-operating owners can influence these operational decisions, and what happens when they exit, is the question addressed by Chapter 3.

How heterogeneity interacts with climate policy

The multiplicity of dimensions along which oil deposits differ implies that the aggregate effect of any climate policy on the oil sector is the result of thousands of individual responses, each depending on a unique combination of extraction costs, carbon intensity, fiscal regime, ownership structure, geological constraints, and investment history. Aggregate supply curves, whether constructed at the country or basin level, inevitably overlook this complexity and can therefore produce misleading predictions about the effects of climate policies on production, revenues, and emissions. These dimensions do not merely add descriptive richness to the picture of the oil sector: they interact with climate policy in ways that can be counterintuitive, and accounting for them is essential to understanding how the oil sector actually responds to environmental regulation.

Institutional heterogeneity shapes the effectiveness of current environmental policies

The political obstacles described above imply that, in the foreseeable future, climate policy in the oil sector will continue to operate largely through partial instruments rather than a comprehensive global carbon tax. Voluntary corporate initiatives, ESG-driven investor pressure, and regulatory constraints imposed by specific jurisdictions on companies headquartered within their borders have emerged as the primary mechanisms through which environmental improvements are pursued. The effectiveness of these mechanisms, however, depends fundamentally on the institutional characteristics of oil assets.

A first reason is that such policies only reach a fraction of the industry. Voluntary initiatives such as the World Bank's Zero Routine Flaring (ZRF) initiative attract primarily large publicly-listed companies that face reputational incentives and investor scrutiny. Similarly, regulatory constraints imposed by European or North American jurisdictions bind companies headquartered in those regions but do not extend to national oil companies or to private firms with weaker reporting obligations. If the assets most in need of environmental improvement are held by firms outside their reach, aggregate effectiveness is limited regardless of how committed the targeted firms are. A second reason is that even within assets affected by such policies, institutional heterogeneity determines whether commitments translate into operational change. When a committed firm is also the operator, it can implement improvements directly. When it is a non-operating equity partner, it must work through governance mechanisms to influence an operator that may not share its priorities. The same corporate commitment can thus have very different environmental consequences depending on the governance context in which it is exercised.

This governance problem gives rise to a voice-versus-exit dilemma, echoing the framework first developed by Hirschman (1972). A committed owner unable to influence operational decisions can either remain on the management board and continue exerting pressure (voice), or divest its stake and reallocate capital toward assets where it has more control (exit). Exit may improve the divesting firm's portfolio in terms of environmental metrics, but it risks worsening actual emissions at the asset if the buyer is not committed. Whether voice or exit prevails depends on institutional characteristics: the committed firm's role as operator or non-operator, the share of equity held by committed companies, and external factors such as the availability of gas valorisation infrastructure. Chapter 3 studies these dynamics using the ZRF initiative as a natural experiment. It shows that committed companies do reduce flaring over time, but that these gains are sharply reversed when the last committed owner divests, even when the exiting firm was not the operator. This "last good owner" effect reveals a significant loophole in voluntary environmental frameworks: without provisions governing divestment to uncommitted parties, such initiatives risk merely reshuffling emissions across ownership structures rather than genuinely reducing them.

Heterogeneity and distributional consequences of climate policy

The limitations of partial instruments reinforce the case for a comprehensive, global carbon pricing mechanism. As explained before, the main political obstacle to such a mechanism is the resistance of oil-producing countries, which fear losing a major source of fiscal revenue. Yet the deposit-level heterogeneity documented in this thesis suggests that this opposition may be less monolithic than commonly assumed. As illustrated in Figures 3 and 4, OPEC-core countries, notably Saudi Arabia, Kuwait, and the UAE, produce oil that is simultaneously among the cheapest to extract and the least carbon-intensive in the world. Under a global carbon tax, these countries should therefore be relatively less affected than high-cost, high-carbon producers such as Canada, Venezuela, or US shale operators. Chapter 2 shows that for low-cost producers, government fiscal revenues follow an inverted U-shaped pattern as the carbon tax rises, similar to the Laffer curve: moderate levels of carbon taxation can fully offset the decline in royalties and income taxes through carbon tax revenues, particularly if producing countries can preempt the taxation of downstream combustion emissions. This resilience is further reinforced by the production flexibility of OPEC-core countries: because they maintain significant spare capacity, they retain the ability to adapt their output in response to the new policy environment rather than being locked into predetermined production paths. A global carbon tax could thus benefit the most influential OPEC members while imposing substantial costs on their high-cost competitors, an insight that could in principle serve as a source of political leverage in international climate negotiations.

The distributional consequences of carbon taxation are, however, far more complex than the simple observation that low-cost, low-carbon producers benefit would suggest. Chapter 2 demonstrates, both theoretically and empirically, that the relationship between a deposit's lifecycle carbon intensity and its change in value under carbon taxation is not straightforward. Using a global simulation model calibrated on over 21,000 deposits, the chapter documents that changes in the social value of oil following an increase in the social cost of carbon are only weakly correlated with lifecycle carbon intensity. Strikingly, the social value of certain relatively high-carbon deposits can actually increase as climate policy tightens, while that of some cleaner deposits declines. These counterintuitive outcomes arise because changes in deposit values reflect the interplay of multiple forces, including shifts in the competitive position of deposits relative to one another, changes in the timing of extraction, and the reallocation of production across heterogeneous assets, none of which can be reduced to carbon intensity alone. The pattern extends to the firm and country levels: while aggregate firms' profits decline as carbon taxes rise, the magnitude of the decline varies substantially across firms and is only weakly correlated with the average carbon intensity of their asset portfolios. These findings reveal that the distributional consequences of carbon taxation among oil deposits are far more complex than a simple "the dirtier the oil, the bigger the loser" intuition would suggest, and that understanding the full picture requires an asset-level approach.

Carbon lock-in and the case for targeted supply-side measures

The prospect of a global carbon tax, however potentially beneficial for low-cost producers, remains distant. The cost of delaying the implementation of such instrument is often underestimated, because it is not limited to the excess emissions that occur during the period of inaction. Each year of delay also gives rise to new irreversible investments in oil production infrastructure, which in turn commit future emissions over the lifetime of the assets they create. This phenomenon is known as infrastructure lock-in (Unruh, 2000), one of three dimensions of carbon lock-in identified by Seto et al. (2016). The mechanism is rooted in the dual cost structure described earlier. Because development costs are sunk and production costs are low relative to revenues, a newly built asset will continue to produce and emit for decades, even if a subsequent carbon tax renders the project socially undesirable. Phasing out a producing asset requires that the tax exceed the much lower production breakeven price, not the full project breakeven. The wider the gap between these two thresholds, the deeper the lock-in. Estimates of committed emissions from existing infrastructure have grown over time (Tong et al., 2019; Trout et al., 2022), but most integrated assessment models only partially capture these dynamics (Grubb et al., 2021). Chapter 1 fills this gap with the first precise, asset-level quantification of infrastructure lock-in in the oil sector.

A critical feature of lock-in is that it builds up progressively, as new infrastructures are built. The emergence of unconventional oil over the past fifteen years has reinforced both the heterogeneity in lock-in dynamics and their importance: US shale wells involve limited capital commitments and create only shallow, short-lived lock-in (Wachtmeister and Höök, 2020), while deepwater offshore projects, expanding rapidly off the coasts of Brazil, Guyana, and in the Gulf of Mexico, require enormous upfront investments and produce over decades, creating deep, long-lasting lock-in.

This accumulation of committed emissions creates a wedge between what policymakers believe they need to compensate for current inaction, and what is actually required. If the carbon tax is calibrated on the basis of the supply curve observed before the delay, ignoring the downward shift induced by new investments, it will be systematically too low. Chapter 1 formalises this gap as the carbon tax lock-in premium, and shows that ignoring it leads to a residual overshoot of 17 to 19 GtCO2eq, nearly 10% of the remaining carbon budget allocated to oil in 2026. Crucially, overcommitted emissions are highly concentrated in a small number of capital-intensive offshore projects. This concentration makes the case for targeted supply-side interventions particularly compelling: a large share of the long-term climate cost of policy delay could be alleviated by acting on a limited number of identifiable developments. This reinforces the political feasibility argument developed in the supply-side literature (Green and Denniss, 2018; Piggot, 2018): precisely targeted policies are easier to mobilise public support around than diffuse consumption-based taxes, as illustrated by campaigns against specific projects such as Keystone XL (Erickson and Lazarus, 2014). These findings further ground the general supply-side policy literature in detailed asset-level evidence. While the international community works toward a comprehensive climate framework, the combination of deposit-level data and an understanding of investment dynamics can already inform the design of targeted, effective, and politically feasible measures to mitigate the growing cost of inaction.

Introduction générale

L'âge de pierre ne s'est pas terminé par manque de pierres,
et l'âge du pétrole ne se terminera pas par manque de pétrole.
Sheikh Ahmed Zaki Yamani, ancien ministre saoudien du Pétrole

Le pétrole est une ressource inépuisable
qui va se faire de plus en plus rare.
Dominique de Villepin, ancien Premier ministre français

Le pétrole demeure la première source d'énergie primaire au monde, représentant environ 30% de l'approvisionnement énergétique mondial. La production quotidienne dépasse 100 millions de barils, alimentant un système industriel qui couvre les transports, la pétrochimie, le chauffage et la production d'électricité. Malgré la croissance des investissements dans les énergies renouvelables et l'accélération de la transition énergétique, la demande mondiale de pétrole n'a cessé d'augmenter, portée en particulier par la croissance des économies émergentes. La place centrale du pétrole dans l'économie mondiale implique que toute politique visant à réduire sa production ou sa consommation emporte des conséquences bien au-delà du secteur énergétique, affectant les balances commerciales, les recettes budgétaires et les équilibres géopolitiques.

Néanmoins, le pétrole n'est pas une matière première homogène extraite d'un gisement représentatif. La production mondiale provient de dizaines de milliers d'actifs individuels (champs ou portions de champs), répartis dans plus de 80 pays et exploités par une grande diversité d'entreprises, des compagnies pétrolières nationales aux petits producteurs indépendants. La figure 5 présente la distribution géographique des actifs pétroliers recensés dans la base de données Rystad UCube, qui couvre plus de 100 000 actifs pétroliers et gaziers dans le monde, dont environ 18 000 produisent activement du pétrole en 2025. Ces actifs diffèrent profondément par leurs caractéristiques géologiques, leurs technologies d'extraction, leurs coûts de production, leur intensité carbone, leurs structures d'actionnariat, et les régimes fiscaux auxquels ils sont soumis. Cette granularité révèle un degré d'hétérogénéité masqué par les statistiques agrégées, qui s'avère central pour comprendre l'interaction entre le secteur pétrolier et les politiques climatiques.

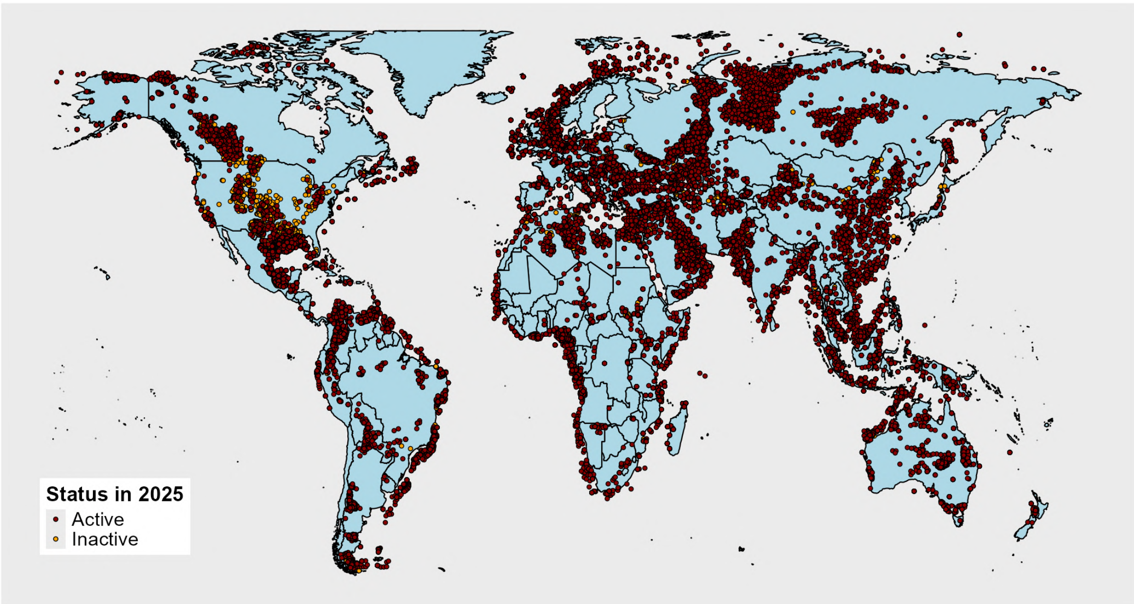

Figure 5 : Actifs pétroliers dans le monde (2025).
Source : base de données Rystad Energy UCube.

L'importance de cette interaction est bien établie. L'extraction, le raffinage et la combustion du pétrole représentent ensemble une part majeure des émissions mondiales de gaz à effet de serre. L'Accord de Paris, signé en 2015, a fixé l'objectif de contenir la hausse des températures mondiales "bien en deçà de 2°C, et de poursuivre les efforts pour la limiter à 1,5°C". Atteindre ces objectifs implique de respecter un budget carbone fini, c'est-à-dire un plafond cumulatif sur la quantité de CO2 pouvant encore être émise. La littérature scientifique a traduit cette contrainte en implications directes pour le secteur pétrolier : McGlade and Ekins (2015) estiment qu'un tiers des réserves de pétrole doit rester sous terre pour respecter un budget compatible avec l'objectif de 2°C, tandis que Welsby et al. (2021) montrent que la contrainte est plus sévère encore sous l'objectif de 1,5°C, avec plus de la moitié des réserves prouvées devant demeurer inexploitées. Extraire, raffiner et brûler l'ensemble des réserves prouvées de pétrole générerait environ 670 GtCO2eq, soit environ 1,5 fois le budget carbone résiduel total pour l'ensemble des sources sous l'objectif de 1,5°C. Les réserves de pétrole sont, en d'autres termes, bien trop abondantes au regard de ce que le climat peut tolérer. Cette contrainte implique d'identifier quelles réserves doivent rester sous terre, mais ses conséquences vont bien au-delà. Parce que les gisements pétroliers sont si hétérogènes, la conception et le timing des politiques climatiques reconfigurent l'ensemble du paysage sectoriel : ils modifient la dynamique d'investissement dans de nouvelles infrastructures de production, redistribuent les revenus entre entreprises et gouvernements, modifient la valeur relative des gisements, et interagissent avec les décisions de gouvernance des entreprises qui détiennent et exploitent chaque actif. Comprendre ces interactions, et le rôle que l'hétérogénéité au niveau des gisements joue dans leur façonnement, est l'objet de cette thèse.

Politiques climatiques pour le secteur pétrolier

L'argument en faveur d'une tarification du carbone

La littérature économique reconnaît largement la taxation carbone comme l'outil le plus efficace afin de réduire l'usage des combustibles fossiles (Nordhaus, 2007; Stern, 2007; Metcalf, 2019). Une taxe carbone mondiale, appliquée uniformément à l'ensemble des producteurs, permettrait d'internaliser l'externalité environnementale associée à la production et à la combustion du pétrole. Elle permettrait d'ordonner l'extraction selon le coût social de chaque baril, défini comme la somme du coût privé d'extraction et du dommage environnemental lié aux émissions associées. Sous un budget carbone contraignant, la taxe carbone dynamiquement efficiente croît au taux d'actualisation social (Lemoine and Rudik, 2017; Dietz and Venmans, 2019), reflétant la raréfaction croissante du budget carbone restant. En principe, une telle taxe allouerait l'effort de réduction des émissions là où son coût est le plus faible, minimisant ainsi le coût total de l'action climatique et de la réussite quant aux objectifs fixés. L'idée d'un prix mondial du carbone appliqué aux combustibles fossiles n'est pas purement théorique : l'Organisation maritime internationale s'est accordée en principe en 2025 sur un mécanisme de tarification carbone pour le transport maritime international, démontrant que des dispositifs coordonnés couvrant des échanges internationaux à grande échelle sont envisageables.

Politiques existantes et subventions implicites

En 2025, l'écart entre cet idéal théorique et les politiques climatiques effectivement en place reste considérable. Certaines juridictions ont mis en œuvre des mécanismes de tarification carbone, comme le montre la figure 6. Toutefois, peu de ces mécanismes couvrent

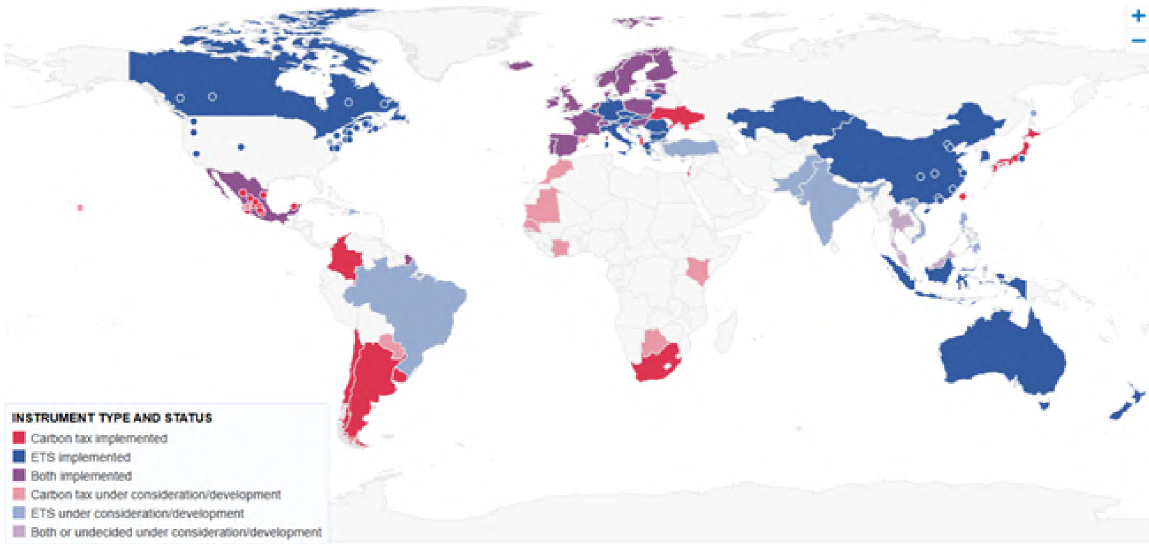

Figure 6 : Mécanismes de tarification carbone en vigueur ou à l'étude (2025).
Source : World Bank Carbon Pricing Dashboard (2025).

les émissions liées à la production pétrolière, et lorsque c'est le cas, les niveaux de taxation existants sont insuffisants pour s'aligner sur les objectifs de l'Accord de Paris (Metcalf, 2019). Parmi les dix plus grands pays producteurs de pétrole, seuls les États-Unis, la Chine et le Canada ont introduit un mécanisme de tarification carbone, et aucun d'entre eux ne couvre de manière exhaustive les émissions d'extraction en amont [3]. L'essentiel de la réglementation climatique cible la combustion en aval, à travers des taxes sur les carburants, des normes d'émission des véhicules ou des subventions aux énergies alternatives, laissant intégralement de côté les émissions d'extraction et de raffinage, ainsi que leur hétérogénéité. Parallèlement, les combustibles fossiles continuent de bénéficier d'importantes subventions implicites à l'échelle mondiale. Comme le documentent Parry et al. (2021), les subventions mondiales aux combustibles fossiles, incluant les transferts explicites et la sous-tarification des externalités environnementales, se sont élevées à environ 7 000 milliards de dollars en 2022, soit 7,1% du PIB mondial. Les subventions implicites spécifiquement destinées aux producteurs de pétrole et de gaz, bien que d'une ampleur moindre, peuvent avoir des effets substantiels sur la production : Erickson et al. (2017) montrent que dans un environnement de prix bas, près de la moitié des projets pétroliers et gaziers américains en dépendent. Ces subventions agissent à l'inverse des incitations qu'une taxe carbone créerait, et bénéficient de manière disproportionnée aux formes de production les plus intensives en carbone.

Obstacles politiques

[3] Voir table B.1 en annexe.

Le manque d'ambition persistant des politiques climatiques dans le secteur pétrolier reflète des contraintes d'économie politique profondes, enracinées dans la dépendance économique des grands pays producteurs vis-à-vis des revenus pétroliers. L'opposition à des politiques climatiques ambitieuses sur les combustibles fossiles est portée par deux grandes coalitions. La première est constituée des pays de l'OPEP+, pour lesquels les revenus pétroliers représentent une part considérable, voire indispensable, des budgets publics et du PIB. Le Fonds monétaire international rapporte que le prix du pétrole d'équilibre budgétaire (*fiscal breakeven price*), c'est-à-dire le prix auquel les pays exportateurs équilibrent leur budget, dépasse 80$/bbl pour l'Arabie saoudite et est encore plus élevé pour plusieurs autres membres de l'OPEP (International Monetary Fund, 2024). Pour ces pays, une taxe carbone mondiale qui réduirait significativement la demande ou les prix du pétrole menacerait directement la stabilité budgétaire. Les pays de l'OPEP se sont ainsi montrés les principaux opposants à tout moratoire international sur l'extraction de combustibles fossiles et le développement de nouveaux gisements (Lujala et al., 2022). La seconde source de résistance provient des États-Unis, devenus au cours de la dernière décennie le premier producteur mondial de pétrole grâce à l'expansion rapide du pétrole de schiste. Au niveau international, comme le note Van Asselt (2021), la Convention-cadre des Nations unies sur les changements climatiques privilégie une comptabilité territoriale des émissions de gaz à effet de serre, centrée sur la consommation plutôt que sur la production de combustibles fossiles, incitant ainsi les pays à concentrer leurs efforts sur des mesures de demande pour respecter leurs engagements internationaux, tout en laissant la production largement libre de contraintes.

La littérature sur les actifs échoués

Ce contexte d'économie politique a mené à l'émergence d'une vaste littérature étudiant les conséquences de politiques climatiques insuffisamment ambitieuses, avec une attention particulière portée sur les réserves de combustibles fossiles qui deviendraient des "actifs échoués" (*stranded assets*) sous différents scénarios de budget carbone. McGlade and Ekins (2015) proposent une décomposition géographique des réserves non exploitables sous un objectif de 2°C, tandis que Welsby et al. (2021) actualisent et resserrent ces estimations pour l'objectif de 1,5°C. D'autres contributions ont étudié les implications financières du risque que représentent les potentiels actifs échoués pour les investisseurs et les entreprises (Mercure et al., 2018; Van der Ploeg and Rezai, 2020). Cependant, cette littérature repose en majorité d'une part sur des modèles théoriques avec des gisements représentatifs, et d'autre part sur des courbes d'offre agrégées construites au niveau du pays ou du bassin (Heal and Schlenker, 2019). Si ces approches sont utiles pour établir des ordres de grandeur, elles masquent inévitablement l'importante hétérogénéité au niveau des gisements qui, comme cette thèse le démontre, est centrale pour comprendre les conséquences réelles des politiques climatiques. En pratique, la politique climatique n'opère pas sur des courbes d'offre agrégées : elle se matérialise à travers les décisions spécifiques d'investissement, de production et de gouvernance prises pour chaque actif individuel.

L'argument en faveur des politiques d'offre

La difficulté persistante à mettre en œuvre des mesures globales agissant sur la demande a conduit un nombre croissant de travaux à plaider en faveur de politiques climatiques agissant sur l'offre, qui contraignent directement la production de combustibles fossiles plutôt que de pénaliser uniquement leur consommation. Erickson et al. (2018) soutiennent que l'action sur l'offre de combustibles fossiles doit être vue comme complémentaire aux efforts existants sur la demande plutôt que comme un substitut. Comme l'expliquent Green and Denniss (2018), plusieurs arguments d'économie politique justifient le recours à de telles mesures : (i) les politiques d'offre ont des effets plus directs et prévisibles sur les émissions, puisque l'interdiction d'un projet spécifique garantit avec certitude une quantité d'émissions évitée, (ii) elles impliquent des coûts administratifs plus faibles, car réguler un petit nombre de producteurs est plus simple que de surveiller des millions de consommateurs, et (iii) elles tendent à bénéficier d'un plus large soutien populaire, parce que la distribution de la charge financière est perçue comme plus équitable et les gains environnementaux plus tangibles que ceux des taxes à la consommation. L'efficacité des mobilisations citoyennes autour de projets fossiles spécifiques a été documenté par Piggot (2018) et Carter and McKenzie (2020), qui montrent que la plupart des interdictions de production adoptées à ce jour sont issues de campagnes citoyennes ciblant des projets identifiables. Plus récemment, Green et al. (2024) ont appelé à l'instauration d'une norme internationale d'"interdiction de tout nouveau projet fossile", arguant qu'une telle norme est à la fois réalisable et efficace. Or, comme l'observent Lazarus and van Asselt (2018), la littérature académique sur les politiques d'offre reste sous-développée relativement à celle portant sur les politiques de demande. Cette thèse contribue à combler ce manque, en quantifiant le *carbon lock-in* (verrouillage carbone) au niveau des actifs pour identifier les mesures d'offre les plus efficaces (Chapitre 1), en documentant les conséquences distributives de la tarification carbone (Chapitre 2), et en examinant l'efficacité d'un cadre volontaire, l'initiative *Zero Routine Flaring* de la Banque mondiale, en tant que forme plus souple de politique d'offre (Chapitre 3).

L'hétérogénéité des gisements pétroliers

De nouvelles sources de données

L'un des défis majeurs dans l'étude de l'interaction entre politiques climatiques et secteur pétrolier au niveau des gisements réside dans la rareté de données globales et cohérentes. Cette thèse s'appuie sur une combinaison de bases de données propriétaires, d'observations satellites et de modèles d'ingénierie pour surmonter cette difficulté. La source de données principale, commune aux trois chapitres, est la base UCube de Rystad Energy, l'un des ensembles de données les plus complets sur la production pétrolière et gazière mondiale. Elle couvre plus de 100 000 actifs pétroliers et gaziers, dont plus de 20 000 produisent activement du pétrole en 2025. Pour chaque actif, la base fournit une ventilation détaillée de la production annuelle par type de pétrole, des coûts par catégorie (coûts d'exploitation, dépenses d'investissement, exploration, prélèvements publics), des réserves exploitables, des calendriers de développement, ainsi que des caractéristiques administratives, géographiques et géologiques. Point essentiel pour le Chapitre 3, la base recense également la structure actionnariale et l'identité de l'opérateur de chaque actif, ainsi que leur évolution dans le temps.

La thèse exploite également des observations satellitaires du torchage (*flaring*) de gaz provenant de l'instrument VIIRS (*Visible Infrared Imaging Radiometer Suite*), collectées par l'*Earth Observation Group* de la *Colorado School of Mines* en collaboration avec la NOAA (Elvidge et al., 2015). Les torches détectées de nuit par l'instrument sont converties en estimations de volumes torchés au moyen de l'algorithme VIIRS Nightfire (Zhizhin et al., 2021). Étant donné que près de 95% des opérateurs ne déclarent pas publiquement les volumes torchés, les observations satellitaires constituent la source d'information la plus fiable sur cette pratique aux conséquences environnementales majeures. Je développe dans cette thèse une procédure de géolocalisation en plusieurs étapes qui attribue plus de 134 000 observations satellitaires enregistrées entre 2012 et 2024 à des actifs individuels de la base Rystad, en utilisant les coordonnées des puits et les surfaces géographiques des actifs. Une fois appariées, ces données sont utilisées dans le chapitre 1 pour estimer les ratios de gaz torché par baril de pétrole produit (*flaring-to-oil ratio*) pour chaque actif, un intrant clé dans l'estimation de leur intensité carbone, et dans le Chapitre 3 comme mesure principale des performances environnementales.

Enfin, la thèse développe des estimations originales de l'intensité carbone sur l'ensemble du cycle de vie de chaque actif, en combinant trois modèles d'ingénierie issus de l'*Oil-Climate Index* : le modèle OPGEE (*Oil Production Greenhouse Gas Emissions Estimator*) pour les émissions en amont, le modèle PRELIM (*Petroleum Refinery Lifecycle Inventory Model*) pour les émissions de raffinage, et le modèle OPEM (*Oil Products Emissions Module*) pour les émissions en aval. Des valeurs issues directment des modèles sont obtenues pour un échantillon de plusieurs centaines de champs pétroliers, puis des modèles de régression prédictive sont calibrés afin d'étendre ces estimations à l'ensemble de l'univers d'actifs Rystad. Cette combinaison d'estimations d'ingénierie, de données satellites de torchage et de modélisation prédictive fournit des estimations d'intensité carbone sur le cycle de vie complet pour l'ensemble des actifs en production et futurs dans le monde. Ces estimations sont utilisées dans les Chapitres 1 et 2, et les méthodes employées sont décrites en détail en annexe A.2.

Hétérogénéité des coûts d'extraction

Les coûts d'extraction varient considérablement d'un gisement à l'autre. Le graphique 7 présente les prix de *breakeven* moyens, pondérés par les réserves, pour les vingt plus grands pays producteurs. Les actifs conventionnels du Moyen-Orient, concentrés au Koweït, en Irak, au Qatar, en Arabie saoudite, en Iran et aux Émirats arabes unis, affichent des prix de *breakeven* moyens inférieurs à 12$/bbl. À l'autre extrémité de la distribution, les gisements canadiens et chinois présentent des prix de *breakeven* moyens supérieurs à 33$/bbl, soit environ trois fois plus. Se concentrer sur les seules moyennes par pays est toutefois trompeur. En effet, la dispersion au sein d'un même pays est considérable : au Canada, les 10% d'actifs les plus coûteux ont des prix de *breakeven* dépassant 55$/bbl, tandis que les 10% les moins chers se situent en dessous de 12$/bbl. Des schémas similaires s'observent dans l'ensemble des grands pays producteurs. Cette hétérogénéité intra-pays est ignorée par les courbes d'offre agrégées construites au niveau du pays ou du bassin, mais est pourtant centrale dans l'analyse menée dans cette thèse.

Au-delà de cette hétérogénéité entre gisements, la structure duale des coûts typiquement associée aux projets pétroliers joue un rôle important dans la manière dont les politiques climatiques affectent le secteur. Les projets pétroliers nécessitent deux types distincts de dépenses : d'une part les coûts de développement, qui correspondent aux in-

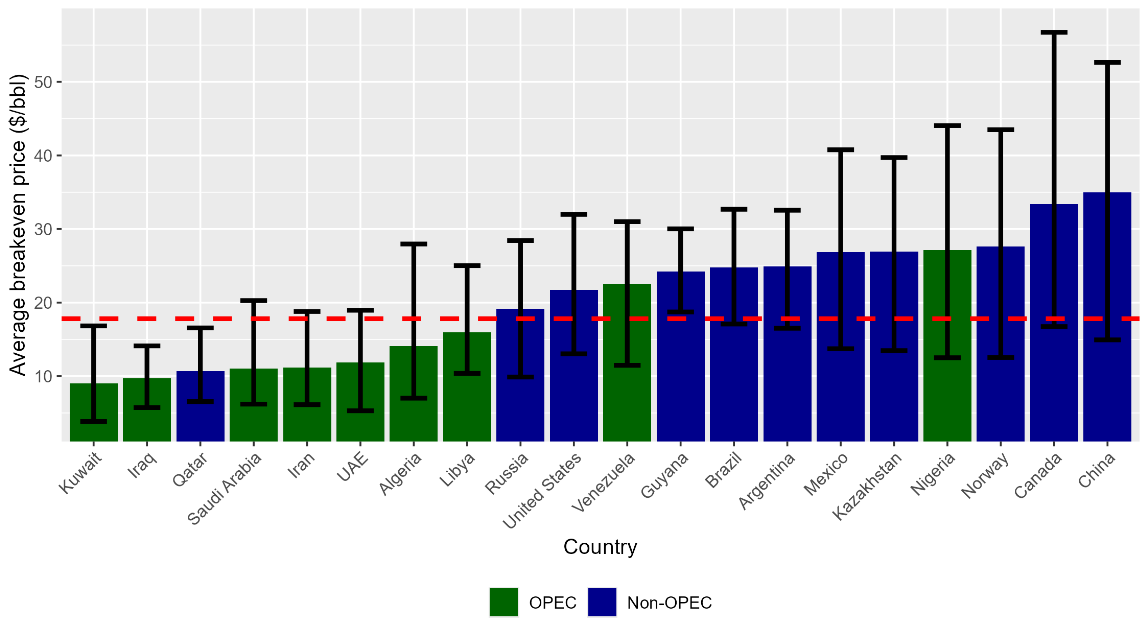

Figure 7 : Prix de *breakeven* par pays ($/bbl) : moyenne pondérée par les réserves en 2025, quantiles à 10% et 90%.

Notes : les dépenses fiscales sont exclues des calculs ; seules les dépenses d'investissement et d'exploitation sont prises en compte.
Source : calculs de l'auteur à partir de la base Rystad Energy UCube.

vestissements initiaux dans la construction des installations de production, le forage des premiers puits et la mise en place des infrastructures avant ou dans les premières années de production ; d'autre part les coûts de production, que sont les charges d'exploitation par baril une fois l'actif en activité. Le poids relatif de ces deux composantes varie considérablement selon les types de gisements. Les projets en eaux profondes, par exemple, nécessitent des investissements initiaux très importants (souvent plusieurs milliards de dollars) mais sont relativement peu coûteux à exploiter une fois les infrastructures en place. À l'inverse, les puits de pétrole de schiste impliquent des coûts de forage individuels comparativement modestes mais font face à des charges d'exploitation par baril élevées et leur production décline rapidement. Cette structure duale s'avère centrale pour comprendre les conséquences du retard dans la mise en œuvre des politiques climatiques, car l'écart entre le prix de *breakeven* global du projet et le prix de *breakeven* de production détermine dans quelle mesure un actif, une fois développé, devient résilient à la taxation carbone.

Hétérogénéité de l'intensité carbone

Tous les barils de pétrole n'ont pas le même coût environnemental. Les émissions de gaz à effet de serre associées au cycle de vie d'un baril, de l'extraction à la combustion finale, peuvent varier d'un facteur de deux ou plus d'un gisement à l'autre. La Figure 8 illustre cette variation pour les vingt plus grands pays producteurs. Alors que le Qatar, le Kazakhstan, les Émirats arabes unis et l'Arabie saoudite affichent des intensités carbone moyennes sur le cycle de vie de l'ordre de 400 à 440 kgCO2eq/bbl, le pétrole canadien et vénézuélien dépasse en moyenne 550 kgCO2eq/bbl. Comme pour les coûts d'extraction, la dispersion au sein d'un même pays peut être considérable, reflétant la diversité des types de pétrole, des méthodes d'extraction et des pratiques de torchage. Les membres de l'OPEP ne forment pas un groupe homogène : les producteurs du Moyen-Orient disposent généralement d'un pétrole plus propre que les pays producteurs de bruts lourds comme le Venezuela ou le Canada, ou les pays à fort torchage comme l'Algérie et le Nigeria.

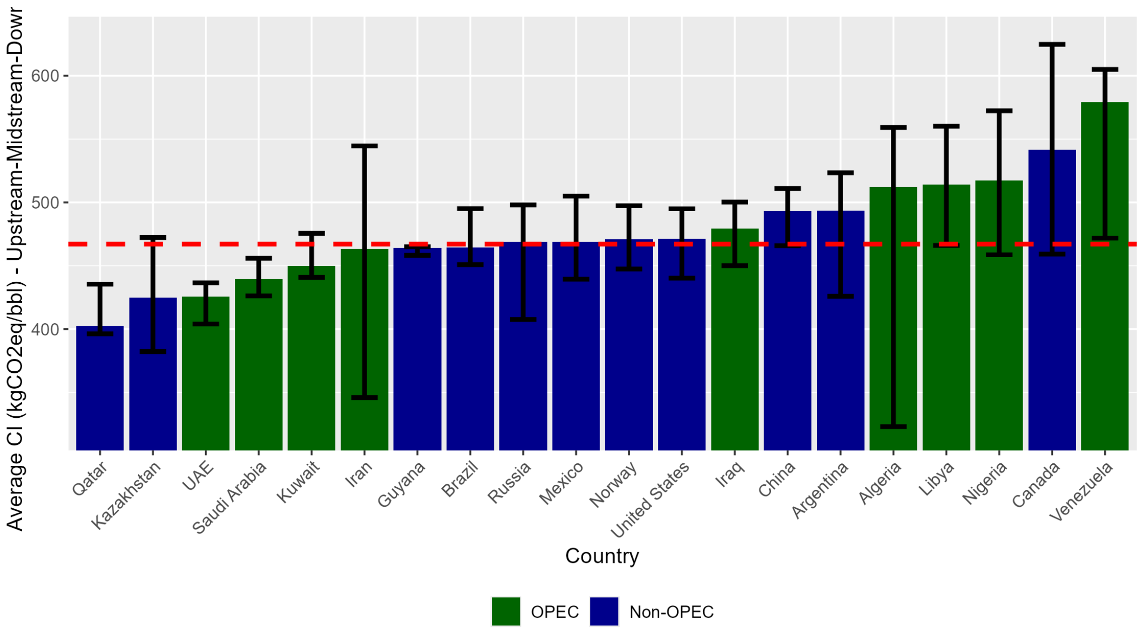

Figure 8 : Intensité carbone sur le cycle de vie par pays (kgCO2eq/bbl) : moyenne pondérée par les réserves en 2025, quantiles à 10% et 90%.

Note : les valeurs présentées sont des intensités carbone sur le cycle de vie complet, incluant les émissions en amont, au raffinage et en aval.
Source : calculs de l'auteur à partir de la base Rystad Energy UCube.

Les émissions sur le cycle de vie du pétrole se décomposent en trois étapes principales : (i) les émissions en amont, associées à l'extraction du brut, (ii) les émissions liées au raffinage, et (iii) les émissions en aval, qui correspondent au transport et à l'utilisation finale des produits raffinés par les consommateurs. Bien que les émissions en aval représentent environ 80% des émissions totales sur le cycle de vie, les trois étapes contribuent à l'hétérogénéité observée entre gisements, avec un rôle prépondérant joué par la première. Comprendre les déterminants de cette hétérogénéité à chaque étape est essentiel, car une taxation carbone appliquée à l'ensemble du cycle de vie affecterait les gisements très différemment selon leurs caractéristiques.

L'intensité carbone en amont est principalement déterminée par deux facteurs. Le premier est lié à la nature du pétrole lui-même et les méthodes d'extraction qu'il requiert. Les pétroles lourds et visqueux, tels que le bitume, nécessitent souvent l'injection de vapeur pour réduire leur viscosité et permettre l'extraction, un procédé particulièrement énergivore qui se traduit par des émissions en amont élevées. Les opérations d'exploitation des sables bitumineux en sont un exemple bien connu : lorsqu'une valorisation sur site en brut synthétique (*bitumen upgrading)* est nécessaire, les émissions en amont peuvent être plusieurs fois supérieures à celles du brut léger conventionnel (Brandt and Farrell, 2007). Le second facteur, et sans doute le plus déterminant, est le traitement du gaz associé. Lors de l'extraction du pétrole, du gaz naturel est généralement ramené à la surface en tant que co-produit. Ce gaz associé peut être vendu sur le marché si l'actif est connecté à une infrastructure gazière, réinjecté dans le réservoir pour maintenir la pression, ou utilisé pour alimenter les équipements sur place. Lorsqu'aucune de ces options de valorisation n'est économiquement viable ou techniquement réalisable, le gaz est soit torché (brûlé sur site), soit éventé (rejeté directement dans l'atmosphère). L'éventage, qui libère le méthane directement, présente l'impact environnemental le plus dommageable et est interdit ou sévèrement restreint dans la plupart des juridictions (World Bank, 2022). Le torchage reste le mode d'élimination par défaut en l'absence de valorisation du gaz, mais il est loin d'être anodin : il représentait plus de 150 milliards de mètres cubes dans le monde en 2024 (World Bank, 2025), soit près de 1% des émissions mondiales de gaz à effet de serre liées à l'énergie[4]. Les actifs à fort torchage peuvent ainsi présenter des intensités carbone en amont plusieurs fois supérieures à la moyenne mondiale. La quantité de gaz associé, la présence ou l'absence d'infrastructures gazières et les choix opérationnels de l'exploitant quant à l'utilisation de ce gaz apparaissent donc comme des déterminants critiques de l'empreinte environnementale d'un actif.

Les émissions liées au raffinage ainsi qu'au transport et à l'utilisation finale, bien que moins variables individuellement que les émissions en amont, contribuent également à l'hétérogénéité de l'intensité carbone des gisements. Les émissions de raffinage dépendent de la configuration de la raffinerie nécessaire pour traiter un type de brut donné : les bruts lourds et riches en soufre nécessitent un traitement plus intensif (par exemple la "conversion profond", ou *deep conversion*), tandis que les bruts légers peuvent être traités par des configurations plus simples (comme l'*hydroskimming*), générant des émissions de

[4]Ce volume annuel représente par ailleurs une source substantielle de gaz naturel non valorisé, et pourrait couvrir plus de trois fois la consommation de l'Afrique Subsaharienne.

raffinage nettement inférieures. Les émissions en aval sont principalement déterminées par la répartition en termes de produits pétroliers issus du raffinage, laquelle dépend à la fois des caractéristiques du brut et de la configuration de la raffinerie. Bien que les émissions en aval soient plus homogènes entre types de brut que les émissions en amont, elles sont loin d'être uniformes, et leurs variations renforcent l'hétérogénéité entre gisements des émissions totales sur le cycle de vie, documentée dans le graphique 8.

Un résultat empirique essentiel, documenté dans le Chapitre 2 à partir de données au niveau des gisements, est que l'intensité carbone sur le cycle de vie et le coût privé d'extraction ne sont que faiblement corrélés. La comparaison des graphiques 7 et 8 fournit un premier indice : le classement des pays par coût d'extraction ne reflète pas le classement par intensité carbone. La raison en est que le torchage et l'utilisation des sols, qui représentent une part importante des émissions en amont, ne sont pas positivement corrélés aux coûts privés de production (Coulomb et al., 2026). Cela a une implication profonde : l'ordre d'extraction efficient en termes de coûts, dans lequel les barils les moins chers sont produits en premier, et l'ordre efficient d'un point de vue climatique, dans lequel les barils les plus propres sont prioritaires, diffèrent substantiellement. Une politique qui ignorerait l'hétérogénéité de l'intensité carbone et s'appuierait uniquement sur les coûts d'extraction pour déterminer quels gisements développer générerait davantage d'émissions que nécessaire pour respecter un budget carbone donné.

Hétérogénéité institutionnelle

Au-delà de leurs caractéristiques physiques et environnementales, les gisements pétroliers diffèrent également par le cadre institutionnel dans lequel ils sont exploités. Deux dimensions de cette hétérogénéité institutionnelle sont particulièrement pertinentes pour cette thèse : les régimes fiscaux et la structure actionnariale.

Chaque gisement opère en étant soumis à un régime fiscal spécifique qui régit la répartition des revenus entre les entreprises privées et le gouvernement du pays hôte. Ces régimes prennent des formes diverses : redevances assises sur les revenus bruts (*royalties*), contrats de partage de production dans lesquels le gouvernement reçoit une part du pétrole produit, impôts sur les bénéfices nets, et diverses taxes liées à la production comptabilisées comme charges d'exploitation. En pratique, les régimes fiscaux sont souvent adaptés à un projet ou un bassin d'extraction spécifique, avec des taux et des structures parfois négociés au cas par cas entre gouvernements et compagnies pétrolières. Cette spécificité au niveau du projet signifie que la charge fiscale effective peut varier considérablement, y compris entre gisements situés dans un même pays. Les régimes fiscaux façonnent les incitations tant des producteurs que des gouvernements : ils affectent la rentabilité des décisions d'investissement, la propension des gouvernements à renégocier les conditions en réponse à l'évolution des marchés, et la répartition des revenus sous taxation carbone. Le Chapitre 1 fait abstraction des prélèvements publics, et fait l'hypothèse qu'en présence d'une taxe carbone mondiale, les gouvernements producteurs renégocieraient les conditions fiscales pour maintenir les actifs viables en activité et continuer à capter une part de la rente. Le Chapitre 2 adopte une approche différente en intégrant des données fiscales au niveau des gisements, couvrant redevances, impôts et parts gouvernementales, afin de décomposer les variations de la valeur du pétrole en profits après impôts des entreprises et revenus des États.

La seconde dimension de l'hétérogénéité institutionnelle est la structure actionnariale et la gouvernance des actifs pétroliers. La production pétrolière implique généralement d'importants besoins en capitaux, un risque géologique et une expertise technique, conduisant fréquemment au recours à des structures de coentreprise (*joint ventures*) dans lesquelles plusieurs entreprises partagent la propriété d'un même actif. L'un des partenaires est désigné comme opérateur, responsable de la gestion opérationnelle du champ et des décisions quotidiennes. Ces dernières sont relatives au forage, aux niveaux de production et aux pratiques environnementales telles que le torchage. Les autres partenaires sont des détenteurs de parts non opérateurs qui partagent les revenus et les coûts de production au prorata de leur participation, mais ne contrôlent pas directement les opérations. Les investissements majeurs, tels que les infrastructures de captage du gaz ou les équipements de réduction du torchage, nécessitent généralement l'approbation de comités d'exploitation conjoints dans lesquels les détenteurs non opérateurs disposent d'une représentation et de droits de vote proportionnels à leur part.

La composition actionnariale des actifs varie considérablement et joue un rôle majeur dans la réponse des opérateurs aux politiques climatiques. Les compagnies pétrolières cotées en bourse (et a fortiori les "*majors*") sont soumises à de fortes pressions en matière de responsabilité environnementale, sociale et de gouvernance (ESG) ainsi qu'à des obligations de transparence imposées par les investisseurs et les régulateurs. À l'inverse, les compagnies pétrolières nationales, qui produisent environ la moitié du pétrole mondial, font face à un contrôle bien moindre de la part des investisseurs (Cahill and Swanson, 2023). Les entreprises privées sans cotation boursière sont soumises à des obligations de transparence minimales et opèrent largement en dehors de la supervision des régulateurs financiers et environnementaux (Matikainen, 2022). Lorsqu'un actif est détenu conjointement par des entreprises aux priorités environnementales hétérogènes, l'efficacité de l'engagement de chaque entreprise dépend de la dynamique de gouvernance au sein du conseil de l'actif. Que l'entreprise engagée soit ou non l'opérateur, la part de capital qu'elle détient et la posture environnementale des autres partenaires déterminent dans quelle mesure les engagements volontaires se traduisent en changements opérationnels concrets. Cette hétérogénéité de gouvernance, et ses conséquences sur l'efficacité des initiatives environnementales, est l'objet central du Chapitre 3.

Le processus d'extraction pétrolière : dynamiques d'investissement et rigidités de production

Les dimensions d'hétérogénéité décrites ci-dessus n'existent pas indépendamment les unes des autres : elles sont évaluées conjointement par les investisseurs lorsqu'ils décident d'investir ou non dans le développement d'un nouveau gisement, et comment le faire. Ce processus d'investissement et de production est ce qui, in fine, traduit les caractéristiques des gisements en production effective, en revenus et en résultats environnementaux.

Lorsqu'un investisseur envisage de développer un nouveau champ, il évalue la valeur actualisée nette attendue du projet, qui dépend des prix futurs anticipés du pétrole, de la structure de coûts du gisement (coûts de développement et de production), du régime fiscal applicable, de l'intensité carbone du gisement et de son exposition potentielle à des politiques climatiques, ainsi que de la structure du partenariat dans lequel le projet sera opéré. Sur la base de cette évaluation, l'investisseur détermine le niveau optimal d'infrastructure à construire, ce qui fixe en retour la capacité de production et les méthodes d'extraction à employer (Venables, 2014; Okullo and Reynès, 2016). Cette décision initiale d'investissement est en grande partie irréversible. Une fois les coûts de développement engagés, l'actif entre en production et continuera généralement à fonctionner tant que les revenus excèdent les coûts de production, indépendamment du fait que le projet dans son ensemble reste ou non économiquement justifié au regard d'éventuels changements de politique.

Au niveau du puits, la rigidité de la production est bien documentée : les contraintes géophysiques et technologiques dictent largement la trajectoire de production au point qu'aucune adaptation significative aux variations du prix du pétrole ne peut être observée au niveau des puits (Anderson et al., 2018). Au niveau du gisement, cependant, certains ajustements restent possibles. Les opérateurs peuvent par exemple choisir de forer des puits de production supplémentaires, de forer des puits d'injection pour augmenter la pression du réservoir et donc la production, ou d'investir dans la récupération assistée (*enhanced oil recovery*) pour prolonger la durée de vie de l'actif. Une exception notable au schéma général de rigidité de la production est fournie par les pays "moteurs" de l'OPEP (Arabie saoudite, Koweït et Émirats arabes unis), qui maintiennent délibérément une capacité de production excédentaire afin de pouvoir ajuster rapidement leur production en réponse aux chocs (Fattouh, 2007; Kellogg, 2024).

Malgré la relative rigidité de la production une fois la décision initiale de développement prise, des choix opérationnels importants subsistent tout au long de la vie productive de l'actif. Outre les décisions relatives au forage de nouveaux puits ou à l'investissement dans la récupération assistée, un choix critique pour les opérateurs porte sur le traitement du gaz associé : investir dans des infrastructures de collecte et de traitement du gaz, connecter l'actif à un marché du gaz, ou continuer à torcher. Ces décisions sont façonnées non seulement par des considérations économiques, mais aussi par la gouvernance de l'actif, et en particulier par le fait que les entreprises siégeant au conseil d'administration aient ou non pris des engagements environnementaux. Dans quelle mesure des détenteurs non opérateurs engagés peuvent-ils influencer ces décisions opérationnelles, et que se produit-il lorsqu'ils se retirent ? C'est la question à laquelle répond le Chapitre 3.

Comment l'hétérogénéité interagit avec les politiques climatiques

La multiplicité des dimensions dans lesquelles les gisements pétroliers diffèrent implique que l'effet agrégé de toute politique climatique sur le secteur pétrolier résulte de milliers de réponses individuelles, chacune dépendant d'une combinaison unique de coûts d'extraction, d'intensité carbone, de régime fiscal, de structure actionnariale, de contraintes géologiques et d'historique d'investissement. Les courbes d'offre agrégées, qu'elles soient construites au niveau du pays ou du bassin, ignorent inévitablement cette complexité et peuvent ainsi produire des prédictions trompeuses sur les effets des politiques climatiques sur la production, les revenus et les émissions. Ces dimensions n'enrichissent pas seulement la précision avec laquelle le secteur pétrolier est décrit : elles interagissent avec les politiques climatiques de manière parfois contre-intuitive, et leur prise en compte est indispensable pour comprendre comment le secteur pétrolier répond effectivement aux politiques climatiques.

Hétérogénéité institutionnelle et efficacité des politiques

Les obstacles politiques décrits précédemment impliquent qu'à court terme, la politique climatique dans le secteur pétrolier semble devoir se reposer principalement sur des dispositifs partiels plutôt que sur une taxation carbone globale. Les initiatives volontaires d'entreprises, la pression des investisseurs liée aux critères ESG, et les contraintes réglementaires imposées par certaines juridictions aux entreprises dont le siège se trouve sur leur territoire constituent les principaux leviers par lesquels des améliorations environnementales sont espérées. L'efficacité de ces leviers dépend toutefois fondamentalement des caractéristiques institutionnelles des actifs pétroliers, ce pour plusieurs raisons.

Premièrement, ces dispositifs n'atteignent qu'une fraction de l'industrie. Les initiatives volontaires telles que l'initiative *Zero Routine Flaring* (ZRF) de la Banque mondiale attirent principalement de grandes entreprises cotées en bourse, actives à l'international, soumises à des incitations réputationnelles et à la surveillance des investisseurs. De même, les contraintes réglementaires imposées par les juridictions européennes ou nord-américaines s'appliquent aux entreprises dont le siège est dans ces régions, mais ne s'étendent ni aux compagnies pétrolières nationales ni aux entreprises privées soumises à des obligations de déclaration moins contraignantes. Si les actifs les plus polluants sont détenus par des entreprises hors de portée de ces dispositifs, leur efficacité globale est limitée, quel que soit le degré d'engagement des entreprises ciblées. Deuxièmement, même au sein des actifs concernés, l'hétérogénéité institutionnelle détermine si les engagements se traduisent en changements opérationnels concrets. Lorsqu'une entreprise engagée est également l'opérateur d'un actif, elle peut mettre en œuvre des améliorations directement. Lorsqu'elle n'en est qu'un partenaire non opérateur, elle doit recourir aux mécanismes de gouvernance pour influer sur un opérateur qui ne partage pas nécessairement ses priorités. Un même engagement d'entreprise peut ainsi avoir des conséquences environnementales très différentes selon le contexte de gouvernance dans lequel il s'exerce.

Ce problème de gouvernance donne lieu à un dilemme entre prise de parole et défection (*voice versus exit*), faisant écho au cadre initialement développé par Hirschman (1972). Une entreprise engagée dans l'initiative ZRF qui éprouve des difficultés à influencer les décisions opérationnelles peut soit rester dans le *board* et continuer à exercer une pression (*voice*), soit céder sa participation et réallouer ses capitaux vers des actifs sur lesquels il dispose d'un plus grand contrôle (*exit*). La stratégie d'*exit* peut améliorer le profil environnemental du portefeuille de l'entreprise, mais elle risque d'aggraver les émissions effectives de l'actif si l'acquéreur n'a aucun engagement environnemental. La résolution de ce dilemme dépend de caractéristiques institutionnelles : le rôle de l'entreprise engagée en tant qu'opérateur ou non, la part de capital détenue par des entreprises engagées, et des facteurs externes tels que la disponibilité d'infrastructures de valorisation du gaz. Le Chapitre 3 étudie ces dynamiques en utilisant l'initiative ZRF comme expérience naturelle. Il montre que les entreprises engagées réduisent effectivement le torchage au fil du temps, mais que ces gains sont brutalement inversés lorsque le dernier détenteur engagé se désengage, y compris lorsque l'entreprise sortante n'était pas l'opérateur. Cet effet du "dernier bon actionnaire" (*last good owner*) révèle une faille importante des cadres environnementaux volontaires : en l'absence de dispositions encadrant la cession à des parties non engagées, ces initiatives risquent de simplement redistribuer les émissions entre structures actionnariales plutôt que de véritablement les réduire.

Hétérogénéité et conséquences redistributives

Les limites de ces dispositifs partiels renforcent l'argument en faveur d'un mécanisme global de tarification du carbone. Comme exposé précédemment, le principal obstacle politique à un tel mécanisme est la résistance des pays producteurs, qui craignent de perdre une source majeure de recettes budgétaires. Or, l'hétérogénéité au niveau des gisements documentée dans cette thèse suggère que cette opposition pourrait être moins monolithique qu'on ne le suppose généralement. Comme l'illustrent les graphiques 7 et 8, les principaux pays de l'OPEP, notamment l'Arabie saoudite, le Koweït et les Émirats arabes unis, produisent un pétrole à la fois parmi les moins chers à extraire et les moins intensifs en carbone au monde. Soumis à une taxe carbone mondiale, ces pays devraient donc être relativement moins affectés que les producteurs à coûts élevés et à forte intensité carbone tels que le Canada, le Venezuela ou les exploitants américains de pétrole de schiste. Le Chapitre 2 montre que pour les producteurs à bas coûts, les recettes budgétaires suivent un profil en U inversé à mesure que la taxe carbone augmente, analogue à la courbe de Laffer : des niveaux modérés de taxation carbone peuvent entièrement compenser la baisse des redevances et des impôts par les recettes de la taxe carbone, sous la condition que les pays producteurs puissent préempter la taxation des émissions de combustion en aval. Cette résilience est encore renforcée par la flexibilité productive des pays de certains pays de l'OPEP : parce qu'ils maintiennent d'importantes capacités de réserve, ils conservent la possibilité d'adapter leur production au nouvel environnement réglementaire plutôt que d'être enfermés dans des trajectoires de production prédéterminées. Une taxe carbone mondiale pourrait ainsi bénéficier aux membres les plus influents de l'OPEP tout en imposant des coûts substantiels à leurs concurrents à coûts élevés, un constat qui pourrait en principe servir de levier politique dans les négociations climatiques internationales.

Les conséquences distributives de la taxation carbone sont cependant bien plus complexes que ne le suggère la simple observation selon laquelle les producteurs à bas coûts et faible intensité carbone en bénéficieraient. Le Chapitre 2 démontre, tant théoriquement qu'empiriquement, que la relation entre l'intensité carbone d'un gisement et la variation de sa valeur en cas de taxation carbone n'a rien de triviale. À l'aide d'un modèle de simulation mondiale calibré sur plus de 21 000 gisements, le chapitre documente la faible corrélation entre variations de la valeur sociale du pétrole consécutives à une hausse du coût social du carbone et intensité carbone. De manière frappante, la valeur sociale de certains gisements relativement intensifs en carbone peut même augmenter lorsque la politique climatique se durcit, tandis que celle de certains gisements plus propres diminue. Ces résultats contre-intuitifs s'expliquent par le fait que les variations de valeur des gisements reflètent l'interaction de multiples forces, incluant des modifications d'avantages concurrentiels entre gisements, des changements dans le timing d'extraction, et la réallocation de la production entre actifs hétérogènes, dont aucune ne peut être réduite à la seule intensité carbone. Ce schéma s'observe aussi aux niveaux de l'entreprise et du pays : si les profits agrégés des entreprises diminuent lorsque les taxes carbone augmentent, l'ampleur de la baisse varie considérablement d'une entreprise à l'autre et n'est que faiblement corrélée à l'intensité carbone moyenne de leur portefeuille d'actifs. Ces résultats révèlent que les conséquences distributives de la taxation carbone entre gisements pétroliers sont bien plus complexes que ne le suggérerait l'intuition simple selon laquelle "plus le pétrole est sale, plus il perd", et que saisir le tableau complet nécessite une approche au niveau des actifs.

Carbon lock-in et politiques d'offre ciblées

La perspective d'une taxe carbone mondiale, aussi avantageuse soit-elle pour les producteurs à bas coûts, demeure lointaine. Le coût de cette attente est souvent sous-estimé, car il ne se limite pas aux émissions excédentaires qui surviennent pendant la période d'inaction. Chaque année de retard donne également lieu à de nouveaux investissements irréversibles dans des infrastructures de production pétrolière, lesquels engagent à leur tour des émissions futures sur la durée de vie des actifs qu'ils créent. Ce phénomène est connu sous le nom d'*infrastructure lock-in* (Unruh, 2000), l'une des trois dimensions du *carbon lock-in* (verrouillage carbone) identifiées par Seto et al. (2016). Le mécanisme repose sur la structure duale des coûts décrite précédemment. Parce que les coûts de développement sont irrécupérables et que les coûts de production sont faibles par rapport aux revenus, un actif nouvellement construit continuera de produire et d'émettre pendant des décennies, même si une taxe carbone ultérieure rend le projet socialement indésirable. Arrêter un actif en production nécessite que la taxe excède le prix de *breakeven* de production, bien inférieur au *breakeven* du projet complet. Plus l'écart entre ces deux seuils est important, plus le verrouillage est profond. Les estimations des émissions engagées par les infrastructures existantes se sont aggravées au fil du temps (Tong et al., 2019; Trout et al., 2022), mais la plupart des modèles d'évaluation intégrée ne capturent que partiellement ces dynamiques (Grubb et al., 2021). Le Chapitre 1 comble cette lacune en proposant la première quantification précise, au niveau des actifs, de l'enfermement infrastructurel dans le secteur pétrolier.

Une caractéristique essentielle du *lock-in* est qu'il s'accumule avec chaque année de retard dans la mise en œuvre des politiques climatiques, à mesure que de nouvelles infrastructures sont construites. L'émergence du pétrole non conventionnel au cours des quinze dernières années a renforcé à la fois l'hétérogénéité des dynamiques de verrouillage et leur importance : les puits de pétrole de schiste américains impliquent des engagements en capital limités et ne créent qu'un verrouillage superficiel et de courte durée (Wachtmeister and Höök, 2020), tandis que les projets en eaux profondes, en expansion rapide au large des côtes du Brésil, du Guyana et dans le Golfe du Mexique, nécessitent d'énormes investissements initiaux et produisent sur des décennies, créant un verrouillage profond et durable.

Cette accumulation d'émissions engagées crée un écart entre ce que les décideurs estiment nécessaire pour compenser l'inaction présente, et ce qui est réellement requis. Si la taxe carbone est calibrée sur la base de la courbe d'offre observée avant le retard, sans tenir compte de la translation vers le bas induit par les nouveaux investissements, elle sera systématiquement trop faible. Le Chapitre 1 formalise cet écart sous le nom de prime de *lock-in* sur la taxe carbone (*carbon tax lock-in premium*), et montre que l'ignorer conduit à un dépassement du budget carbone de 17 à 19 GtCO2eq, soit près de 10% du budget carbone résiduel alloué au pétrole en 2026. De manière cruciale, les émissions sur-engagées sont fortement concentrées dans un petit nombre de projets offshore requérant de larges investissements initiaux. Cette concentration rend l'argument en faveur de mesures d'offre ciblées particulièrement convaincant : une part importante du coût climatique à long terme du retard politique pourrait être atténuée en agissant sur un nombre limité de projets identifiables. Cela renforce par ailleurs l'argument de faisabilité politique développé dans la littérature sur les politiques d'offre (Green and Denniss, 2018; Piggot, 2018) : un ciblage précis des mesures suscite plus aisément la mobilisation de l'opinion publique que des taxes diffuses sur la consommation, comme l'illustrent les campagnes contre des projets spécifiques tels que Keystone XL (Erickson and Lazarus, 2014). Ces résultats ancrent davantage la littérature générale sur les politiques d'offre dans des données détaillées au niveau des actifs : tandis que la communauté internationale s'efforce d'élaborer un cadre climatique global, la combinaison de données au niveau des gisements et d'une compréhension des dynamiques d'investissement peut d'ores et déjà éclairer la conception de mesures ciblées, efficaces et politiquement réalisables pour atténuer le coût croissant de l'inaction.

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Chapter 1

The Price of Delay: Infrastructure Lock-in and the Carbon Tax Lock-in Premium in the Oil Sector

Keywords: Climate change, fossil fuels, oil, carbon mitigation, carbon tax, climate policy delay, carbon lock-in, infrastructure lock-in, climate policy credibility.

JEL Codes: Q35, Q52, Q54, Q58, L71.

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1.1 Introduction

Renewed states commitments towards lowering carbon emissions lead carbon-intensive companies to expect future ambitious climate policies, especially in the oil and gas sector. In 2015, the Paris Agreement set a common objective for states parties to hold the rise in global temperatures to "well below 2°C [...] and pursuing efforts to limit [it] to 1.5°C". In order to remain within these limits, significant portions of existing fossil fuel reserves must remain unextracted (McGlade and Ekins, 2015; Welsby et al., 2021). To achieve this goal, the economic literature widely recognises carbon taxation as the most effective tool for curbing fossil fuel use in the coming century. However, despite this recognition, as of 2023, only a few regions or countries worldwide have implemented such taxes, and their levels fall far short of what is required to align with the objectives outlined in the Paris Agreement framework (Metcalf, 2019).

This lack of stringency in current climate policies leads to two consequences. First, it results in overproduction from existing production facilities, as studies have shown that low carbon taxes exert minimal influence on current oil and gas production (Heal and Schlenker, 2019). Second, it does not act as a sufficient deterrent to change producers' investment strategies. A large share of oil and gas firms use carbon shadow pricing in order to assess their investments to account for future climate policies (Fried et al., 2021). However, Ahluwalia (2017) shows that these shadow prices are usually low, although higher than observed carbon prices, ranging from \$23 to \$80 per metric ton[1], leading oil companies to only marginally adapt their investments to climate change. Low observed shadow prices used by oil companies highlight the poor credibility of States when they commit, and have long-lasting impacts on the path of oil investments and supply. The long term impact of new investments in fossil fuels is due to the inertia of carbon emissions associated with it and is referred to as carbon lock-in. Once investments are made into new producing infrastructure, carbon emissions from the associated deposits can become locked-in for long time periods, and render a low-carbon energy transition more difficult. Failing to account for these dynamics can lead to underestimate the real cost of delaying climate action, and more importantly underestimate the level of carbon tax required in the future in order to meet climate objectives.

In this paper, I use detailed field-level data on oil production facilities to study the long-term consequences of delaying carbon taxation in the oil production sector from 2016 to 2030. I show that policy delay leads to a substantial overshoot of the carbon budget set in 2016, driven not only by excess production during the delay but also by long-lived investment decisions that lock in future emissions. Crucially, I show that these investment dynamics create a wedge between a naïve tax, computed under the assumption of fixed

[1] Prices higher than \$60 are only used to stress test some highly-exposed investments

extraction costs, and the optimal tax required after the delay. I call this gap the carbon tax lock-in premium. Ignoring lock-in thus leads to systematically underestimating carbon prices and translates into additional post-tax emissions overshoot. Finally, I identify the new oil projects most responsible for these long-term costs and assess how targeted supply-side policies can mitigate both overproduction and lock-in.

Climate policy delays and the feasibility of climate objectives have been extensively studied within the integrated assessment models (IAMs) literature. Most studies conclude that delaying ambitious climate policies increases substantially mitigation costs, but a delay up to 2030 still allows for the 450ppm or 2°C objectives to be attained (Eom et al., 2015; Riahi et al., 2015; Kotlikoff et al., 2021). Making up for the inaction until 2030 renders strong and rapid adaptation necessary in the 2030-2050 period. However, as noted by Van Vuuren et al. (2016), the feasibility of the expected decarbonisation rates in this period is far from assured, and the potential of carbon capture and storage (CCS) might be overestimated in these models. A strong focus is placed on the green paradox[2], while carbon lock-in effects and path dependency remain understudied. Studying a large variety of IAMs, Grubb et al. (2021) show that the majority only partially takes into account the dynamic nature of investments and production processes, and considers abatement costs in each period to be independent from prior abatement. Path dependency, inertia, and induced innovation are largely disregarded.

Among these models, only IMACLIM-R, REMIND, and WITCH incorporate these three features, and can thus result in unattainable climate objectives following delays in climate action. In Riahi et al. (2015)' study, only the IMACLIM-R model leads to the conclusion that the transition is infeasible with a delay up to 2030. Similarly, using these three models, Jakob et al. (2012) find that delaying the implementation of ambitious climate policies until 2030 makes it impossible to attain the 450ppm goal. Luderer et al. (2013) and Bertram et al. (2015) find similar results using the same models. Assumptions on the lifetime of infrastructures are crucial. Carbon lock-in in these studies is probably overstated because the models assume that infrastructure built is used to the end of its lifetime. Concurrently, Insley (2017) shows that early retirement of carbon-intensive infrastructure is to be expected if carbon policy is implemented. Nevertheless, despite capturing some carbon lock-in, the complexity of these models limits the understanding of the mechanisms underlying lock-in effects and the identification of specific investments leading to it (Grubb et al., 2021). Additionally, IAMs' modelling of fossil fuel supply often produces unrealistic responses to changes in demand (Höök and Tang, 2013). Despite accounting for heterogeneity across fields or regions, production costs are often considered constant at the field level, and computed as the breakeven price of the project (as in Heal

[2]Coined by Sinn (2008) and extensively studied afterwards (see e.g. van der Ploeg and Withagen (2015)), the green paradox states that a delay in implementing carbon policy can encourage fossil fuels producers to overproduce during the delay (as compared to a "no policy" scenario) instead of losing their resource after the policy is implemented.

and Schlenker (2019)). Investors' logic and geophysical constraints are largely disregarded, leading to strong adaptation in the allocation of the supply. In reality, when choosing to develop a field, investors determine the optimal level of infrastructure needed, which in turn determines the ultimate recovery rate of the field and its production rates (Venables, 2014). Production rates once the field has started its production cannot be adjusted at will and are mainly dictated by geophysical and technological constraints, to the point that no adaptation to changes in oil price can be observed at the well level (Anderson et al., 2018).

An increasing body of literature has shifted its focus towards carbon lock-in. Carbon lock-in refers to the phenomenon whereby investments in carbon-intensive infrastructures and technologies induce long-term dependence on high carbon emissions and hinder the transition to low-carbon alternatives. Developed by Unruh (2000), this concept includes three main dimensions identified by Seto et al. (2016) that are mutually intertwined and contribute to the inertia of carbon emissions: infrastructural, institutional and behavioural. This study focuses on the first dimension, through which newly developed oil deposits and associated infrastructures can render more difficult the phasing-out of oil supply and associated emissions. Studying carbon lock-in in this setting implies a change in perspective from annual emissions to committed emissions over the future associated with each infrastructure (see Davis and Socolow (2014) for more details on the commitment accounting of carbon emissions). Several papers tried to estimate the quantity of carbon emissions embedded in existing carbon-intensive infrastructures. Unsurprisingly, the later a study is published, the more embedded emissions are found to jeopardise climate change mitigation objectives (Smith et al., 2019; Tong et al., 2019), with Trout et al. (2022) finding that current fossil fuels producing infrastructure would warm the world beyond 1.5°C.

Additional papers adopt a dynamic perspective to assess carbon lock-in from carbon-intensive infrastructures (see Fisch-Romito et al. (2021) for a review of this literature). Lazarus et al. (2015) offers an initial assessment, though partial, of the carbon lock-in arising from fossil fuel supply infrastructure. Wachtmeister and Höök (2020) compare the investment and production cycles of conventional and unconventional tight oil to assess the consequences of delayed climate action[3]. In both these studies, the underlying logic is that once capital investments are incurred, the decision to produce or not does not depend on initial expenses. The decision to invest relies on the expected net present value of the project, whereas the decision to produce only depends on extraction costs. Oil projects are on average costly to develop but cheap to operate (Lazarus et al., 2015; Venables, 2014), making newly developed projects a burden for future climate action. Fossil fuel supply results from dynamic, discrete and long-lasting investment choices which determine

[3]The characteristics of each source of oil has tremendous impacts on the associated dynamics of investment, and affect in turn the potential lock-in effects emerging from it.

a path of production for every project. In this sense, newly developed projects represent committed emissions over the future, and lead to a carbon-intensive path dependency. Consequently, the academic literature has seen a rise in support for supply-side policies (Erickson et al., 2018; Green and Denniss, 2018; Fæhn et al., 2017), with an emphasis put on banning new oil projects (Green et al., 2024; Erickson and Lazarus, 2018; Van Asselt, 2021). Addressing the supply is more and more perceived as complementary to demand-side policies. On the one hand, it provides numerous advantages in terms of feasibility: low administrative costs, higher abatement certainty, higher public support. On the other hand, it allows to mitigate the cost arising from carbon lock-in risks (Green et al., 2024). Nevertheless, as noted by Lazarus and van Asselt (2018), the literature on demand-side policies far outweighs the one on supply-side policies, despite its apparent benefits.

To date, no study has offered precise estimates of the emissions consequences resulting from carbon lock-in in the oil production sector, taking into account the dynamic nature of investments. Additionally, while the literature on carbon lock-in emphasizes the persistence of emissions due to long-lived infrastructure, it has not translated these effects into implications for the level of carbon taxation following policy delays. This paper introduces the concept of a carbon tax lock-in premium, defined as the additional carbon price required to offset committed emissions resulting from delayed policy implementation and associated over-investments.

This paper aims at filling these gaps in the literature by investigating the consequences of a 14 years delay, between 2016 and 2030, in the implementation of carbon taxation in the oil producing sector. Both a constant carbon tax and a carbon tax increasing at a 2% discount rate are assessed. I find that for a 100$/tCO2eq constant carbon tax (respectively 57$/tCO2eq increasing one), in line with the updated oil carbon budget set by Welsby et al. (2021), the delay induces a 33% overshoot of the budget (respectively 19%). Two thirds of the overshoot occurs at the intensive margin, the rest originating from fields developed during the delay. From a carbon lock-in perspective, results suggest that overcommitted emissions represent more than half of total committed emissions during the period, and are concentrated in a small number of specific oil projects (mainly deepwater offshore). Overcommitted emissions from these assets appear resilient to higher taxes. As a result, achieving the same carbon budget after a policy delay requires a systematically higher carbon tax than suggested by a naïve calibration ignoring investment-driven lock-in. Failing to take into account this lock-in effect leads to a residual overshoot of the budget ranging from 15 GtCO2eq to 20 GtCO2eq, more than 10% of the remaining budget. I refer to this additional required tax as the carbon tax lock-in premium, and assess its implications for policy stringency over time as lock-in builds up.

The remainder of the paper is organized as follows. Section 1.2 presents the data and methodology used to estimate the magnitude of the carbon budget overshoot and to identify investments responsible for carbon lock-in effects. Section 1.3 presents results for (i) yearly overemissions stemming from the delay, (ii) the role played by carbon lock-in and (iii) the carbon tax lock-in premium required to compensate lock-in effects. Section 1.4 tests the sensitivity of my results to a large array of alternative assumptions, including the inclusion of time-varying tax pass-through from producers to consumers. In section 1.5, the implications of the results are discussed and policy recommendations are provided to mitigate the cost of carbon lock-in. Finally, Section 1.6 concludes.

1.2 Data and methods

1.2.1 Oil and gas field data

This study relies primarily on proprietary micro-level data from Rystad Energy. Rystad's UCube database[4] covers the vast majority of oil and gas production in existing fields worldwide since 1900. It contains as well informations on fields that are expected to be developed and discovered up to 2100. The analysis is conducted at the "asset" level, the smallest geographic scale available in the database, which typically corresponds to a field or to a portion of a field. The data comprises more than 100,000 oil and gas producing assets, among which 20,322 are producing in 2025, when data was extracted. For each of these assets, the data set contains a breakdown of annual production by oil types and of costs by category, as well as various administrative, geographic and geological characteristics. Annual costs include operating expenses (disaggregated into production, transportation and SG&A OPEX), capital expenses (facility, well), exploration expenses, and government take (income taxes, royalties, taxes reported as OPEX, bonuses, etc)[5].

1.2.2 Field-level carbon intensities.

In addition to the Rystad database, I estimate lifecycle carbon intensities (CI) for each asset by combining upstream, midstream, and downstream emissions. This section provides an overview of the methods used to estimate these CI. Further details about the methods are described in depth in Appendix A.2.

For upstream CI, I use data obtained from Masnadi et al. (2018) for 737 oil fields, matched to 1,136 Rystad assets, for which I compute CI using the Oil Production Greenhouse Gas Emissions Estimator (OPGEE)[6]. A key driver of upstream CI variability is flaring intensity, for which information is scarcely available. I thus use estimates of flar-

[4]Information on this database is available at https://www.rystadenergy.com/services/upstream-solution.

[5]See appendix A.1 for a description of each category of cost.

[6]The OPGEE model is openly accessible at: https://eao.stanford.edu/research-project/opgee-oil-production-greenhouse-gas-emissions-estimator

ing intensity derived from satellite observations (2012–2024) observed by NOAA's VIIRS instruments Elvidge et al. (2015), converted into GHG emissions using their algorithm. These observations are geographically matched to Rystad assets following the procedure detailed in Ghosh et al. (2025). For assets without direct flaring data, I predict flaring-to-oil ratios (FOR) using a multi-stage matching procedure and assign reserve-weighted averages for unmatched observations. I then fit a regression model incorporating FOR, steam-to-oil ratio (SOR), gas-to-oil ratio (GOR), offshore status, and oil type to predict upstream CI for all remaining assets, including not-yet-developed ones.

For midstream and downstream CI, I use the Petroleum Refinery Life-Cycle Inventory Model (PRELIM)[7] and the Oil Products Emissions Module (OPEM)[8]. PRELIM estimates refining emissions based on crude assay characteristics, primarily API gravity and sulfur content (Abella and Bergerson, 2012), while OPEM accounts for transport and end-use emissions. I match 493 PRELIM crude assays publicly available within the two models to 392 Rystad crude streams (covering 64.4% of 2024 production) and predict CI for unmatched assets using API gravity, sulfur content, and refinery configuration (deep conversion, hydroskimming, or medium conversion). For assets lacking API gravity and/or sulfur content data (12% of 2016 reserves), I assign reserve-weighted averages from similar fields (e.g., same reservoir, sub-basin, or oil type).

Finally, I sum upstream, midstream, and downstream CI to generate a unique lifecycle CI estimate for each asset. This approach, combining direct OPGEE/PRELIM/OPEM computations, satellite-derived flaring data, and predictive modeling, ensures robust CI estimates across all assets, even those with incomplete data. In 2024, the production-weighted average carbon intensity is 466 kgCO2eq/bbl, with a weighted standard deviation of 112 kgCO2eq/bbl. Figure A.1 reports cumulative reserves in 2016 ordered by lifecycle carbon intensity. A large share of reserves are situated around, with a small tail to the left with cleaner assets, and a larger tail to the right with dirtier assets. The graph splits cumulative reserves according to assets development timing. No significant difference exist between assets developed during the delay and assets already producing in 2016.

1.2.3 Data selection and definition of costs

Assets selection. A selection on the data was performed so as to keep only oil-producing assets (gas is disregarded in this analysis). Among oil-producing assets, only those which reserves are discovered by 2025[9]. This leaves me with 28,327 assets with

[7]The PRELIM model is openly accessible at: https://ucalgary.ca/energy-technology-assessment/open-source-models/prelim

[8]The OPEM model is openly accessible at: https://ociplus.rmi.org/methodology

[9]Since Rystad fully models production, costs, and reserves characteristics of not-yet discovered assets, I prefer to exclude them from the analysis. Sensitivity analysis includes a specification in which not-yet

available oil reserves in 2016, among which 16,309 already producing in 2016.

Reserves and oil definition. Reserves used in this study corresponds to oil resources as defined by Rystad, i.e. all economically viable recoverable oil over the lifetime of the asset. Quantities are typically larger than 2P reserves because it includes reserve growth over time, generally due to geological uncertainty declining as production takes place. Oil is defined as any type of crude oil (bitumen, extra heavy, heavy, regular, light or sour), as well as condensate and natural gas liquids (NGL). The last two hydrocarbons are assimilated to oil in reason of their liquid nature and the fact that they are sold as light types of oil (and often mixed with other crude oils before being sold in order to reduce their viscosity). As explained in the next subsection, although NGL is sold at a much lower price than oil, its price is primarily dictated by oil price and not by gas price. I thus include it in my definition of oil. A given asset can produce only one type of crude oil, but can also produce condensate and/or NGL as well.

Development costs. Following Rystad's disaggregation of costs, only expenses imputed to oil operations are kept. Capital expenses related to wells drilling and the building of facilities during the pre-production period and in the early years of production correspond to the development cost of the assets. Following the definition given by Rystad, the threshold before which all capital expenses are assimilated to development ones is set at two years after production startup. For modelling purposes, development expenses occuring once production has started are gathered in the year preceding production startup (and discounted accordingly in order to leave projects NPV unchanged).

Production costs. As stated above, only production costs imputed to oil are taken into account. I define as production costs all operating expenses (production opex, transportation opex, and selling, general and administrative opex), as well as all non-development capital expenses. This implies that expenses directed at building new facilities and new wells after the development phase of the asset occured (i.e. later than two years after production startup) will be included in productions costs.

Government take. Government takes are disregarded in the analysis, to the exception of taxes reported as OPEX by companies (variable "Taxes Opex" in appendix A.1). This last category of taxes is reported as operating costs by producers, and are not directly linked to the companies revenues or any asset's profitability. I thus include it in production costs. Among other types of government takes, some are based on profits from an asset (income tax), or expensed before exploration occurs (bonuses). I assume that these taxes do not affect investors in their decision to develop or not an asset, since the former affects the magnitude of the asset net profitability (after carbon taxation if

discovered assets are kept, and the associated exploration costs treated as development costs.

there is any) but not its sign, and the latter is already expensed before development decisions are made. Fiscal regimes based on royalties could render a project unprofitable since its tax base is gross revenues. However, I assume that in presence of a worldwide carbon tax, governments would renegotiate and adapt royalty rates in order to keep assets active and continue extracting a share of the rent. Nevertheless, Ahlvik et al. (2022) and Stefanski et al. (2025) show that oil taxes affect substantially investments and production decisions. Given the complex and interdependent role of fiscal regimes on investment decisions (Erickson et al., 2017; Ahlvik and Harding, 2025), this study focuses on market-driven costs and carbon pricing mechanisms, excluding government takes from the analysis.

1.2.4 Modelling framework

The model of oil extraction in this paper relies on a sequence of decisions made by investors at each period (year) for each asset. Decisions made for each asset are independent of other assets. At each period, investors thus decide to invest or not in the development of the asset for undeveloped ones, and to produce or not for an active one.

Investment decision. I assume investment decisions made by investors to be primarily dictated by the expected net present value (NPV) of the project. Even though discounted cash-flow analysis is not the only criterion used in the oil industry to determine which field to develop (risks are also assessed regarding political, financial, and safety hazards), it remains the cornerstone of project-based decision-making (Bailey et al., 2000; Jahn et al., 2008). Moreover, as noted by Erickson et al. (2017) and Wood Mackenzie (2013), it is the method used by the oil industry itself in their analysis. I use a 10% discount rate to compute these NPVs, which is the most common discount rate used in the oil and gas sector (Bailey et al., 2000), and is the one used by Rystad to compute breakeven prices, i.e. the price of oil above which the NPV of a project is positive. Robustness checks include the use of a 5% and a 15% discount rates.

In T_0, the decision to invest or not in an asset i depends on the anticipations of investors regarding the introduction of a carbon tax τ, and is thus governed by the equation:

$$\sum_{t=T_1}^{T} \frac{P(1-d_i)q_{it} \; - c_i}{(1+\delta)^{t-T_0}} - C_{iT_0} - \sum_{t=T_\tau}^{T} \tau \; \theta_i \; q_{it} \frac{(1+r)^{t-2016}}{(1+\delta)^{t-T_0}} \geqslant \; 0 \tag{1.1}$$

where T, and T_τ stand respectively for the expected last year of production and anticipated year of tax implementation. The production of the asset q_{it} is sold at a constant individual price $P(1-d_i)$, which corresponds to a reference oil price minus a discount specific to the asset. The costs are divided between development costs C_{iT_0}, assumed to be expensed 1 year before production startup, and production costs c_{it}. The carbon tax paid is equal to the carbon intensity of the asset θ_i multiplied by an initial 2016 carbon tax τ increasing at rate r. Finally, future production revenues and costs are discounted at a rate δ equal to 10%.

Rearranging equation (1), the maximum anticipated level of tax for the asset to be developed is as follows:

$$\bar{\tau} \; = \; max_{\{T\}} \; \frac{P(1-d_i) \; - \; BEP_i^{Project}(T)}{\theta_i \; Q_i^{\tau}(T)} \tag{1.2}$$

with $BEP_i^{Project}(T) = \frac{C_{iT_0} + \sum_{t=T_1}^{T} c_{it} \; (1+\delta)^{-t}}{\sum_{t=T_1}^{T} q_{it} \; (1+\delta)^{-t}}$ the breakeven price of the project including development costs up to the year T, and $\theta_i \; Q_i^{\tau}(T) \; = \; \theta_i \; \frac{\sum_{t=T_\tau}^{T} \frac{q_{it} \; (1+r)^{t-2016}}{(1+\delta)^{t-T_0}}}{\sum_{t=T_1}^{T} \frac{q_{it}}{(1+\delta)^{t-T_0}}}$ the anticipated share of discounted emissions taxed in 2016 tax terms.

Investors choose T, the last year of production, in order to maximise the net present value of the asset. In the absence of tax, T is always the last year of production predicted by Rystad. However, the set of anticipations affects this maximisation and can lead investors to expect an asset lifetime different from what will effectively be the case once production has started and uncertainty about the tax has resolved. $\bar{\tau}$ thus corresponds to the maximum carbon tax consistent with a non-negative NPV, given anticipated taxation and optimal production duration.

In some observed cases, development does not occur in one year, and can sometimes span over multiple years, or stop at some point and resume later. This last situation is common for assets which development started in 2019: after the breakout of the COVID-19 pandemic and the subsequent drop in oil prices, many developments were stopped, and before resuming years later. This has an impact on the development status of an asset in each scenario. Therefore, the maximum level of tax for the development of the asset to be economically viable is computed for *each* development stage, with C_{iT_0} defined as the *remaining* amount of development costs. The asset is developed if and only if the anticipated level of tax is below the maximum tax $\bar{\tau}$ at each development stage. This is straightforward for assets which are developed before the end of the delay, since $\bar{\tau}$ increases as C_{iT_0} gets lower. However, development expenses can occur and suddenly stop when the tax is implemented, as uncertainty resolves and $Q_i^{\tau}(T)$ increases to 1 (or more in the case of a increasing tax). The development costs already incurred for these assets are lost, and the corresponding infrastructures built are abandoned.

Production decision. Once an asset is developed, the quantities produced are determined by comparing production costs with the oil price minus the tax (which is equal to zero during the delay). Production costs are computed as breakeven prices by stage of production, and are strictly increasing in time. Rystad predicts long lifetime for assets because it depends on a minimisation of the breakeven price of the full project, including development expenses. When removing these expenses, the breakeven price of production can be minimum with a final year of production earlier than forecasted by Rystad. This last characteristic stems from the fact that (i) operating costs increase over time, and (ii) as the production rate declines, capital expenses per barrel increase. Therefore, I identify for each asset stages of production with strictly increasing breakeven prices. In order to do so, I compute the last year of production that minimises the breakeven price, which corresponds to the end of stage 1, and repeat the operation for remaining years to identify subsequent stages. For assets developed after 2015, the average number of production stages is 5.8, and the average duration of a production stage is 5.2 years, stage 1 usually being the longest.

Investment and production paths. The stream of investments and production assessed for each asset is the one observed up to 2025 and predicted by Rystad for the future. The rationale behind this assumption is twofold. Regarding the shape of the path, production rates (plateau and decline) mainly depends on initial investments and cannot be changed at will afterwards because of geological and technological constraints (Anderson et al., 2018; Venables, 2014; Cairns and Smith, 2019). Changing the production path would thus entail modifying development expenses, as well as all subsequent costs. Regarding the timing of the decisions, (i) during the delay, carbon taxation is not or only partially anticipated by investors, making the "business-as-usual" scenario forecasted by Rystad relevant, and (ii) once the tax is implemented the year of development does not matter, since I focus on cumulative production up to 2100.

One could argue that investors anticipating a carbon tax to be implemented, *a fortiori* an increasing one, would be incentivised to push their production and investments earlier in time, giving rise to a Green paradox effect. This valid concern would imply higher investments in earlier years, i.e. during the delay, and higher cumulative production for a given level of tax. This means that this study provides a lower bound of the cost of delaying climate policy.

Price assumptions. The model only focuses on supply-side responses, and therefore ignores the effect on oil prices of the implementation of a carbon tax. The level of tax assessed in the paper can thus be interpreted as the level of tax borne by producers. Section 1.4 includes a sensitivity analysis of my results incorporating carbon tax initially partially borne by consumers and progressively passed on to producers. I use exogenous predicted evolutions of the pass-through rates by carbon tax increase rates drawn from Heal and Schlenker (2019).

The reference oil price (Brent) is assumed to be constant and equal to \$80/bbl.

Two rationales justify this choice. First, changing the level of price would only change the optimal level of carbon tax in my computations, without significantly changing the results in terms of carbon budget overshoot. Only the overall pool of economically viable assets (assets with breakeven prices lower than the oil price) is affected by this assumption. Yet, only few projects require prices higher than $80/bbl to be developed. Robustness checks include an oil price of $100/bbl and $60/bbl. A second rationale is the coherence of a $80/bbl price as compared to oil markets dynamics. The stability of the oil market is mainly guaranteed by the 3 core countries of OPEC, namely Saudi Arabia, Kuwait and United Arab Emirates (Kellogg, 2024). For these countries, the IMF reports average past and projected (in 2025) fiscal breakeven prices of $80/bbl (past) and $84/bbl (future) for Saudi Arabia, and below for the two other countries (International Monetary Fund, 2024). Fiscal breakeven prices corresponds to the required oil price for oil exporting countries to balance their budget. Given the influence exerted by core OPEC countries on oil price, using their expected fiscal breakeven price as a reference for future oil price seems appropriate. Overall, using a $80/bbl oil price allows me to model and capture invested infrastructure decisions and lock-in effects without biasing results toward extreme low-price environments where investment incentives collapse.

One might argue that the introduction of a carbon tax would alter long-term oil price forecasts. While this is true in principle, I argue that Middle Eastern countries would be significantly less affected by a carbon tax than other regions, due to their relative cheap costs and low carbon intensity, even more so when it comes to OPEC-core countries. Should the carbon tax be levied by producer States, OPEC-core countries would thus see their revenues stemming from oil production only moderately reduced by the tax. Results in figure A.3 support this rationale.

Price volatility is also disregarded in the analysis because investment decisions are mainly based on expected oil prices in the long-term, especially for long-lived investments which are the key focus of this study. This assumption is further motivated by results from the literature. Pierru et al. (2018) shows that OPEC's use of its spare capacities significantly reduces oil price volatility and reinforce trust from investors in a stable market. Saudi Arabia acts as a swing producer, keeping significant spare capacities in order to adjust its production level quickly in response to demand shocks (Fattouh, 2007). Additionally, Berntsen et al. (2018) show that effects of price volatility on oil and gas investments in Norway are insignificant, while Tang et al. (2017) argue from a theoretical setting that it does not affect the decision to invest or not but could rather induce earlier investments.

Asset-specific discounts. While oil prices generally refer to reference prices such as Brent, WTI or WCS, each asset sells its oil at a unique price corresponding to a reference price minus a discount (which can be a premium in certain cases). These discounts or premiums depend on the crude oil quality (API Gravity, sulphur content, etc) and can vary through time. The price discounts used in the model are the long term discounts forecasted by Rystad and the reference oil price is the Brent price. Table A.1 displays the reserve weighted average discount by oil type. All types are sold on average at a discount as compared to the Brent price. This can be explained by the high quality of the oils constituting the Brent crude stream. The average 12% discount across all reserves implies that while using a reference oil price of \$80/bbl, selling prices are on average closer to \$70/bbl in the model. Heavier oils being more difficult to refine in order to yield the same products, their discounts are relatively higher. NGL is a particular case, being sold at approximately half the price of oil.

1.2.5 Counterfactual scenarios

Main scenarios. The analysis relies on the comparison between two scenarios: one in which the carbon tax is implemented in 2016 (the "*Baseline*" scenario), and another in which the tax is delayed until 2030 without being anticipated by investors (the "*No anticipation*" scenario). In terms of investment and production decisions, this second scenario is equivalent to a "business-as-usual" until 2030, when the regime suddenly changes and the tax is implemented.

Alternative anticipations by investors. Additional policy delay scenarios are assessed in this study. These scenarios are designed to span a plausible range of investor beliefs about the timing and stringency of future carbon pricing. In the "*Tax in 2030*" scenario, investors anticipate the correct level of tax to be implemented in 2030, and thus anticipate the social planner behaviour correctly. In the "*5 years after startup*" scenario, investors anticipate the correct level of tax to be implemented after 5 years of production in the asset. This is one of the methods used in the oil and gas sector to screen projects before investing in it. In the "*75% of the tax*" and "*50% of the tax*" scenarios, investors anticipate the tax to be implemented directly after investing in an asset, thus being imposed on the totality of its lifetime production, but underestimate its level by 25% and 50%. This approach seems to closely match reality: Ahluwalia (2017) shows that most oil and gas companies verify that their projects could sustain the introduction of a tax, but usually use tax rates far lower than the ones needed to be in line with the 1.5°C or 2°C objectives. The 50% underestimation appears close to observed investors' behaviour.

Heterogeneous firms' beliefs Finally, four scenarios are added, in order to assess the effect of potential heterogeneity across firms in terms of anticipations. For two specific subsets of assets, investments are assumed to be screened assuming an immediate implementation of the full tax (thus matching the "Baseline" scenario), or an implementation of the tax in 2030 (similar to the Tax in 2030 scenario). The first subset of assets correspond to oil projects in which European firms own at least 10% of the shares. This is justified by the relatively higher pressure put on firms based in European countries in terms of environmental responsibility. The second subset is all assets owned at 10% by a firm publicly listed. Publicly listed companies can face higher pressure by investors to display greener behaviour and prove resilient to carbon taxation.

1.2.6 Key metrics

Carbon budget overshoot. The first objective of the paper is to estimate the carbon budget overshoot stemming from the delay in the implementation of the carbon tax. The overshoot is computed by comparing cumulative emissions in one of the delay scenarios with the one that would be observed in the "Baseline" scenario. Once the overshoot is computed for each scenario and level of tax, I disentangle between both the intensive and the extensive margins. The approach there is straightforward: overproduction is considered to happen at the intensive margin if the corresponding asset produces any oil in the "*Baseline*" scenario or produced any oil before 2016. It is considered as the extensive margin if the asset does not produce any oil in the "*Baseline*" scenario. Since the set of anticipations only affect investment decisions, the results found with the different scenarios only vary at the extensive margin.

A reference level of tax used throughout the paper is the tax necessary to match the carbon budget for oil set by Welsby et al. (2021). To adapt it to the framework of this paper, the 747Gbbl budget for oil over the 2018-2100 period is adapted to the 2016-2100 period by adding observed production in 2016 and 2017. It is then converted into a carbon budget by multiplying it by the average downstream oil carbon intensity used in most models (including the TIAM-UCL used in Welsby et al. (2021)), derived from IPCC (2006). The resulting carbon budget for oil in 2016 is equal to 350.3 GtCO2eq. A shortcoming of this study is that one could legitimately expect the carbon budget dedicated to oil to be reduced should carbon taxation be imposed on oil, and leak towards other fossil fuels. Results are thus to be interpreted as if similar policies were applied to other fossil fuels production in parallel. In order to assess the potential impact of this assumption on my results, I include in robustness checks scenarios in which the carbon budget is reduced by 10% and 25%.

Infrastructure lock-in. Assessing lock-in effects requires to switch to a commitment accounting perspective. The objective is to identify for each asset the year of investment that locks it in into full development. For a given asset and set of anticipations, the year of investment responsible for the locking-in of the asset can vary depending on the level of tax. In order to identify the investments responsible for the lock-in of emissions, I compute for each asset the maximum amount of remaining development costs $\hat{C}_{iT_0}(\tau)$ for the asset to be irreversibly developed. Rearranging equation 1.1:

$$\bar{C}_{iT_0}(\tau) \ = \ max_{\{T\}} \sum_{t=T_1}^{T} \frac{P(1-d_i)q_{it} \ - c_i}{(1+\delta)^{t-T_0}} \ - \sum_{t=T_\tau}^{T} \tau \ \theta_i \ q_{it} \frac{(1+r)^{t-2016}}{(1+\delta)^{t-T_0}} \tag{1.3}$$

The year of investment responsible for the lock-in of the asset is the year during which this threshold is crossed, i.e. the investment that makes remaining development expenses lower than this threshold. This year of investments "responsible" for the lock-in means that even if the tax was to be implemented immediately after, the asset would still be developed all the way. Note that since $\bar{C}_{iT_0}(\tau)$ depends on τ, the year of investments "responsible" for the asset development can vary depending on the level of tax, meaning that for different levels of tax, the overcommitted emissions from one asset can be attributed to different years. This also means that investments made during the delay can lock-in assets which production will start after the tax implementation, sometimes multiple years later.

(Over)committed emissions. Committed emissions refer to the cumulative future emissions that are expected to occur from an asset over its remaining lifetime once it has become locked-in. The quantity of committed emissions depends on the level of carbon tax, since it determines the timing of phasing-out of the asset. Overcommitted emissions are the subset of committed emissions attributable to lock-in effects induced by the policy delay: they correspond to the committed emissions from assets whose development would have been deterred under timely policy implementation, but which, once the lock-in threshold is crossed during the delay, will be developed and produce even if the tax is implemented immediately afterwards. These overcommitted emissions can span over multiple decades if the "depth" of the lock-in, i.e. the gap between the tax necessary to deter the development of the asset and the tax required to phase out production once it has started, is large.

Tax lock-in premium and "naïve" overshoot. I define the naïve overshoot as the residual carbon budget overshoot that arises when the delayed tax is calibrated using the pre-delay supply curve, ignoring the reduction in breakeven costs induced by new investments during the delay. The carbon tax lock-in premium is the wedge between this naïvely calibrated tax and the higher tax needed in the implementation year to offset the full extent of that overshoot.

1.3 Results

1.3.1 Overshoot of the budget due to the delay.

I first examine the cumulative carbon budget overshoot (relative to the 2016 benchmark) for a given level of carbon tax delayed from 2016 to 2030. Figure 1.1 depicts the corresponding carbon budget for each level of tax implemented in 2016 (baseline emissions), as well as the overshoot resulting from the delay at both the intensive and the extensive margins. The total overshoot increases slowly with the tax up to \$50/tCO2eq, before increasing steadily at a faster rate once the tax reaches higher levels. At low tax levels, cumulative production and in turn carbon budget overshoots are only marginally affected, highlighting the high levels of rent perceived by oil producers.

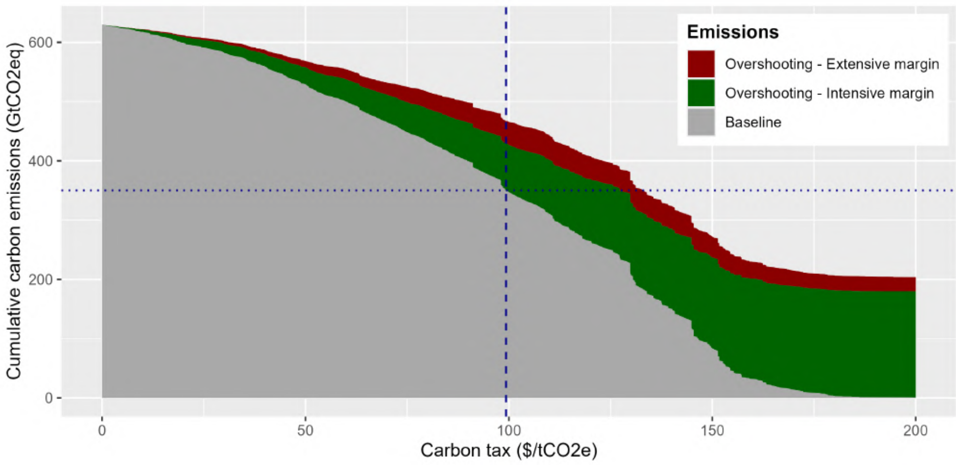


Figure 1.1: Oil carbon budget overshooting when delaying carbon taxation from 2016 to 2030.

Notes: The x-axis reports levels of carbon tax. The grey area corresponds to the "Baseline" scenario" scenario, i.e. the cumulative emissions if the tax is implemented in 2016. The green and red areas depict the overshoot resulting from delaying the tax until 2030 at the intensive and extensive margins respectively. The dotted horizontal line indicates the oil carbon budget set by Welsby et al. (2021). The dashed vertical line is the corresponding carbon tax needed in 2016 in order to constrain cumulative emissions within this budget. The scenario used to compute the overshoots is the "no anticipation" one. All computations are made assuming a reference Brent oil price of \$80/bbl.

The magnitude of the overshoot at each margin does not evolve in the same way with the carbon tax. As expected, it increases at the intensive margin as the tax gets higher. This is a mechanical effect due to the fact that the higher the tax, the larger the share of production between 2016 and 2030 that would be driven out of the market in the presence of the tax. However, the effect at the extensive margin does not display such a simple behaviour. It increases steadily as the tax goes higher, and reaches a plateau once the tax exceeds \$100/tCO2eq, near its required level in order to respect the 1.5°C carbon budget. At these levels, overemissions at the extensive margin fluctuate around 40 GtCO2eq. This feature can be explained by the relatively flat nature of the supply curve around the cheapest oil sources. For an asset to be included in the overshoot at the extensive margin, it must be that its project breakeven price is too high to enter the market in presence of the tax, but its production breakeven price is low enough so as to start production if it is developed. In addition to high-capital-intensity assets, will also be included assets whose project breakeven prices are just above the threshold to enter the market. These assets are thus more numerous when the threshold is situated around the levels of price where the supply curve is flatter, i.e. for levels of tax driving out a significant share of the market. Any ambitious carbon taxation delayed would thus lead to a significant overshoot at the extensive margin.

The dashed and dotted blue lines in figure 1.1 depict the oil carbon budget set by Welsby et al. (2021) updated as of 2016 (350.3GtCO2eq), and the corresponding carbon tax necessary in 2016 in order to constrain cumulative emissions within it ($99.3/tCO2eq). For this level of tax, the magnitude of the overshoot amounts to 78.7 GtCO2eq at the intensive margin and 38.4 GtCO2 at the extensive one, which combined represent 33% of the initially set carbon budget. To provide a better perspective, it is worth noting that a 117 GtCO2eq overshoot is equivalent to nearly three years' worth of global emissions as estimated in 2020 (Climate Watch, 2023). $99.3/tCO2eq (hereafter approximated to $100/tCO2eq in the paper) is the level of tax used as reference in the remaining of this paper when assessing the timing of overproduction and the effect of carbon lock-in.

Supply source heterogeneity. I further decompose in figure 1.2 the overshoot by oil supply segment (main types of oil sources) in order to determine which types of assets contribute the most to each margin. First, it is worth noticing that in the "Baseline" scenario" scenario, a $100 tax drives out of the market most unconventional productions: oil sands and shale oil represent together 0.3% of total cumulative supply. The high operating costs of oil sands and shale assets entail that most of their overproduction happens at the intensive margin. The higher carbon intensity of oil sands further reinforces this effect. More generally, the overshoot at the intensive margin is heavily concentrated in onshore assets. On the opposite end, deepwater assets have high development costs but are relatively cheap to operate, meaning that their overproduction is largely due to additional assets being developed during the delay. Similar conclusions can be drawn regarding midwater assets, to a lesser extent. This translates into offshore assets being responsible for two third of the overshoot at the extensive margin, with around 60% of it from deepwater assets alone.

Regional heterogeneity. This heterogeneity in terms of oil sources has implications for the geographical distribution of the overshoot. Figure A.3 displays its decomposition by world region. Comparing North and South America provides a striking illustration of the regional heterogeneity in terms of overshoot. Because North America concentrates most shale and oil sands production, it alone is responsible for half of the

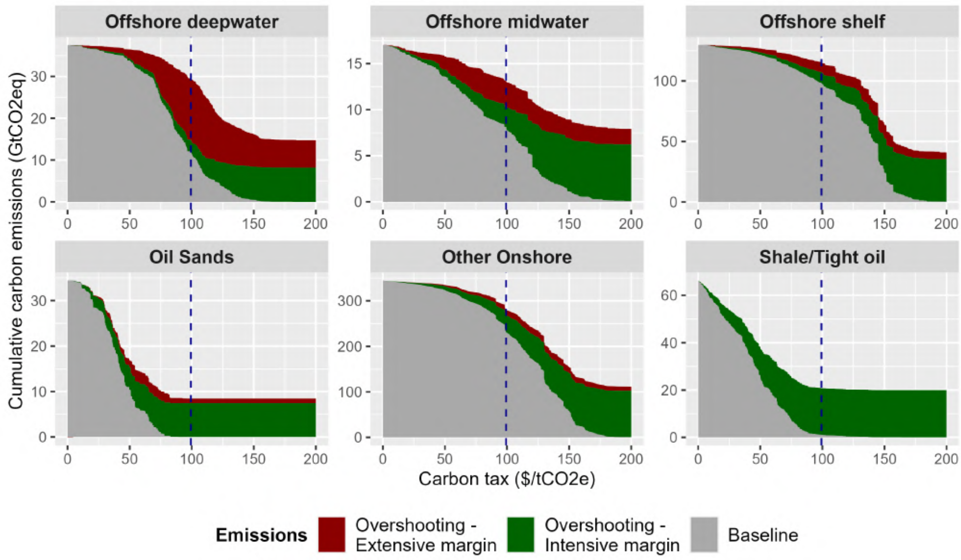


Figure 1.2: Oil carbon budget overshooting by supply segment.

Notes: Each panel corresponds to an oil supply segment. For each panel, the x-axis is the level of carbon tax. The grey area corresponds to the "Baseline" scenario" scenario, i.e. the cumulative emissions if the tax is implemented in 2016. The green and red areas depict the overshoot resulting from delaying the tax until 2030 at the intensive and extensive margins respectively. The dotted horizontal line indicates the oil carbon budget set by Welsby et al. (2021). The dashed vertical line is the corresponding carbon tax needed in 2016 in order to constrain cumulative emissions within this budget. The scenario used to compute the overshoots is the "no anticipation" one. All computations are made assuming an oil price of $80/bbl.

overshoot at the intensive margin. Conversely, a third of the overshoot at the extensive margin is attributable to South America alone, due to the numerous and massive deep-water projects currently being developed along the coasts of Brazil and Guyana. In other regions, each margin participates in similar proportions to the overall overshoot, although its magnitude vary greatly across regions. Middle Eastern production is not heavily affected by either the tax or the delay in implementation, while North and South American production greatly depends on the delay and would virtually disappear in presence of the tax. Asian, European and Russian productions are also greatly affected by the tax. This pattern may partly explain the focus put by Occidental countries on demand-side policies rather than supply-side ones.

Overshoot at the extensive margin by scenario of anticipations. Ahluwalia (2017) showed that most oil and gas producers anticipate some taxation to be implemented at some point, despite the levels of tax anticipated being low. I investigate the impact of a variety of anticipations scenarios on the overshoot at the extensive margin. I focus on this sole margin because anticipations only play a role in the decision to invest or not in a new project, thus only impacting the extensive margin. Figure 1.3 displays the overshoot resulting from each scenario aggregated by 5-years time period. A first observation is that anticipations of future tax by producers do not lead to a massive decrease in the magnitude of the overshoot.

The first two alternative scenarios investigate the effect of anticipating the correct

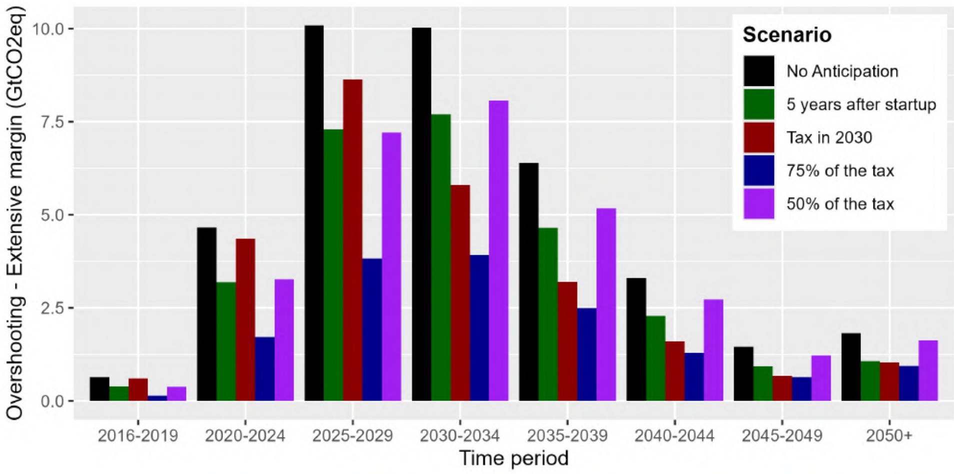


Figure 1.3: Yearly oil carbon budget overshooting at the extensive margin by scenario.

Notes: 5-years periods are on the x-axis. "2050+" corresponds to the 2050-2099 period. The y-axis corresponds to the quantities of oil overproduced by period and by scenario of anticipation compared with the "Baseline" scenario . The level of tax used in the graph is $99.3/tCO2eq. All computations are made assuming a reference Brent oil price of $80/bbl.

level of tax to be implemented later in time. When investors anticipate it to be implemented in 2030 (thus perfectly predicting the delay) or 5 years after production starts, the overshoot at the extensive margin still represents around 70% of the one in the "no anticipation" scenario. While not greatly affecting the total magnitude of the overshoot, the two sets of anticipations lead to a different time distribution: anticipating a tax in 2030 phase out most investments in new assets in the 2025-2029 period, thus leading to a smaller overproduction after the end of the delay. Conversely, the effect of anticipating the tax to be implemented 5 years after production startup affects investment decisions uniformly during the whole delay, leading to a reduction of the overshoot more or less constant over time.

The two following scenarios assume that investors expect the tax to be implemented any time, but underestimate its level. I provide results for taxes equivalent to 50% and 75% of the optimal one. Underestimating the tax by only 25% reduces overemissions at the extensive margin significantly (60% decrease compared with the no anticipation scenario), although the decrease in not proportional to the tax relative to its optimal level. This is confirmed when assuming a 50% underestimation of the tax. In this case, less than 23% of overemissions at the extensive margin disappear. While these two last scenarios replicate the behaviour of investors as observed by Ahluwalia (2017), the 25% underestimation is probably the least realistic one. Only few producers screen the viability of their projects in the presence of a tax higher than $60/tCO2eq, and the sole purpose of these high tax screening is to stress-test existing assets, not guide investments in new assets. Most carbon shadow prices used in the oil industry to screen new projects lie in the $23-$60 range. The 50% underestimation scenario is therefore much more aligned with investors anticipations described by Ahluwalia (2017). This explains the relative continuity of oil investments observed despite more and more firms adopting internal carbon pricing schemes.

Heterogeneous beliefs across firms. Figure 1.4 reports the same results considering various scenarios of heterogeneous behaviour across firms. I test a first scenario in which only a subset of investors (European firms or public firms) screen their investments with the full tax, as in the "Baseline" scenario, and another one in which the same subset of investors anticipate the full tax to be implemented in 2030, as in the "Tax in 2030" scenario. For reference, approximately 20% of assets and reserves are at least partially owned by European firms, while the share of assets with publicly-listed firms shareholders represents two thirds of the total. Unsurprisingly, screening projects with a $100 tax reduces greatly the overshoot at the extensive margin, by 40% for the Europe-owned subset and a striking 84% for the public subset. However, in line with results shown in figure 1.3, anticipating the tax to be put in place in 2030 only slightly affects its magnitude.

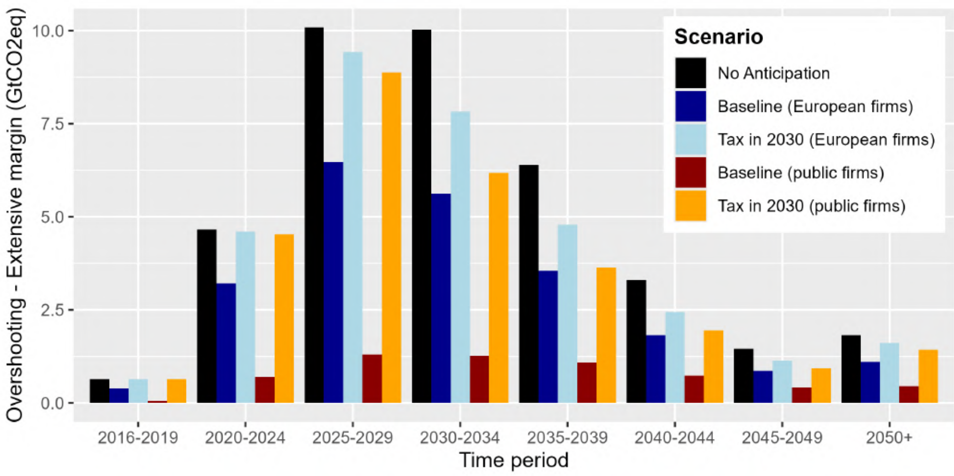


Figure 1.4: Yearly oil carbon budget overshooting at the extensive margin by scenario of firms heterogeneous beliefs.

Notes: 5-years periods are on the x-axis. "2050+" corresponds to the 2050-2099 period. The y-axis corresponds to the quantities of oil overproduced by period and by scenario compared with the baseline. The level of tax used in the graph is $99.3/tCO2eq. All computations are made assuming a reference Brent oil price of $80/bbl.

Over-emissions post-2030. A key result is that overproduction remains substantial after the delay: depending on the scenario, 47% to 64% of the overshoot at the extensive margin occurs after the tax is implemented, with 13% to 27% still remaining in 2040, 10 years after its implementation. Focusing on overproduction until the right policy is put in place when studying the delay does not allow to capture this future overproduction, thus leading to an underestimation of the full extent of its cost. The accumulation of investments in new oil assets should not be viewed solely as an increase in production capacity, but rather as committed emissions from infrastructures costly to phase out once they are built. Next subsection approaches overproduction at the extensive margin through this concurrent perspective, in order to better understand the consequences of climate inaction in terms of overinvestment rather than overproduction.

1.3.2 The role of infrastructure lock-in and the carbon tax premium

In this section, I shift my analysis from examining the annual emissions associated with each scenario to focusing on carbon lock-in: the cumulative emissions committed by investments in new oil infrastructure during the delay. Unlike annual overshoot, lock-in quantifies future emissions that become unavoidable once assets are developed, even if policies tighten later. This approach allows for a more comprehensive assessment as it quantifies the future emissions attributed to each year of delay. To evaluate the quantities of committed emissions, I define the lock-in threshold as the cumulative investment beyond which an asset's development becomes irreversible under a given tax level. The year in which this threshold is reached is considered to be the year of investment responsible for the lock-in of this asset. The analysis reveals that more than half (51%) of all committed emissions during the delay are overcommitted (i.e. attributable to the delay itself) and that these emissions are disproportionately concentrated in a small number of mainly offshore projects. This highlights how few capital-intensive projects drive most carbon lock-in effects, with major implications for targeted climate policies.

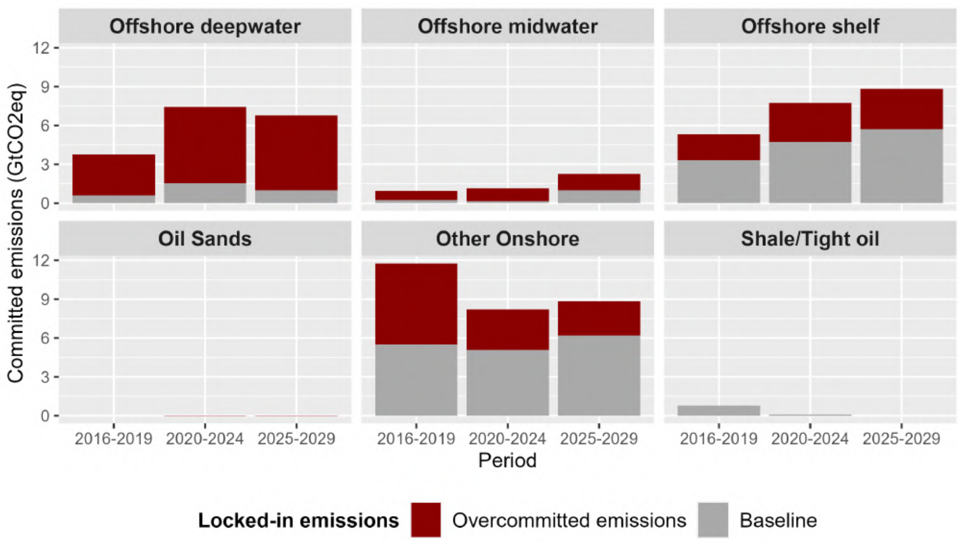


Figure 1.5: Overcommitted emissions by oil supply segment.

Notes: Each panel corresponds to an oil supply segment. Each bar corresponds to the amount of committed emissions from investments in the development of new oil assets in the corresponding segment.The grey share of the bar depicts committed emissions that are emitted in the "Baseline" scenario. The red share corresponds to committed emissions from assets that are not developed in the "Baseline" scenario, thus representing the cost arising from carbon lock-in effects. Committed emissions from a given asset are attributed to the year from which the full development of this asset becomes inevitable, which depends on the level of tax. The level of tax used for this graph is \$100/tCO2eq. The scenario used for this graph is the "No anticipation" one. All computations are made assuming a reference Brent oil price of \$80/bbl.

Figure 1.5 reports committed emissions by supply source, split between committed emissions existing in the "Baseline" scenario and overcommitted emissions. As expected, virtually no new unconventional project is developed during the delay. 70% of overcommitted emissions are concentrated in offshore assets, with deepwater assets representing the majority of it. More than 80% of committed emissions in these projects are overcommitted ones, compared with around 40% for onshore and offshore shelf projects.

The depth of lock-in. Reported overcommitted emissions in figure 1.5 are relative to a single tax level of $100/tCO2eq. However, it does not give information on the depth of the lock-in. If assets responsible for overcommitted emissions only required a slightly higher tax in order to be phased out, increasing the tax to compensate for it would not entail a redistribution of stranded assets across the globe. However, if these assets are resilient to high taxes, the necessary tax increase to compensate for the delay would involve stranding assets that would not have been affected by the tax should it be implemented in 2016. Figure 1.6 maps projects responsible for overcommitted emissions, aggregated by province. The color of the dots reports the average level of required tax to phase out assets in the province. While a small number of assets, mainly in China, can be phased-out with taxes below $120/tCO2eq, most overcommitted emissions would remain even at high tax levels (more than 86% at $120/tCO2eq and still 41% at $140/tCO2eq). Delaying climate policies is thus not neutral in terms of geographical distribution of stranded assets.

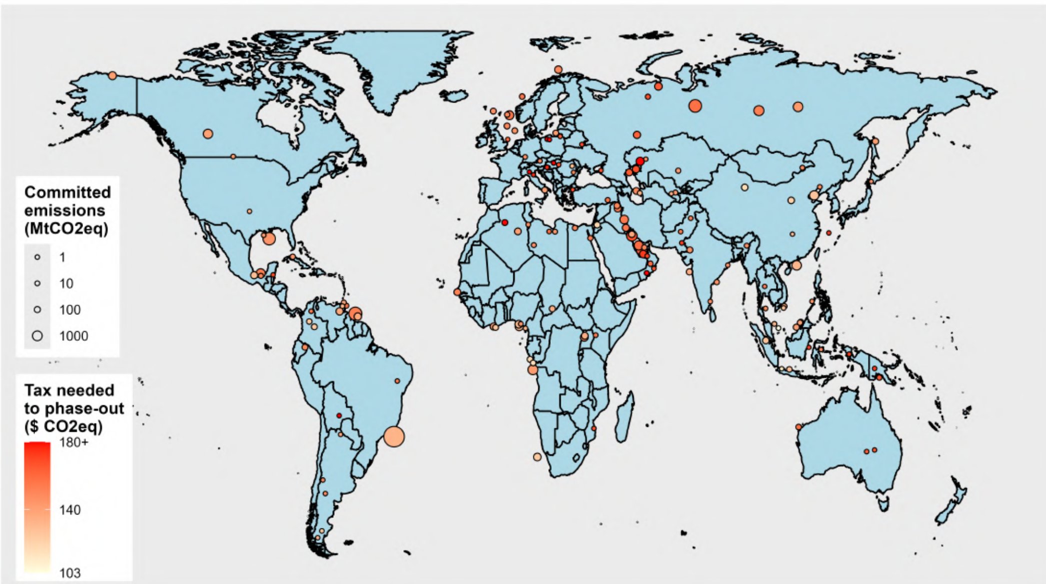


Figure 1.6: Carbon tax required to phase-out projects responsible for overcommitted emissions (aggregated by province).

Notes: Each circle correspond to one province in which overcommitted emissions exist. A province is a level of disaggregation available in Rystad's data situated between the asset-level and the country-level. Circle sizes depend on the quantities overcommitted and colors reflect the average tax needed to phase-out committed emissions right after commitment. The scenario used for this graph is the "No anticipation" one. All computations are made assuming a reference Brent oil price of $80/bbl.

Lock-in premium and naïve overshoot. The extensive overproduction occurring during the delay entails sharp year-to-year increases in the optimal carbon tax needed to meet the 1.5°C objective. Beyond this direct overproduction effect, cumulative investments in new production facilities lead to a downward shift in the supply curve and higher locked-in emissions for any given level of tax. Accounting for investment dynamics therefore further increases the tax required to constrain emissions within the carbon budget. I call this gap between the "naïve" carbon tax and the optimal one the *carbon tax lock-in premium*. It is defined as the difference between (i) the carbon tax that would be sufficient to meet the carbon budget if this drop in asset breakeven prices was ignored, and (ii) the higher tax required once investment-driven lock-in effects are accounted for.

The lock-in premium depends on the initial state of oil production facilities, and builds up as new projects are developed (and their breakeven price decreases). Ignoring the change in the supply curve induced by these new investments, the naïve tax needed in 2030 in order to remain within the carbon budget is set at \$130/tCO2eq. However, implementing this tax from 2030 onwards would still lead to overshooting the budget by more than 18 GtCO2eq (12.5% of the remaining budget in 2030), due to overcommitted emissions from assets developed during the delay. Resolving this residual overshoot caused by infrastructure lock-in requires adding a lock-in premium of \$3/tCO2eq to the tax in order to effectively meet the budget, resulting in an optimal 2030 carbon tax of \$133/tCO2eq[10]. Ignoring infrastructure lock-in thus leads policymakers to systematically underprice carbon, even when the carbon budget itself is correctly identified.

Table 1.1 shows year-to-year change in the carbon tax required to meet the carbon budget, increasing from \$99.3/tCO2eq in 2016 to \$133/tCO2eq in 2030. The carbon tax lock-in premium is indicated for each year, as well as the overshooting of the budget resulting from the implementation of the naïve tax (the percentage between parenthesis is relative to the remaining carbon budget). Results are provided for two periods of delay and investments: the "Baseline" scenario 2016-2030, as well as the 2025-2030. The initial state of existing facilities and production costs in the second period is thus the one observed in 2025.

As outlined in the table, the magnitude of the lock-in premium does not necessarily reflect the real climate cost of implementing the naïve tax. Its maximum level of \$6.4 is reached in 2026 whereas overemissions associated with the naïve tax rank higher in 2030. This can be explained by the fact that the supply curve can be locally flat, with large reserves becoming locked-in at similar tax rates[11].

[10]Note that implementing such tax in 2016 would lead to cumulative emissions equal to 183 GtCO2eq, roughly 48% below the 1.5°C carbon budget. This underscores the strikingly high cost of delaying carbon taxation, even for a 14 years period.

[11]A striking example of this is shown in the case of the 2025-2030 period. The required tax premium in 2028 is close to 0 while being associated with a 7.3GtCO2eq naïve overshoot. This is generally caused by a unique asset with particularly large locked-in reserves.

Tax year	Remaining budget ($GtCO_2eq$)	Lock-in adjusted carbon tax ($\$/tCO_2eq$)	From 2016		From 2025	
			Tax premium ($\$/tCO_2eq$)	Naïve overshoot ($GtCO_2eq$)	Tax premium ($\$/tCO_2eq$)	Naïve overshoot ($GtCO_2eq$)
2016	350.3	99.3	–	–	–	–
2018	321	105	2.6	7.3 (2.3%)	–	–
2020	291.8	109.7	2.4	9.6 (3.3%)	–	–
2022	264.6	112.7	2.1	12.4 (4.7%)	–	–
2024	236	118	3.5	13.3 (5.6%)	–	–
2026	206.9	125.5	6.4	16.4 (7.9%)	0.6	2.7 (1.3%)
2028	177	129.8	2.6	15.5 (8.7%)	<0.1	7.3 (4.1%)
2030	146.7	133	3	18.4 (12.5%)	2.2	10.1 (6.9%)

Table 1.1: Lock-in premium and naïve overshoot by tax implementation year (constant tax).

Notes: The table reports the remaining carbon budget and the optimal carbon tax required to respect it for each implementation year. The tax premium corresponds to the additional carbon tax required due to investment-driven lock-in. Overshoot reports the implied carbon budget overshoot in $GtCO_2eq$, with percentages expressed relative to the remaining budget. "From 2016" and "From 2025" indicate the baseline year from which delay is measured.

The buildup of lock-in. The cost of not accounting for infrastructure lock-in increases at rapid pace, especially over the first few years. The relatively lower buildup afterwards is due to the fact that additional overcommitted emissions from new investments are partly compensated by past overcommitted emissions being effectively emitted. This explains the fast increase in the naïve overshoot when starting from 2025. Overlooking only five years of investments would entail exceeding the budget by more than 10 GtCO2eq, almost 7% of the remaining budget.

Crucially, the overcommitted emissions constituting this remaining overshoot are defined under a much more restrictive counterfactual than the ones presented in the former section. In both cases, overcommitted emissions originate from assets whose development is dependent on the $100/tCO2eq tax being delayed to 2030. However, while committed emissions in the former section corresponded to locked-in emissions in a $100 tax world, the overshoot computed in table 1.1 corresponds to locked-in emissions in a world where the much higher naïve tax is implemented. It is a direct consequence of the deep gap between full project breakeven prices and production breakeven prices outlined in figure 1.6.

1.3.3 Increasing tax

This section presents the main results outlined in the precedent section, this time considering an increasing carbon tax. Under the constraint of a carbon budget, the dynamically efficient carbon tax increases at the discount rate (Lemoine and Rudik, 2017;

Dietz and Venmans, 2019). Additionally, the marginal damage of emissions increases over time (Greenstone et al., 2013). A third argument in favour of such tax is its higher acceptability by the public and the industry, giving it time to adjust. The tax assessed in this section increases at a 2% social discount rate[12]. The choice of a discount rate lower than market rates is justified by uncertainty about the far-distant future growth and interest rates (Stern, 2007; Weitzman, 2007). The declining discount rate literature points to social discount rates ranging from 1% to 2% (Weitzman, 1998; Gollier, 2009), while Nordhaus (2007) justifies using a higher rate, around 3%. I adopt a social discount rate of 2% which lies between these different approaches. Different rates ranging from 1% to 6% are included in the sensitivity analysis.

Overshoot due to the delay. Figure 1.7 reports the overshoot at both the intensive and extensive margins by initial carbon tax level. The carbon tax consistent with the budget from Welsby et al. (2021) is implemented in 2016 at 57.1$/tCO2eq[13]. For this level of tax, the delay entails a significant overshoot of the budget, though smaller than the one computed with a constant tax, amounting to 65 GtCO2eq GtCO2eq in total (18.6% of the budget), of which 36% happens at the extensive margin. This smaller amount can be explained by the relatively low level of tax during the delay, reducing mechanically the overshoot at the intensive margin. The extensive margin is also reduced because most assets developed during the delay would be economically viable in a low tax environment, with the high rentability of oil projects allowing to compensate for development costs in only few years of production. Assets developed during the delay are also phased-out earlier than in a constant tax environment, when the tax gets too high for operations to continue, further reducing the post-2030 overshoot at the extensive margin. Yet, delaying would still entail exceeding the carbon budget by a considerable margin, and allow for the development of many unnecessary oil projects.

Heterogeneity across sources. Figures A.5 and A.5 report the overshoot by supply segment and regions respectively. Most conclusions drawn in the case of a constant tax stay valid when assessing an increasing one. Oil sands and shale concentrate a large share of overproduction at the intensive margin, while deep offshore projects remain a major source of overshoot at the extensive one. North and South America still play complementary roles in explaining the total overshoot, with North American assets overproducing during the delay and South American ones after.

A notable difference is the relatively smoother reduction in emissions induced by increasing the level of tax. All oil and gas producers are affected more uniformly, mainly

[12] All tax levels in this section are expressed in initial 2016 level, regardless of the timing of implementing.

[13] For reference, the tax amounts to $75/tCO2eq in 2030, reaches $100/tCO2eq in 2045 and $200/tCO2eq in 2080.

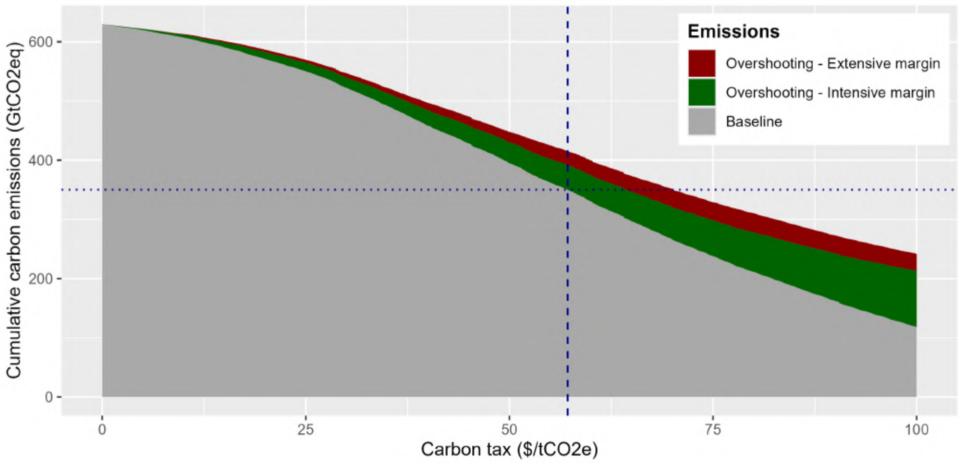


Figure 1.7: Oil carbon budget overshooting from the delay - 2% increasing tax.

Notes: The x-axis is the 2016 level of carbon tax, increasing annually at a 2% rate. The grey area corresponds to the "Baseline" scenario" scenario, i.e. the cumulative emissions if the tax is implemented in 2016. The green and red areas depict the overshoot resulting from delaying the tax until 2030 at the intensive and extensive margins respectively. The dotted horizontal line indicates the oil carbon budget set by Welsby et al. (2021). The dashed vertical line is the corresponding carbon tax needed in 2016 in order to constrain cumulative emissions within this budget. The scenario used to compute the overshoots is the "no anticipation" one. All computations are made assuming a reference Brent oil price of $80/bbl.

in later years of the period when the tax reaches high levels. This translates into a higher reduction in cumulative emissions from regions only marginally affected by the constant tax, such as Middle Eastern countries. Yet, Middle Eastern oil projects do not appear to give rise to overproduction at any margin. A probable explanation is their low production cost and carbon intensity. Most Middle Eastern assets are resilient to tax levels during the delay and only get affected by the policy in later years.

Lock-in premium and naïve overshoot. Table 1.2 reports the carbon tax lock-in premium by year as well as the overshoot associated with implementing the naïve tax. Ignoring investment-driven lock-in from 2016 results in a residual overshoot of the budget of 17.1 GtCO2eq, comparable to the one found with a constant $100/tCO2eq tax. The tax premium required to offset it amounts to 3.7$/tCO2eq, expressed in 2016 levels (5.1$/tCO2eq in 2030). Ignoring the next five years of investments leads to a similar cost to the one computed for a constant tax, around 10 GtCOeq. Overall, although the tax increase rate affects the distribution of stranded assets, the magnitude of the cost resulting from ignoring lock-in effects remains unchanged.

Despite reducing the lifetime of assets, in turn reducing the quantity of committed emissions of newly developed assets, the naïve overshoots remains consequent when considering a 2% increasing tax. A consequence of the increasing nature of the tax is the fact that almost all assets responsible for overcommitted emissions will have a share of cumulative emissions locked-in once the tax is in place. The tax being relatively low in the 2030-2039 decade, most locked-in assets will continue producing, and increasing marginally the level of tax can only induce slightly earlier phase-out. Consequently, this

Tax year	Remaining budget ($GtCO_2eq$)	Lock-in adjusted carbon tax ($\$/tCO_2eq$)	From 2016 Tax premium ($\$/tCO_2eq$)	From 2016 Naïve overshoot ($GtCO_2eq$)	From 2025 Tax premium ($\$/tCO_2eq$)	From 2025 Naïve overshoot ($GtCO_2eq$)
2016	350.3	57.1	–	–	–	–
2018	321	58.3	0.4	2.8 (0.9%)	–	–
2020	291.8	59.3	0.7	5.1 (1.7%)	–	–
2022	264.6	60.4	0.9	8.2 (3.1%)	–	–
2024	236	62.6	2	10.9 (4.6%)	–	–
2026	206.9	64.6	2.5	13.8 (6.7%)	0.4	1.9 (0.9%)
2028	177	67.2	3.2	15.1 (8.5%)	1.3	5.6 (3.2%)
2030	146.7	70	3.7	17.1 (11.6%)	2	10.4 (7.1%)

Table 1.2: Lock-in premium and naïve overshoot by tax implementation year (increasing tax).

Notes: The table reports the remaining carbon budget and the optimal carbon tax required to respect it for each implementation year. Carbon taxes are reported as their 2016 level and increase annually at a 2% rate. The tax premium corresponds to the additional carbon tax required due to investment-driven lock-in, again in 2016 level. Overshoot reports the implied carbon budget overshoot in $GtCO_2eq$, with percentages expressed relative to the remaining budget. "From 2016" and "From 2025" indicate the baseline year from which delay is measured.

requires to assess the depth of lock-in using a different and more restrictive approach compared with the one presented in figure 1.6.

The map in figure 1.8 displays the size of overcommitted emissions from assets undeveloped in the "Baseline" scenario" scenario, aggregated by province, as well as a measure of this depth of lock-in. Points colour denotes the 2016 level of carbon tax required to reduce by 10% the quantity of overcommitted emissions. The geographical distribution of overcommitted emissions is relatively similar to the case of a constant tax. However, the lock-in appears to be even deeper entrenched in the case of an increasing tax. To the exception of Canadian oil sands and a few relatively marginal onshore projects, the vast majority of locked-in emissions are resilient to carbon taxes higher than 70$/tCO2eq. Implementing a 70$/tCO2eq tax (equivalent to the lock-in adjusted tax necessary in 2030) would leave 82% of overcommitted emissions unchanged, and a 80$/tCO2eq one 67%. Reserves are even more locked-in in deep offshore projects with 86% and 75% of overcommitted emissions resilient to these two levels of tax.

1.4 Sensitivity

The central quantitative results presented in section 1.3, a naïve overshoot ranging from 17 to 19 GtCO2eq and a tax lock-in premium of 3 to 4$/tCO2eq, rest on a specific combination of assumptions regarding investor behaviour, cost structure, carbon intensity

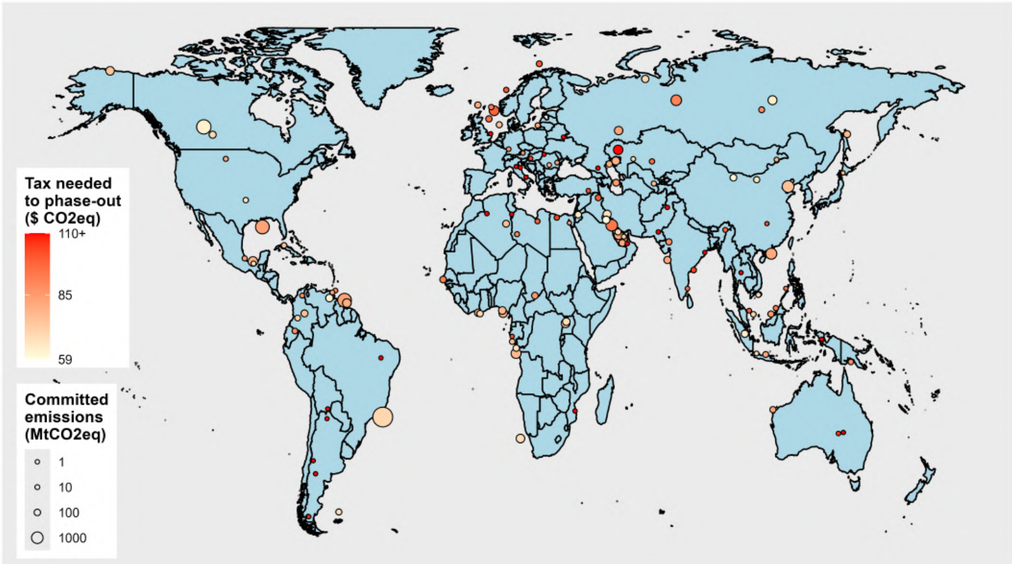


Figure 1.8: Carbon tax required to reduce overcommitted emissions from locked-in projects by 10% (aggregated by province).

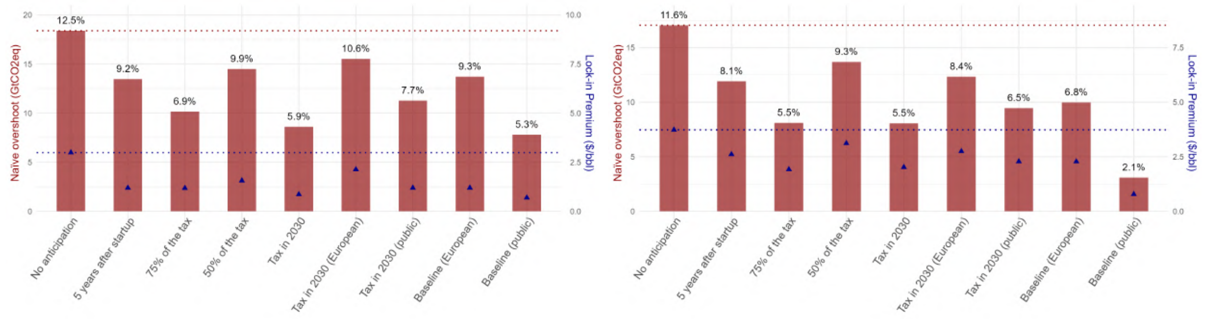
Notes: Each circle correspond to one province in which overcommitted emissions exist. A province is a level of disaggregation available in Rystad's data situated between the asset-level and the country-level. Circle sizes depend on the quantities overcommitted and colors reflect the province average of the 2016 increasing carbon tax required to reduce locked-in emissions in a 57.1$/tCO2eq world by 10%. The scenario used for this graph is the "No anticipation" one. All computations are made assuming a reference Brent oil price of $80/bbl.

and carbon tax design. This section evaluates the sensitivity of these results to a variety of assumptions. Results are reported for both the constant and increasing tax cases, and are separated by category of assumptions. The first subsection focuses on the effect of anticipation scenarios, the second on assumptions on costs definition and the computation of NPV, and the third on assumptions on carbon intensities. In the fourth subsection, the impact of various carbon tax increase rate is studied. Finally the last subsection incorporates a time-varying pass-through of the carbon tax between producers and consumers. Across all configurations, the naïve overshoot in 2030 remains significant, and the lock-in premium is positive throughout.

1.4.1 Scenarios of anticipation

Figure 1.9 reports the consequences of various investors anticipations scenarios on the 2030 tax premium and naïve overshoot. As expected, investors anticipating a tax to be implemented at some point always reduces the cost of ignoring lock-in effects in both the cases of a constant tax (panel a) as well as an increasing one (panel b). Large heterogeneity exists across scenarios, but the overall results follow the logic highlighted in figures 1.3 and 1.4. Screening fully projects but with an underestimated level of tax does drive out of the market a non-negligible share of projects that would otherwise participate in the naïve overshoot of the budget. The scenario closest to the observed behaviour of firms ("50% of the tax") still leads to a sizeable residual naïve overshoot, close to 15 GtCO2eq whatever the tax increase rate. Expecting the tax to be implemented in 2030 dramatically reduces investments realized in later years of the delay, thus reducing significantly the cost of ignoring lock-in effects.

Interestingly, testing heterogeneous beliefs across firms leads to differing conclusions whether the tax increases or not. Although the relative effect of each scenario on the naïve overshoot remains the same, its magnitude is consistently reduced when considering an increasing tax.



(a) Constant tax (b) Increasing tax

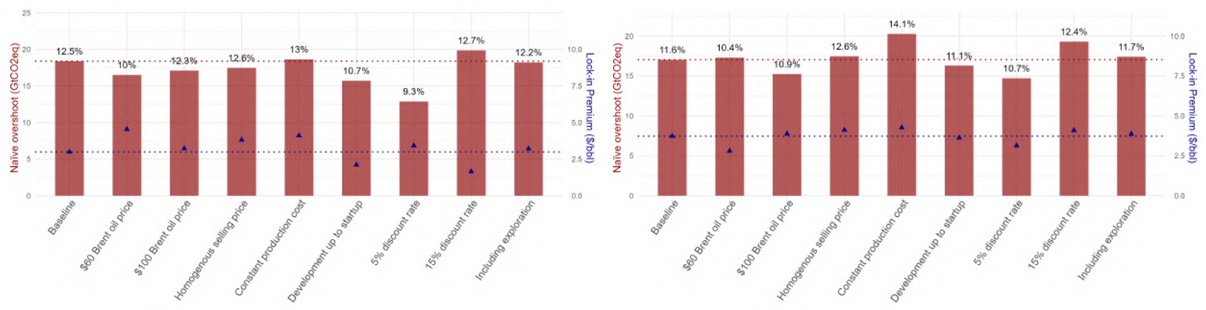
Figure 1.9: Sensitivity to scenarios of anticipations.

Notes: Each red bar reflects post-2030 overemissions resulting from the implementation in 2030 of the naïve carbon tax. Above each bar is indicated the corresponding share of the remaining carbon budget in 2030. The red dotted line extends the baseline results. Points correspond to the carbon tax lock-in premium, with the blue dotted line extending the baseline lock-in premium.

1.4.2 Costs and NPV calculations

I shift my sensitivity analysis to focus on alternative assumptions on development and extraction costs, as well as discount rates. I first test the sensitivity of my results to different selling prices: I use a \$60/bbl and \$100/bbl reference Brent oil price instead of \$80/bbl, and assume in a third configuration that all oils produced are sold at an homogeneous price (thus ignoring asset-specific discounts). I then use a constant production price, defined as the breakeven price of production over the asset full lifetime (still excluding development costs). I then reduce the amount of capital expenses flagged as development ones by excluding facility and well capex over the first two years of production, otherwise included in development expenses. The next two configurations explore the impact of using a 5% and 15% discount rates when computing the net present value of projects. Finally, the last configuration includes all not-yet-discovered assets in 2025, and assimilate their exploration expenses to development ones.

Figure 1.10 reports the results for each of these configurations in the case of a constant and increasing tax. Overall, the level of the naïve overshoot remains consistent with my baseline results for each of these alternative setups. Unsurprisingly, increasing the discount rate strengthens the gap between breakeven prices including and excluding development expenses, strengthening in turn the lock-in premium and the naïve overshoot. Similarly, reducing the amount of capital expenses included in development expenses mechanically reduces this gap, thus reducing the magnitude of the results. One could expect the inclusion of exploration costs and not-yet-discovered assets in the analysis to increase the total cost of lock-in effects, yet the results do not appear to change significantly. A major reason is that most assets discovered after 2025 will not be developed before the end of the delay, and would therefore be stopped by the introduction of the tax in 2030. They can thus not participate in the buildup of the naïve overshoot.



(a) Constant tax

(b) Increasing tax

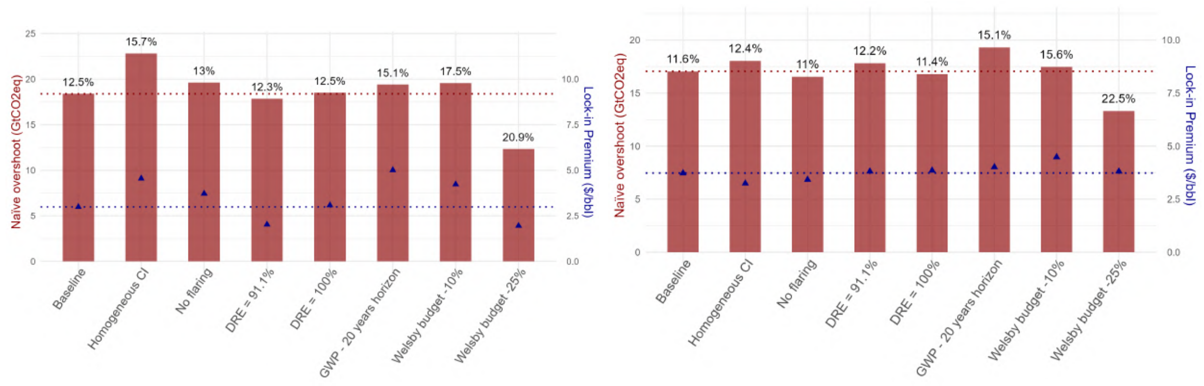
Figure 1.10: Sensitivity to assumptions on costs and NPV calculations.

Notes: Each red bar reflects post-2030 overemissions resulting from the implementation in 2030 of the naïve carbon tax. Above each bar is indicated the corresponding share of the remaining carbon budget in 2030. The red dotted line extends the baseline results. Points correspond to the carbon tax lock-in premium, with the blue dotted line extending the baseline lock-in premium.

1.4.3 Carbon intensities

I further investigate the sensitivity of my results to changing assumptions on the carbon intensity of assets. In the first set of CI, I assume homogeneous CI across assets and assign to all of them a reserve weighted average CI equal to 468 kgCO2eq/bbl. The second set of CI assumes that all assets stop flaring at no cost. The third and fourth sets of CI assume levels of flaring efficiency (DRE, i.e. destruction removal efficiency, see appendix A.2) of 91.1% and 100% instead of the baseline 96.1%. This greatly affects the quantity of methane being emitted from flaring assets. In the next set of carbon intensity, I use global warming potentials assuming a 20-year horizon instead of the baseline 100-years horizon. This implies sharp increases in CI as well as reordering across assets due to the higher global warming potential of methane associated with this assumption. Finally, I keep the baseline carbon intensity but use a reference carbon budget reduced by 10% and 25%.

Panel 1.11 reports the results associated with the differing sets of CI and carbon budgets. Results remain in line with my baseline computations for all sets of CI. Interestingly, homogenizing CI across assets leads to higher tax premium and locked-in emissions. This could indicate that on average, assets becoming locked-in solely because of the delay in taxation have lower carbon intensity than other assets. Regarding carbon budgets, reducing the reference one from Welsby et al. (2021) by 10% leaves my results unchanged. A larger 25% reduction leads to smaller overcommitted emissions in 2030. A possible explanation is the relatively high level of tax needed in 2030 to meet the budget, largely reducing the absolute value of overcommitted emissions per asset. This reduction is not proportional to the reduction of remaining budget in 2030, with a naïve overshoot representing more than 20% of the remaining budget, much higher than in the baseline.



(a) Constant tax

(b) Increasing tax

Figure 1.11: Sensitivity to assumptions on carbon intensities.

Notes: Each red bar reflects post-2030 overemissions resulting from the implementation in 2030 of the naïve carbon tax. Above each bar is indicated the corresponding share of the remaining carbon budget in 2030. The red dotted line extends the baseline results. Points correspond to the carbon tax lock-in premium, with the blue dotted line extending the baseline lock-in premium.

1.4.4 Carbon tax increase rate

Although results presented in section 1.3 seem to indicate that the magnitude of the naïve overshoot does not differ significantly whether the policy instrument is a constant or an increasing one. However, one would expect the rate of increase to affect the results in two separate ways. On the one hand, an increasing tax implies a relatively lower tax environment in early periods, which discourage less investments as compared to the higher constant tax scenario. This translates into more similar investments realized in the baseline scenario as compared to the "No anticipation" one, reducing the cost from lock-in effects. On the other hand, the increase rate of the tax affects the quantities of emissions committed by newly developed projects: assets phased-out shortly after the delay in the constant tax scenario can continue producing longer (as long as the increasing tax is below the constant one), and other assets are phased-out earlier as the tax gets higher in later periods.

Figure 1.12 reports the tax lock-in premiums and naïve overshoots for carbon tax increasing at different rates, from 0% annually to 6%. Going from 0% to 1% increases the cost from lock-in effects, meaning that the second effect is positive and dominates the first one. For all other configurations, the naïve overshoot and lock-in premium decrease as the increase rate gets higher, being more than halved for a 6% increasing tax as compared to the constant one. The first effect dominates: when the social discount rate is high, the carbon tax starts at a level low enough to only marginally reduce investments realised in new oil-producing projects.

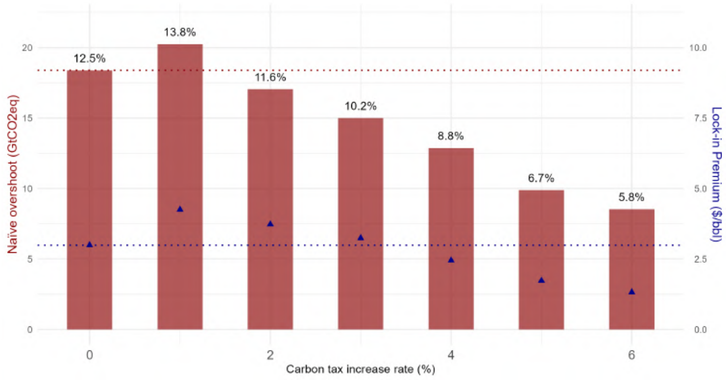


Figure 1.12: Sensitivity to carbon tax increase rates.

Notes: Each red bar reflects post-2030 overemissions resulting from the implementation in 2030 of the naïve carbon tax. Above each bar is indicated the corresponding share of the remaining carbon budget in 2030. The red dotted line extends the baseline results. Points correspond to the carbon tax lock-in premium, with the blue dotted line extending the baseline lock-in premium.

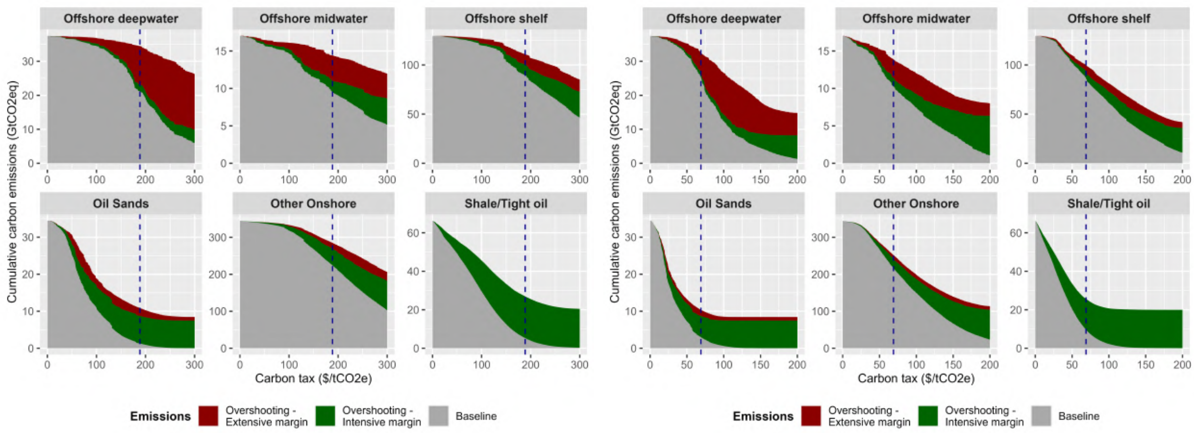
1.4.5 Time-varying pass-through

The carbon tax assessed up to now was assumed to be fully borne by producers. This entails that the underlying assumption was that there is no pass-through from producers to consumers, or that this pass-through is constant over time. This assumption is relaxed in this subsection, by integrating time-varying pass-through rates into the model. I use the evolution of pass-through rates computed by Heal and Schlenker (2019) in the case of a constant tax and in the case of a 2% increase tax. While the model used in their paper is different from mine, the data on oil fields used is the same. Authors report in both cases an initial high share of the tax being supported by consumers, the cost being progressively passed on to producers over around 60 years. The initial share of the tax borne by producers is 30% in the case of a constant tax and 60% when considering a tax increasing exponentially at a 2% rate. I thus use these initial shares of the tax borne by producers, and assume they increase exponentially at a rate such that 60 years after the tax is introduced, producers bear its full cost. Equation 1.2 is thus adapted such that:

$$\theta_i \; Q_i^\tau(T) = \theta_i \; \frac{\sum_{t=T_\tau}^{T} \frac{q_{it} \;\; (1+r)^{t-2016}}{(1+\delta)^{t-T_0}} \;\; min(\Lambda_{T_\tau} \;\; (1+\lambda)^{t-T_\tau},1)}{\sum_{t=T_1}^{T} \frac{q_{it}}{(1+\delta)^{t-T_0}}}$$

with Λ_{T_τ} the initial share of the tax borne by producers when it is implemented in T_τ, and λ its increase rate such that after 60 years the tax is fully borne by producers (around 2.03% in the case of a constant tax and 1.16% for a 2% increasing one). I assume the share of tax borne by producers can not exceed 100%.

This setup differs from the case of an increasing tax for one main reason. The pass-through rate and its evolution starts *when the tax is introduced*, and not in 2016. Consequently, the level in 2030 of a delayed tax is not directly comparable to the one of a tax implemented in 2016. The share of the tax borne by producers in 2030 will be much lower in the first case as compared to the second. This adds a second dimension to the cost of delaying taxation, since it delays the shifting of its burden to producers further into the future.



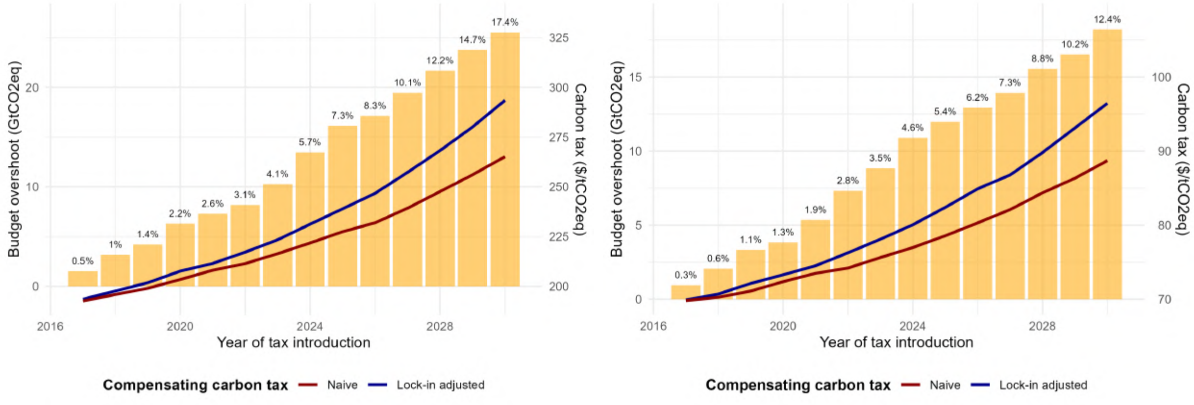
(a) Constant tax (b) Increasing tax

Figure 1.13: Overshoot by supply segment accounting for time-varying pass through.

Notes: Each panel corresponds to an oil supply segment. For each panel, the x-axis is the level of carbon tax. The grey area corresponds to the "Baseline" scenario, i.e. the cumulative emissions if the tax is implemented in 2016. The green and red areas depict the overshoot resulting from delaying the tax until 2030 at the intensive and extensive margins respectively. The dotted horizontal line indicates the oil carbon budget set by Welsby et al. (2021). The dashed vertical line is the corresponding carbon tax needed in 2016 in order to constrain cumulative emissions within this budget. The scenario used to compute the overshoots is the "No anticipation" one. All computations are made assuming an oil price of $80/bbl.

Figures 1.13 shows the overshoot of the budget by supply segment for both the constant (panel a) and the increasing tax (panel b). Overall, overshoots are higher by around 13% at both margins compared with results presented in section 1.3. This is mainly explained by the fact that the passing through of the tax starts later in time, thus creating a difference in emissions committed by the same projects developed during the delay.

This translates into a higher cost stemming from the lock-in of new infrastructures. Figure 1.14 shows the progressive buildup of the naïve overshoot, as well as the tax premium (thus corresponding to tables 1.1 and 1.2). As expected, the levels of necessary carbon tax to meet the 1.5°C budget are particularly high. The lock-in premium is also massive, reaching 28.4$/tCO2eq in 2030 in the constant tax case. The magnitude of the premium needs to be put into perspective given the role played by the delayed passing through: whether is starts in 2016 or in 2030, the same naïve tax, equal to 265$/tCO2eq in 2030, translates into a 25.8$/tCO2eq difference in what is actually borne by producers. Accounting for this fact, the magnitude of the lock-in premium is reduced to around 3.6$/tCO2eq, consistent with results found in other configurations. As for the naïve overshoot, its magnitude is consistently higher than in other configurations.



(a) Constant tax

(b) Increasing tax

Figure 1.14: Naive overshoot and tax premium accounting for time-varying tax pass-through.

Notes: For each year reported in x-axis, the size of bar reflects the residual overshoot resulting from implementing the naïve tax (left y-axis), with above its corresponding share of the remaining budget. The red and blue lines report the levels of the naïve and lock-in adjusted carbon taxes (right y-axis) required for each year to meet the carbon budget set by Welsby et al. (2021). The difference between the two lines corresponds to the lock-in premium. The scenario used to compute the overshoots is the "No anticipation" one. All computations are made assuming an oil price of $80/bbl.

1.5 Discussion and policy implications

Delaying induces more stranded assets. Findings presented in section 1.3 highlight the rapid pace of transition away from oil needed to meet the 1.5°C objective, and the massive cost resulting from delaying the implementation of the required ambitious climate policies. Compensating for the overproduction during the delay will translate into a vast number of stranded assets. Continued investments in new facilities further reinforces this. Among newly built assets, some will constitute stranded assets (and in turn losses for firms) when climate policies will finally be put in place. However, as shown in figure 1.8, a large share of these assets exhibits deep lock-in and will therefore continue producing once the tax is in place. This leads to additional stranded assets since cumulative emissions from these assets will need to be compensated through the early phase-out of already existing infrastructures.

The case for supply-side policies. Targeted policies aiming at alleviating this source of additional cost could thus be justified in terms of efficiency. The main findings of this paper align with the literature advocating for supply-side carbon policies. As Erickson et al. (2018) argue, supply-side policies can act as complement to demand-side policies, and provide major benefits among which its direct and certain impact on fossil fuel supply and subsequent use. A major argument raised by Green and Denniss (2018) in favour of such policies is its greater potential to mobilise public support, in reason of its higher perceived benefits and distributional fairness as compared to consumption based policies. The methods used in this paper support this claim by showing the feasibility of precise accounting of expected benefits from supply-side policy, as well as the identification of potential winners and losers of the tax at the project level.

Results provide additional arguments to the growing literature supporting bans on new oil projects (Green et al., 2024), showing that targeting deepwater projects specifically could prove particularly beneficial. Indeed, worldwide carbon taxation imposed on oil producers as envisioned in the paper seems highly unlikely in the near future. At the international level, the role played by international organisations does not play in favour of supply-side policies in addressing climate change. As Van Asselt (2021) puts it, the "United Nations Framework Convention on Climate Change (UNFCCC) territorial accounting for greenhouse gas emissions emphasizes fossil fuel consumption over supply, so countries are incentivized to pursue demand-side policies to fulfil international agreements". However, this shortcoming is not the end of the road for production-based policies: a substantial share of the cost arising from infrastructure lock-in could be alleviated through targeted action.

Carbon pricing and fiscal regimes in producing countries. A further complication for supply-side policy design is the interaction between carbon pricing and existing fiscal regimes in producing countries. This paper abstracts from government take by assuming that, in the presence of a worldwide carbon tax, producing governments would renegotiate royalty structures to avoid stranding assets from which they collect a rent. Yet, producing-country governments face a Ramsey taxation problem: the optimal extraction tax balances the social return to resource rents against the production distortion it creates (Daubanes and Lasserre, 2023).

In principle, a government already at its Ramsey optimum could respond to a worldwide carbon tax by adjusting fiscal regimes so as to leave the equilibrium unchanged, substituting carbon tax revenue for royalty revenue. However, fully restoring the pre-tax equilibrium would in some cases require subsidising development or production. Assets that become uneconomical under the carbon tax (either because the project does not generate sufficient revenues to offset development costs or because production costs become too high towards the end of their lifetime) could only be kept active if the State offsets the carbon tax beyond what its own rent allows. Fully absorbing the tax in this way would therefore imply higher cumulative extraction and development than estimated in this paper. Conversely, contractual rigidities in existing production-sharing agreements and concession contracts may prevent renegotiation in the short term, causing the carbon tax to pile on top of the existing fiscal regime and further constrain investment and production beyond what the model predicts.

The assumption made in this paper implicitly lies between these two extremes: governments are assumed to absorb the carbon tax to the extent necessary to keep viable assets in production (potentially capturing a higher rent in the process) without subsidising uneconomical ones, so that assets not resilient to the carbon tax are either phased out earlier than in a no-tax world or simply remain undeveloped. Under this assumption, the overcommitted emissions identified in the paper represent an intermediate scenario, with the full-absorption case providing an upper bound and the full-rigidity case a lower bound on the true extent of infrastructure lock-in. These interactions further suggest that supply-side bans on new field development may in some cases be a more administratively tractable instrument than relying on carbon pricing to overcome the fiscal inertia of producing States.

Targeted policies. My results show that carbon lock-in is much deeper in assets concentrated in specific regions and/or using specific extracting technologies, making potential national or sub-national actions effective at mitigating the cost of delaying global action. Figure 1.15 shows post-2030 overcommitted emissions, aggregated by province, resilient to the 2030 2% increasing naïve tax, originating from investments that would not have been carried out should the 57.1\$/tCO2eq tax be implemented in 2016. With this restrictive definition, the total 15 GtCO2eq overcommitted emissions originate from 548 assets scattered around the globe. Despite large over-investments in the past 10 years, future investments over the next five years are crucial. As highlighted in the figure, the cost from carbon lock-in has yet to become fully locked-in in many regions. In terms of distribution, carbon lock-in effects occur in most oil producing countries, with offshore projects representing more than 75% of the total (half of which comes from deepwater projects). Identifying specific assets responsible for it, I find that it is highly concentrated, with only 31 assets making up for 50% of the total. Deepwater offshore projects play a major role: assets along the Brazilian and Guyanese coasts, as well as in the Gulf of Mexico, represent alone more than 30% of total overcommitted emissions. Issuing targeted bans towards these problematic projects could go a long way in avoiding some of the cost arising from infrastructure lock-in.

The effectiveness of public action. Convincing large producing countries to issue such bans might prove difficult, with OPEC countries being the leading opposition to any moratorium on fossil fuel extraction and new field development (Lujala et al., 2022), but smaller initiatives could prove effective. As Piggot (2018) and Carter and McKenzie (2020) explain, most bans issued thus far have originated from public mobilisation

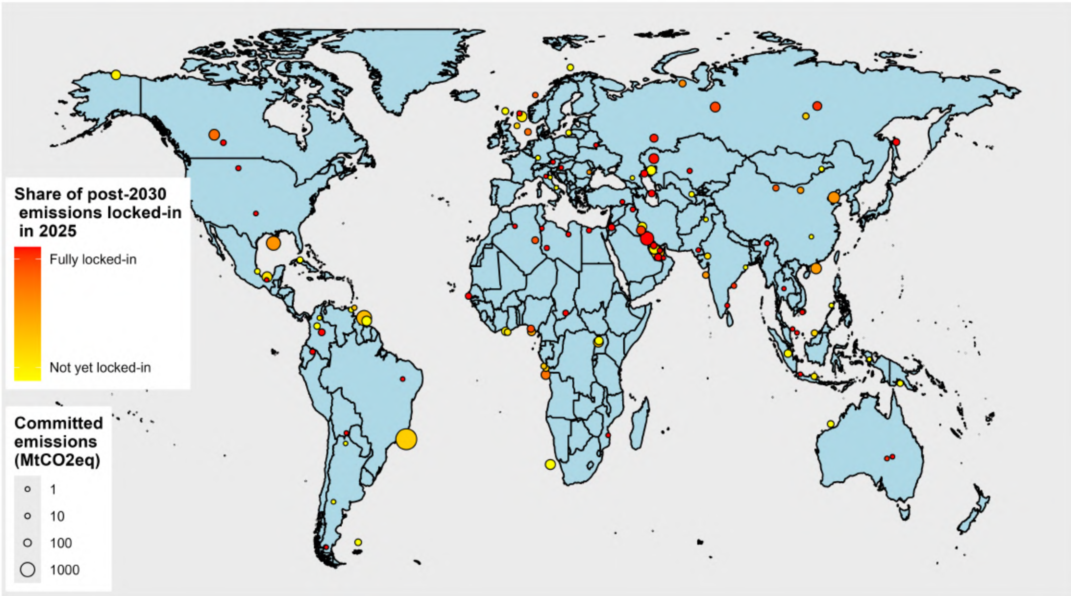


Figure 1.15: Overcommitted emissions aggregated by province (increasing tax).

Notes: Each circle correspond to one province in which overcommitted emissions exist in the "no anticipation" scenario. A province is a level of disaggregation available in Rystad's data situated between the asset-level and the country-level. Circle sizes depend on the quantities overcommitted and colors depend on the degree of commitment (share of the reserves committed) of the associated quantities as of 2025. Committed emissions from a given asset are attributed to the year from which the full development of this asset becomes inevitable. The level in 2016 of the delayed tax used to identify projects responsible for overcommitted emissions is 57.1\$/tCO2eq. The level of 2030 tax used to compute the quantities of overcommitted emissions per project is 66.3\$/tCO2eq. The scenario used for this graph in the "no anticipation" one. All computations are made assuming a reference Brent oil price of \$80/bbl.

efforts focused on specific projects. These targeted initiatives have demonstrated higher ability of public action to more directly influence political decisions. Given the important concentration of overcommitted emissions in a small number of projects, actions by the public directed to specific projects could have tremendous impact. Among the more recent successful public mobilisations is the definitive cancelling of the Keystone XXL pipeline. Erickson and Lazarus (2014) previously estimated that the building of the pipeline could embed up to 110 MtCO2eq of annual emissions in reason of increased Canadian oil sands production. On the opposite end, an unsuccessful instance of public initiative can be found with the Alaskan Willow project, extensively discussed in 2022 in the American public debate. Finally approved by the Biden administration in 2023, this project embeds 238 MtCO2eq of overcommitted emissions over the next 25 years, resilient to the higher levels of tax required in 2030 to meet the carbon budget. It is responsible for 1.4% of the 2030 naïve overshoot computed in table 1.2.

Public action on larger oil projects could prove even more effective: the deepwater offshore Buzios project in Brazil concentrates 8.3% of the naïve overshoot. More than 60% of deepwater projects developed during the delay are majoritarily owned and operated by majors (BP, Chevron, Eni, ExxonMobil, Shell and TotalEnergies), and their development is often heavily dependent on technologies mastered by these firms. Developed countries could impose a ban on these specific projects, and incentivise major firms to redirect the important financial means required to develop these projects towards cleaner development opportunities for the corresponding countries. This leverage is further reinforced by the results from section 1.3: screening projects with the full carbon tax reduces the overshoot at the extensive margin by a striking 84% for the subset of assets with publicly-listed firm shareholders, highlighting the potential of investor pressure as a complementary channel.

Energy transition and equity. Results also provide insights on the path towards an equitable and fair transition. As explained by Pye et al. (2020), “prioritising equity criteria could result in high-cost reserves and those that require new infrastructure being developed, while low-cost reserves and those with infrastructure in place are phased out”. Focusing on the key sources of overshoot at the intensive and extensive margins could allow to resolve this tension between equity and cost-efficiency.

On the one hand, overproduction during the delay is mainly due to North American production, in particular from shale and oil sands. Early action from the more developed countries could be justified in terms of global distributive justice (Lenferna, 2018), and would reap important returns in terms of carbon emissions abatement. Localised actions in specific areas (e.g. Alberta shale and oil sands in Canada, Texas and North Dakota shale in the US) would lead to a significant reduction in overemissions during the delay while being constrained to limited jurisdictions. On the other hand, the deep lock-in concentrated in offshore projects largely owned and operated by major firms headquartered in developed countries creates a complementary lever: redirecting the considerable financial resources currently committed to these projects towards renewable development in oil-dependent countries would address both the efficiency and the distributional dimensions of the transition simultaneously.

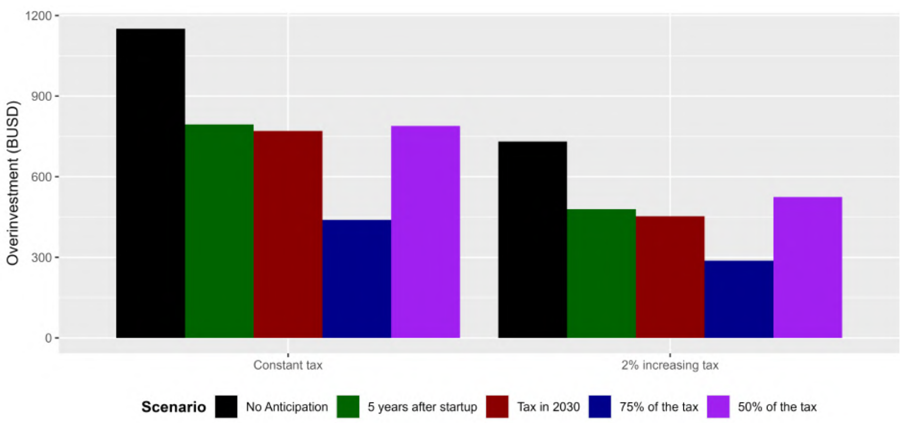


Figure 1.16: 2025 discounted total over-investment by scenario.

Notes: Each bar reports the 2025 discounted sum (2% discount rate) of investments in oil projects dependent on the tax being delayed. The level of 2016 delayed tax used to identify over-investments is 99.3$/tCO2eq. All computations are made assuming a reference Brent oil price of $80/bbl.

The opportunity cost of lock-in. Not only does over-investment caused by the delay result in more stranded assets, and in turn losses for oil and gas firms as well as governments, but it also uses considerable financial resources which could be otherwise used to develop renewable energy. Figure 1.16 reports the sum of these inefficient investments carried out because of the delay, discounted as of 2025. In the case of a constant tax, over-investments represent more than 80 BUSD annually, for a total of 1150 BUSD. For reference, this level of annual inefficient investments worldwide is equivalent to 75% of total investments in renewables in Europe in 2023 (International Energy Agency, 2024). In the case of an increasing tax, the carbon tax is lower during the delay, making more assets economically viable even in presence of the tax. Total over-investment is reduced but remains considerable, amounting to 730 BUSD. Alternative sets of anticipations by investors can reduce this figure by up to 60% (scenario of tax underestimated by 25%). Nonetheless, directing these financial resources towards building renewables capacities, especially in oil-dependent countries, could greatly accelerate the pace of the transition.

1.6 Conclusion

This study quantifies the climate consequences of delaying carbon taxation in the oil sector, revealing that postponing the implementation of a constant 99.3$/tCO2eq tax (respectively a 2% increasing 57.1$/tCO2eq), aligned with the 1.5°C target, from 2016 to 2030 results in a 33% overshoot (respectively 19%) of the oil carbon budget. The overshoot arises from two distinct mechanisms: on the one hand, overproduction occurs during the delay, concentrated in high-cost, high-emission assets such as Canadian oil sands and U.S. shale. On the other hand, long-term infrastructure lock-in builds up, driven by capital-intensive offshore projects that are costly to develop but cheap to operate once developed. Overcommitted emissions from locked-in assets are highly concentrated, with deepwater projects along the coasts of Brazil, Guyana and in the Gulf of Mexico contributing disproportionately. These findings entail important policy implications. While global supply-side climate policies face significant opposition, my results indicate that targeted interventions focusing on specific technologies (such as deepwater projects) or specific regions could yield substantial emissions reductions. Strategies aiming at counteracting the key sources of over-emissions can be aligned with fairness and equity considerations in transitioning away from fossil fuels.

The analysis introduces the concept of carbon tax lock-in premium: the additional tax premium required to offset the downward shifts in the supply curve due to cumulative investments in new production facilities. Ignoring these lock-in effects leads policymakers to systematically underprice carbon, despite correctly defining the remaining carbon budget. For a constant tax as well as an increasing one, implementing a naïve tax in 2030 would result in a 17 to 19 GtCO2eq overshoot of the budget, originating in emissions taking place after 2030. Crucially, this residual overshoot tends to buildup at a rapid pace: failing to account for only the next five years of cumulative investments in new production facilities would result in overshooting the carbon budget by over 10 GtCO2eq. Even short-term delay in policy action translates into costly long-term commitments. By highlighting the irreversible nature of infrastructure lock-in and the cumulative cost of policy delays, this study highlights the necessity of supply-side climate policies, not designed as a substitute for carbon pricing, but as a complementary tool to mitigate long-term carbon lock-in effects.

Several limitations of this analysis point to directions for future research. First, the paper focuses exclusively on the oil sector, and the results should be interpreted under the implicit assumption that comparable policies are applied simultaneously to gas and coal. To the extent that a carbon tax on oil alone induces substitution toward other fossil fuels, the budget overshoots documented here represent a lower bound on the true cost of delay. Second, the model abstracts from the Green Paradox mechanism: producers anticipating future taxation may be incentivised to invest earlier in new producing facilities, thereby amplifying overproduction during the delay and reinforcing lock-in. The results should therefore be read as conservative estimates of the full cost of policy inaction. Additionally, the model abstracts from the Ramsey taxation problem faced by resource-owning governments: in practice, producing States optimise their fiscal regime to capture extraction rents, and the interaction between this optimal fiscal structure and an externally imposed carbon price is non-trivial. Exploring the joint determination of optimal resource taxation and carbon pricing in a world of delayed policy remains an important direction for future work. Finally, while the model quantifies the long-term cost of delayed policy implementation, it does not endogenize the political economy mechanisms that generate such delays. Findings suggest that ignoring lock-in effects likely leads to underestimating the costs of postponing the carbon policy. Future research could usefully compare the magnitude of the additional cost induced by carbon lock-in to the political constraints explaining the limited ambition of currently observed climate policies.

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Chapter 2

The Climate-wise Values of Oil

This chapter was co-authored with Renaud Coulomb (CERNA, Mines-PSL) and Fanny Henriet (CNRS, AMSE)

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2.1 Introduction

Achieving global climate goals requires a rapid decline in oil production and consumption (Meinshausen et al., 2009; McGlade and Ekins, 2015; Welsby et al., 2021), with major implications for the value of reserves, firms' profits, and governments' fiscal revenues.[1] While these aggregate supply-side effects are well recognized, much less is known about the profound heterogeneities they generate: some deposits, firms, and countries lose heavily, while others are comparatively less affected.

This paper addresses three central questions: i) How does restricting the amount of extractable carbon influence the social value of oil across different deposits? ii) Which oil deposits should be extracted or left undeveloped under an optimal policy? iii) What are the implications of carbon pricing for oil firms and resource-dependent governments?

To answer these questions, we calibrate a global simulation model using deposit-level data from Rystad Energy, covering approximately 21,031 oil deposits. The model relies on microdata on deposits' finite reserves, private extraction costs, estimates of carbon intensities, and capacity constraints. At the core of our approach is the recognition that oil deposits are highly heterogeneous in both extraction costs and carbon intensities. The social cost of extracting oil has two dimensions: direct production costs borne by firms and the environmental cost associated with GHG emissions from extraction. Both direct production costs and emissions vary substantially across deposits due to differences in oil quality, geological conditions, and extraction technologies (Masnadi et al., 2018).[2] Our contribution is to document, quantify, and explain the heterogeneous distributive effects of climate policies on the supply-side of oil, combining unprecedented microdata with a theoretical framework that clarifies the underlying mechanisms.

Our analysis documents these heterogeneous effects across three distinct scales and oil value metrics. First, we study deposit-level heterogeneity in a first-best social planner framework. We find that the impact of carbon pricing on the social value of oil is non-monotonic with respect to deposits' carbon intensity. Paradoxically, carbon taxation does not uniformly penalize the most polluting assets; instead, some high-intensity deposits experience valuation gains or attenuated losses, while cleaner substitutes may face significant depreciation.

Second, using a market equilibrium approach, we look into firm-level exposure to carbon pricing. While aggregate industry profits decline, the impact is sharply differentiated by the composition of company portfolios of deposits.

[1] These shifts are particularly consequential for oil-producing countries, where resource rents constitute a major economic pillar. For instance, in 2014, oil revenues represented 23.5% of GDP in Middle Eastern and North African countries.

[2] For example, producing a barrel of oil from Canadian bitumen emits approximately twice the GHGs of a barrel from Saudi light crude.

Finally, we analyze country-level fiscal outcomes. Fiscal impacts remain highly divergent even under a "producer-favorable" regime where oil-exporting nations capture global carbon revenues associated to their oil consumption. For some, tax revenues can offset—or even exceed—the decline in royalties and income taxes under moderate carbon prices, while for others, losses remain substantial. This heterogeneity reflects differences in extraction costs, carbon intensities, and fiscal structures across producers.

These results suggest that the energy transition does not merely induce a transfer of wealth from producers to consumers; rather, it reconfigures the internal hierarchies of the global oil sector. By identifying the emerging "winners" and "losers", our framework elucidates the political economy of climate policy. Understanding the distributional impacts of carbon pricing is crucial for assessing the incentives of countries in climate negotiations, firms' lobbying strategies, and broader debates on compensatory mechanisms for resource-dependent economies, such as OPEC nations.

Our analysis follows four steps.

We first analyze how changes in the social cost of carbon (SCC), required to meet a given carbon budget, affect the social value of oil across deposits in a first-best framework. We find substantial heterogeneity in the effects of increasing the SCC, with some deposits experiencing increased social value while others—sometimes cleaner ones—see declines. Importantly, we document that variations in social value are weakly correlated with carbon intensity.

Second, to explain these paradoxes, we develop a stylized model of intertemporal extraction that highlights two forces in tension: a competition effect, which favors cleaner deposits, and a timing effect, which can instead mitigate losses—or generate gains—for some dirtier deposits when taxation slows extraction. Building on the frameworks of Chakravorty et al. (2006, 2008) and Coulomb and Henriet (2018), our model incorporates deposit-level heterogeneity in both carbon intensity and extraction costs. We find theoretical conditions under which, counterintuitively, the social value of oil from certain deposits may increase under higher carbon taxes, while the value of cleaner deposits may decline. This framework provides the micro-foundations necessary to interpret our simulation results.

Third, we take a market-oriented approach to analyze how carbon pricing influences corporate profits and government fiscal revenues. Beyond carbon taxation, at least two key factors drive production away from a first-best framework: distortive taxes, such as production levies, and market power.

To capture these dynamics, we compile a comprehensive dataset of global oil tax regimes, detailing royalty rates, government take, income taxes, and other production-related levies for each deposit worldwide. We integrate these pre-existing production taxes into our model to account for their distortive effects on market equilibrium. This allows us to move beyond an idealized first-best policy framework and instead examine a more realistic market response to carbon taxation in the presence of fiscal distortions. Additionally, to reflect OPEC's strategic supply management, we incorporate capacity constraints that represent potential voluntary production limits. These constraints are calibrated based on observed production levels.

We begin by validating our model's ability to replicate the historical production split across countries. We then compute how revenues are distributed between governments and corporations under varying tax scenarios, assuming that existing distortive tax structures remain unchanged. We find that carbon taxation affects after-tax corporate profits in complex ways, echoing our findings regarding the impact of carbon cost on oil social value. For instance, while some high-carbon deposits see increased profits after taxation, others cleaner deposits face steep declines, complicating the policy incentives for both companies and governments. Yet, with large tax increases, such as from \$0 to \$200/tCO_2eq, deposits' after-tax profits generally decline as taxes rise. Eventually, at the company level, most carbon-tax hikes translate into profit reductions due to company deposits' portfolios that exhibit significant variance.

Fourth, we explore the fiscal implications of carbon taxation for oil-producing countries. Our results suggest that carbon tax revenues can, under moderate taxation levels, offset declines in royalties and income taxes—particularly if oil-exporting countries preemptively impose consumption-based taxes. Revenue patterns often follow an inverted U-shape, akin to the Laffer curve. Low-cost producers, such as Saudi Arabia and the UAE, see stable or rising revenues under carbon taxation, while high-cost producers, such as China, Venezuela, and Canada, experience revenue declines due to reduced demand and lower fiscal receipts. This highlights the resilience of OPEC nations compared to the fiscal challenges faced by high-cost producers under stringent carbon taxation. However, when combustion-related carbon taxes are imposed by entities other than the producing countries themselves, domestic revenue losses tend to be much more severe across all producing regions.

Our findings contribute to the broader debate on the distributive effects of climate policies and provide two key policy insights. First, we highlight the heterogeneity of incentives within the oil industry, which complicates coordinated lobbying efforts. Second, we show that the incentives faced by firms and governments depend heavily on the specific policy chosen to achieve a given objective, whether through alternative climate policies or the implementation of an optimal carbon tax.

Our work contributes to the growing literature on supply-side climate policy and fossil-fuel supply restrictions, including theoretical analyses (e.g., Harstad 2012; Eichner and Pethig 2017; Eichner et al. 2023) and quantitative modeling (e.g., Fæhn et al. 2017; Prest 2022; Ahlvik et al. 2022; Coulomb et al. 2025). In contrast to the literature, we focus on the within-industry distributional aspects of climate mitigation policies instead of outcomes such as pollution, cumulative productions or future discoveries.

We contribute to the literature on stranded assets by building on research highlighting the existence of unburnable fossil fuels (Meinshausen et al., 2009; McCollum et al., 2014) and their uneven distribution (McGlade and Ekins, 2015; Welsby et al., 2021). Recent studies (McGlade and Ekins, 2014; Brandt et al., 2018) acknowledge the importance of carbon-intensity heterogeneity in mitigating emissions. However, these works do not explore the trade-off between private extraction costs and emission reductions. By leveraging granular deposit-level data, we document heterogeneity in stranded asset risk, highlighting cases where certain oilfields may become more attractive due to carbon taxation, thereby increasing their social value and profitability.

Our paper connects to the recent literature that studies the impact of OPEC market power in the context of carbon mitigation. Market power is often viewed as beneficial for resource conservation (Hotelling, 1931; Solow, 1974) and thus for pollution mitigation. Asker et al. (2024) quantify the environmental benefits of OPEC's production restrictions, demonstrating that, despite the welfare losses from market distortions, restricting supply has reduced emissions over the 1970–2025 period. In contrast, Benchekroun et al. (2020) use a two-resource model to demonstrate that OPEC's strategies may inadvertently accelerate pollution by facilitating the earlier market entry of higher-cost, higher-emission producers. In the same line of research, De Cannière (2025) analyzes how carbon taxes make the cartel shift production forward in time to avoid asset devaluation, in a cartel-fringe setting with granular data on deposits. Unlike these studies, our analysis focuses on the valuation of oil deposits under carbon taxation and distributional impacts of carbon taxation. Although our baseline model does not explicitly formalize market power, we account for OPEC supply restrictions by calibrating field-level capacity constraints using observed production data. This approach captures the systematic under-utilization of technical capacity among certain OPEC members and thus provides a plausible approximation of market equilibrium and welfare implications arising from producer market power. However, in this preliminary study, our approach does not capture OPEC reaction to carbon taxation.

Our contribution is also to clarify the theoretical mechanisms behind the potential counterproductive changes in oil social value after a tax increase. Extending the *Grey Paradox* (Coulomb and Henriet, 2018) to a setting where multiple exhaustible resources exist, we show that owners of a carbon-intensive resource may benefit from carbon taxation, and that owners of a cleaner deposits may actually suffer from it. This finding diverges from conventional literature, which typically suggests that carbon mitigation reduces the use of polluting resources while increasing reliance on cleaner alternatives (Van der Ploeg and Withagen, 2012). Our numerical analysis that relies on micro-level data verifies that for some deposits the Grey Paradox phenomenon occurs. While multiple-resource extraction has been extensively studied (Chakravorty et al., 2008; Van der Ploeg and Withagen, 2012; Michielsen, 2014; Fischer and Salant, 2017; Coulomb and Henriet, 2018), limited research has explored optimal sequencing of heterogeneous exhaustible resources considering both private extraction costs and pollution levels. We demonstrate that the interplay between these factors yields counterintuitive distributional outcomes of carbon taxation.[3]

The literature has explored the impact of global carbon regulations on fossil fuel prices and revenues, yet no consensus has emerged. Most energy-economic model simulations conclude that OPEC would incur losses from Kyoto Protocol-like regulations (see Barnett et al. 2004 for a review), with some exceptions (Persson et al., 2007). Unlike prior studies, we investigate the distributional consequences of both first-best and alternative carbon policies within a framework that explicitly accounts for producers' intertemporal trade-offs in resource extraction and heterogeneity in oil deposits. Our numerical analysis is complemented by a stylized theoretical model that: (i) establishes conditions under which fossil fuel rents increase as regulations tighten, and (ii) determines when the value of a polluting deposit rises more (or declines less) than that of a cleaner resource.

The remainder of the paper proceeds as follows. Section 2.2 describes the data and estimation of deposit-level carbon intensities and extraction costs. Section 2.3 details the numerical model and calibration. Section 2.4 discusses the distributional impacts of carbon taxation on social value. Section 2.5 presents our theoretical model to provide intuition on our simulations findings. Section 2.6 presents results on corporate profits, and fiscal revenues. Section 2.7 offers concluding remarks.

2.2 Oilfield data, extraction costs, fiscal regimes and carbon intensities

This section presents our oilfield data. It explains the methodology for calculating field-level private extraction costs, modeling of capacity constraints and estimating carbon intensities. Additionally, it outlines the fiscal regimes to which deposits are subjected, and that we used to assess the distortive effects of taxation on production and the distribution of oil revenues and profits between governments and private firms.

[3] The theoretical literature on the distributional effects of carbon regulation on resource owners includes Bergstrom 1982, Brander and Djajic 1983, Rubio and Escriche 2001, and Liski and Tahvonen 2004, which primarily focus on rent capture. In our setting, the social planner remains neutral regarding resource ownership.

2.2.1 Oil-deposit data

The Rystad Upstream dataset. Our empirical analysis uses data from the Rystad UCube Database (Rystad, afterwards), one of the most comprehensive datasets of oil fields. This covers most of World oil production, with 16,667 active assets in 2024. It includes precise field-level data on oil production, exploitable reserves, discoveries, capital and operational expenditures from exploration to field decommissioning as well as production taxes, current governance (e.g., ownership and operators), asset-development dates (license, discovery, approval, start-up, and production end), oil characteristics (e.g., oil type, density and sulfur content), and reservoir information (e.g., water depth, basin and location). In certain cases, an asset might refer to a single oil well. However, in the majority of cases, especially for onshore oil fields, a field consists of multiple oil wells. We restrict the sample to oilfields discovered before 2025, with positive reserves in 2025 and production starting before 2100 in Rystad's forward scenario.

The Rystad dataset does not contain information on fields' upstream carbon intensities but records key variables that influence emissions from extraction or refining, such as oil type (e.g., bitumen or light), API gravity, gas-to-oil ratio, and sulfur content.

Additional data. Two extraction techniques that affect emissions from extraction—methane flaring and steam injection—are not recorded precisely in Rystad. Flaring consists in the on-site burning of the co-produced gas. As only a minority of countries and companies collect and publish data on flared gas, this information is missing in Rystad data. We complement the data using the geocoded flaring volumes calculated by the Earth Observation Group (EOG), within the Payne Institute for Public Policy at the Colorado School of Mines, in collaboration with the National Oceanic and Atmospheric Administration (NOAA). Satellite data is collected from the Visible Infrared Imaging Radiometer Suite (VIIRS) embarked on satellites launched in 2012 and 2017. Heat emitted by gas flares are observed at night and corresponding flared volumes are computed using the VIIRS Nightfire algorithm (VNF) developed by the EOG (Elvidge et al., 2016; Zhizhin et al., 2021). We use the 141,841 positive annual flaring volumes recorded by the EOG between 2012 and 2024 and match it geographically to Rystad oil and gas fields.

Steam injection is a thermal Oil Enhanced Recovery (EOR) technique employed in some fields—mostly those producing heavy oil—to facilitate extraction. Rystad data identify the use of steam injection only for bitumen fields. We add steam-injection data from the International Energy Agency (IEA, 2018).

Appendix B.1 provides a detailed description of our data.

2.2.2 Modeling deposit private extraction costs and extraction capacities

Computing marginal costs. In modeling costs, we first use the most detailed level of data we have, an asset as the basic unit to compute average extraction costs. In our main specification, we measure annual costs as the sum of "well" and "facility" CAPEX, and the "selling, general and administrative", "transportation", "production" OPEX. These annual costs are deflated using deflating factors contained in Rystad's database. In most of our analysis, we compute private extraction costs as average costs that are: $\hat{c}_d = (\sum_{1901}^{2099} Exp_{dt})/(\sum_{1901}^{2099} x_{dt})$, where x_{dt} and Exp_{dt} denote respectively the annual deposit production and *total* cost.

Modeling annual capacity constraints. The amount of resources extracted annually from a deposit is constrained by its extraction capabilities. The extraction process at the field level typically unfolds in two main stages: an initial steady output phase, known as the plateau, followed by a decline phase. The "plateau" constraint forces a field's production to be below k_d, that we compute as the average production rates during the top-10% highest producing years of a field. Following the literature, we assume the decline rate is constant over time (Fetkovich, 1980; Robelius, 2007; Höök and Aleklett, 2008; Höök et al., 2014; Jackson and Smith, 2014). The "decline" rate constraint forces a field's extraction rate to be below α_d, that we compute as the highest decline rate observed in the data for each asset up to 80% depletion rate. More details on the calibration of these constraints are provided in Appendix B.2.2.

2.2.3 Tax data

Fiscal regimes. Rystad analysts collect detailed information on fiscal regimes of each deposit from governments and Oil & Gas operators. There are 857 distinct fiscal regimes that apply to the current global oil supply. Production Sharing Contracts (PSCs) and royalty schemes are the primary frameworks governing oil revenue sharing between governments and oil firms. Under PSCs, the government receives a share of the profit oil, known as government profit oil. In royalty schemes, the government receives a fixed royalty, generally a percentage of gross revenue. Additionally, governments may levy other taxes on oil firms to capture further fiscal revenue such taxes opex that relate to production and income taxes that relate to net profits. Each fiscal regime provides information on tax bases and rates that relate to four groups of taxes: "Taxes opex", "Royalties", "Government profit" and "Income taxes".

Tax Bases and reverse-engineered tax rates. These fiscal regimes are used by Rystad analysts to model each stream of paid taxes, for each deposit over the 2025–2100 period. Using paid tax amounts and their tax bases, we then reverse-engineer the tax rates of each tax component as follows.

Opex taxes (TO_i) are gross taxes reported as opex by oil and gas companies. We compute its average value per barrel produced in deposit i as $TO_i = \frac{\sum_{2025}^{2100} \text{Taxes Opex}(t)}{\sum_{2025}^{2100} \text{Production}(t)}$.

Royalties (RO_i) are based on gross revenues. The asset-specific royalty rate is estimated as $RO_i = \frac{\sum_{2025}^{2100} \text{Royalties}}{\sum_{2025}^{2100} \text{Gross Revenues}}$. Gross revenues are computed by multiplying production with an asset-specific selling price, equal to the Brent reference price corrected with an average asset-specific discount (which depends on the field's oil characteristics) retrieved from the data.

For deposits where government profit oil applies, we compute the government profit share as $GP_i = \frac{\sum_{2025}^{2100} \text{Gov't Profits}}{\sum_{2025}^{2100} (\text{Gross Revenues}-\text{CAPEX}-\text{OPEX}-\text{Royalties}-\text{Taxes Opex})}$.

Finally, income tax rate is calculated by
$IT_i = \frac{\sum_{2025}^{2100} \text{Income Tax}}{\sum_{2025}^{2100} (\text{Gross Revenues}-\text{CAPEX}-\text{OPEX}-\text{Royalties}-\text{Taxes Opex}-\text{Gov't Profits})}$.

Tax-Induced Supply Distortions. Some taxation channels can distort producers' supply away from the competitive allocation. Therefore, in some exercises aiming at exploring market reaction to a carbon tax, we add production taxes to private costs to compute production tax-included private costs to better understand private incentives of extractors in response to a carbon tax. Specifically, in some exercises, we augment the per barrel private production cost of the cost of taxes OPEX and royalties as follows: writing $\hat{c}_i$ the private cost of deposit i, we compute the tax-augmented production cost, $\tilde{c}_i = (\hat{c}_i + TO_i)/(1 - RO_i)$. With this approach, we make the assumption that royalties will be levied on gross revenues net of carbon taxes. In some robustness checks, we assume that carbon taxes are not deductible from Royalties.

Revenue Allocation Analysis. For a given price and quantity vectors $(p(t))$, $(q_d(t))$, we use our estimated tax rates for the four taxes described above, to then compute for field d the revenue distribution between firms and governments. See details in Appendix B.1.2. We call $PaidTaxes_d$ the total taxes paid in field d for a given vector of future prices and quantities (p_t), $(q_d(t))$, taxes not directly related to carbon taxation. The total tax burden consists of Taxes Opex, Royalties, Government profit share, and income taxes. Pre- and post-tax profits are given by:

$$\Pi_i^G = \sum_0^\infty x_i(t) \left[p(t) - \theta_i \mu(t) - \hat{c}_i\right] / (1+r)^t, \quad \Pi_i^N = \Pi_i^G - \text{Paid Taxes}_i.$$

2.2.4 Measuring deposit life-cycle carbon intensities

Upstream carbon intensity. Before oil extraction starts, GHG emissions are generated from field exploration and the setting up of injecting and extracting wells.[4] After production begins, activities such as well maintenance, oil extraction and surface processing, as well as transport to the refinery inlet, emit greenhouse gases.

Field-level carbon intensities are assumed to be exogenous to carbon policies and time-invariant in our main approach: this reflects the role of exogenous factors, such as oil viscosity, density and content, in emissions from extraction and refining. Emissions are also linked to extraction techniques, which are largely tied to oil type. For example, lifting heavy oils requires a more intensive use of Enhanced Oil Recovery (EOR) techniques, such as thermal EOR or Gas-EOR. Another example is flaring: methane flaring is typically associated with high carbon intensity, and is partly determined by exogenous factors such as the reservoir's gas-to-oil ratio and the distance to a significant consumer market for natural gas.[5]

We use the *Oil Production Greenhouse Gas Emissions Estimator* (OPGEE3) of the Oil-Climate Index (Carnegie Endowment for International Peace) to estimate upstream emissions. We proceed as follows: we first run the OPGEE using data on 737 deposits, formatted to be used as model inputs and publicly available from Masnadi et al. (2018). These represent 38.3% of 2024 World production. We then match these deposits to those in the Rystad dataset, and select the estimation model that best explains OPGEE carbon intensities using the variables in the Rystad dataset and supplementary sources (IEA and NOAA-VIIRS). The explanatory variables are selected based on the scientific literature (Brandt et al., 2015; Gordon et al., 2015; Masnadi et al., 2018). We find that field upstream carbon intensity varies by oil type (e.g., regular or heavy), the gas-to-oil and flaring-to-oil ratios, and the use of steam injection. Offshore location also plays a role, but to a lesser extent. Finally, we predict the carbon intensities of the remaining fields in the Rystad dataset using this model.

Appendix A.2 describes OPGEE, the matching procedure and results, the estimation model and the robustness checks in more detail.

Midstream carbon intensity. When reaching a refinery, crude oil from different fields is combined and refined into petroleum products, such as gasoline and other fuels.[6]

[4]We will use the terms carbon emissions, pollution and CO_2-equivalent (CO_2eq) to refer to GHG emissions, similarly carbon intensity must be understood as GHG intensity. Local (air/water/soil) pollution from oil extraction, transportation and refining is ignored.

[5]Less flaring could, however, in theory be implemented by operators. Because of data limitations on the field-level costs of abating flaring emissions (Malins et al., 2014), and the difficulty in relating less flaring to true GHG-emission reductions due to the possibility of an increase in venting,—the direct release of methane into the atmosphere—accompanying flaring reduction (Farina, 2011; Calel and Mahdavi, 2020), our main specification assumes fixed flaring-to-oil ratios.

[6]Our approach assumes that refineries are fixed, so that heterogeneity in pollution due to installations

Additionally, a given field production can be split up between different crude streams in order to be blended with a variety of other crudes. We employ the OCI *Petroleum Refinery Life-Cycle Inventory Model* (PRELIM) to assess midstream carbon intensities that mostly consist of CO_2, CH_4 and N_2O. This engineering-based model requires very detailed information on the physical and chemical properties of crudes ("crude assays"). As detailed oil properties are only partially available in Rystad, we proceed as follows. We first run PRELIM with the 657 assays of major oil crudes (from companies, specialized websites and past research) that are publicly available with PRELIM. We associate these crudes to crude streams available in Rystad dataset using name fuzzy matching as well as further manual identification for unmatched crudes. Weighting by 2024 production, 64.4% of midstream carbon intensities come directly from the PRELIM model.

We then estimate PRELIM carbon intensities using the Rystad variables related to oil characteristics and known to impact refining carbon intensity: the API gravity and the sulfur content. Indeed, based on these two variables, a default refinery configuration is chosen by PRELIM that determines the level of processing intensity, which is the main driver of refining carbon intensity. Last, we predict refining carbon intensities for the rest of the crudes/fields in the Rystad dataset at the most precise level possible.

Downstream carbon intensity. The downstream sector represents the transport of refined products to end consumers, and their use (mostly combustion). We use the *Oil Products Emissions Module* (OPEM) model from the Oil-Climate Index (https://oci.carnegieendowment.org/) to compute downstream carbon intensity of petroleum products for each crude assay. The OPEM model using results from the PRELIM model as direct inputs, we then match these crude assays with Rystad fields using a matching approach identical to that used for midstream emissions, and employ a similar estimation model to predict downstream carbon intensities for unmatched Rystad fields.

2.2.5 Descriptive evidence

This section provides descriptive evidence that: (i) the life-cycle carbon intensity of oil barrels differs significantly across deposits; (ii) life-cycle carbon intensities and private extraction costs are not strongly correlated; and (iii) proven oil reserves exceed climate-wise future demand.

or country particularities (from local air pollutant regulations, for example) can be ignored. We here focus on those refining emissions that are tied to the nature of the extracted oil, i.e. that could be affected by a change in deposit extraction. Assuming that refining technologies could change, or new refineries could be set up with the best-in-class climate-wise process, would add another lever of carbon mitigation and bring overall larger emission reductions (Jing et al., 2020). Our approach also assumes that refineries that treat heavy oil can be reconfigured at low cost to refine lighter oil. This aligns with the need of complex configurations for refining heavy oil, which also include units for refining lighter oil. (see e.g., U.S. Energy Information Administration, 2020).

Carbon intensity varies across oil deposits. From our estimation, the average upstream carbon intensity (CI) of oil in 2021 was 56.2 kgCO_2eq/bbl, while average midstream and downstream carbon intensities were 31 kgCO_2eq/bbl and 378.3 kgCO_2eq/bbl respectively. The distribution of upstream carbon intensities across deposits has considerable variance: 25% of the upstream CI distribution is under 40.1 gCO_2eq/MJ, 50% under 45.9, and 75% under 55.3. Midstream carbon intensities display some variance as well: 25% of the midstream CI distribution is under 23.7 gCO_2eq/MJ, 50% under 33.4, and 75% under 34.5. There is less variance in downstream carbon intensity: 25% of the downstream CI distribution is under 362.8 gCO_2eq/MJ, 50% under 368.8, and 75% under 399.5.

The combined extraction-refining-end use carbon intensities of oil reserves vary by oil type (Figure 2.1a). Unconventional oil, such as bitumen, is about 25% more polluting than conventional oil, such as light oil. Most of the heterogeneity comes from the extraction and refining stages of the oil life-cycle, with bitumen upstream-midstream average carbon intensity being twice as large as the one of light oil. Flaring and steam injection also play a role, and partly explain the large variation in carbon intensity within oil categories.

As countries have different kinds of oil, the average carbon intensity of an oil barrel varies by country of extraction (Figure 2.1b): oil extracted in Canada and Venezuela generates about 20% more emissions than the average barrel pumped in Kuwait, Qatar or Saudi Arabia. OPEC members (the grey bars in the figure) are not a homogeneous group in terms of oil carbon intensity. Middle Eastern countries tend to have much cleaner oil than heavy oil producing countries such as Venezuela and high flaring countries (Algeria, Nigeria). There exists significant within-country heterogeneity: for instance, Canada is host to very different types of oils (oil sands, shale oil, conventional oil), whereas some other countries, such as Saudi Arabia or Kuwait, have more homogeneous oil located in only a few fields.

Little correlation between carbon intensities and private extraction costs. Private extraction costs vary with oil type (Appendix Figure B.2a, top panel), which translates into some countries, e.g. Saudi Arabia, having average private extraction costs around three times lower than those in countries with the most expensive oil, e.g. Canada (Appendix Figure B.2b, bottom panel). These figures are consistent with the rankings based on country of extraction or/and oil types in IEA (2008) and Wood Mackenzie (2019).

At the fine level of disaggregation of our data, there is only little correlation between carbon intensity and private extraction cost. We then conjecture that cost-effective carbon mitigation in the oil industry implies a very different extraction path to that under a pro-competition policy that ignores pollution. The reason why carbon intensities and

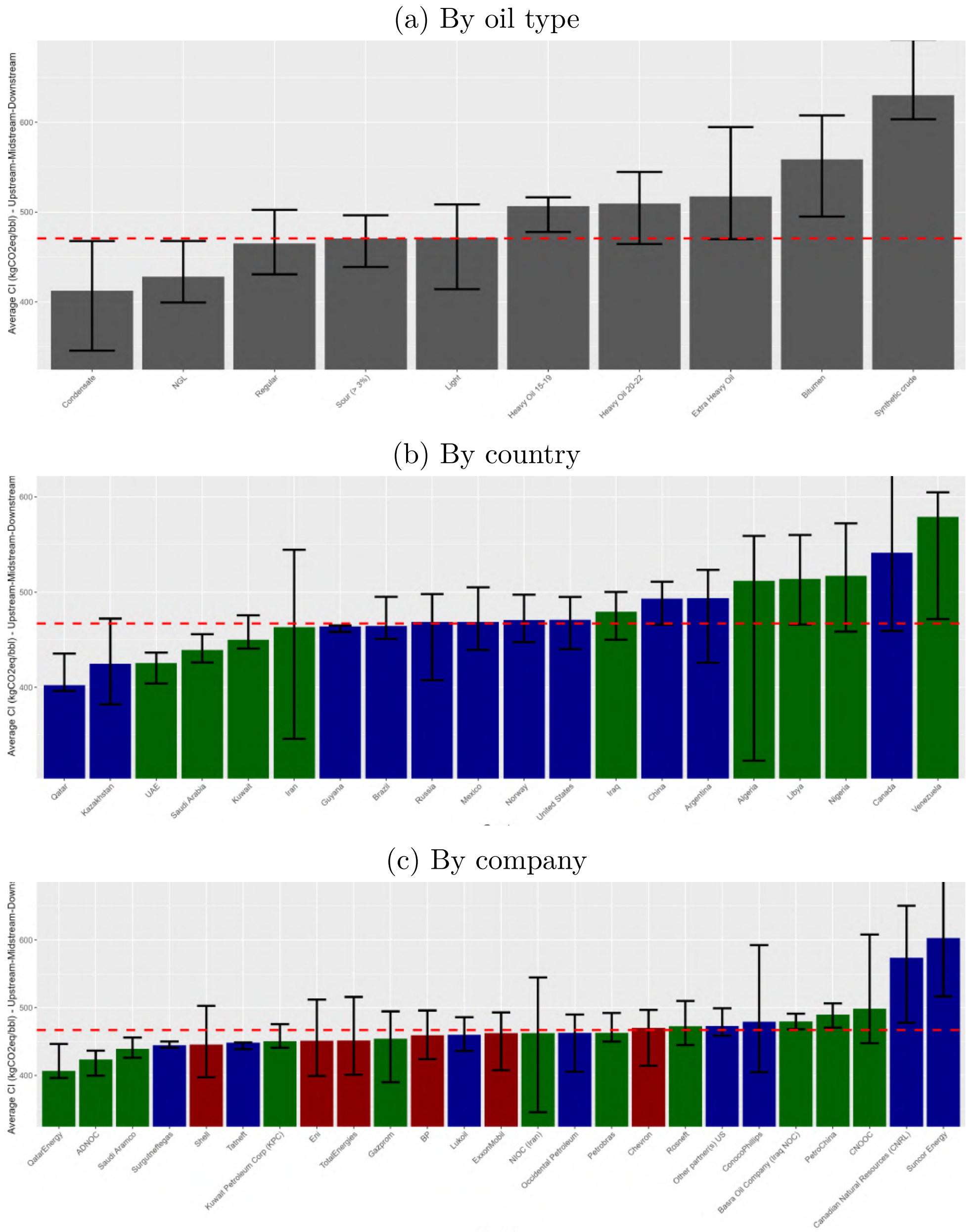


Figure 2.1: Heterogeneity in life-cycle emissions of 2025 oil reserves.

Notes: Panel (a), Panel (b) and Panel (c) represent the life-cycle GHG emission (in kgCO_2eq) per barrel of 2025 reserves by oil type, producing country, and company, respectively. The bar height represents the average (weighted by 2025 reserves), and the extremities of the lines the 10% and 90% deciles. The red dashed line corresponds to the World average figure. Panel (b): Only the top 20 oil producers over the period are represented, and OPEC country carbon intensity bars appear in blue. OPEC, as of 2025, included Algeria, Angola, Congo, Ecuador, Equatorial Guinea, Gabon, Iran, Iraq, Kuwait, Libya, Nigeria, Saudi Arabia, the UAE, Venezuela, and the Neutral Zone shared by Kuwait and Saudi Arabia. Section 2.2.4 and Appendix B.3.2 provide detailed information on the estimation of carbon intensities. Panel (c): NOCs', Majors', other firms' carbon intensity bars appear in green, red, and blue, respectively.

extraction costs are poorly correlated are twofold: first, flaring, venting and land use are important contributors of upstream GHG emissions (about 50% of the upstream emissions according to OPGEE, the rest being mostly emissions related to energy use), and they are not positively correlated to private production costs. Second, GHG emissions related to energy use are more correlated with private extraction costs, yet the correlation is not perfect since the carbon footprint of a given level of energy consumption varies with the type of energy used on site (electricity, oil, gas from in situ sources or imported) and some oil and reservoir characteristics require more capital expenditures per unit of energy use for extraction.

Too much oil and carbon mitigation. The scientific literature has highlighted that oil assets are too abundant, given the estimated carbon budgets, to curb global warming to 1.5 or 2°C throughout the century (Meinshausen et al., 2009; McCollum et al., 2014; McGlade and Ekins, 2014; Welsby et al., 2021). Using our carbon intensity estimates, extracting, refining and burning all proven reserves of oil (as recorded in Rystad) would generate about 669.4 $GtCO_2eq$, which is about 1.5 times the average *total* remaining carbon budget (that encompasses emissions from oil, gas and coal, and land transformation) required to keep the temperature increase below 1.5°C (IEA, 2021).

2.3 General Model of Global Oil Extraction

This section introduces the social-planner welfare maximization framework and model calibration.

2.3.1 The social-planner's objective

The social planner's goal is to maximize discounted social surplus from extraction, considering pollution costs. The gross surplus from energy consumption, $u()$, is twice continuously differentiable, strictly increasing and concave, with $\lim_{e\to\infty} u'(e) = 0$ and $\lim_{e\to 0} u'(e) = +\infty$. We write $p(t)$ the endogenous energy price, that corresponds marginal utility of energy consumption at each date.

Energy can be provided by oil deposits and a costly, carbon-free backstop that provides a perfect substitute for oil at a constant marginal cost, c_c, per barrel-equivalent energy amount. We define I as the universe of available oil deposits, representing proven oil reserves as of 2025. Each barrel is characterized by a private extraction cost, c_i, and an associated external cost. This external cost is the product of the barrel's carbon intensity, θ_i, and the shadow price of carbon. Consistent with a carbon-budget approach, this shadow price is assumed exogenous and grows at the social discount rate, r. This formulation is also consistent with the Social Cost of Carbon (SCC) derived from a damage function with a threshold. Therefore, we refer to the shadow price of carbon and the SCC interchangeably throughout the paper.

Writing μ the 2025 marginal carbon cost with year 2025 the year zero, the current-value SCC reads:

$$\mu(t) \quad = \mu e^{rt} \quad \text{for all } t. \tag{2.1}$$

Environmental costs are only a function of accumulated emissions, and the timing of pollution does not matter.[7]

Oil extraction is subject to feasibility constraints: each deposit's extraction is capped by reserves, delayed until its availability year, and restricted by extraction capacities. Most fields follow a two-phase extraction with plateau and decline stages (see Section 2.2.2). All fields in our sample have been discovered before 2025 but we include development lags after a field discovery before production can start as in Arezki et al. (2017); Bornstein et al. (2017). For oil types other than shale and tight oil, extraction can begin as early as five years post-discovery or after observed production, whichever is sooner; for shale and tight oil, a one-year lag applies.

Our approach is anchored in a first-best framework in which the social planner fully internalizes the costs of greenhouse gas emissions. As shown in Coulomb et al. (2025), within such a framework, exploration generates only limited welfare gains, and only a small fraction of new discoveries is socially optimal to exploit, given the abundance of low–carbon-intensity deposits in current proven reserves. Accordingly, we abstract from exploration for new oil deposits. Introducing additional resources would tend to reduce rents across all oil deposits, with heterogeneous effects depending on deposit characteristics. Yet, ...

Let $x_c(t)$, the energy flow from the carbon-free backstop in year t, $x_i(t)$ denote the annual production of deposit i in year t, $k_i(t)$ its capacity during plateau, α_d its maximal extraction rate during decline, t_i the availability year, and R_i its reserves in 2025. Taking

[7]The scientific literature shows that temperature increase is proportional to cumulative carbon emissions at a given time horizon, and independent of emission pathways (Allen et al., 2009; Matthews et al., 2009; Meinshausen et al., 2009; **?**; Stocker et al., eds, 2013; Allen, 2016). This evidence serves as a rationale for mitigation targets formulated in terms of carbon budgets (Rogelj et al., 2019).

2025 as the starting and reference year, the social welfare maximization problem is:

$$\mathcal{P}_1(\mu(t)): \quad \int_0^\infty e^{-rt}\Big(u\big(\sum_{i\in I\cup\{c\}} x_i(t), t\big) - \sum_{i\in I\cup\{c\}} (c_i + \theta_i\mu(t))\cdot x_i(t)\Big)dt$$

s.t.

$$\int_0^\infty x_i(t)dt \leq R_i \text{ for all } i \in I \tag{2.2}$$

$$0 \leq x_i(t) \leq k_i \text{ for all } t,\, i \in I \tag{2.3}$$

$$x_i(t) \leq \alpha_i R_i(t) \text{ for all } t,\, i \in I \tag{2.4}$$

$$x_i(t) = 0 \text{ for all } t < t_i,\, i \in I \tag{2.5}$$

where (2.2) ensures cumulative production does not exceed reserves, (2.3) caps annual production by plateau capacity, (2.4) limits decline-phase production, and (2.5) prevents extraction from starting before the year the field is available for extraction. The time path of $\mu(t)$ is set by (2.1).

2.3.2 Calibration

Utility is consistent with a demand function that is assumed iso-elastic and calibrated to 2019[8]. We assume price elasticity to be 0.1 in 2025 in line with Uria Martinez et al. (2018) and Kilian (2022). To capture that in the long term, demand becomes more elastic because consumers and industries have more time and resources to respond to price changes by finding substitutes or reducing consumption, price elasticity is assumed to increase by 0.02 per year in the first 10 years, by 0.01 per year in the following 20, and by 0.005 in the following 20 years until reaching a maximum of 0.6 in 2072.

The production cost c_i corresponds either to the direct production cost, $\hat{c}_i$, when computing the first-best allocation, or to the tax-augmented production cost, $\tilde{c}_i$, when analyzing the market equilibrium.

Similarly, the capacity constraint equals $\hat{k}_i$ in the first-best allocation—i.e., the true physical capacity—or $\tilde{k}_i$ in the market equilibrium, where capacities are inferred from observed production levels and incorporate reductions due to OPEC outages.

Private extraction costs, capacity constraints, carbon intensities, and reserves, are specified in Section 2.2 and Appendices B.3.2–B.3. The clean-backstop price is set at \$174 per barrel-equivalent. The social discount rate is 3%.

[8]The year 2019 was chosen in order to avoid capturing market effects of the COVID pandemic.

2.4 Results: the social value of oil across deposits

In this section, we first examine how the social value of oil in each deposit in the World is affected by variations in the social cost of carbon within a first-best framework.

2.4.1 Heterogeneity in social value changes

We begin by examining how the social value of oil in each deposit evolves when the social cost of carbon (SCC) varies. Table 2.1 records our findings. When the pollution cost increases from 0 to \$200/tCO_2eq, we observe an overall decrease in social value of oil across deposits of about 62 trillion. But decreases in oil value are not spread uniformly across deposits. We observe sharp heterogeneity across deposits.

When we decompose the jump in the SCC into smaller increments, we observe that even relatively small pollution cost changes can have substantial effects on the social value of oil. Increasing the internalized pollution cost from \$0 to \$50 reduces the social value by about 27 percent for the average field in the top 5 percent of the distribution of social value changes.

For some increases in pollution costs, the effect on social value is even more heterogeneous. When the cost rises from \$150 to \$200, the average field in the top 5 percent of the distribution experiences only a limited decline in social value of about 3 percent, whereas the average field in the bottom 5 percent of the distribution experiences a decrease in social value of about 94 percent.

	SCC hikes				
	\$0–\$200	\$0–\$50	\$50–\$100	\$100–\$150	\$150–\$200
Median	-90.2	-36.3	-46.7	-44.7	-31.1
Top 5%	-63.8	-26.7	-29.4	-16.6	-2.9
Bottom 5%	-100.0	-65.3	-91.1	-99.1	-93.5

Table 2.1: Distribution of Social Value Changes of Oil Across Deposits after SCC Increases

Notes: The table reports the median change in oil social value (in %), the average change for deposits in the bottom and top 5% of the distribution (weighted by each deposit's initial reserves) of social value changes.

Changes in social value of oil contained in a deposit can come from changes in per barrel's oil social value, and change in the deposit's aggregate production. The interaction between these effects determines the direction and magnitude of the change in oil social value of a deposit. Appendix Figures B.3a and B.3b depict the changes in social value per barrel (price effect) and cumulative production (quantity effect), when the SCC changes. As evident from the figures, both the price and quantity effects play a role in shaping the overall impact of a tax change on the total discounted social value of oil across deposits.

2.4.2 Carbon intensity and social value changes

Figure 2.2 illustrates the changes in the total social value of oil across deposits as the carbon cost increases, plotted as a function of barrels' life-cycle carbon intensity.[9] As shown in the figure, for some increase in SCC, the discounted total social value of oil contained in certain deposits can increase with a higher carbon tax (panel (d)). This outcome is generally observed for deposits with relatively lower carbon intensities. However, the relationship between changes in social value and carbon intensity is non-monotonic: some deposits with average carbon intensity experience an increase in social value, while others with lower carbon intensity may see a decrease. We provide theoretical insights in the mechanisms at stake and the intuition between these results in Section 2.5.4. More precisely, we theoretically show that—i). oil social value in some deposits can increase with the SCC, and ii). that of high-carbon resource can increase while that of a lower-carbon resource may decrease.

As illustrated in Appendix Figures B.4 and B.5, changes in social values after an SCC hike are not monotonic with deposits' private extraction costs or deposits' social cost, illustrating the complex interplay between changes in social value and resource characteristics when intertemporal trade-offs are taken into account.

The following section introduces a simplified dynamic model of resource extraction. Its purpose is to clarify the mechanisms at work and to provide intuition for the counterintuitive results observed in the simulations.

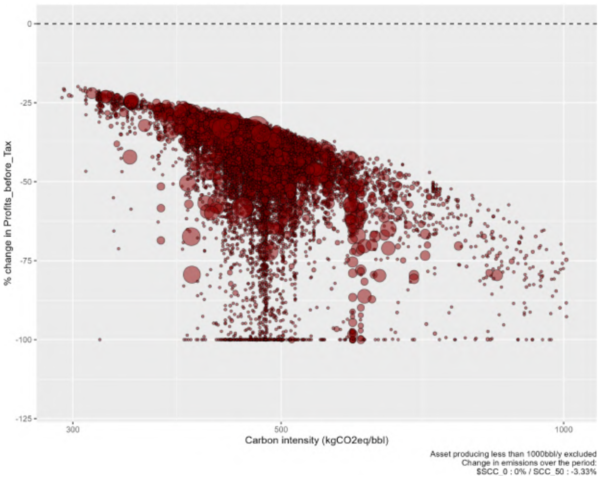
2.5 Theoretical insights: Winners and Losers from carbon mitigation

To provide insights on the impact of carbon pricing on the social value of oil deposits, we consider in this section a simpler problem in which three oil grades compete with a carbon-free alternative.

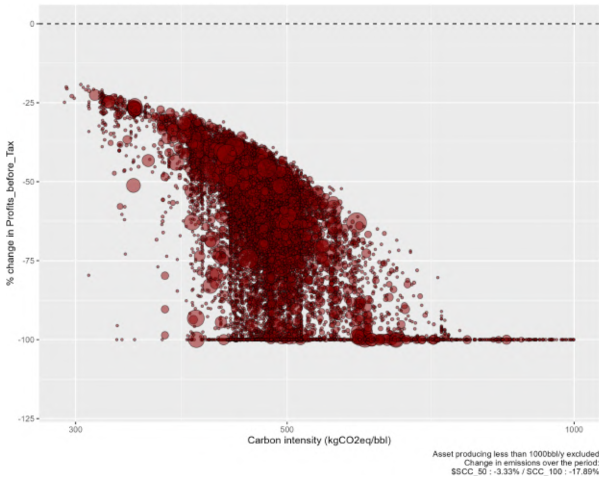
The key takeaway is that oil deposits compete not only against the alternative but also with one another. However, while lower-carbon-intensity deposits can benefit from carbon pricing, this is not always the case. In some instances, cleaner deposits may gain less or lose more than dirtier ones.

[9] Specifically, we compute the two production sequences that maximize social welfare, corresponding to the solutions of $\mathcal{P}_1(\mu(t))$, under varying carbon costs. We then analyze the variation in the social value of oil in a deposit i, defined as $\sum_0^\infty x_i(t)\,(p(t) - \theta_i \mu(t) - \hat{c}_i)\, e^{-rt}$, when the carbon cost varies.

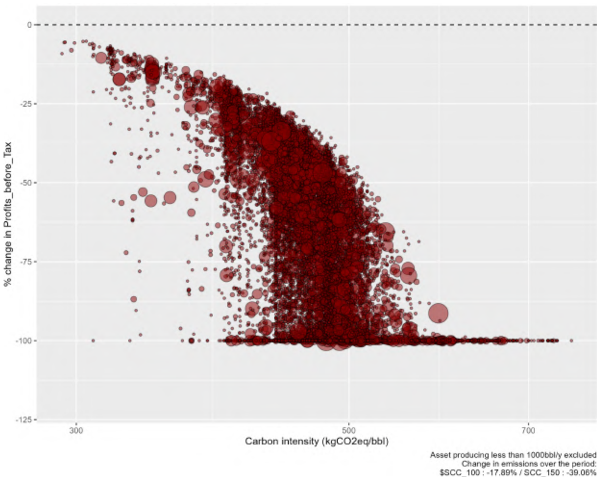
(a) Carbon tax increase: \$0 to \$50/tCO_2.



(b) Carbon tax increase: \$50 to \$100/tCO_2.



(c) Carbon tax increase: \$100 to \$150/tCO_2.



(d) Carbon tax increase: \$150 to \$200/tCO_2.

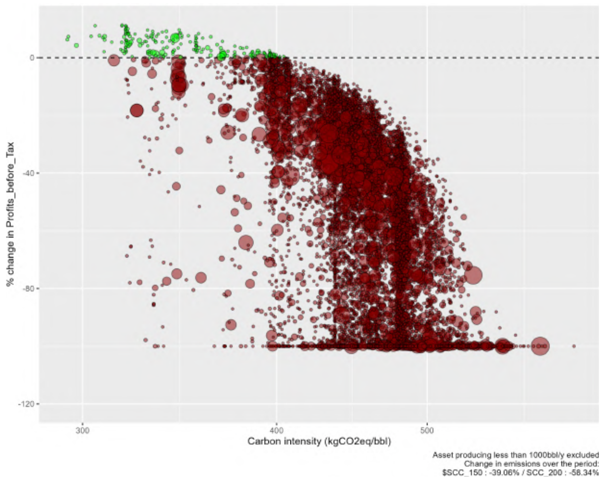


Figure 2.2: Changes in deposits' social value of oil.

Notes: Each circle represents an oil deposit, with the circle size an indication of the deposit volume. Deposits producing fewer than 1,000 barrels per year are omitted for visibility, as are outliers. For assessment of deposits' carbon intensity, measured in kgCO_2eq per barrel, refer to Section 2.2.4 and Appendix B.3.2. The vertical axis indicates the relative change in deposits' total discounted social value of oil after the SCC increases across oil deposits. For each deposit i, the total discounted social value of oil is computed as $\sum_0^\infty x_i(t)\,(p(t) - \theta_i\mu(t) - \hat{c}_i)\,e^{-rt}$.

2.5.1 Simplified model: Three oil deposits, scarcity constraints are not binding

Two exhaustible oil deposits are available (labeled as deposits a and b or R_a, R_b), as well as an even more polluting non-exhaustible deposit (labeled as the dirty backstop, or R_d), and a non-exhaustible clean backstop (R_c). For each resource i, $i \in \{a, b, c, d\}$ we note $x_i(t)$, its extraction flow, c_i, its private extraction cost, and θ_i its GHG intensity. We assume that $0 < c_a < c_b < c_d < c_c$ and $0 < \theta_a < \theta_b < \theta_d$ and $\theta_c = 0$. The dirty and the clean backstops are available in infinite quantity at cost c_c and c_d, respectively. These four resources are all perfect substitutes in demand. The optimal extraction path maximizes the discounted social surplus of extraction that factors in pollution. The consumer gross surplus associated with oil consumption, $u()$, is twice continuously differentiable, strictly increasing and strictly concave. We also assume that $\lim_{g \to \infty} u'(g) = 0$ and $\lim_{g \to 0} u'(g) = +\infty$. The decreasing energy (inverse)-demand function, associated to $u()$, is $D(.)$, and p_i, $i \in \{a, b, c, d\}$, denotes the resource prices.

The social planner wishes to establish the extraction $\{x_a(t), x_b(t), x_c(t), x_d(t)\}$ which maximises the net discounted social surplus—the sum of the consumer surplus and the producer one—under the environmental constraint:

$$\int_0^\infty e^{-rt} \left(u\Big(\sum_{i \in \{a,b,c,d\}} x_i(t) \Big) - \sum_{i \in \{a,b,c,d\}} c_i x_i(t) \right) dt$$

s.t. deposits' scarcity constraints and the carbon-budget constraint:

$$\dot{X}_i(t) = -x_i(t), X_i(t) \geq 0, \forall i \in \{a, b\} \tag{2.6}$$

$$x_i(t) \geq 0, \forall i \in \{a, b, c, d\} \tag{2.7}$$

$$\int_0^\infty \sum_{i \in \{a,b,d\}} \theta_i x_i(t) dt \leq \bar{Z} \tag{2.8}$$

Let $\lambda_i(t)$ be the shadow value of the remaining stock of the exhaustible resource, $i \in \{a, b\}$ and $-\mu(t)$ that of the cumulative emissions, $Z(t)$. The transversality conditions are given by:

$$\lim_{t \to \infty} \lambda_i(t) e^{-rt} X_i(t) = 0, \forall i \in \{a, b\} \tag{2.9}$$

$$\lim_{t \to \infty} \mu(t) e^{-rt} (\bar{Z} - Z(t)) = 0. \tag{2.10}$$

Eq. 2.9 simply states that an exhaustible resource must be exhausted in the long run if its initial scarcity rent is strictly positive. Similarly, Eq. 2.10 indicates that the carbon budget is exhausted in the long run if the initial shadow cost of pollution is strictly positive.

2.5.2 The optimal extraction path and deposits' social value

We use control theory to solve this problem. Appendix B.4.1 provides more details on the solution, and discusses its existence and uniqueness.

The co-state variable $\lambda_i(t)$, $i \in \{a, b\}$, represents the current value of the scarcity rent of the exhaustible resource i. As in Hotelling (1931), this increases at the rate of the discount rate r: the discounted net marginal surplus of extraction must be constant. Along the optimal path, extracting a supplementary unit must be equivalent to saving it for later use. Writing $\lambda_i \equiv \lambda_i(0)$, $i \in \{a, b\}$, we have:

$$\lambda_i(t) \quad = \quad \lambda_i e^{rt}.$$

The costate variable $\mu(t)$ represents the current value of the carbon budget, it increases at the rate of the interest rate. The intuition is similar to that of the Hotelling rule, as emitting CO_2 can be seen as exhausting the carbon budget.

$$\mu(t) \quad = \quad \mu e^{rt}.$$

Finally, along the optimal extraction path, the marginal utility from resource consumption must equal the true social cost of each energy resource that factors in scarcity rent and pollution costs, if any:

$$u'(\sum_{j \in \{a,b,c,d\}} x_j(t)) = c_i + \lambda_i(t) + \theta_i \mu(t) - \gamma_i(t) \tag{2.11}$$

where $\gamma_i(t) \geq 0$ and $\gamma_i(t) x_i(t) = 0$, $i \in \{a, b, c, d\}$, and $\lambda_i(t) = 0$, $i \in \{c, d\}$.

Assume that the exhaustible deposits a and b are exhausted and that the dirty backstop is used to exhaust the carbon budget, which is the case iff $\bar{Z} > \theta_a X_a + \theta_a X_b$. Along the optimal path, resources are used sequentially and follow Herfindahl principle (Herfindahl, 1967); a more expensive resource cannot be used before all less expensive resources have been exhausted. Once all exhaustible deposits are exhausted, the dirty backstop is used, and the corresponding social cost is $c_d + \theta_d \mu e^{rt}$. We call t_1 and t_2 the dates at which resource a and resource b are exhausted respectively and t_3, the date at which the clean backstop starts to be used, which corresponds to the date at which the carbon budget is exhausted.

Finding the optimal extraction path requires to determine the initial scarcity rents,

λ_a and λ_b, the dates of resource switches t_1, t_2, and t_3 that satisfy the following conditions:

$$
\begin{aligned}
c_a + \lambda_a e^{rt_1} + \theta_a \mu e^{rt_1} &= c_b + \lambda_b e^{rt_1} + \theta_b \mu e^{rt_1} && (2.12)\\
c_b + \lambda_b e^{rt_2} + \theta_b \mu e^{rt_2} &= c_d + \theta_d \mu e^{rt_2} && (2.13)\\
c_d + \theta_d \mu e^{rt_3} &= c_c && (2.14)\\
\int_0^{t_1} u'^{-1}(c_a + \lambda_a e^{rt} + \theta_a \mu e^{rt})dt &= X_a && (2.15)\\
\int_{t_1}^{t_2} u'^{-1}(c_b + \lambda_b e^{rt} + \theta_b \mu e^{rt})dt &= X_b && (2.16)\\
\theta_a X_a + \theta_b X_b + \theta_d \int_{t^2}^{t^3} u'^{-1}(c_d + \theta_d \mu e^{rt})dt &= \bar{Z} && (2.17)
\end{aligned}
$$

Eq. 2.12–2.14 reflect the continuity of the marginal surplus between phases. Eq. 2.15 and 2.16 shows the exhaustion of resources a and b at dates t_1 and t_2, respectively. Eq. 2.17 shows the carbon budget is exhausted at date t_3.

The extraction path is composed of four phases as displayed in Appendix Figure B.6: R_a is used first, then R_b is used once R_a is exhausted, then R_d is used once R_b is exhausted; R_c is used once it became more competitive than the dirtiest oil type which corresponds to the date the carbon budget is exhausted.

Total and marginal social values of oil in a deposit. The discounted social value of oil in deposit i, where $i \in \{a, b\}$, is given by $\Pi_i = \int_0^\infty \lambda_i(t)x_i(t)e^{-rt}dt$. Assuming exogenous initial reserves, since $\lambda_i()$ growing at the interest rate, we directly derive that: $\Pi_i = \lambda_i X_i$. Thus, analyzing the impact of a shift in the carbon budget on the total discounted social value of a deposit reduces to assessing its effect on the marginal social value of oil in a deposit i.

Carbon budget, social cost of carbon, tax and decentralization. We have formulated the problem as a welfare maximization problem with the environmental constraint taking the form of a carbon budget. The problem can be reformulated as a welfare-maximization problem, where the social cost of each deposit includes its external cost $\theta_i \mu(t)$. Throughout the paper, when discussing a tightening of the environmental constraint, we either refer a decrease in the carbon budget or an increase in the SCC, both being equivalent.

When producers operate under perfect competition and hold accurate expectations about the tax trajectory, and in the absence of additional distortions, such as production taxes, that could discourage market participation by some producers, the optimal allocation can be decentralized by setting the carbon tax equal to $\mu(t)$. This tax uniquely ensures the optimal decentralization of extraction, when multiple deposits are utilized, i.e., provided the carbon budget satisfies $\bar{Z} > \theta_a X_a$. In this decentralized economy, the marginal profit for resource owners aligns with the scarcity rent of the resource.

2.5.3 Owners of oil deposits can benefit from carbon pricing

In this section, we analyze the impact of a fall of the carbon budget on the social value of oil in the two exhaustible deposits or equivalently on the profits of owners of this resource in a decentralized economy.

From (2.13), it is straightforward to derive:

$$\lambda_b \quad = (c_d - c_b)e^{-rt_2} + (\theta_d - \theta_b)\mu \tag{2.18}$$

Thus, when μ increases following a decrease in carbon budget, two effects counteract each other. On the one hand, resource b is less polluting than resource d, benefiting from taxation through an increase in $(\theta_d - \theta_b)\mu$ ("competition effect"). On the other hand, the quantity of oil consumed at each date decreases, postponing the date at which an extra unit from reserves b could replace one unit from reserve d in consumption. This postponement reduces the social value of increasing the reserves of the less-expensive-to-extract resource b to replace the dirty backstop d and reduce extraction costs. Due to discounting, $(c_d - c_b)e^{-rt_2}$ indeed decreases when t_2 increases ("timing effect"). When demand is inelastic, the timing effect vanishes, and the social value of oil in deposit b increases with taxation. Similarly, when extraction costs c_d and c_b are very close, the reduction in the social value of increasing the reserves of the less-expensive-to-extract resource b to replace the dirty backstop d due to discounting is small, as this value is already minimal.

Similarly using (2.12) and (2.13), we get:

$$\lambda_a = (c_d - c_b)e^{-rt_2} + (c_b - c_a)e^{-rt_1} + (\theta_d - \theta_a)\mu$$

When demand is inelastic, all timing effects disappear and the social value of resource a also increases with taxation.

Overall, when the demand is inelastic, $\frac{d\lambda_a}{dZ} < 0$ and $\frac{d\lambda_b}{dZ} < 0$: a fall of the ceiling or equivalently an increase in the social cost of carbon increases the scarcity rent of resources a and b. This simple example underlines the role of the elasticity of demand in the outcome. Without more information on this elasticity, we can still show that the following proposition holds:

Theorem 1 *If along the optimal extraction path both exhaustible resources are exhausted and the dirty backstop is used, an increase in the social cost of carbon increases the value of the exhaustible resource i if any of the following holds:*

(i) The elasticity of demand is small enough.

(ii) The pollution cost of R_i is low enough.

(iii) Its extraction cost is close enough to that of the dirty backstop.

This proposition generalizes some results of **?** to the case where two exhaustible resources, instead of one, compete with a dirtier abundant resource and a clean backstop (See proof Appendix B.4.2).

Endogenous reserves. Furthermore, we can verify that were deposits' initial reserves be function of early exploration efforts as in Gaudet and Lasserre (1988)[10], the sign of variation of the scarcity rent of resources a and b following an increase in social cost of carbon would be the same as in the exogenous reserves case. See proof in Appendix B.4.7.

2.5.4 Owners of a high-carbon deposit can benefit more or suffer less than owners of low-carbon deposits

We now investigate how a change in $\bar{Z}$ impacts the difference in rents, $\lambda_a - \lambda_b$, to determine who benefits the most or loses the least from tightening carbon regulation.

From (2.12), it is straightforward that:

$$\lambda_a - \lambda_b \;\; = (c_b - c_a)e^{-rt_1} + (\theta_b - \theta_a)\mu \tag{2.19}$$

If $\theta_b > \theta_a$, increasing μ has two counteracting effects on the difference in rents. First, as $\theta_b > \theta_a$, resource a is less polluting than resource b, benefiting from taxation (competition effect). Second, the quantity of fossil fuel consumed at each date decreases, so the social value of replacing one unit of the relatively more expensive resource b with the less-expensive-to-extract reserves a decreases due to discounting (timing effect).

If θ_b is close to θ_a, the competition effect is weak, and an increase in μ, which delays the transition date between resource a and b, t_1, decreases $\lambda_a - \lambda_b$: the social value of having one extra unit of a cheap resource a in place of one unit of the more expensive resource b decreases due to discounting.

Specifically, in the limit case where both deposits a and b exhibit the same carbon intensity, an increase $d\mu$ leads to: $d\lambda_a(t_1) = d\lambda_b(t_1)$ (from (2.12)), i.e. $d\lambda_a - d\lambda_b = r(\lambda_b - \lambda_a)dt_1$. When the tax increase, t_1 increases, and the right-hand part of the equation is negative since $(\lambda_b - \lambda_a) < 0$ (due to the assumption that resource a is cheaper than resource b, $c_a < c_b$) it follows that the left-wing part is also negative and thus:

$$\frac{d\lambda_a - d\lambda_b}{d\bar{Z}} > 0$$

[10] Reserves of resources a and b result from separate exploration activities. Exploration costs are increasing and convex in discovered reserves, and the marginal cost of exploration when reserves are nil is zero.

After a decrease of the carbon budget, the social value of resource b will increase more (or decrease less) than that of resource a.

We show in Appendix B.4 that:

Theorem 2 *If both resources a and b are exhausted and the dirty backstop is used, and if the pollution content of resource b is close enough to that of resource a, an increase in the social cost of carbon increases more (or decreases less) the scarcity rent of resource b (the dirtier resource) than that of resource a (the cleaner resource).* $\forall c_a < c_b \leq c_d$, $\forall \theta_a < \theta_d$, $\exists \theta^* \in]\theta_a, \theta_d[$

$$\left.\begin{cases} \theta_b \leq \theta^* \\ \bar{Z} > \theta_a X_a + \theta_b X_b \end{cases}\right\} \Longrightarrow \frac{d\lambda_a - d\lambda_b}{d\bar{Z}} > 0$$

Note that this implies that if $\theta_b \leq \theta_a$, then $\frac{d\lambda_a - d\lambda_b}{d\bar{Z}} > 0$.

Thus, owners of the least polluting resource are not necessary those who gain the most or lose the least as the carbon regulation is tightened. It is in particular true for $\theta_a = 0$, as long as θ_b is small enough: owners of a clean resource can win less than owners of a polluting resource! The marginal value of R_a is always higher than the one of R_b: this is true for all carbon budget, as long as both resources are used. The reason is that R_a is cheaper and less polluting than R_b, so the former resource is always preferred to the latter. However carbon policy can modify the value of the resources in an unexpected way.

The intuition is easy to grasp. Substituting a cheaper resource for a more expensive one is generally advantageous. However, if extraction is slowed due to a carbon tax, the social value of replacing one unit of an expensive resource with a cheaper one diminishes because of discounting. This is because the additional unit of the cheaper resource would only be used in the long run to replace the expensive one, as the total quantity consumed each year is reduced. When the cheapest resource is also the cleanest, there is an added benefit of replacing one unit of an expensive and dirty resource with a clean and cheap one. Yet, if the carbon intensities of the two resources are similar, the initial effect due to discounting predominates. Consequently, the social value of adding one extra unit of the cheap resource increases less (or decreases more) than the social value of increasing the reserves of the most expensive and dirtiest resource, after an increase in carbon tax.

On the other hand, when demand is inelastic, the timing effect vanishes, and the social value of reserves a increases more with taxation than the social value of reserves b. Similarly, when extraction costs c_a and c_b are very close, the social value of resource a also increases more with taxation than the social value of b as the tax on carbon differentiates deposit a from its competitor deposit b. In Appendix B.4, we show the following proposition holds:

Theorem 3 *If both resources a and b are exhausted and the dirty backstop is used, an increase in the social cost of carbon increases more (or decreases less) the scarcity rent of resource a than that of resource b if any of the following holds:*

(i) the elasticity of demand is small enough;

(ii) the extraction costs of the two exhaustible resources are close enough.

Finally, is it possible that λ_a decreases whereas λ_b increases after a tightening of the carbon budget? Using (2.18) and (2.19), we verify that in the extreme case where $c_d = c_b > c_a$ and $\theta_d = \theta_b = \theta_a$, it is straightforward to see that such a situation can occur. Using the implicit function theorem and a continuity argument, we show in Appendix B.4 that this result generalizes for $\theta_d = \theta_b > \theta_a$, as long as θ_a is close enough to θ_b.

Theorem 4 *If along the optimal extraction path both exhaustible resources are exhausted and the dirty backstop is used, an increase in the social cost of carbon may increase the value of the more expensive and dirtier exhaustible resource, and decrease the value of the cheaper and cleaner exhaustible resource. This holds in particular with $c_d = c_b > c_a$, $\theta_d = \theta_b$, and $\theta_a < \theta_b$ but close enough to θ_b.*

Endogenous reserves. Making deposits' initial reserves endogenous as in Gaudet and Lasserre (1988), there are situations where owners of high-carbon deposits do win, while some lower-carbon oil don't. Indeed, we show in Appendix B.4.7 that with endogenous reserves, the sign of variation of the scarcity rent of resources a and b following an increase in social cost of carbon is the same as in the exogenous reserves case.

This is an important consideration for within-deposit oil reserve extension: a carbon tax can favor Enhanced-Oil Recovery or development expenses in favor of relatively high-carbon deposits.

Adding exogenous capacity constraints. With capacity constraints, we know that the solution to the social planner welfare maximization problem becomes complicated: Coulomb et al. (2026) shows that the Herfindahl (1967)'s least-cost-first principle of extraction[11] does not hold in the presence of capacity constraints and when some oil deposits are not exhausted.

With intertemporal trade-off. We show below that the phenomena of interest described above,—that are i) resource owners can benefit from carbon regulation, ii) owners of an oil deposit can benefit more than owners of cleaner oil deposits from a tightening

[11]This principle amended to account for exogenous capacity constraints can be formulated as: in any year, using a deposit implies that the capacity or reserve constraints bind for all cheaper deposits that year.

of the carbon regulation,—can still occur when exogenous capacity constraints are added to the model. Imagine that capacity constraints of resources a and resource d are not binding, but capacity constraints of resource b, denoted $\bar{x}_b$, are binding. In this case, as long as the carbon budget is not too stringent, the extraction is composed of four phases: first resource a is used, then resource b is used until exhaustion together with resource d; then resource d only is used, and finally the clean backstop is used after the carbon budget is exhausted.

In this case we have:

$$\lambda_a = (c_d - c_a)e^{-rt_1} + (\theta_d - \theta_a)\mu$$

The two previous effects (competition and timing effects) exist. In particular, if the increase in t_1 is large enough, resource a can lose from regulation. It is straightforward that this is the case in particular of $\theta_a = \theta_b = \theta_d$. Using a continuity argument, this remains true for $\theta_a = \theta_b - \epsilon/2 = \theta_d - \epsilon$ with ϵ small enough.

For resource b on the other hand, the total discounted profit is

$$(c_d - c_b)e^{-rt_1}\bar{x}_b(1 - e^{-rX_b/\bar{x}_b})/r + (\theta_d - \theta_b)\mu X_b$$

Imagine that $c_d = c_b$, in this case, the only effect of a small increase $d\mu$ is the positive competition effect $(\theta_d - \theta_b)d\mu X_b = \epsilon/2 d\mu X_b > 0$.

Without intertemporal trade-off. Now consider the case where a and b both face binding capacity constraints at each date from date 0 onwards and are exhausted before the carbon budget is depleted. This pattern happens as long as the carbon budget is not too stringent. Resource d serves as the marginal resource used along the optimal path as long as the carbon budget is not exhausted. In this scenario, there is no intertemporal trade-off governing resources a and b extraction, and the dynamic problem reduces to a series of separate static problems. Resource profits that resemble Ricardian rents at each date are determined by the gap between the marginal resource price and the resource's private costs, augmented by the carbon costs. As the marginal resource is more polluting, the infra-marginal resources a and b benefit from a tightening of the carbon ceiling, provided that their extracted quantities remain unchanged, i.e., as long as resource d remains the marginal resource. It also follows that a tightening of the carbon budget cannot make one resource owner better off than another resource owner with a lower carbon intensity oil.

So far, our analysis has been framed as a first-best approach to oil extraction, focusing on changes to the *social value* of oil across deposits after a change in the social cost of carbon. In the next section, we take a more positive approach, examining how markets could react to carbon taxes and analyzing the subsequent distribution of oil revenues between governments and private companies.

2.6 Results: corporate profits and fiscal revenues

We now adopt a positive approach to analyze the market impacts of carbon taxation, with a particular focus on its distributional consequences across deposits, companies and countries. Specifically, we study how carbon taxation influences firm profits, revenues, and government oil-related fiscal outcomes.

2.6.1 Market approach

The transition from analyzing the social value of oil to assessing corporate revenues requires accounting for two key sources of distortion that push real-world outcomes away from the first-best scenario: the role of existing and potentially distortive production taxes and the influence of market power.

Accounting for existing oil production taxes unrelated to carbon emissions. Production taxes, such as royalties or various local taxes associated with production (known as "Taxes Opex") can distort resource allocation away from the optimal allocation if these taxes are not designed to address externality concerns. In what follows, to compute the equilibrium sequence solution of $\mathcal{P}_1(\mu(t))$, we now include oil production taxes in the private extraction costs, as described in subsection 2.2.3. With this approach, we deviate from a first-best policy and instead focus on a more plausible market reaction to carbon taxation that accounts for distortive effects of production taxes that preexist to carbon taxation.

Capacity constraints as a proxy for market power. Although our model assumes perfect competition, it indirectly captures market power effects through capacity constraints calibrated on observed production levels. Notably, OPEC producers face tighter capacity constraints, which limit their production capacity relative to non-OPEC producers. While these constraints are exogenous in our model, they act as a structural limitation on output that approximates the effects of market power. In reality, firms with market power, such as OPEC members, strategically withhold supply to influence prices. Although our model does not allow firms to make explicit strategic production decisions, the calibrated constraints reflect the fact that some producers systematically operate below their full potential. This provides a reasonable first approximation of market power, as it ensures that observed supply patterns—including the sustained underutilization of capacity by OPEC producers—are reflected in the model's equilibrium outcomes.

We verify that observed OPEC outages (i.e., the underuse of their true production capacities) increase their profits in a laissez-faire scenario with no carbon tax. This supports the idea that observed production restrictions are profit-enhancing for OPEC, and therefore we can calibrate extraction capacities based on observed production levels in order to partially capture OPEC's behavior.

Furthermore, we show that OPEC profits tend to increase with OPEC outages even across a wide range of carbon tax levels, suggesting that, even in the presence of a carbon tax, our physical capacity constraints reflect OPEC's strategic underuse of their production capacities.

How well does the market model perform?

We begin by validating that our model, which incorporates capacity constraints calibrated on observed production levels, can replicate the historical production split across countries. Specifically, we solve the program $\mathcal{P}_1(0)$ using private extraction costs augmented with royalties and operational taxes, while excluding any carbon tax. The optimization process is initialized in 2017. We first verify that predicted World production over the 2017–2019 period corresponds to observed production: we found that the predicted production is about 0.3% smaller than the observed one.

We then predict production by country over the 2017–2019 period and compare these productions with observed data. Results, presented in Appendix Figure B.7, demonstrate that our model reproduces well the historical distribution of production across countries. However, some discrepancies remain, stemming primarily from two sources. First, external shocks, such as wars and embargoes, have affected production in Iraq, Iran, and Libya. While these events are partially captured in the calibration of capacity constraints, they are not fully addressed. Second, deviations for major producers—Saudi Arabia, Russia, and the United States—stem partially from our model's limited ability to capture the market power effects of OPEC. OPEC's market power over the past years led to reduced productions in low-cost OPEC countries such as Saudi Arabia, and a production increase in US high-cost fields compared to a cost-effective supply. As capacity constraints are calibrated over observed productions, they partly account for distortions induced by market power. Yet, our model predicts slightly lower production in the U.S. than observed.

Looking ahead, we verify that our model's projections under a business-as-usual scenario (i.e., no carbon tax) are consistent with established literature on future oil demand. Specifically, by 2100, our demand forecast reaches 1353.6Gbbl to be compared to 1505.8Gbbl predicted by Rystad analysts over the 2025–2100 period.

2.6.2 Distributional Impacts on Corporate Revenues

Companies' after-tax profits. In what follows, we study how extraction-related after-tax profits,—i.e., profits once carbon taxes and production taxes have been paid to governments—vary after a carbon tax increase. Overall, a jump in tax from 0 to 200 per tCO_2 wipes about 27 trillion of cumulative corporate profits.

At the company level, it is not immediately clear whether the average carbon intensity per barrel plays the most significant role in explaining corporate gains or losses from carbon taxation. Indeed, as illustrated in Figure 2.1c, the distribution of assets by company, in terms of GHG intensity, varies across companies but there exists significant variations in each company's portfolio of deposits. This figure underscores the difficulty of predicting whether the deposit-level effects presented above will persist at the company level.

The aggregation from deposits to companies is based on ownership rather than operational control. This reflects the fact that companies derive profits from holding extraction leases, rather than from their status as operators when ownership and operational control are distinct. For large fields, it is often the case that the operator is one of many owners. In these instances, the operator is typically compensated at zero margin, merely covering operational costs, while the owners capture the resource rent.

Table 2.2 provides evidence of significant heterogeneity in changes in corporate profits as the tax increases. However, this heterogeneity is substantially less pronounced than at the asset level, due to firms' portfolio compositions, which incorporate deposits with diverse characteristics.

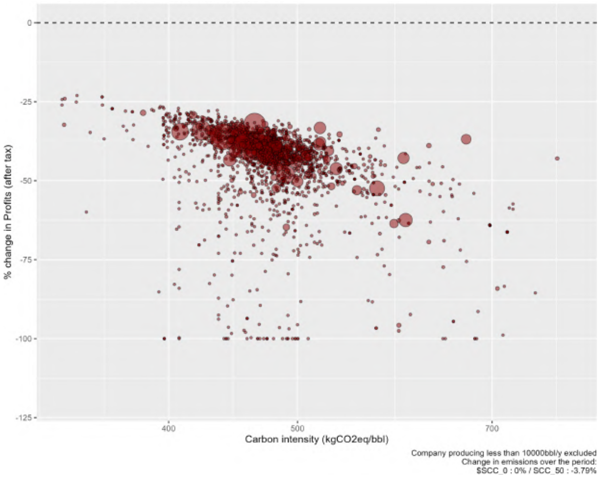
	Tax hikes				
	\$0–\$200	\$0–\$50	\$50–\$100	\$100–\$150	\$150–\$200
Median	-86.7%	-36.6%	-46.0%	-37.8%	-32.5%
Top 5%	-76.7%	-31.8%	-37.9%	-28.1%	-22.9%
Bottom 5%	-98.4%	-47.9%	-67.4%	-65.2%	-51.3%

Table 2.2: Distribution of Changes in Companies' Profits (Percentage)

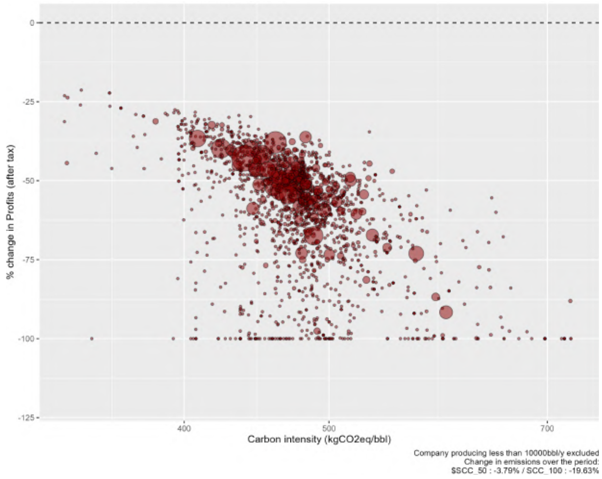
Notes: The table presents the distribution of percentage changes in aggregate after-tax profits across companies following a carbon tax hike. Details on the measurement of changes in after-tax profits are provided in the notes to Figure 2.3.

Figure 2.3 illustrates corporate after-tax profits, representing the residual profits accruing to companies after accounting for royalties, income taxes, tax-deductible operating expenses (OPEX), and government profit-sharing. As carbon taxes increase from a baseline of zero, most companies experience losses from tax hikes. However, we observe substantial heterogeneity across firms, with profit losses being only weakly correlated with the carbon intensity of their portfolios. Yet, on average, low-CI firms are less impacted by tax hikes. This evidence suggests that while firms could theoretically mitigate regulatory risks by rebalancing their portfolios toward lower-carbon deposits—which tend to be less vulnerable to carbon regulation—companies with lower overall carbon intensity do not necessarily demonstrate greater resilience.

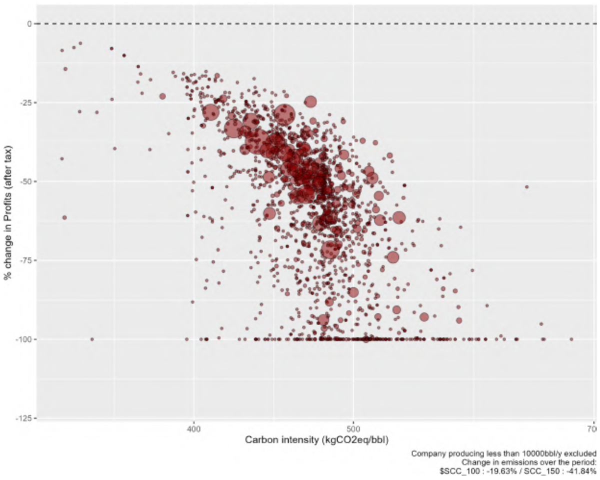
(a) Carbon tax increase: \$0 to \$50/tCO_2.



(b) Carbon tax increase: \$50 to \$100/tCO_2.



(c) Carbon tax increase: \$100 to \$150/tCO_2.



(d) Carbon tax increase: \$150 to \$200/tCO_2.

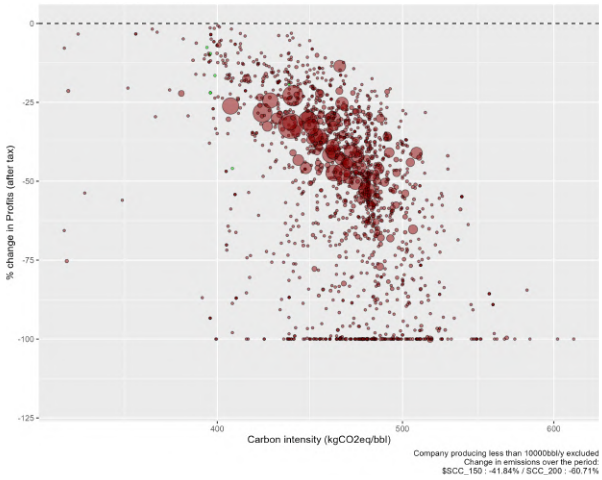


Figure 2.3: Changes in companies' after-tax profits: Various tax changes.

Notes: Each circle represents a company, with the circle size an indication of the company resource size. Outliers are omitted for visibility. The vertical axis indicates the change in company' net profits after a change in tax. For each company, net profits are computed as $\sum_{i \in C} \sum_0^\infty x_i(t)\,(p(t) - \theta_i \mu(t) - \hat{c}_i)\, e^{-rt} - \sum_{i \in C} PaidTaxes_i$ where C is the portfolio of deposits of the company. For assessment of deposits' carbon intensity, measured in kgCO_2eq per barrel, refer to Section 2.2.4 and Appendix B.3.2.

In Table 2.3, we report changes in after-tax profits across company types under different tax hikes. The results show substantial profit reductions for all company types. Heterogeneity across firms translates into only small differences across these groups, with National Oil Companies (NOCs) in the Middle East and Majors experiencing less sharp decreases in profits.

We complement this analysis by looking at profits received by the top-18 largest oil companies for carbon tax levels between 0 and \$200/tCO_2. Overall, as clear from Appendix Figure B.10, while all companies experience a negative impact on profits from an increase in carbon taxes, the sensitivity of profits to taxation varies significantly across

	Tax hikes				
	\$0–\$200	\$0–\$50	\$50–\$100	\$100–\$150	\$150–\$200
Major	-87.9%	-37.1%	-48.1%	-43.0%	-34.9%
NOC (Middle East)	-82.6%	-34.4%	-41.6%	-35.4%	-29.5%
Other NOCs	-91.0%	-40.2%	-53.3%	-48.8%	-36.9%
Other firms	-91.9%	-40.6%	-53.6%	-50.0%	-41.0%

Table 2.3: Changes in Companies' Profits by Type (Percentage)

Notes: The table presents percentage changes in aggregate after-tax profits for Oil Majors (BP, Chevron, Eni, ExxonMobil, Shell, TotalEnergies), Nationally Owned Companies (NOC) in the Middle East and in the rest of the World, and for the rest of firms (mostly independent firms) following a carbon tax hike. Details on the measurement of changes in after-tax profits are provided in the notes to Figure 2.3.

firms. This reflects heterogeneity in the composition of current portfolios of oil and gas deposits. Yet, most national oil companies (NOCs), with the exceptions of PetroChina and Venezuela (not included in the figure), display relative rigidity in profit levels under carbon taxation. Majors and companies tied to expensive and high-carbon oil such as Canadian Natural Resources (CNRL), experience a more severe decrease in profits.

The analysis above assumes that oil companies have no capacity to preempt carbon tax revenues. Appendix Figure B.10 shows profits and carbon tax revenues by companies, assuming companies can preempt carbon taxation on combustion-related emissions. Those revenues exhibit an inverted U-shape as the tax rate increases for both major international oil companies (IOCs) and Middle Eastern national oil companies (NOCs). A strategy to preempt carbon taxation could be advantageous for these firms, allowing them to internalize carbon revenues. The figure raises important questions regarding firms' incentives to coordinate around internal carbon pricing mechanisms—such as those implemented by TotalEnergies or BP—to restrict the global supply of oil and preempt external carbon taxation. By doing so, firms could potentially internalize carbon revenues while maintaining a comparable level of global oil production and greenhouse gas (GHG) emissions relative to a scenario where governments impose and collect carbon taxes. Such coordination could mitigate the decline in profits by offsetting it with increased carbon-related revenues.

2.6.3 Countries Revenues

Countries' Revenues. Countries may perceive oil extraction differently from companies, as oil-related taxes constitute a critical component of government budgets in most top oil-producing nations. We now turn to analyze how revenue streams of a country—oil production taxes (royalties, profit oil and tax opex, income tax), national oil company (NOC) domestic and foreign after-tax profits, and carbon taxes collected on all life-cycle emissions associated to extracted barrels—change in response to the carbon tax for the major oil-producing countries. We display country fiscal revenues sensitivity to carbon taxation in Figure 2.4.

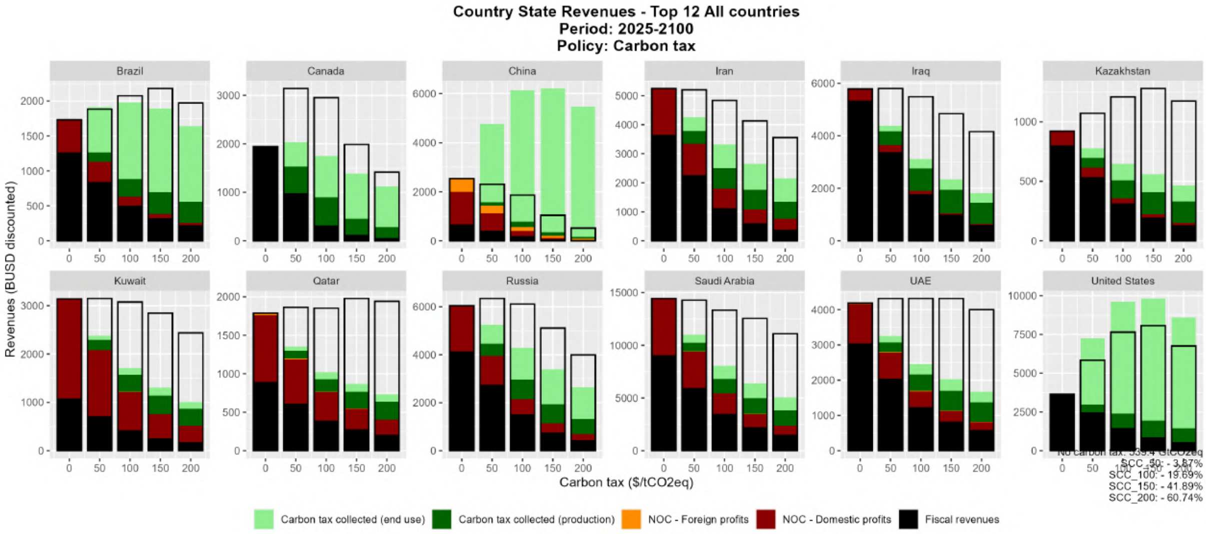


Figure 2.4: Countries fiscal revenues.

Notes: The figure displays discounted government revenues in billions USD (2025 present value) of top-18 countries (in terms of 2025 reserves) when the carbon taxes vary from 0 to \$200/tCO_2eq. Revenues are split as follows: non-carbon fiscal revenues (royalties, tax opex, income tax, profit sharing agreement), NOC domestic and international profits, and carbon revenues raised on domestic production and refining (dark green), domestic end-use (light green) and all end-use (domestic and international, rectangle with no color). Private companies profits are excluded.

Our analysis indicates that carbon tax revenues can, in some cases, offset reductions in royalties and income taxes, at least for moderate tax levels if oil-producing countries can fully preempt downstream emissions taxation, allowing them to impose taxes on the consumption of their oil globally, regardless of where it is consumed (light green in the figure). For many countries, revenues follow an inverted U-shaped curve as carbon tax rate increases, similar to the well-known Laffer curve. In countries with low carbon intensity (CI) and low private extraction costs, such as Saudi Arabia and the UAE, total revenues remain relatively stable and even increase across a broad range of carbon tax hikes. However, in countries with higher carbon intensities and more expensive petroleum production—like China, Venezuela, and Canada—total revenues tend to decrease as the carbon tax rises, even with small increases. In China, for instance, revenues decline with increasing carbon taxes, driven by reduced fiscal income and diminished profits for national and international operations of Chinese state-owned companies, despite gains from carbon tax revenues. This illustrates the varying degrees of resilience in oil-producing nations to changes in global carbon taxation and underscores the fiscal challenges faced by high-cost producers in adapting to higher carbon taxes, while also highlighting the relative advantage of OPEC nations in maintaining revenue stability under stringent taxation policies. Under an alternative assumption where downstream carbon taxes are not levied by oil producers, not even on their domestic consumption (dark green in the figure), tax revenues are significantly affected.

The relatively smaller declines in OPEC revenues highlight their resilience, attributable to lower extraction costs and carbon intensity.

2.7 Concluding remarks

This paper examines the distributional impacts of carbon taxation on oil deposits, corporate profits, and government revenues, emphasizing the heterogeneity of oil extraction costs and carbon intensities. Our theoretical model highlights the complex, and at times counterintuitive, effects of carbon pricing on the social value of oil. While carbon prices generally reduce the valuation of oil resources, certain high-carbon deposits can gain, while some cleaner deposits experience declines. We calibrate our model using detailed deposit-level data and find that i). the oil social value of certain deposits can increase as the carbon cost rises, and ii). carbon taxation's impact on oil valuation is weakly correlated with carbon intensity as predicted by our theoretical model.

Our numerical analysis further decomposes the fiscal and corporate consequences of carbon taxation, revealing that while most oil companies face declining after-tax profits, the magnitude varies significantly based on their asset portfolios. Governments' fiscal revenues exhibit an inverted U-shape response to carbon taxation, with low-cost oil producers showing greater resilience to tax hikes. Yet, if producer countries are limited in their ability to preempt carbon taxes on final use, corporate revenues are mostly decreasing when the tax rises. Conversely of course, if these countries are able to preempt final-use taxes by adjusting their own fiscal regimes, the tax incidence shifts toward the demand side, resulting in a more significant reduction in consumer surplus.

We also assess alternative policies, such as uniform per-barrel taxes and extraction bans, demonstrating that these policies impose significant welfare costs compared to optimal carbon pricing and exhibit distinct distributive effects. These results reveal that some firms or countries may choose to be involved in policy design to favor these alternative policies that benefit them even at the expense of society.

Despite its contributions, this paper has three key limitations. First, our model does not explicitly account for the evolution of OPEC's market power in response to carbon taxation. While capacity constraints are calibrated on historical production constraints that reflect OPEC's limited productions, future strategic behavior remains uncertain and could influence the trajectory of oil valuation and fiscal revenues. Second, our analysis assumes oil tax regimes remain constant over time, yet governments may adjust fiscal policies in response to carbon taxation. Future research should explore dynamic tax policy responses and strategic interactions among oil-producing nations to refine the understanding of optimal carbon taxation in a transitioning global energy market. Third, when policies are designed to ensure that producer surplus does not decrease significantly, the tax burden is effectively shifted to the end-user. This preservation of corporate and fiscal revenues comes at the expense of consumer surplus, which may jeopardize the public's support for the energy transition. Future research should explore how to balance these distributive impacts to ensure both industry participation and broad social buy-in.

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Chapter 3

Voice Until the Last Exit: Ownership Structure and the Limits of Voluntary Environmental Commitments

This chapter was co-authored with Sanjana Ghosh (Northwestern University) and Pritam Saha (University of Geneva)

Keywords: Environmental regulation, corporate social responsibility (CSR), corporate governance.

JEL Codes: D22, K23, K32, L71, M14, O13, Q35, Q52, Q53.

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3.1 Introduction

Oil and gas production is a major contributor to global greenhouse gas emissions, accounting for approximately 15% of total energy-related emissions worldwide (Agency, 2023). Gas flaring, the burning of natural gas associated with oil extraction, is particularly significant and was responsible for nearly 1% of total energy-related greenhouse gas emissions in 2022 (IEA, 2023). Beyond its climate impact, flaring represents a substantial waste of energy resources. In Sub-Saharan Africa alone, approximately 35 billion cubic meters of gas are flared annually, equivalent to half of the continent's total power consumption. The problem is especially acute in regions with weak governance and limited gas infrastructure, where the lack of pipeline networks and processing facilities makes flaring economically rational despite its environmental and social costs.

Oil and gas companies face increasing pressure to reduce emissions from both regulatory and market sources. Recent regulatory initiatives include the U.S. EPA's 2024 rule to curb methane emissions from oil and gas operations under the Clean Air Act and the EU Methane Regulation of 2023, which mandates leak detection and flaring limits for energy producers. Alongside these mandatory regulations, companies are increasingly adopting voluntary climate commitments. As of 2023, 80% of the largest U.S. oil and gas companies voluntarily report their Scope 1 and Scope 2 greenhouse gas emissions (Ernst & Young LLP, 2024). Many also subscribe to voluntary disclosure and decarbonization initiatives. A prominent example is the World Bank's Zero Routine Flaring by 2030 initiative, launched in 2015, which has attracted endorsements from 60 oil and gas companies committing to eliminate routine flaring by the end of the decade.

Yet these commitments are made at the level of firms, while emissions occur at the level of individual oil and gas assets. Ownership of these assets frequently changes through mergers, acquisitions, and divestments, often transferring stakes between companies with very different environmental incentives and governance structures. As a result, firms that adopt environmental commitments may reduce emissions while they operate an asset, but these improvements may not persist once ownership changes. This dynamic raises a broader question about how environmental commitments affect real emissions outcomes: do firms reduce pollution through voice, by influencing operational and investment decisions while they remain owners, or through exit, by divesting from assets that are costly to decarbonize? While divestment may improve the environmental performance of the divesting firm's portfolio, it does not necessarily reduce emissions from the underlying assets, which may instead be transferred to owners with weaker environmental incentives. Findings from Malek et al. (2022) point toward a recent increase in the number of oil and gas deals whereby firms with net-zero commitments move assets away toward companies with no environmental commitment.

In this paper, we examine how voluntary climate commitments affect environmental outcomes when firms can both influence operations and divest from assets. We frame this question using Hirschman (1972) distinction between exit and voice as responses to organizational decline. In Hirschman's original formulation, members of an organization who are dissatisfied with its performance can either leave (exit) or attempt to change it from within (voice), and the relative effectiveness of each strategy depends on the institutional context. We adapt this framework to the setting of jointly owned oil and gas assets, where committed equity owners face an analogous choice: they can exercise voice by using their governance rights to push for emissions reductions, or they can exit by divesting their stakes in assets that are costly to decarbonize. Because voluntary environmental initiatives rely largely on reputational incentives and stakeholder pressure, their effectiveness may depend critically on how firms navigate this tradeoff. If committed firms reduce emissions primarily through voice, improvements should persist as long as they remain involved in asset governance. By contrast, if firms rely on exit by selling high-emission assets, emissions reductions at the firm level may not translate into reductions at the asset level once ownership changes. We study these mechanisms in the context of the World Bank's Zero Routine Flaring by 2030 initiative, under which oil and gas companies pledge to eliminate routine flaring.

The global oil and gas industry provides a particularly suitable setting to study this trade-off between voice and exit. Oil and gas fields are typically owned by multiple firms through joint venture arrangements, where one company operates the asset while other partners participate as non-operating equity owners. Although operators manage day-to-day activities, non-operating owners retain governance rights over major investment and operational decisions that can affect environmental outcomes, including flaring. At the same time, ownership stakes in oil and gas assets are frequently traded, allowing firms to divest assets that are costly to decarbonize. This combination of shared governance and active asset markets provides a natural setting to examine whether committed firms reduce emissions through influence while they remain owners or instead through divestment.

To study this, we assemble an asset-level dataset combining ownership and production data from Rystad Energy's UCube database with satellite-based measurements of gas flaring from the Visible Infrared Imaging Radiometer Suite (VIIRS). The data track nearly 20,000 oil and gas assets across more than 120 countries between 2012 and 2024 and provide detailed information on asset ownership structures, including the identity and equity shares of each partner as well as the operator responsible for managing the field. We combine this information with satellite observations of flaring to measure emissions at the asset-year level. We focus on voluntary climate commitments under the World Bank's Zero Routine Flaring by 2030 (ZRF) initiative, which commits companies to eliminate routine flaring. Our empirical strategy exploits two sources of variation in committed ownership: when firms endorse the initiative and when committed owners subsequently divest their stakes in assets. Using an event-study framework with asset and operator fixed effects, we examine how flaring evolves when assets become owned by committed firms and when committed ownership subsequently disappears.

We report two main findings. First, when at least one equity owner commits to the ZRF, flaring at that asset declines gradually but persistently, reaching approximately 45% below its pre-commitment levels after nine years. This decline reflects the exercise of environmental voice by committed owners. Committed operators reduce flaring quickly by leveraging direct control over day-to-day field practices, while committed non-operators exert influence more slowly over the long run. The pattern also varies with asset characteristics: in assets already connected to gas markets, committed owners exploit existing infrastructure immediately, while in assets with no prior gas market connection, reductions are slower but ultimately more ambitious, suggesting that committed owners successfully advocate for the capital investments necessary to connect these assets to gas networks. Across both asset types and ownership roles, the evidence points to a common mechanism: committed equity owners exercise active governance voice, and that voice produces real and sustained reductions in emissions.

Second, exit by the last committed owner reverses these gains entirely. When the final committed equity partner divests, leaving no committed presence on the management board, flaring rises sharply and persistently, increasing by roughly 40% within two years and over 90% within six years relative to assets that remain under committed ownership. This is the paper's central result, which we term the "last good owner effect". It is not simply a divestment effect: when a committed firm exits while at least one other committed equity partner remains, flaring does not increase. The environmental discipline associated with committed ownership requires at least one committed voice on the board, not merely a history of committed ownership. Notably, this result holds even when the exiting committed firm was a non-operator, so that the same operator continues managing the field before and after the transaction. The removal of committed governance presence is sufficient to trigger deterioration, independent of any change in operational management. These findings suggest that committed firms can improve their portfolio-level emissions metrics by divesting high-emitting assets to uncommitted buyers, while actual emissions at those assets persist or worsen, a form of greenwashing through governance exit that is invisible in firm-level emissions data.

Our paper contributes to three strands of literature. First, we contribute to the literature on exit versus voice in corporate governance. Admati and Pfleiderer (2009) and Edmans (2009) formalize this tradeoff for blockholders in public equity markets, while Edmans and Manso (2011) extend these insights to settings with multiple blockholders. Broccardo et al. (2022) and Gollier and Pouget (2022) apply this framework to socially responsible investors, demonstrating theoretically that voice generally dominates exit when a majority of investors have prosocial preferences. A growing empirical literature examines how institutional investors affect corporate environmental behavior, including through climate risk engagement (Krueger et al., 2020), demand for climate risk disclosure (Ilhan et al., 2023) influence over corporate environmental policies (Kim et al., 2019), and the relative effectiveness of engagement versus divestment in reducing carbon emissions (Kahn et al., 2023). This literature, however, studies voice and exit through the lens of public equity ownership, where influence operates primarily through stock prices, proxy voting, and shareholder proposals. We provide direct empirical evidence that this tradeoff has real environmental consequences in jointly owned assets, where voice operates not through stock price pressure but through board representation and capital expenditure approval rights. Critically, we show that even non-operating minority equity owners exercise meaningful governance voice over emissions, and that their exit removes a binding environmental constraint independently of any change in operational management.

Second, we contribute to the literature on whether corporate environmental commitments translate into real emissions reductions. Bolton and Kacperczyk (2025) study firm-level climate commitments and find that committing firms subsequently reduce their emissions, though the overall effect on aggregate emissions has been small. Raghunandan and Rajgopal (2024) find that signatories to the Business Roundtable's stakeholder commitment exhibited worse environmental and labor records than peers, with no improvement after signing. Yang et al. (2021) show that mandatory emissions disclosure reduced US power plant emissions. In the oil and gas sector, Wen et al. (2023) evaluate the aggregate effectiveness of the ZRF initiative using synthetic control methods, and Chang et al. (2025) study the same initiative in African onshore fields and find that flaring declines substantially in continuously operated blocks, with larger reductions where preexisting governmental oversight is weakest. We extend this literature by showing that the persistence of commitment-driven emissions reductions depends critically on the ownership structure of the underlying assets. While commitments do reduce flaring, these gains are fully reversed when the last committed owner exits, a result that is invisible in firm-level emissions data. Our findings highlight a structural limitation of voluntary initiatives that existing studies, which typically evaluate commitments at the firm or country level, are not designed to detect.

Third, we contribute to the literature on greenwashing through asset divestitures.Duchin et al. (2025) and Chlond and Germeshausen (2025) show that divestitures of pollutive assets do not reduce pollution at the divested facilities, while Shive and Forster (2020) and Ben-David et al. (2021) document that asset transfers to less scrutinized or less regulated owners may worsen environmental outcomes. Our setting allows us to go further in isolating the governance mechanism. We show that the environmental consequences of divestiture depend on the ownership structure that remains after the transaction. When at least one committed equity owner stays on the board, divestiture by another committed owner has no effect on flaring. Environmental deterioration occurs only when all committed governance oversight is removed, a form of greenwashing through governance exit that is invisible in firm-level emissions data.

3.2 Background

3.2.1 Flaring and the World Bank's Zero Routine Flaring Initiative

Oil extraction typically brings up associated natural gas as a co-product, mixed with the oil in the reservoir and separated at the surface along with other fluids. Once at the surface, producers face a choice regarding how to handle this gas. They can capture it and sell it to market through pipelines or liquefaction facilities, reinject it into the reservoir to maintain pressure and enhance oil recovery, or use it on-site to power field operations. When none of these valorisation options are economically viable or technically feasible, the remaining alternatives are to flare the gas (burn it) or vent it (release it unburned into the atmosphere). Venting is considerably more harmful than flaring because unburned methane is a far more potent greenhouse gas than the carbon dioxide produced by combustion. For this reason, routine venting is prohibited or severely restricted in most oil-producing jurisdictions (World Bank, 2022). Flaring is therefore the default disposal method when gas cannot be valorised.

Gas flaring thus represents both a significant environmental challenge and an enormous waste of valuable energy resources. In 2022, it accounted for nearly 1% of total global energy-related greenhouse gas emissions (Agency, 2023). The problem is especially acute in developing regions: Sub-Saharan Africa alone flares approximately 35 billion cubic meters of natural gas annually, equivalent to half of the continent's total power consumption, highlighting the opportunity cost of this practice where energy access remains limited.

Policymakers and international organizations have pursued multiple approaches to reduce gas flaring in oil production. Traditional regulatory approaches establish mandatory limits on flaring volumes or rates, enforced through penalties, fines, or production restrictions, as exemplified by the US EPA's 2024 methane rule under the Clean Air Act and the EU's 2023 Methane Regulation. Market-based mechanisms such as carbon pricing and emissions trading schemes create economic incentives for flaring reduction, while technological solutions like improved gas capture infrastructure can make flaring economically unnecessary, though such investments require substantial capital and may be uneconomical in remote locations. Recognizing the limitations of mandatory approaches in contexts with weak regulatory capacity or where international coordination is needed, voluntary commitment frameworks have emerged as an alternative strategy, relying on reputational incentives, stakeholder pressure, and corporate social responsibility to drive behavioral change.

In response to this dual challenge of environmental harm and resource waste, the World Bank launched the Zero Routine Flaring by 2030 (ZRF) initiative at the 2015 Paris Climate Conference (COP21), adopting a voluntary commitment approach. The initiative represents a voluntary commitment framework designed to eliminate routine flaring from oil production operations worldwide by 2030. Under the terms of the commitment, endorsing companies pledge to immediately cease routine flaring at all new oil field developments and to eliminate routine flaring at existing oil fields by the 2030 deadline. Signatories are also required to report annually on their progress and flaring volumes, creating a mechanism for transparency and public accountability. Since its launch, the initiative has attracted 60 oil and gas companies as endorsers, with commitments made on a staggered basis. Table C.1 lists the companies that have endorsed the ZRF policy since it was announced.

Unlike mandatory flaring regulations enforced through penalties or legal sanctions, the ZRF relies primarily on reputational incentives and stakeholder pressure to drive compliance. The World Bank publishes annual progress reports documenting each signatory's performance, creating public visibility around companies' environmental commitments. However, there are no binding enforcement mechanisms or financial penalties for companies that fail to meet their pledges. This institutional design reflects both the practical challenges of implementing binding international environmental agreements across diverse jurisdictions and a theory of change that emphasizes the power of public commitments, investor pressure, and corporate reputation in shaping firm behavior. Whether such soft governance mechanisms can effectively reduce emissions in practice, particularly in contexts with weak domestic regulatory capacity, remains an empirical question.

3.2.2 Ownership Structure and Governance in Global Oil and Gas Industry

Oil and gas extraction typically involves significant capital requirements, geological risk, and technical expertise, leading to widespread use of joint venture structures where multiple companies share ownership of individual assets or fields. In these arrangements, equity ownership is distributed among several partners, each holding a percentage stake in the asset. One partner is designated as the operator, responsible for managing day-to-day field operations, making operational decisions, and coordinating activities on behalf of all equity owners. In many cases, the operator is also the majority stakeholder. However, operator selection may also be based on technical expertise, regulatory requirements, or contractual arrangements. The remaining partners are non-operating equity owners who share in production revenues and costs proportional to their ownership stakes but do not control operational decisions. This separation between operational control and equity ownership creates a principal-agent relationship where the operator's decisions may not fully align with the preferences of non-operating partners (Hart, 1995; Shleifer and Vishny, 1997).

The operator's authority extends to critical decisions affecting environmental performance, including flaring practices. Operators determine day-to-day operational practices, establish field-level environmental management procedures, and decide when flaring is operationally necessary. However, major capital investments, such as gas capture infrastructure or flare reduction equipment, typically require approval from joint operating committees where non-operating equity owners have representation and voting rights proportional to their ownership stakes. Non-operators can leverage these governance mechanisms to influence flaring decisions, including by building consensus around reduction targets, exercising approval rights over budgetary decisions, and advocating for emissions reduction investments (Clean Air Task Force, 2025). While industry reports recognize the potential for non-operator influence, existing academic research has largely focused on operator behavior, with limited empirical evidence on whether non-operating equity owners can effectively constrain emissions through governance mechanisms.

The potential for governance conflicts becomes particularly acute when joint venture partners have heterogeneous environmental commitments or face different stakeholder pressures. A publicly-listed operator facing intense investor scrutiny over ESG performance may prioritize flaring reduction even when non-operating partners, such as National Oil Companies or private equity firms with weaker disclosure requirements, might prefer to minimize costs. Conversely, a committed non-operating equity owner may struggle to influence an uncommitted operator's environmental practices, especially if flaring reduction requires capital investments that reduce short-term profitability. These conflicts are further complicated by the fact that infrastructure investments in gas capture benefit all equity owners collectively, creating potential free-rider problems where individual partners may prefer others to bear the costs of emissions reduction (Olson, 1965; Grossman and Hart, 1980).

Ownership changes adds more complexity to these governance dynamics. When a committed company divests its stake in an asset, the environmental implications depend critically on who acquires that stake and whether any remaining equity owners maintain climate commitments. Strategic divestitures may allow companies to improve their portfolio-level environmental metrics while actual emissions at divested assets persist or increase under new ownership. If the exiting company was the operator, operational control transfers to a new entity that may have different environmental priorities. Even if the divesting company was a non-operating equity owner, its departure may shift the balance of influence within joint management committees. These ownership transitions create natural experiments for examining whether voluntary climate commitments reflect genuine environmental preferences or merely respond to external pressures. If non-operating committed owners exercise meaningful governance influence over emissions, their exit from joint ventures may lead to deteriorating environmental performance at the divested assets.

3.3 Data

In this section, we describe our data sources and how we combine it to create a novel dataset. We also present descriptive evidence on committed and uncommitted asset ownership, flaring and other key indicators in the oil and gas industry.

3.3.1 Oil and gas assets

Our primary data source is the UCube database on upstream oil and gas industry obtained from Rystad Energy (hereafter Rystad). It reports annual data on over 25,242 active commercial assets (i.e., a specific area or field where companies have the rights to explore or produce) spanning over 129 countries from 2012 to 2024. The dataset contains information on annual production sold and outages by oil and gas type. We define oil as any of the following: crude oil, NGL, and condensate. Since volumes reported corresponds to production reaching the market, no recorded gas volumes does not necessarily mean no associated gas was produced. It rather means that the asset is not connected to the local gas market or does not possess gas liquefying plants that would allow selling the gas.

The dataset also reports annual cost data disaggregated into categories[1]. The types of cost we use are well CAPEX and facility CAPEX. The former corresponds to "capitalized costs related to well construction, including drilling costs, rig lease, well completion, well stimulation, steel costs and materials", while the latter includes "costs to develop, install, maintain and modify surface installations and infrastructure".

From the same source, we also retrieve information on the ownership structure and operatorship of each asset, and its evolution through time. The operator of an asset is generally a single company among the owners, although in certain cases joint ventures are created between different owners to operate the asset. Only one operator is reported per year, corresponding to the company having operated the asset for the largest share of the year in the case of changing operatorship. Participation per company is reported as a percentage interest in the asset. For years during which a transaction occurs, the value reported is proportional to the share of the year during which the company held the percentage interest in the asset. This allows us to identify the exact timing of the transaction within the year.

[1] Available categories are: production OPEX, transportation OPEX, selling, general & administrative OPEX, well CAPEX, facility CAPEX, exploration CAPEX, taxes reported as OPEX, royalties, government take in nature, and income tax.

On top of volumes, economic and ownership data, we need geographical information about assets in order to match flaring satellite observations to assets responsible for it, as described in section 3.3.3. We thus complement data retrieved through Rystad's UCube with geographical data and information on wells drilled obtained in 2022 through the WellCube, from Rystad as well. This dataset contains the number of wells drilled per year per asset and the purpose of the well (production, injection, or exploration). The associated shapefile contains geographical shapes of assets when available, and a generic one when not (i.e. a circle). For assets existing in the production dataset described above absent from the shapefile, a generic shape was created based on the centroid of the asset shape (available in the first dataset). Finally, we use data made available by Rystad on wells locations to improve the precision of the matching of satellite flares observations. It contains the geographical location of more than 1,700,000 wells drilled since 2000. In some countries, the locations reported correspond to the centroid of the asset shape, while for others such as the US and Canada, each well has a unique location[2].

3.3.2 Commitments by companies

The World Bank's Zero Routine Flaring by 2030 provides the list of companies that endorsed the initiative. By using the Wayback Machine, we search through all snapshots between the commencement of the program and the end year of our analysis to identify when the companies committed to the initiative. We then scrape through the past versions of the website and note the year in which the company's name appears for the first time, and we use this as the year the company endorsed the program.[3] We link this to the Rystad database using the company names and use various sources to ensure that the names are consistent across both the data sources. Table C.1 lists the companies and their endorsement years. Manual corrections were performed to account for joint ventures in which committed companies were involved.

3.3.3 Flaring

We obtain the flaring data from the Colorado School of Mines' dataset on Global Gas Flaring Observed from Space. The data is collected by the National Oceanic and Atmospheric Administration (NOAA) using the Visible Infrared Imaging Radiometer Suite (VIIRS) instrument aboard three satellites. It can record a wide range of electromagnetic spectral bands between the wavelengths $0.412\mu m$ and $12.01\mu m$. Elvidge et al. (2015);

[2] Onshore US (except Alaska) and Canada only containing generic shapes, this data on wells precise location is crucial.

[3] The same information can be found on the World Bank's ZRF reporting dashboard, however, it appears that only a subset of the companies are reported here. The current list of endorsers can be found at https://www.worldbank.org/en/programs/zero-routine-flaring-by-2030/endorsers, while the dashboard can be accessed from https://www.worldbank.org/en/programs/zero-routine-flaring-by-2030/reporting.

Zhizhin et al. (2021) develop a methodology that exploits specific shortwave infrared wavelengths during nighttime satellite observations, commonly used for fire detection, to detect active flares and derive estimates of flared volumes. Using the coordinates of the flaring observations, we link this to the Rystad data using the location of the wells or the shape of the assets. The methodology in order to match satellite observations follows a five step process allowing to attribute each flaring observation to a unique asset when possible, or to split flared volumes among nearby active assets when needed. The precise methodology used to perform this matching is described in appendix. Among 141,841 observations over the 2012-2024 period, 134,580 were matched to at least one asset (representing 95.3% of total volumes flared), of which 56% were attributed to a unique asset.

3.3.4 Identifying divestments

From the ownership data described above, we leverage the structure of the data to identify all transactions occurring for each asset and their characteristics. The key feature of the data allowing us to retrieve this information is the proportional nature of participation interests reported. For each company within each asset, we compute the percentage interest sold or bought when possible, or derive it from other companies involved in the transaction if possible. Same goes for the timing of the transaction. Manual corrections were made in three particular cases. For assets located in 2 blocks in Nigeria ("OML 011" and "OML 017"), no transaction was recorded in the data, but a transaction was recorded in older ownership data from the same source extracted a year before. After verification, the older data was used for these assets. Verifying that the operating company was either a company holding shares of the asset or a joint venture led us to manually verify and correct the operator in specific years for 9 assets (7 in Congo, 2 in Gabon). Finally, apparent inconsistencies in the computations of implied shares and transaction timing brought manual verification and correction for 9 assets.

Overall, among the 24,159 assets active during the period, 13,155 experienced at least one transaction. For 747 assets, a large number of simultaneous and/or consecutive transactions made the computation of the share bought and sold by each company unachievable. We drop these assets from our sample. For all other assets, we verify that the computed shares sold and bought by each company, and the computed timing for each transaction, implicates a total percentage available of exactly 100%. 615 assets did not pass this verification process for at least one year, and are therefore dropped from our sample. Finally, restricting to assets that produced and sold any oil during the period leaves us with a sample size of 19,968 assets.

For the remaining sample, commitment to the ZRF was defined as the commitment by any company owning interest in the asset, or the buying by any committed company of an interest in the asset. The commitment treatment lasts as long as any share of the asset is held by a committed company. Conversely, the divestment treatment is defined as the selling of all shares owned by one or several committed companies in an asset. This means that in the case of a progressive divestment (either a committed company progressively selling its shares, or one committed company fully divesting before the other divests), only the last committed owner exiting leads to a treatment. The divestment treatment lasts as long as no committed company holds any share of the asset. This implies that if a company owning shares in the asset commits later, after the divestment, the asset will stop being considered as treated. We also define an operator divestment treatment, which happens once an operator commits to the ZRF or when the operatorship of the asset is transferred to a committed company.

From these three main treatment variables, we define a variety of refined treatment variables allowing us to better understand the dynamics of divestment and its consequences. We first define an "incomplete" divestment variable, identifying full divestments from committed companies in assets where at least one committed owner stays. We then divide the divestment variable into two separate cases: divestment in assets in which the exiting company is also the operator and transfers operations to an uncommitted firm, and divestment in assets that were already operated by an uncommitted firm. Finally, we build a placebo treatment to test the robustness of our findings by identifying all other possible divestments: from uncommitted owners to other uncommitted owners, from uncommitted to committed ones, and finally from committed to committed firms.

Table C.2 displays summary statistics for the different samples used and main treatment groups. The first half of the table reports averages for relevant variables for the complete sample, comprising all oil producing assets, while the second half restricts the sample to assets with no recorded gas sales (indicating no connection to any gas market). For each sample, our treatment group of interest consists of assets whose owners (at least one) committed to the ZRF initiative before divesting to uncommitted owners: 672 assets fall into this category, of which 325 are not connected to gas markets before commitment. Two opposite control groups are then defined: assets that became committed and then remained committed through at least one shareholder until the end of the period (hereafter always-committed assets), and assets that never had any committed shareholder (hereafter never-committed assets).

For both production levels and flared volumes, the distribution is heavily skewed to the right in all groups, with a small number of assets being responsible for most oil production and flared gas. Unsurprisingly, assets unconnected to the gas market seem to produce less oil on average but flare larger quantities of associated gas, highlighting the key benefit of having access to an existing gas market easily.

Although no group seems to differ significantly from one another, it is worth noting that always-committed assets tend to be larger on average than other assets. Two factors can explain this discrepancy: first, larger companies are on average more often committed than smaller ones, and tend to have working interests in larger oil fields. Secondly, large assets ownership is more often split between different firms, increasing the probability that one of them is committed. Contrary to always-committed assets, the never-committed sample display average age, production levels and flared volumes remarkably close to that of divested assets, making it an ideal control group in our setting.

3.4 Empirical Strategy

3.4.1 Event Study Approach

We aim to measure how flaring volumes change when there is a shift in an asset owner's commitment to reducing flaring. We use an event study approach to show the effects on flaring in a series of events when the share of ownership by committed firms of an asset change. We exploit two primary sources of change in the share owned by committed firms: if any of the owners endorse the Zero Routine Flaring initiative, or if any of the owners who are already endorsed sells their share of the asset to a company who is not endorsed. The former examines the effect of commitment, focusing on how the endorsed company seeks to reduce flaring—either by lowering emissions from assets it continues to own or by divesting from them. The latter captures the effect on flaring from assets that committed owners have sold, and which are subsequently owned by less committed or entirely uncommitted owners. However, to purely capture the effects of committed owners, we need to eliminate the effect of potential confounders by omitted variables. We use a combination of high-dimensional fixed effects to account for different sources of unobserved heterogeneity at the asset, year, continent, and operator levels.

We include asset fixed effects to control for time-invariant characteristics that may influence flaring levels. These characteristics include, for example, the initial oil-to-gas ratio, the type of oil extracted, and geological attributes of the reservoir such as depth and rock type. We include continent-by-year fixed effects to absorb time-varying shocks or influences that are common to all countries within a continent in a given year—for example, the effects of global oil price movements or disruptions to international supply chains that may impact continents differently. Ideally, we would prefer to use country-by-year fixed effects to account for country-specific policy shocks. However, doing so would risk absorbing the actions of national oil companies (NOCs) in countries where they own most assets, either directly or through joint ventures. Lastly, operator fixed effects control for the efficiency and environmental outlook of the company in charge of managing the asset.

The advantages of using an event study approach are that it allows us to estimate the dynamic effects of committed owners, as well as the pre-trends, testing whether the outcomes of the assets divested under our specified treatment are likely to have evolved similarly in the absence of the treatment. For asset i at year t, we estimate using the following event study specification:

$$Y_{it} = \sum_{\substack{k=-K \\ k \neq -1}}^{K'} \beta_k \, D_{i,t+k} + \gamma' X_{it} + \alpha_i + \delta_t + \theta_{it} + \varepsilon_{it} \tag{3.1}$$

Here, α_i denotes the asset fixed effect, δ_t the year fixed effect, θ_{it} the operator fixed effect, and X_{it} a vector of controls with conformable coefficients γ'. The scalar ϵ_{it} represents an unobserved shock that is not correlated with the event and the error term is clustered at the asset level. Y_{it} is the outcome of interest capturing emissions, or other economic variables that may help explain the change in emissions. $\sum_{k=-K,\, k\neq -1}^{K'} \beta_k D_{i,t+k}$ captures the dynamic effects of the treatment, and allows us to see the effect on the outcome from $K \geq 0$ years before t and $K' \geq 0$ years after t. The values of K and K' are constrained by the intersection of the data period, the timing of ownership commitments, and the year in which the asset begins or ceases production; thus, making it an unbalanced panel. The estimates $\{\beta_k\}_{k=-K}^{K'}$ are graphically summarized in the event-study plots and are normalized at $k = -1$.

Our main outcome variable is the volume of flaring, measured at the asset–year level in million standard cubic feet (mmscf/y), which is the standard reporting unit. To explore potential mechanisms, we also use several additional outcomes. These include annual capital expenditures on facility infrastructure and on drilling new wells, both measured in million U.S. dollars, the total number of wells in an asset in a given year, as well as annual oil production from the asset measured in barrels per year (bbl/y). The outcome variables contain a substantial share of zero values, with the exception of oil production, since we restrict the analysis to assets that are actively producing. Oil production is also included as a control variable in the specifications where flaring is the main outcome.

Our preferred method for estimating the event-study coefficients is Poisson pseudo-maximum likelihood (PPML). PPML accommodates zeros in the dependent variable and allows for a proportional analysis that can account for the substantial variation in differences of baseline characteristics arising from differences in oil-field size, firm size and production (Silva and Tenreyro (2011)). It is also the natural choice when the dependent variable is non-negative, as it avoids the issues associated with natural logarithm and inverse hyperbolic sine transformations (Correia et al. (2020)). This is particularly relevant when the dependent variable is flaring, investment or production, whereas for the number of wells PPML is especially appropriate because the outcome is a count variable.

Another advantage of PPML is that it performs well in the presence of high-dimensional fixed effects (Correia et al. (2020); Bergé (2018)). Under this estimator, the coefficients of interest can be interpreted as semi-elasticities, that is the percentage change in the outcome induced by a change in ownership structure.

3.5 Results

3.5.1 Commitment and Flaring

We first examine whether voluntary commitments to the World Bank's Zero Routine Flaring initiative translate into actual flaring reductions. Figure 3.1 presents event study estimates comparing flaring trajectories at assets where at least one equity owner commits to ZRF (treatment group) to assets that are never owned by committed companies during our study period (hereafter "never committed" assets). The treatment year is defined as the first instance when at least one owner of an asset endorses the ZRF initiative, or when a committed owner buys into the asset. The specification includes asset fixed effects to control for time-invariant field characteristics, year-by-continent fixed effects to absorb regional shocks and trends, and operator fixed effects to account for systematic differences in operational practices across companies. We control for oil production to ensure that changes in flaring intensity reflect operational decisions rather than changes in production volumes. Standard errors are clustered at the asset level to account for serial correlation within fields over time.

The pre-period coefficients provide conclusive evidence of flat pre-trends. This flat pre-trend indicates that assets whose owners would eventually commit to ZRF were not on systematically different flaring trajectories than never-committed assets, conditional on our fixed effects and controls. Following commitment, flaring begins to decline gradually but persistently. The reductions continue to grow over time, and they are economically meaningful reductions in emissions intensity that persist through the end of our observation window. The magnitude of the last coefficient indicates that around 45% of flaring is abated after 9 years[4]. The persistence and magnitude of these effects indicate that voluntary climate commitments can drive substantial emissions reductions when companies maintain continuous ownership of assets.

As explained in section 3.4, our preferred specification includes (log) production of oil as control variable. Since our model is multiplicative, it entails that we are implicitly assessing the flaring intensity, i.e. the amount of gas flared per barrel of oil produced.

[4]PPML coefficients correspond to semi-elasticities, which differ from percentage change for large coefficients. The -0.6 coefficient associated with the last point estimate is thus to be interpreted as a 45% decrease in flaring.

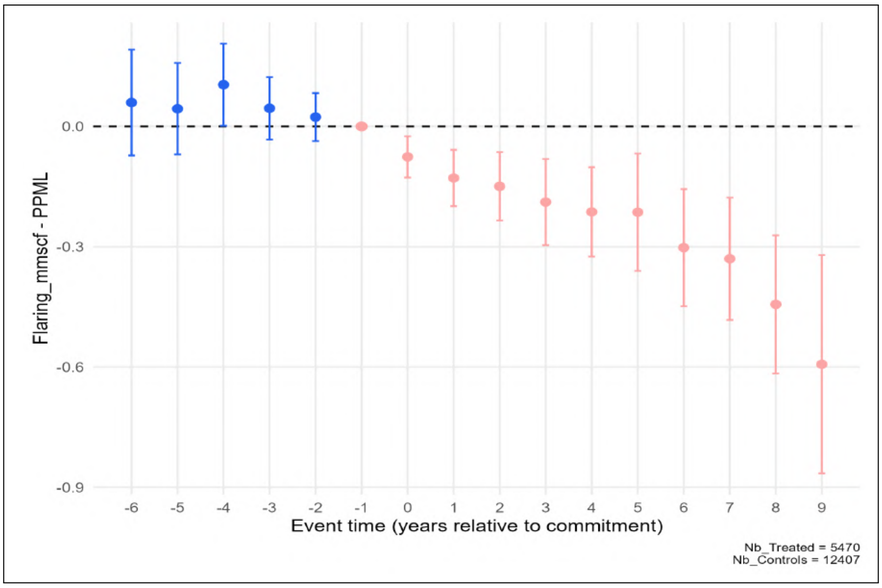


Figure 3.1: Event Study: World Bank ZRF commitment on flaring

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is exp(β) - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

However, our data only allows us to observe sold quantities of oil, which can differ from the quantities actually extracted in the asset. This explains why in numerous instances, we observe flared volumes while no oil production is recorded. These observations would be dropped if we were to use the flaring-to-oil ratio (FOR) directly as the dependent variable in our regressions. Figure C.1 displays event study estimates using this specification. 330 treated assets are entirely dropped because of the exclusion of observations with no recorded oil production. Overall, estimates are particularly noisy, with most confidence intervals spanning from +2 to -2, and they do not appear to follow a specific trend.

We thus decide to keep our preferred specification, including oil production as control variable and flared volumes as dependent variable. While the observed decline in flaring observed in committed assets seems steady and sustained, it hides large disparities across treated assets that we explore below.

Pre-existing connection to gas markets. One could assume that owners of an asset already connected to gas markets could more easily abate gas flaring when required. Restricting our treated group to assets already active before becoming committed, we identify assets with prior connection to gas market as all assets in which any gas sale was recorded before commitment. Figure 3.2 reports event study results for both groups (connected and non-connected before commitment), as well as estimates from the whole sample. As expected, estimates confirm our intuition: the magnitude and significance of the reduction in flaring appear much larger for assets with prior connection to gas markets over the first 4 years of commitment. Interestingly, this apparent advantage only seems play a role in the short run. Assets with no prior connection tend to catch up after five years, and to abate flaring more ambitiously afterwards. After nine years, the magnitude of the estimate is twice as large for assets with no prior connection, translating into an average decrease of 50% as compared to 30% for assets connected to gas markets before connection. Once the owner of an asset becomes committed, it appears to start using immediately existing gas selling infrastructures when there are any to reduce flared quantities, but do not seem to further invest to fully abate flaring. On the contrary, committed owners of non-connected assets require more time to invest in new infrastructures, but their actions seem on average to aim at fully abating flaring.

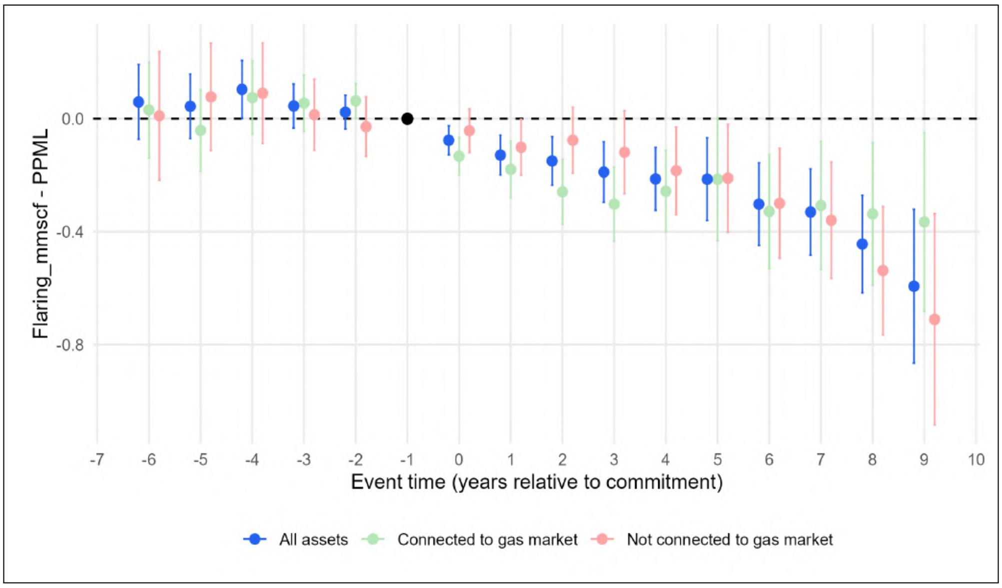


Figure 3.2: ZRF commitment on flaring by pre-commitment gas market connection status.

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is exp(β) - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

The role of operational control. Apart from assets characteristics, the dynamics among equity owners also play a role in the pace at which changes can happen within an asset. The firm operating the asset often have more authority over day-to-day practices and more importantly over any new investments required to phase out flaring. Figure 3.3 displays result from the event study splitting treated assets between those in which the company committing is also the operator, and those in which the operator remains uncommitted. The difference in the short run is once again striking: committed operators tend to reduce flaring by around 15–25% immediately after commitment while committed shareholders with no operating control do not appear to affect flaring practices over the first five years of commitment. Coefficients for assets with uncommitted operator and committed shareholders eventually catch up, although late, and reflect an even larger decrease in the longer run.

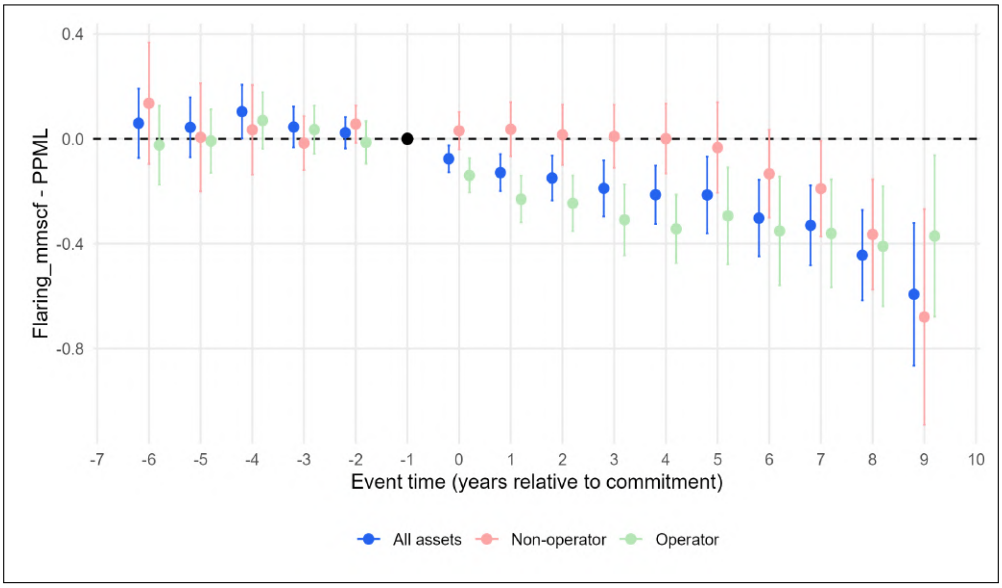


Figure 3.3: ZRF commitment on flaring by operator commitment status.

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is exp(β) - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

Commitment intensity. Decision making within the board is not only influenced by who operates the asset, but also by the relative weight of each actor. Although not operating the asset, a firm owning 50% of an asset probably has higher influence over investment decisions than an owner owning 5%. We explore this dimension by introducing in our specification an intensity of treatment, reflecting the shares owned by committed companies. Figure 3.4 displays the results associated with interacting treatment variables with this intensity. The magnitude of the reduction is increased, by 10% to 50% depending on the point estimate considered. Yet, the overall evolution of flaring in committed assets is unchanged. We further look into the influence of the shares owned by committed firms by splitting our treated sample between committing firms owning the majority of the asset vs minority shares. Results in figure C.2 do not allow to conclude that any significant difference exist between the two groups. Majority owners do not necessarily appear to exert more influence over the operations conducted in active assets, contrary to operating firms.

In figures 3.2 and 3.3, the specific trends and catching up by non-operators and

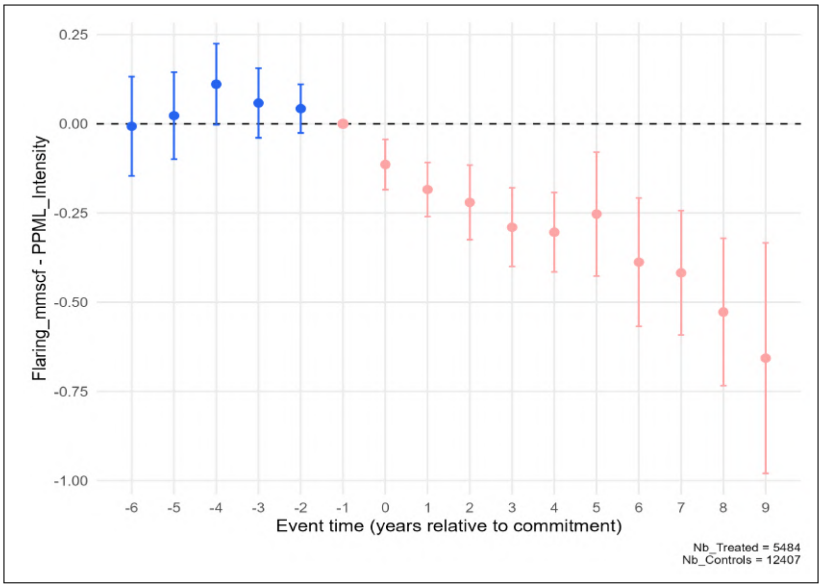


Figure 3.4: ZRF commitment on flaring: treatment intensity.

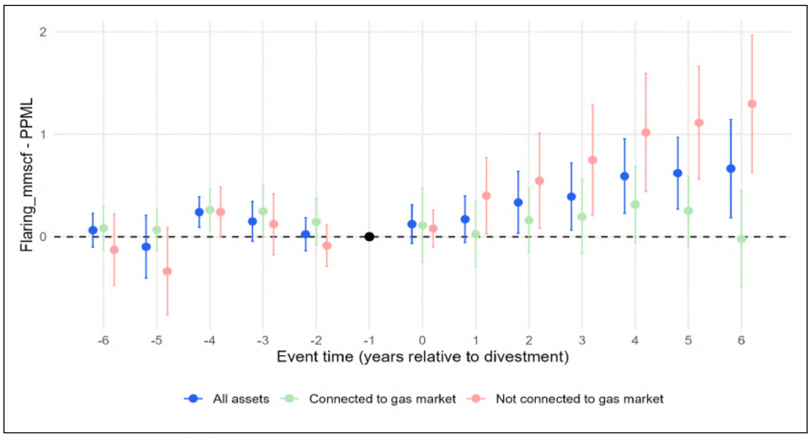
Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta)$ - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

assets with no prior connection to gas markets partly results from a selection of assets. Assets in which flaring abatement is more difficult to implement probably have a higher probability of being divested by committed owners. The results presented in this section raise important questions about what happens to flaring when committed companies divest their stakes. If the observed reductions reflect genuine changes in corporate environmental preferences and governance priorities, we might expect flaring to increase when committed owners exit and transfer control to uncommitted parties. We examine this possibility in the following subsection 3.5.2. We further investigate assets characteristics as well as surrounding environment influence the decision to divest in section 3.5.3.

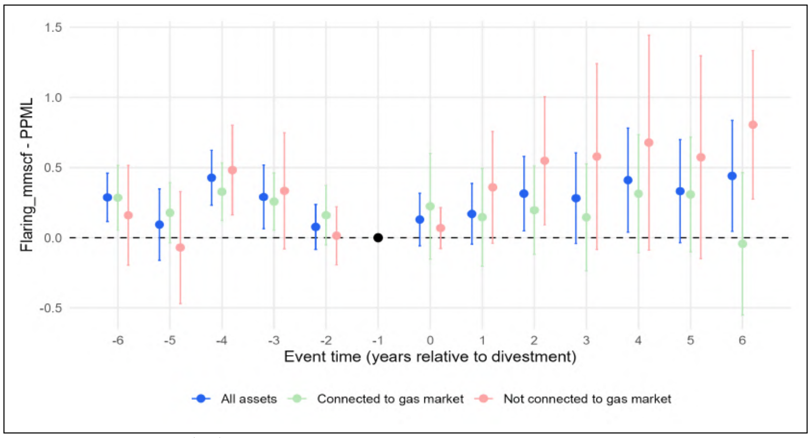
3.5.2 Divestment and flaring

Divestment by the last committed owner. We now examine what happens to flaring when committed companies completely divest their stakes in oil and gas assets, leaving no other committed equity owners on the management board. If committed companies were truly constraining emissions through their equity ownership and governance influence, their complete exit should lead to increased flaring under new, completely uncommitted management board. Figure 3.5 presents event study estimates comparing flaring trajectories at assets where the last committed owner divests (treatment group).

Results are presented using two distinct control groups: assets owned by committed companies that remained in the board until the end of the period (panel A: always committed) and assets that were never owned by committed companies (panel B: never committed). The treatment year is the year when the *last* committed owner divests. The specification includes asset fixed effects, year-by-continent fixed effects, and operator fixed effects, with standard errors clustered at the asset level. Each coefficient represents the difference in flaring intensity between treated assets and control assets at a given event time, relative to one year before divestment, which is normalized to zero.



(A) Control group: always committed



(B) Control group: never committed

Figure 3.5: Divestment on flaring by pre-divestment gas market connection status

Notes: This figure plots event study coefficients estimating the effect of complete divestment by committed companies on flaring intensity. The treatment happens when the last committed equity owner divests, leaving no committed companies on the management board. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before divestment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta)$ - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

The results provide strong evidence that committed owners do keep in check fields operations regarding flaring. In the years prior to divestment, no specific trend exists between divested assets and always committed ones. When comparing to never committed assets, this pre-trend seems to go downward. This is expected, since assets with committed owners reduce flaring after signing onto the ZRF initiative. Following divestment, however, flaring increases sharply and persistently for the treated assets, as compared to both always committed and never committed assets (blue series). Since committed assets tend to see their flaring decrease progressively, it is not surprising that the magnitude of the increase observed in divested assets is sharper when comparing to always committed assets. The magnitude of this effect is economically substantial. Comparing to the always committed control group, the point estimates in years post-divestment suggest flaring increases of 40% after two years and over 90% after 6 years relative to the pre-divestment mean, indicating that the presence of committed equity owners had been meaningfully constraining emissions even after accounting for operator identity and other asset characteristics.

To put this estimate in perspective, 9% of committed assets are eventually divested, but their size is on average smaller than those staying committed, as suggested in table C.2. In 2024, divested assets account for 7% of the flaring originating from oil assets committed at any point in time. This entails that the cumulative environmental gains obtained from the ZRF initiative in assets remaining committed far outweigh the adverse effects from divested assets. However, this conclusion can not be drawn locally: in divested assets, the increase in flaring post-divestment is often larger than the initial decrease post-commitment. This is due to the fact that on average, divested assets stay committed 2.7 years before being divested, limiting the magnitude of the reduction in flaring during commitment.

The role of connection to gas markets. Across both control groups, the post-divestment increase in flaring is positive but statistically insignificant for assets connected to the gas market at the time of divestment (green dots). In these assets, incoming owners have little incentive to stop gas sales, as doing so would amount to renounce to a direct revenue stream. Conversely, assets with no gas market connection at the time of divestment experience a particularly sharp increase in flaring, exceeding 200% five years after divestment when comparing to always committed assets (as suggested by the +1.1 point estimate). Moreover, evidence in the preceding subsection suggested that over the first few years of commitment, flaring is only moderately reduced in these assets. Since the average duration of commitment in divested assets is 2.7 years, our findings suggest that for this subset of assets, the net environmental impact of the ZRF initiative is overwhelmingly negative, and can result in substantially higher quantities of gas flared as compared to a situation with no committed firm ever owning shares.

The last committed owner effect. The sharp increase in flaring following complete divestment by committed companies raises an important identification concern: are we capturing the effect of removing all committed oversight, or simply the effect of any divestment by a committed company. If our results merely reflect selection into divestment, with committed companies divesting their worst-performing or highest-emitting assets, we should observe flaring increases even when some committed companies divest while leaving other committed equity owners on the management board. Figure C.3 addresses this concern by examining incomplete divestments where a committed company exits but at least one other committed equity owner remains. The specification and control groups remain identical to the complete divestment analysis. If the previous results simply captured committed companies divesting problematic assets, we should observe similar post-divestment flaring increases.

The pre-period coefficients show that treated assets exhibit lower flaring than control assets in the years before divestment, with this gap narrowing over time. The negative pretrend coefficients, which become progressively less negative approaching the divestment event, suggest that assets experiencing incomplete divestment had been outperforming never-committed assets in emissions intensity, but this advantage was eroding. This pattern is consistent with committed companies beginning to divest from assets where their governance influence was becoming less effective or where operational conditions were deteriorating. Estimates across all post-divestment periods hover around zero or are slightly negative using both always and never committed control groups. This finding provides validation for the "last good owner effect" interpretation. In contrast to the previous result on full divestment by committed owners, this result shows that flaring either remains flat or reduces slightly post-divestment by the committed owner. Hence, flaring increases occur specifically when all committed oversight is removed from the management board, not simply when any committed company divests. The continued presence of even a single committed equity owner appears sufficient to stop the deterioration in flaring and maintain emissions discipline through governance mechanisms. The stark contrast between complete and incomplete divestments indicates that committed non-operators exercise meaningful influence over environmental outcomes, and that this influence persists until the final committed owner exits.

The role of operational control. The main divestment results demonstrate that flaring increases when the last committed owner exits. A critical question remains: whether these effects are driven entirely by the exit of committed operators who directly control field operations, or whether committed non-operators also constrain emissions through governance mechanisms. If the baseline results simply capture the transition from committed to uncommitted operational management, we should observe effects only when committed operators divest. However, if committed non-operators exercise meaningful influence through joint operating committees, approval rights over capital expenditures, and board representation, their exits should also increase flaring even when the operator remains unchanged. To test this, we split the last committed divestment sample by whether the exiting committed company was the operator or a non-operator. The non-operator subsample provides a particularly clean test: here, the same operator remains in control before and after divestment, isolating the effect of removing committed oversight from the management board while holding operational management constant. If committed non-operators constrain emissions, we should observe flaring increases even in this subsample where operational control does not change hands.

Panel A in figure C.4 presents these heterogeneous effects. All specifications use the "always committed" control group. The results reveal no significant difference in timing and magnitude. While operatorship appears to lead to a faster decrease in flaring post-commitment, it does not seem to play a role in explaining the observed increase in flaring post-divestment. This result indicates that committed non-operators are actively constraining uncommitted operators through governance mechanisms, and their exit allows rapid operational changes even without any change in operational management. To ensure that operator fixed effects do not explain this result, Panel B reports estimates omitting these fixed effects in the model. While the magnitude of the increase in flaring decreases, excluding operator fixed effects does not change our finding: the increase in flaring post-divestment is not driven by committed operators divesting.

3.5.3 Potential Mechanisms

How committed owners exercise voice. We now turn to potential mechanisms that could explain why flaring decreases when an asset is at least partially owned by a committed company, and increases when it is divested. To do so, we examine how operational behaviour changes after commitment and divestment. In particular, we focus on investment and production decisions that may affect the asset's flaring levels. Three mechanisms are investigated: investment in gas abating facilities, restriction on gas production and drilling of new wells, and connection to gas markets.

Investment in gas flaring abating technologies. We do not observe investment broken down into categories that would allow us to distinguish "green" from "brown" expenditures. However, we can analyse investment at a more aggregated level, using two broad categories: spending on facility infrastructure other than wells, and spending specifically on drilling new wells. One could expect the latter to increase after commitment, and to decrease after divestment, mirroring investments in gas abating technologies, such as gas reinjection, or gas pipelines in order to find a use for the gas. Panels A and B in figure C.6 report the evolution of facility capital expenses after commitment and divestment respectively. The results indicate no meaningful change in facility-infrastructure investment following commitment or divestment, as reflected in the statistically insignificant coefficients. Overall, the point estimates are particularly noisy. This could be explained by the fact that facility capex aggregates a wide range of capital expenses, among which only a small share would correspond to expenses aimed at phasing-out gas flaring. Moreover, many oil and gas companies do not report these expenses, translating into a modelised distribution of expenses by Rystad.

Moderating oil production. A second channel through which committed owners may reduce flaring is by moderating oil production and constraining investment in new wells. Two distinct mechanisms link production decisions to flaring intensity. In assets connected to gas markets, higher oil production translates mechanically into larger volumes of co-produced gas, which can exceed the capacity of existing midstream infrastructure. Agerton et al. (2023) and Allen et al. (2024) document this mechanism in the United States, showing that the majority of flaring in the Bakken and Permian basins arises not from a complete absence of infrastructure, but from existing gathering, processing, and transmission capacity being insufficient to handle gas volumes. In assets not connected to gas markets that cannot reinject gas, the mechanism differs. Produced gas is typically used to fuel on-site equipment, with volumes exceeding operational energy requirements flared (Schulz et al., 2020). Since on-site energy demand is largely invariant to production levels, higher oil output increases associated gas volumes more than proportionally relative to on-site consumption, thereby raising flaring intensity. Finally, the drilling of new wells in remote locations can increase flaring intensity at the asset level: wells drilled at the frontier of existing gathering networks may not be connected to gas infrastructure at the time production begins, as operators frequently start oil extraction before completing pipeline connections (Schulz et al., 2020; Agerton et al., 2023).

Figure C.5 investigates this mechanism following commitment and divestment. For both events, the figure reports the evolution of oil production (panels A and D), well capital expenses (panels B and E), and number of wells drilled annually (panels C and F). Divestment results use never committed assets as control groups. In panels A to C, upward pre-trends indicate that production was increasing when assets became committed through one of their owner. The production tends to stabilise after commitment, as well as investment in new wells, with well capital expenses even decreasing slightly. Despite limited magnitudes, these findings corroborate the intuition of a constraining of oil production following commitment, in order to limit excess co-produced gas.

Event studies around divestment further reinforce the importance of this mechanism. Decreasing pre-trends highlight efforts made by committed owners to limit gas flaring through this channel. Soon after divestment, wells drilling and production increase sharply, with magnitudes of the point estimates more than five times larger than the ones observed around commitment. This pattern indicates that uncommitted acquirers actively invest in production capacity following divestment, expanding drilling programs rather than focusing on flaring reduction. Combined with our main flaring results, this finding suggests that increased emissions following committed owner exits reflect deliberate reallocation of operational priorities, with new uncommitted owners massively investing in production-enhancing activities and disregarding concerns about the use or disposal of co-produced gas.

Connection to gas markets. An additional channel through which committed owners can actively reduce gas flaring is connecting the asset to gas markets. In our data, around 40% of assets active before commitment have not sold any gas at time of commitment, implying they are not connected to any gas market. In these assets, the most direct way for committed owners to voice their interests is to convince the board to invest in gas processing and gathering infrastructure in order to sell it. Initially non-connected assets that start selling gas during commitment should thus be associated with a much lower probability of being divested by committed owners. In table 3.1, the last column provides insights on the probability for an asset to be divested according to its gas market connection status. The regression uses cross-sectional variation across committed assets active before commitment and uses as dependent variable a dummy equal to one if the asset is eventually divested after commitment. From the coefficients, assets connected to gas markets are much less likely to become divested, with a coefficient associated with connection during commitment twice as large as the one associated with prior connection. For reference, since 9% of committed assets are eventually divested in our sample, a 5.3 percentage point decrease in the probability of being divested implies that assets in which the connection to gas market occurs while a committed owner is in the board have a 60% lower probability of being divested than assets remaining non-connected. When gas market connections outcomes align with the voice of committed owners, the likelihood of exiting the asset is dramatically reduced.

Internal governance and divestment decisions. This last channel highlights the voice-versus-exit trade-off faced by committed owners: whether to remain on the management board and exert influence over flaring decisions, or to divest. The first two columns in table 3.1 provide insights on how internal governance characteristics are associated with the probability of exit by committed owners. In each regression, the dependent variable is a dummy equal to one if the committed asset is divested in year t, regressed on elements of internal governance: operator commitment status on one side, commitment intensity (i.e. the share of the asset owned by committed companies) and the number of distinct non-committed owners simultaneously on the other. Since divestment affects each of these variables, a one-year lag is introduced to reflect pre-divestment choice conditions. For reference, in each year t, approximately 1.5% of committed assets are divested.

As expected, a committed operator can exert its influence more easily than other owners, reducing the probability of exit. The coefficient associated with the variable is significant, and remarkably close to the proportion of divested assets per year, implying that exits by committed operators are scarce. Regarding the second regression, the coefficient associated with commitment intensity is significant as well, and its magnitude indicates that assets fully owned by committed owners are only rarely divested. This aligns with the idea that when committed owners own most of an asset shares, voicing their interests is easier making exit an unlikely outcome. However, the number of distinct non-committed owners is not correlated with a higher probability of divesting, as one could expect. The sign of the coefficient is even negative, but the absence of statistical significance is consistent with no meaningful effect on the probability of exit.

	Divestment in year t		Divestment in any year
	(1)	(2)	(3)
Committed operator ($t-1$)	$-1.3\text{e-}2^{***}$ (1.5e-3)		
Commitment intensity ($t-1$)		$-1.4\text{e-}2^{***}$ (2.6e-3)	
Nb. non-committed owners ($t-1$)		$-9.8\text{e-}4$ (6.8e-4)	
Connection to gas			
Before commitment			$-2.7\text{e-}2^{***}$ (1.1e-2)
While committed			$-5.3\text{e-}2^{**}$ (2.5e-2)
Continent × Year FE	✓	✓	
Continent FE			✓
Observations	36,938	36,938	6,117
Adj. R^2 (within)	0.003	0.001	0.002

Table 3.1: Mechanisms: determinants of divestment probability (within assets)

Notes: Standard errors in parentheses. The sample is restricted to asset-year observations in which the asset is committed (i.e. owned by at least one ZRF signatory), including the year of divestment. $t-1$ indicates that the variable is lagged. Column (2) includes both commitment intensity and number of non-committed owners simultaneously. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

As explained above, column (3) complements this picture by showing that connection to a gas market is associated with a significantly lower probability of divestment. Since gas market access is itself partly the product of committed owners successfully advocating for infrastructure investment, this result can be interpreted as evidence that effective voice reduces the incentive to exit. Conversely, assets where no gas connection was achieved during the commitment period may reflect governance environments in which voice proved ineffective, making divestment the more rational outcome.

External opportunities and divestment decisions. Beyond internal governance, external environment may or may not provide opportunities to facilitate the valorisation of co-produced gas, thereby influencing the decision to divest or not. Table 3.2 explores this dimension through three separate regressions.

In the first regression, the decision to divest in year t is regressed on the lagged share of assets in a 100km radius that have committed owners. We also include this variable interacted with the number of assets in a 100km radius. Since gas valorisation infrastructure such as gas export pipelines typically requires the aggregation of multiple assets' production, a higher share of committed assets in an area could facilitate efforts to invest in new gas infrastructures. Results indicate that a higher share of surrounding committed assets is associated with a significantly lower probability of divestment, consistent with the idea that a denser local network of committed owners facilitates collective investment in gas valorisation infrastructure, reducing the incentive to exit. The coefficient on the interaction term is negative, as expected, but statistically insignificant, suggesting that the absolute number of surrounding assets does not meaningfully amplify this effect. This null result could reflect the relatively large geographic aggregation implied by a 100km radius, which likely captures assets that are too dispersed to plausibly coordinate on shared infrastructure investments.

	Divestment in year t	Divestment in any year	
	(1)	(2)	(3)
Nearby assets (100km radius)			
Share committed ($t-1$)	$-1.8\text{e-}2^{***}$		
	(3.7e-3)		
N $\times$ Share committed ($t-1$)	$-4.9\text{e-}5$		
	(3.4e-5)		
Gas infrastructures dummy (50km radius)			
Operating		$-3.4\text{e-}2^{***}$	
		(1.1e-2)	
Future		$-3.1\text{e-}3$	
		(9.9e-3)	
Cancelled		$3.8\text{e-}2^{***}$	
		(7.7e-3)	
Gas infrastructures distance score			
Operating			$-6.8\text{e-}4^{***}$
			(1.5e-4)
Future			$-7.0\text{e-}5$
			(1.4e-4)
Cancelled			$6.7\text{e-}4^{***}$
			(1.3e-4)
Continent $\times$ Year FE	✓		
Continent		✓	✓
Observations	37,055	7,457	7,457
Adj. R^2 (within)	0.001	0.005	0.007

Table 3.2: Mechanisms: determinants of divestment probability (environment)

Notes: Standard errors in parentheses. The sample is restricted to committed assets, including the year of divestment. Distance scores are strictly positive and equal to $100-$Distance (in km). $t-1$ indicates that the variable is lagged. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

We further investigate the role of external opportunities by exploiting geographical variation in gas infrastructure from the Global Gas Infrastructure Tracker (Global

Energy Monitor, 2025), which records the location of operating gas export pipelines, LNG terminals, and natural gas power plants worldwide, as well as projects currently under construction, under discussion, shelved, or cancelled. We aggregate these into three categories: operating infrastructure, future infrastructure (under construction or under discussion), and cancelled infrastructure (shelved or cancelled). Since the dataset provides a cross-sectional snapshot as of 2025 rather than a panel, we cannot exploit time variation in infrastructure availability. Columns (2) and (3) of table 3.2 therefore use a cross-sectional specification in which the dependent variable is a dummy equal to one if the asset is eventually divested by its committed owners, and zero if it remains under committed ownership throughout the sample period.

Column (2) includes dummies indicating the presence of each infrastructure category within a 50km radius of the asset, while column (3) replaces these with distance scores (computed as 100 - Distance in km to the nearest infrastructure of each type), capturing proximity more continuously. Results are consistent across both specifications. The presence of (respectively proximity to) operating gas infrastructure is associated with a significantly lower probability of divestment, consistent with the idea that nearby gas infrastructure renders voice a more attractive strategy rather than exit. By contrast, proximity to cancelled infrastructure is associated with a significantly higher probability of divestment, likely reflecting environments where prior attempts to develop gas markets failed, signalling unfavourable local conditions for gas valorisation. The coefficient on future infrastructure is insignificant in both specifications, which may reflect uncertainty about whether planned projects will ultimately be completed.

Interestingly, the magnitudes of the two significant coefficients are roughly symmetric in absolute value. A natural interpretation is that in locations where both operating and cancelled infrastructure coexist, a configuration frequently observed in capacity-constrained basins such as those in the United States, the two effects approximately cancel out, leaving the probability of divestment close to that of assets with no nearby infrastructure at all. What matters for committed owners is not merely the presence of gas infrastructure, but whether that infrastructure has sufficient capacity to absorb co-produced gas. Where existing capacity has proven insufficient, as evidenced by the existence of discussed and subsequently cancelled projects, opportunities to export and/or use co-produced gas are inexistent and exit becomes comparably likely.

3.6 Robustness Checks

3.6.1 Placebo test: divestment from uncommitted to uncommitted

A key concern with our main results is whether we are capturing the specific effects of losing committed ownership, or simply generic divestment dynamics that would occur regardless of the environmental commitments of the transacting parties. If asset divestments inherently lead to operational disruptions, management turnover, or reduced capital investment that mechanically increases flaring, our estimates would conflate these generic divestment effects with the specific role of committed ownership. To address this concern, Figure C.7 examines divestments where uncommitted companies sell assets to other uncommitted companies. If our baseline results merely reflect generic divestment dynamics rather than the removal of committed oversight, we should observe similar post-divestment increases in flaring in this placebo sample.

The results provide strong validation for our main interpretation. Throughout both the pre and post-divestment periods, the estimated coefficients hover around zero with wide confidence intervals that consistently include zero. In the years following divestment, point estimates fluctuate between small positive and negative values with no clear directional pattern. Even in the subsample of assets with no gas sales infrastructure (red series), where operational constraints make flaring reduction most challenging, we observe no systematic increase in flaring following uncommitted-to-uncommitted transactions. This contrasts sharply with our baseline results, where the exit of the last committed owner led to immediate and persistent flaring increases of 40–90%. The absence of meaningful effects in this placebo sample indicates that generic divestment dynamics do not drive substantial changes in flaring intensity. Rather, the increases we document in Section 3.5.2 are specifically attributable to the removal of committed governance oversight from the management board.

3.6.2 Alternative measures of flaring

As explained in appendix C.3, flared volumes used as outcome variable throughout this study is the result of a matching between satellite observations and geographical data on oil and gas assets. This matching relies on several assumptions. We test the sensitivity of our results to the two most influential ones: the distance thresholds used to allocate observations into the different stages of the matching algorithm, and the matching exponent λ. Figure C.8 displays the evolution of flaring after commitment and divestment for three separate robustness checks. In panels A and B, the distance thresholds are all divided by 2. This change in assumption implies relying less on wells location data when matching satellite observations and relatively more on assets geographical shapes data. In panel C and D, a matching exponent of 1 is used instead of 4/3. Finally, in panels E and F, a matching exponent of 2 is used.

Point estimates in the different panels suggest that our main results regarding the evolution of flaring following commitment and divestment by assets owners are remarkably consistent across assumptions on flaring matching. The post-commitment decrease in flaring is virtually identical across the different panels. Similar conclusions can be drawn for post-divestment increases in flaring, to the exception of one point estimate, corresponding to the six years post-divestment change in flaring in panel A. The sudden drop in the coefficient, and the larger error bars can be explained by the lower number of observations participating to this point estimate. This definition of flaring can lead to allocating unique (and sometimes large volume) flaring observations to an entirely different subset of assets, and can result in introducing more noise in the data. The overall trend in post-divestment flaring practices remains unchanged excluding this single point estimate.

3.7 Conclusion

This paper studies how ownership structure shapes the effectiveness of voluntary environmental commitments in the global oil and gas industry. Using asset-level data on ownership and production from Rystad Energy matched with satellite-based flaring measurements from VIIRS, we track nearly 20,000 oil and gas assets across more than 120 countries between 2012 and 2024. We exploit variation in the timing of firm endorsements of the World Bank's Zero Routine Flaring by 2030 initiative and in subsequent ownership transactions to estimate how committed ownership affects flaring at the asset level.

We report two main findings. First, assets with at least one committed equity owner experience gradual and persistent reductions in flaring, reaching approximately 45% below pre-commitment levels after nine years. Committed operators reduce flaring quickly through direct control over field practices, while committed non-operators exert influence more slowly through governance channels. Assets without prior connection to gas markets require more time to reduce flaring but ultimately achieve larger reductions, consistent with committed owners advocating for the capital investments necessary to connect these assets to gas networks.

Second, these reductions are fully reversed when the last committed owner divests. Following complete exit by committed equity partners, flaring increases sharply, rising by roughly 40% within two years and exceeding 90% within six years relative to assets that retain committed ownership. This result, which we term the "last good owner effect," is specific to the removal of all committed governance oversight. When a committed firm exits while at least one other committed partner remains on the board, flaring does not increase. The effect also holds when the exiting committed firm was a non-operator and the same uncommitted operator continues managing the field, indicating that committed non-operators exercise meaningful governance influence over emissions independently of operational control. A placebo test using transactions between uncommitted parties confirms that these patterns do not reflect generic divestment dynamics.

The mechanisms we document are consistent with committed owners exercising active governance voice. During their tenure, committed owners constrain production expansion and drilling activity. Uncommitted acquirers reverse these constraints rapidly after divestment, investing in production-enhancing activities while disregarding the disposal of co-produced gas. Connection to gas markets during the commitment period substantially reduces the probability of subsequent divestment, suggesting that when voice succeeds in aligning asset operations with environmental objectives, the incentive to exit diminishes. Conversely, proximity to cancelled gas infrastructure projects increases the probability of divestment, reflecting environments where prior attempts to develop gas markets failed.

These findings have implications for the design and evaluation of voluntary environmental initiatives. Standard assessments of such programs typically measure outcomes at the firm or country level, where portfolio rebalancing through divestment can improve reported emissions without reducing actual pollution. Our asset-level analysis shows that this form of greenwashing through governance exit is quantitatively important: in divested assets, the post-divestment increase in flaring often exceeds the reductions achieved during the commitment period. While the cumulative environmental gains from the ZRF initiative across assets that remain under committed ownership outweigh the adverse effects from divested assets in aggregate, this conclusion cannot be drawn locally. For the subset of assets whose committed owners eventually exit, the net environmental impact of the initiative can be negative.

More broadly, our results suggest that ownership structure is a first-order determinant of whether voluntary commitments produce durable environmental improvements. Initiatives that rely on firm-level pledges without accounting for the governance structure of jointly owned assets risk creating incentives for committed firms to exit rather than invest, transferring pollution to less scrutinized owners. The evidence presented here indicates that retaining at least one committed voice on the management board is both necessary and sufficient to sustain emissions discipline, a finding that should inform both the design of voluntary frameworks and the regulatory oversight of asset transactions in emissions-intensive industries.

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Appendix A

Appendix to: The Price of Delay

A.1 Additional figures and tables

Oil Type	Average discount	2016 reserves (Gbbl)
Regular	-4.5%	919
NGL	-50.6%	253
Light	-2.1%	248
Condensate	-5.7%	120
Sour (> 3%)	-4.4%	55
Bitumen	-29%	47
Heavy Oil 15–19	-14%	34
Heavy Oil 20–22	-10.4%	22
Extra Heavy Oil	-12.3%	21
Synthetic crude	-7.4%	19
Total	-12%	1738
excluding NGL	-5.4%	1485

Table A.1: Discounts relative to Brent oil prices by oil type.

Notes: The table reports the reserve weighted average of discounts relative to the reference Brent oil price for each specific oil type. Discounts are computed as average discounts observed and predicted by Rystad Energy over the 2016-2100 period.

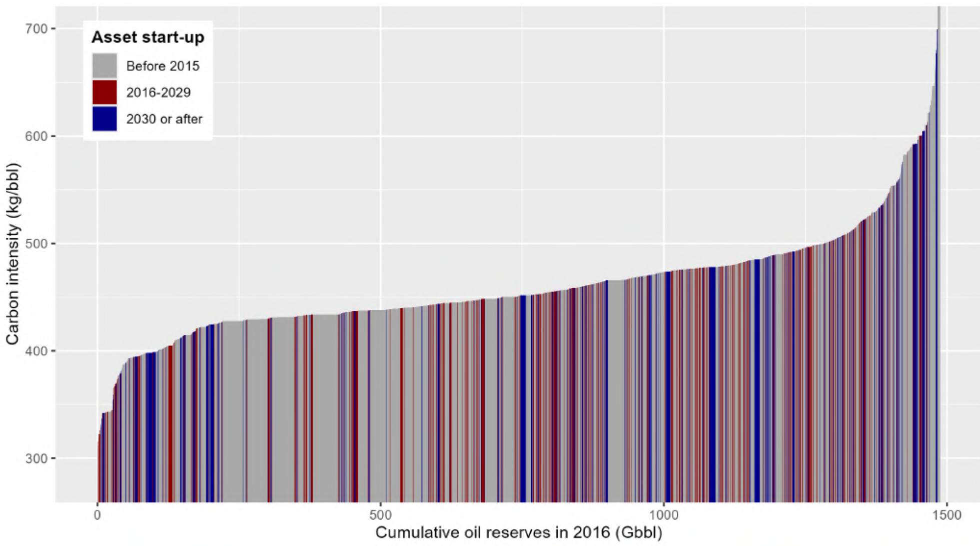

Figure A.1: Cumulative oil reserves in 2016 ordered by carbon intensities.

Notes: The y-axis corresponds to lifecycle carbon intensity, thus including upstream, midstream and downstream emissions. Cumulative reserves are split between assets whose production started before 2016, assets developed during the delay (2016-2029), and assets yet-to-be developed in 2030.

Economics Category	The Economics Category splits the Economics variable (=Revenue) into higher granularity than Economics Group, e.g. capital expenses is subdivided into facility and well expenses. The sum of all items equals the revenue. Use Economics Category to analyze breakdown of cost models or specific cost elements.
Free Cash Flow	FCF is the incurred cash flow available to investors and creditors, thus largely used in DCF valuation analysis. FCF = revenue-capex-opex-gvt take. FCF can be compared with cash from operations and investments activity in the cash flow statement from companies (only upstream). The present value of (all discounted) FCF equals NPV (Net Present Value also called Gross Asset Value) that can be compared with Enterprise Value (=Market cap+Debt for upstream activities), or compared with asset transaction costs. The discount factor in UCube is 10% flat for all assets.
Government Profit Oil	Gvt PO is the PSA equivalent to petroleum taxes, but paid in kind. Gvt PO reduces the company's entitelement production and is thus treated as a royalty effect in company reports.
Royalty effects	Royalty effects is the sum of all gross taxes, including royalty and oil and export duties.
Income Tax	Income tax is a sum of all profit based taxes, ie corporate taxes, where the tax rate is equal to the country's corporate tax rate, and special petroleum taxes. The tax base is the net sales revenue minus opex and depreciation. For petroleum taxes there may be applied an additional deduction called uplift. In many cases, some profit based taxes can be deducted in the tax base for other profit taxes. The income taxes in the UCube assumes there is no interest expenses.
Bonuses	Signature bonus for the license at award, as officially reported.
Well Capex	Well capex is capitalized costs related to well construction, including drilling costs, rig lease, well completion, well stimulation, steel costs and materials.
Facility Capex	Development capex except well construction costs, includes costs to develop, install, maintain and modify surface installations and infrastructure. As reported by operators, field partners or officials, or modelled. Distribution of costs over time is largely modelled, one exception is Norway, where all historic costs are as reported in national budgets.
Exploration Capex	*Costs incurred to find and prove hydrocarbons: seismic, wildcat and appraisal wells, general* engineering costs, based on reports and budgets or modelled.
Abandonment cost	Abandonment cost represents the costs associated with shutting down and dismantling the surface and subsea facilities such as surface equipment, platforms, subsea structures and pipelines. Activities included is topside and jacket removal as well as pipeline cleaning and decommissioning, onshore disposal, surveys and monitoring, and project management costs. Note that the Abandonment cost only includes the remaining P&A work for wells, and not full cycle P&A work. Abandonment cost is mostly modelled and assumed to be incurred and expensed over couple of years after the production has ceased. The estimate is proportional to initial greenfield capex.
Production Opex	Represents operational expenses directly related to the production activity. The category includes materials, tools, maintenance, equipment lease costs and operation related salaries. Depreciation and other non-cash items are not included.
Transportation Opex	Represents the costs of bring the oil and gas from the production site/processing plant to the pricing point (only upstream transportation). The category includes transport fees and blending costs.
SG&A Opex	Represent operation expenses not directly associated with field operations. The category includes administrative staff costs, office leases, related benefits (stocks and stock option plans) and professional expenses (legal, consulting, insurance). Only E&P related SG&A are included.
Taxes Opex	Represent gross tax reported by companies under operational costs and applies for US and Brazil. For US the category represent ad valorem taxes (county based) and severance taxes (state based).

Figure A.2: **Rystad economic data - Split items**: Descriptions of the data are provided by Rystad alongside with the UCube database.

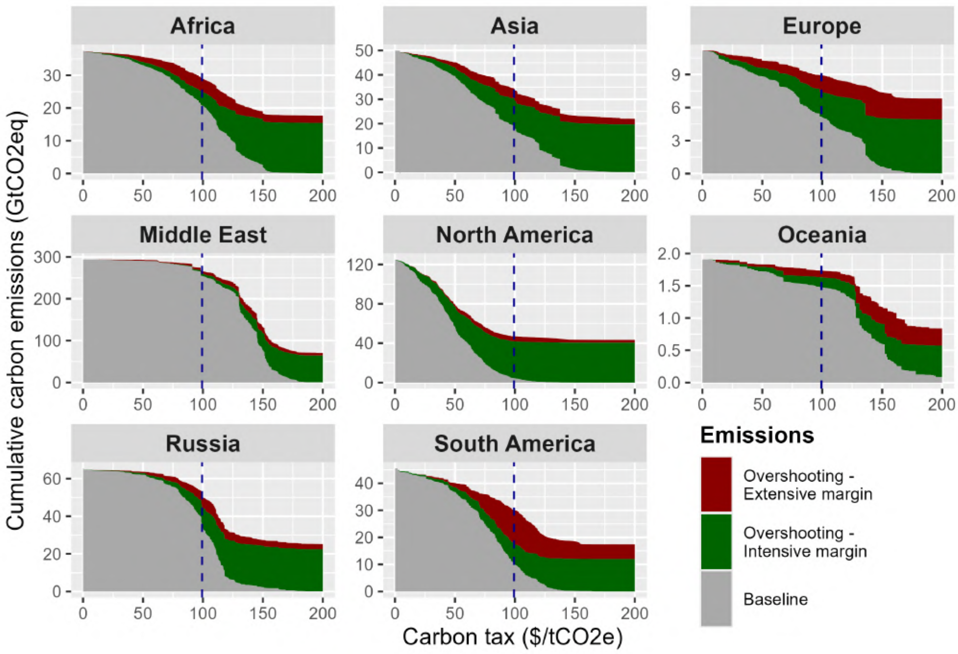

Figure A.3: Oil carbon budget overshooting by world region.

Notes: Each panel corresponds to a region of the world. For each panel, the x-axis is the level of carbon tax. The grey area corresponds to the baseline scenario, i.e. the cumulative emissions if the tax is implemented in 2016. The green and red areas depicts the overshoot resulting from delaying the tax until 2030 at the intensive and extensive margins respectively. The dotted horizontal line indicates the oil carbon budget set by Welsby et al. (2021). The dashed vertical line is the corresponding carbon tax needed in 2016 in order to constrain cumulative emissions within this budget. The scenario used to compute the overshoots is the "no anticipation" one. All computations are made assuming an oil price of $80/bbl.

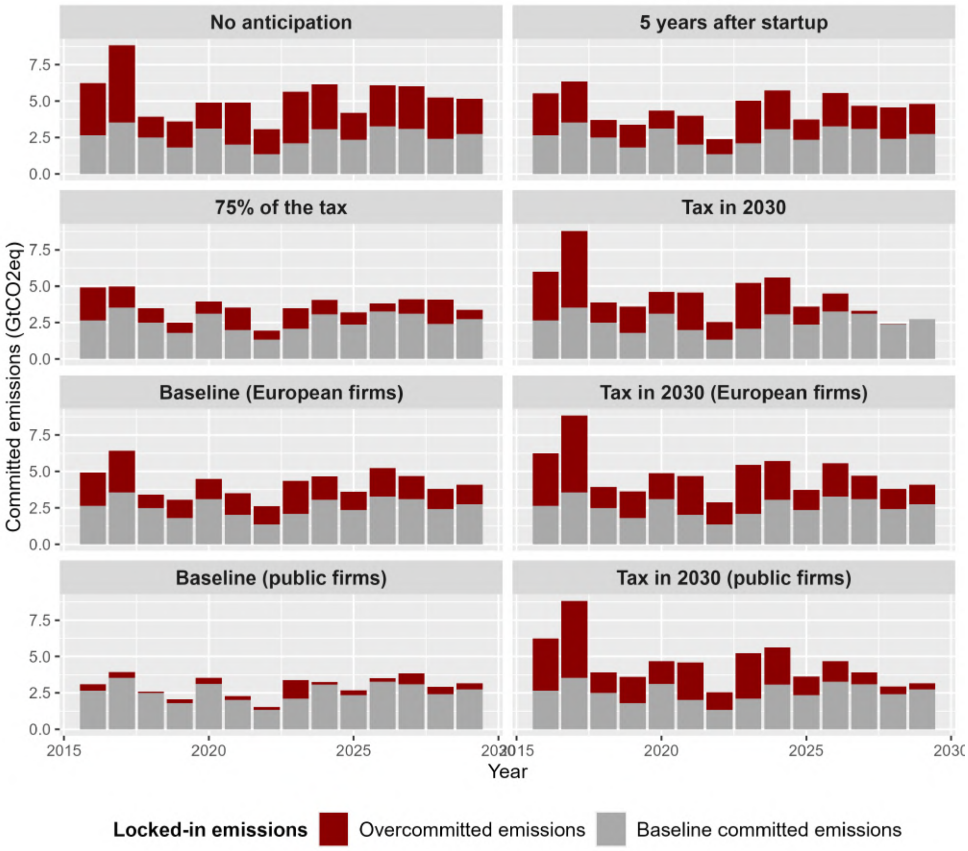

Figure A.4: Yearly committed emissions by scenario.

Notes: Each panel corresponds to a scenario. Each bar corresponds to the amount of committed emissions from investments in the development of new oil assets. The grey share of the bar depicts committed emissions that are emitted in the baseline scenario. The red share corresponds to committed emissions from assets that are not developed in the baseline scenario, thus representing the cost arising from carbon lock-in effects. Committed emissions from a given asset are attributed to the year from which the full development of this asset becomes inevitable, which depends on the level of tax. The level of tax used for this graph is \$100/tCO2eq. All computations are made assuming a reference Brent oil price of \$80/bbl.

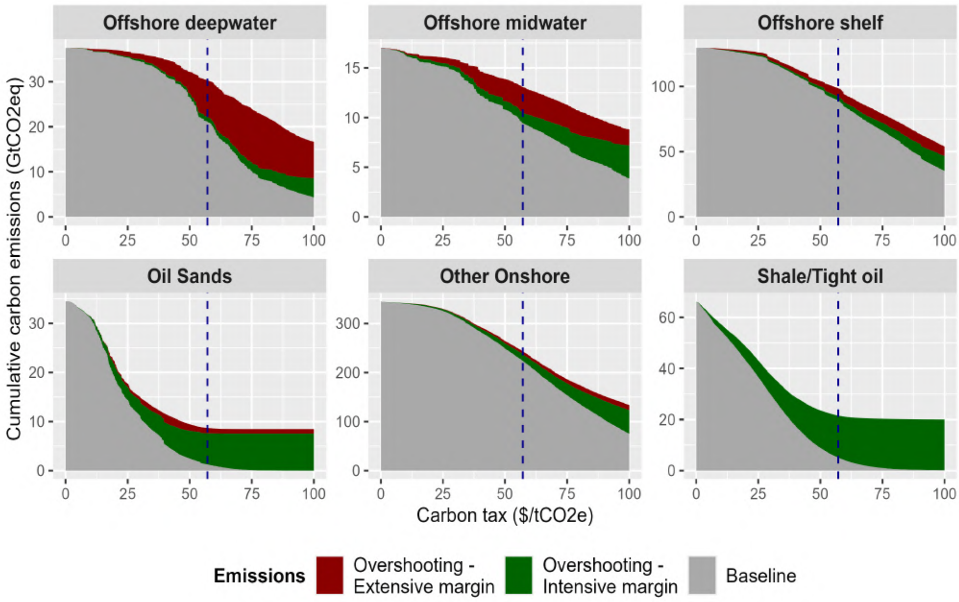

Figure A.5: Oil carbon budget overshooting by supply segment.

Notes: Each panel corresponds to an oil supply segment. For each panel, the x-axis is the level of carbon tax. The grey area corresponds to the baseline scenario, i.e. the cumulative emissions if the tax is implemented in 2016. The green and red areas depicts the overshoot resulting from delaying the tax until 2030 at the intensive and extensive margins respectively. The dotted horizontal line indicates the oil carbon budget set by Welsby et al. (2021). The dashed vertical line is the corresponding carbon tax needed in 2016 in order to constrain cumulative emissions within this budget. The scenario used to compute the overshoots is the "no anticipation" one. All computations are made assuming an oil price of $80/bbl.

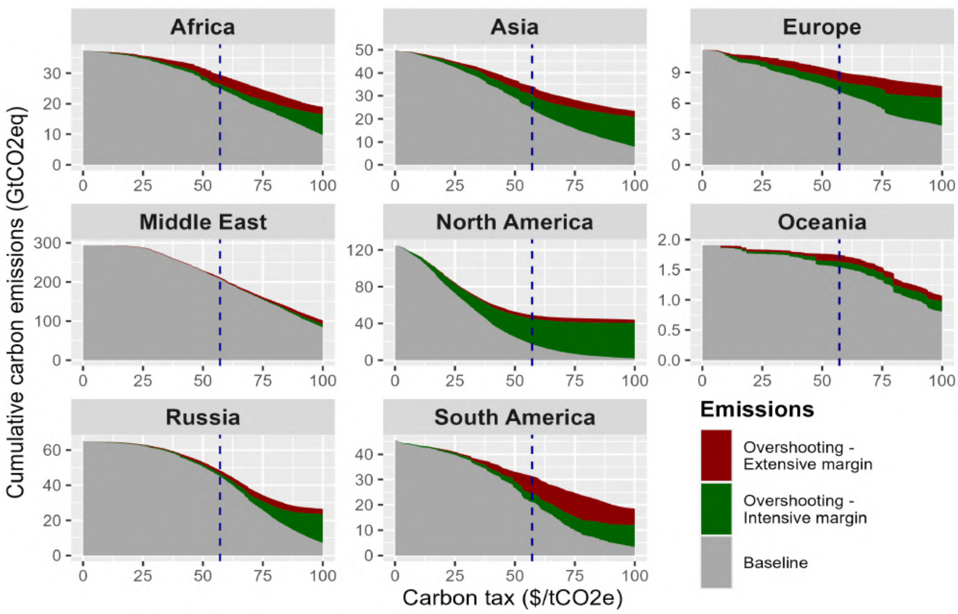

Figure A.6: Oil carbon budget overshooting by supply segment.

Notes: Each panel corresponds to an oil supply segment. For each panel, the x-axis is the level of carbon tax. The grey area corresponds to the baseline scenario, i.e. the cumulative emissions if the tax is implemented in 2016. The green and red areas depicts the overshoot resulting from delaying the tax until 2030 at the intensive and extensive margins respectively. The dotted horizontal line indicates the oil carbon budget set by Welsby et al. (2021). The dashed vertical line is the corresponding carbon tax needed in 2016 in order to constrain cumulative emissions within this budget. The scenario used to compute the overshoots is the "no anticipation" one. All computations are made assuming an oil price of $80/bbl.

A.2 Field level carbon intensities

While Rystad's databases provide detailed asset-level data on oil production, they lack comprehensive carbon intensity (CI) estimates. This appendix fills this gap by combining engineering-based models (OPGEE, PRELIM, OPEM), satellite-derived flaring data (VIIRS), and predictive regressions to estimate lifecycle CI for all assets, including those not yet in production. Carbon emissions related to oil can be split into 3 stages (Gordon et al., 2015). The upstream stage includes all emissions arising from extraction processes to the transport of crude oil to the refinery inlet. The midstream stage corresponds to refining processes and are heavily dependent on the refinery configuration. Finally, the downstream stage comprises emissions from the transport of oil products to their end-user and its final use. While downstream emissions usually represents around 80% of total emissions, the upstream stage features important variability from one field to the other.

For each of these stage, I rely on a state-of-the-art model developed by the Oil-Climate Index on a sample of currently produced oil to estimate their carbon intensity. I then fit a regression model for each stage to assign a carbon intensity to all other Rystad assets.

A.2.1 Upstream CI

A.2.1.1 Flaring data

Upstream emissions exhibit substantial heterogeneity across assets, and the major contributor to this heterogeneity is the flaring-to-oil ratio (FOR). Flaring corresponds to the combustion on-site of co-produced gas[1], and represents more than 150 billion cubic meters worldwide in 2024 (World Bank, 2025). To compute precise estimates of upstream CI, I use satellite data on flares to derive field-specific flaring-to-oil ratios.

Satellite flares observations and imputed emissions. Rystad UCube database does not contain information on gas flared volumes at the asset level. This is mainly due to the fact that most oil and gas operators do not publicly report flared volumes. Moreover, under-reporting by fields operators remains an important issue making the use of self-reported flaring less reliable.

To fill this gap, I use geocoded flaring volumes calculated by the Earth Observation Group (EOG) at the Payne Institute for Public Policy, within the Colorado School of

[1] Co-produced gas is often used to power equipments on-site, or sold, if the asset is connected to the local gas market or possess equipments to liquefy and ship it. If not, excess gas is then either reinjected, to increase pressure in the reservoir, flared, or directly vented. Given the higher global warming potential of methane, flaring is preferable to venting, and venting is banned in most countries except for rare emergencies. Gas reinjection is often costly, making flaring an easier decision in many oil fields.

Mines. The EOG analyzes satellite data from the Visible Infrared Imaging Radiometer Suite (VIIRS), on board of satellites launched in 2012 and 2017, which detects heat from gas flares at night. Flaring volumes are then estimated using the VIIRS Nightfire algorithm (VNF), as detailed by Elvidge et al. (2015) and further refined by Zhizhin et al. (2021). I use the 141,841 observations with positive flared volumes between 2012 and 2024 to assign them to specific assets from Rystad's database.

Rystad assets geographic information. In order to assign flaring observations to Rystad assets, I need geographic information on these assets. I complement data from the UCube with two sets of geocoded data obtained from the same company, one in 2022 and the other in 2025.

First, geocoded boundaries were shared by Rystad in 2022 for 41,762 assets, among which 16,801 were generic (i.e. a 500m radius circle). Since the universe of Rystad assets has evolved between 2022 and 2025, 1,916 generic shapes were added for assets missing in the shapefile, based on their geocoded centroid.

Secondly, data was obtained on the location of wells completed between 2000 and 2025. This dataset contains the latitude and longitude of 1,733,698 producer, injector or wildcat wells, from 39,494 assets. In some regions, the location of wells correspond to the asset centroid, meaning that Rystad probably does not have its precise location. In total, 857,900 unique locations exist in the dataset. Crucially, wells locations are precisely defined for onshore US and Canada assets, where all assets boundaries are generic, making this dataset a useful complement to the first one.

Matching satellite observations to Rystad assets. The assignment of satellite observations to Rystad assets relies on both assets' eligibility (the existence or not of surface wells, excluding for example subsea tiebacks and not yet completed wells) and geographic distances to existing wells and assets' boundaries. For each observation, distances to all completed wells and assets in a 20km radius were computed. A score reflecting the probability of a detected flare to originate from each nearby asset is then computed. For observations with existing wells in a 2km radius, only information on wells was used to compute this score. For other observations, distances to both assets and wells were incorporated in the score. When scores were too close to allow to decide between assets, flared volumes were assigned based on each eligible asset's score and production capacity. Appendix C.3 provides more detailed information on the 5-stage matching procedure used to assign flaring observations to Rystad assets.

Overall, all observations with at least one eligible well or asset in a 20km radius are matched, leaving 7,261 observations unmatched (corresponding to 4.7% of total flared volumes between 2012 and 2024). Among matched observations, around two thirds were assigned to a unique asset (70.4% of matched flared volumes) and the rest split between multiple assets.

Computing flaring-to-oil ratios (FOR). Once the matching procedure is done, total flared volumes over the 2012-2024 period are computed and divided by total oil production to obtain asset-specific FOR. For not-yet-active assets over this period, an average FOR is assigned. Two categories of assets producing over the period were also attributed averages: assets with at least 75% of years in the sample with positive production outages, mainly due to wars in Lybia and Syria (109 assets), and assets which production started less than 2 years before the end of the period (911 assets). The average FOR is computed at the lowest scale possible, for asset of similar size if possible[2], and always by oil type. Averages are capped at 12,000 scf/bbl, corresponding to twice as more gas flared than oil produced in barrel of oil equivalent. Table A.2 reports the number and type of average FORs assigned to Rystad assets, and the corresponding oil reserves.

FOR origin	Number of assets	Share of oil reserves in 2016
From satellite matching	20,571	64.3%
Area x Oil Type x Field Size	5,133	4.2%
Province x Oil Type x Field Size	7,350	7.3%
Province x Oil Type	10,497	8.9%
Country x Oil Type x Field Size	6,662	3.3%
Country x Oil Type	4,250	4.8%
Region x Oil Type x Field Size	2,948	3.4%
Region x Oil Type	3,136	3.1%
World x Oil Type x Field Size	946	0.9%
World x Oil Type	2	0.0%

Table A.2: Origin of FOR values.

Table A.3 shows the binned distribution of computed FOR. Most assets display relatively low levels of flaring with around a third of them having a FOR equal to 0 (representing around 22% of oil reserves). In these assets, the 0 value can be explained by either complete use of co-produced gas or particularly low gas-to-oil ratio, but it can also be due to flared volumes so low (or so intermittent) that it falls below the satellite detection threshold. Among assets producing mainly oil (assets with gas-to-oil ratios below 10,200 scf/bbl), the reserves weighted average FOR is 104.4 scf/bbl, and the median is 13.4 scf/bbl.

[2] Assets are split in 10 categories depending on their reserve size, with thresholds at 1, 3, 10, 30, 100, 300, 1000, 3000 and 10000 MMboe.

FOR (kscf/bbl)	Assets count	Reserves (Gbbl)
0	11,651	345.6
0–0.1	10,820	748.7
0.1–1	10,974	435.4
1–10	1,069	21.0
>10	291	2.5

Table A.3: Distribution of flaring-to-oil ratios.

A.2.1.2 OPGEE model

The Oil Production Greenhouse Gas Emissions Estimator (OPGEE) is an engineering-based model developed by the Oil-Climate Index team (Gordon et al., 2015)[3]. It measures GHG emissions from extraction, processing and transport of crude oil in grams of CO2 equivalent per megajoule of oil produced.

Data inputs and matching with Rystad assets. Running the model requires a wide range of inputs for each field, from extraction methods to gas composition. When information is missing, the model computes smart default values (deduced from other variables when possible). I recover publicly available input data from Masnadi et al. (2018) on 958 oil fields worldwide, and match it to Rystad assets manually. Among the 958 original fields, 737 were matched to 1,136 assets in the Rystad database. Most unmatched fields correspond to marginal heavily-depleted oil fields in the US. The difference between the number of OPGEE fields and matched Rystad assets comes from differing level of aggregation. In many instances, a unique OPGEE field is split into multiple assets in Rystad data. In rare occasions, multiple OPGEE fields were matched to a unique Rystad asset.

After completing the matching procedure, I complement the 737 fields dataset with information on 6 variables from Rystad data in order to improve the accuracy of my estimates. These variables are: production level, field age, gas-to-oil ratio, flaring-to-oil ratio, number of producer wells and number of injection wells. Since Rystad databases do not contain information on the quantity of gas reinjected, used on site, or vented, I assume the amount of gas produced to be equal to the sum of flared and sold gas in order to compute gas-to-oil ratios. Finally, extraction methods reported in Masnadi et al. (2018) input data were adjusted if necessary to match information retrieved from Rystad[4].

Computing CI. The resulting input dataset was fed into the OPGEE model to export results. Two distinct carbon emissions accounting methods are available within

[3]Latest versions of the model are available at: https://eao.stanford.edu/research-project/opgee-oil-production-greenhouse-gas-emissions-estimator

[4]This is for example the case for water reinjection. If Rystad data reports existing injection wells, at least water or gas injection must be indicated in OPGEE input data. When this was not the case, water reinjection was assumed to happen, since it more common in oil fields than gas reinjection.

the model: energy-based allocation (EBA) and carbon product displacement (CPD). In the EBA approach CI are defined such that $CI_{EBA} = \frac{TotalEmissions_{field}}{Prod_{oil}+Prod_{gas}}$, with $Prod_{oil}$ and $Prod_{gas}$ in MJ. This means that total emissions from the oil field are computed, and allocated to oil and gas sold on the market according to their energy content. In the CPD approach, the definition becomes $CI_{CPD} = \frac{TotalEmissions_{field} - Prod_{gas}.\overline{CI}_{gas}}{Prod_{oil}}$. Co-produced gas gives rise to carbon credits corresponding to the energy content of gas produced multiplied by $\overline{CI}_{gas}$, a worldwide average carbon intensity of gas production computed from the CA-GREET model. The latter approach can lead to negative oil CI, which is why the retained approach was the EBA one.

Figure ?? displays CI computed using the model. The production-weighted average carbon intensity is equal to 59 kgCO2eq/bbl (indicated by the dashed red line on the figure). The variance among fields is considerable, with 10% and 90% weighted deciles ranging from 35 kgCO2eq/bbl to 113 kgCO2eq/bbl. Three key features influencing CI are indicated on the graph:

- **FOR**: as expected, large CI values often originate in large quantities of co-produced gas flared. The influence of FOR on CI largely depends on the assumptions made on destruction removal efficiency (DRE). This value measures to the efficiency of flares, i.e. the share of gas effectively combusted and thus transformed into CO2. Since the global warming potential of methane is 28 times larger than the one of carbon dioxide, assumptions on DRE greatly affects computed CI. I keep the default value of 96.1% retained by the model to compute baseline CI, and provide in sensitivity analysis results using values of 91.1% and 98% to cover the range of estimates found in the literature[5](Plant et al., 2022; Jackson and Smith, 2014; Gvakharia et al., 2017; Zavala-Araiza et al., 2021).

- **Steam injection**: Heavier oils (API below 20) tend to be thicker than lighter ones, and often require the injection of steam in order to lower its viscosity and allow its extraction. This is captured by the steam-to-oil ratio (SOR), corresponding to the number of barrels of vaporized water injected to extract one barrel of oil. High SOR values imply larger quantities of liquids extracted, and therefore larger energy consumed to extract oil.

- **Upgrading into synthetic oil**: some particularly viscous oil (bitumen or extra heavy) are extracted without necessitating reducing its viscosity first (for instance using steam injection). This is mainly the case for Canadian oil sands, which are often directly extracted in open air mines. However, they are often too viscous to flow through pipelines and be handled by existing refineries. These oils are therefore

[5] While playing a much smaller role than FOR, the quantity of co-produced gas (GOR) can still influence extraction emissions through fugitive emissions of methane related to its processing and transport on site.

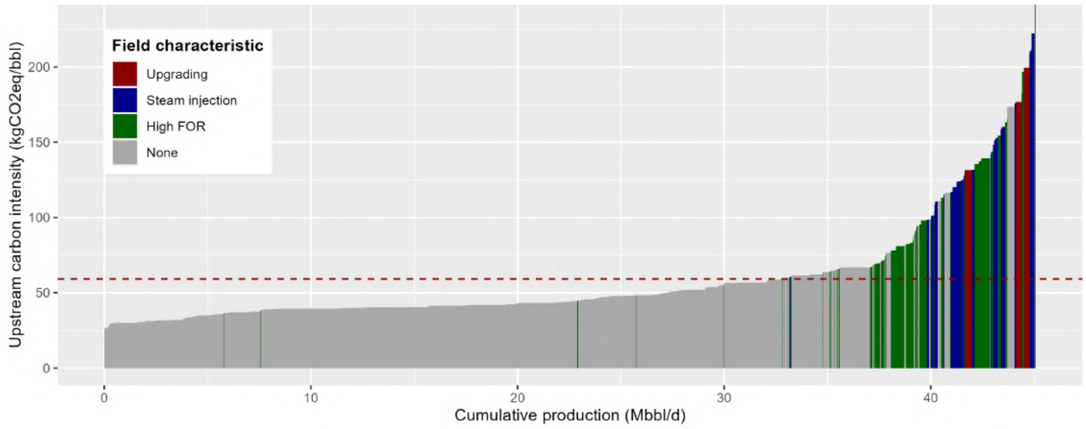

Figure A.7: OPGEE-computed CI.

Notes: The figure depicts CI computed in the OPGEE model, with cumulative production in 2015 on the x-axis. The horizontal dashed red line indicates the production-weighted average.

"upgraded" into synthetic crude on site. This requires using chemical treatments and/or distillation in order to break long hydrocarbon chains into shorter ones, thus reducing viscosity. These processes are often very energy intensive, translating into large CI for fields using it (Brandt and Farrell, 2007). Produced synthetic crude is however easier to refine, translating into relatively lower midstream CI[6].

A.2.1.3 Estimation model

Model and sample restrictions. Once the 737 OPGEE-computed CI are attributed to the corresponding 1,136 Rystad assets, I fit a regression model to explain across-field variations. Following the logic of the EBA approach, the dependent variable in the model thus consists of the field carbon intensity divided by its "energy denominator" $ED_f = \frac{Prod_{oil,f}}{Prod_{oil,f}+Prod_{gas,f}}$. The retained model is:

$$\begin{aligned} CI_f^{OPGEE} * ED_f \quad &= \textstyle\sum_0^{10} \beta_i^C ShareOilType_f + \beta_{11} Offshore_f + \beta_{12} SOR_f \\ &+ \beta_{13} GOR_f + \beta_{14} FOR_f + \epsilon_f, \end{aligned} \tag{A.1}$$

where f denotes a Rystad asset, $ShareOilType_f$ the share of each oil type in its production[7], $Offshore_f$ a dummy indicating if the field is offshore, SOR_f its steam-to-oil ratio (in bbl of water/bbl of oil), GOR_f its gas-to-oil ratio (in scf/bbl) and FOR_f its flaring-to-oil ratio (in scf/bbl).

[6]Midstream CI for synthetic crudes are on average 30% lower when compared to other extra heavy or bitumen oils, and still 8% lower when compared to all types of oil.

[7]An asset produces a unique type of crude oil, among 8 possible (synthetic, bitumen, extra heavy, heavy (15-19), heavy (20-22), regular, light and sour). It can however produce simultaneously condensate and/or natural gas liquids (NGL), which are sold as oil, similar to light crude.

Oil types variables, together with the offshore dummy, capture various aspects of the extraction methods and facility characteristics on site. This is especially relevant for synthetic crude, where upgrading processes such as distillation or chemical treatments significantly increase CI. SOR, GOR and FOR have a more straightforward and linear impact on carbon emissions, and explain most of the remaining variance. SOR data used for the estimation is the one used as input in the OPGEE model since precise values are not available in Rystad data.

The sample used in the regressions was restricted to fields with water-to-oil ratio below 40 bbl of water/bbl of oil, and gas-to-oil ratio below 40,000 scf/bbl. The first restriction excludes fields almost fully depleted, characterized by exponentially increasing WOR[8]. The second one is motivated by the fact that my focus is on oil producing fields. Oil fields are defined by Rystad as reservoirs with GOR inferior to 10,200 scf/bbl. I exclude fields with GOR higher than four times this threshold in order to keep all fields producing non-negligible oil quantities. Altogether, these two restrictions exclude 20 Rystad assets, making the final sample 1,116 assets (associated to 719 OPGEE fields). Weights were added to the regressions in order to keep each unique OPGEE-computed values equivalent. If a unique OPGEE value was matched to multiple Rystad assets, weights were computed based on oil reserves size and sum up to 1.

Regression results. Table A.4 shows coefficient estimates from the regression for two sets of CI: the first corresponds to my baseline estimates, using a 100-years horizon for global warming potential, and the second one use a 20-years horizon (associated with higher global warming potential for methane as compared to carbon dioxide, due to its shorter lifetime). R-squared values are close to 1, and predicted R-squared are also quite close to 1, indicating that the model's predictive power is relatively high.

As expected, synthetic crude production is associated with large CI, and coefficients for heavier crudes tend to be larger than the ones of lighter crudes. While coefficients associated with bitumen and extra heavy crudes are not significantly higher than others, production of these two types often implies the use of steam injection, which means that most of the variation is captured by the SOR variable. The negative coefficient associated with the offshore dummy reflects the absence of land-use transformation, source of GHG emissions. The three remaining variables explain most of the variance. Steam injection greatly increases emissions (the average steam-to-oil ratio is 3 bbl of water/bbl of oil). While having a relatively low magnitude, GOR still plays an important role in explaining the variance in carbon intensity across fields. Fields producing gassy oils, even when selling their co-produced gas, are subject to higher quantities of fugitive emissions. Finally, FOR plays the most crucial role in explaining the variance across fields, as noted by Masnadi

[8]I exclude these fields because Rystad data does not contain information on this ratio. The threshold of 40bbl/bbl is particularly high and only excludes 7 fields from the sample.

et al. (2018).

Regression using 20-years horizon CI as dependent variable display similar results. By construction, reducing the time horizon can only increase CI: while carbon dioxide emissions are not affected, methane emissions lead to higher global warming potential. Consequently, variables associated at least partially with methane emissions see their associated coefficients increase in large proportions. This is particularly visible when comparing the SOR, GOR and FOR variables. While steam injection does not entail methane emissions, gas production leads to fugitive methane emissions and flaring does not fully convert methane into carbon dioxide (depending on the retained DRE value as explained above). The large increase in the coefficients associated to GOR and FOR, along with the SOR coefficient remaining unchanged is thus consistent with this mechanism.

Sensitivity analysis. Figure A.8 highlights the strong correlation between OPGEE-computed CI and CI predicted using the model for the same assets sample. To verify the influence of outliers, I re-estimate coefficients of equation A.1 excluding each observation one by one. Figure A.8 displays estimated coefficients, with the baseline estimate indicated in red. While heavier oils types coefficients can vary due to the low number of assets producing it, results indicate that my estimates are overall robust to changes in the sample. Finally, I test the sensitivity of estimated coefficients to the exclusion of the most influential observations. I compute the Cook's distance of each observation, defined by $CD_i = \sum_{j=1}^{N} \left(\widehat{y}_j - \widehat{y}_{j(i)}\right)^2 / ps^2$, where $\widehat{y}_j$ and $\widehat{y}_{j(i)}$ are the fitted response values from using the full sample and after excluding field i, respectively; s^2 is the mean squared error of the regression model, p the number of coefficients, and N the number of observations. Table A.5 displays re-estimated coefficients excluding the 1, 10 and 50 most influential observations. Heavier oil types see their estimate vary slightly, due to the small number of assets producing it in my OPGEE sample, while SOR, GOR and FOR coefficients are particularly stable. Predicted R-squared stay high, even when removing the 50 most influential observations. Overall, re-estimated coefficients remain consistent with my original estimates using the full sample.

A.2.1.4 Predicting upstream carbon intensity for all Rystad assets

Using estimated coefficients, predicted upstream CI were computed for each Rystad asset. Data on reserves by oil type, offshore status, and GOR was directly available in Rystad databases.Asset-specific FOR were either derived from satellite observations or predicted for not-yet-producing assets, as described in the Flaring data section. Values for other needed variables were assigned as described below:

- SOR: on top of assets matched to OPGEE fields, assets using steam injection were identified and assigned a SOR value of 3, in line with both the default value in the

Variables	EBA - AR6 100y	EBA - AR6 20y
Oil type: Synthetic	157.096***	192.358***
	(6.514)	(9.943)
Oil type: Bitumen	52.854***	78.442***
	(4.555)	(6.952)
Oil type: Extra Heavy	48.122***	78.368***
	(4.169)	(6.363)
Oil type: Heavy (15-19)	54.778***	96.841***
	(3.73)	(5.693)
Oil type: Heavy (20-22)	58.861***	97.629***
	(3.561)	(5.436)
Oil type: Regular	46.522***	79.708***
	(1.105)	(1.687)
Oil type: Light	46.939***	77.204***
	(1.862)	(2.842)
Oil type: Sour (> 3%)	43.308***	71.654***
	(5.14)	(7.846)
Oil type: Condensate	43.535***	72.446***
	(4.235)	(6.465)
Oil type: NGL	48.05***	83.603***
	(6.641)	(10.137)
Offshore	-6.048***	-8.435***
	(1.273)	(1.943)
SOR (bbl water/bbl oil)	26.317***	27.369***
	(0.983)	(1.501)
GOR (kscf/bbl)	4.601***	8.7***
	(0.154)	(0.236)
FOR (kscf/bbl)	83.908***	146.98***
	(0.533)	(0.813)
Corrected R^2	0.970	0.977
Predicted R^2	0.936	0.943

Table A.4: Regression results from upstream emissions estimation

Notes: The table displays coefficients from the regressions using equations A.1, with standard errors indicated between parenthesis. R^2 are corrected in order to account for the absence of intercept, as well as predicted R^2.

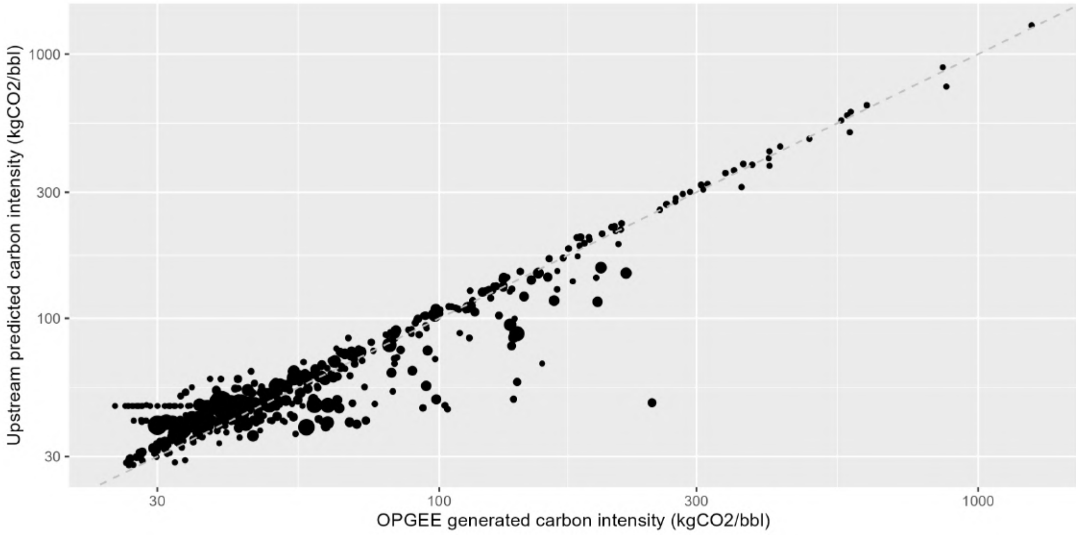

Figure A.8: Predicted vs OPGEE-computed upstream CI.

Notes: The figure displays upstream CI computed using the OPGEE model on the x-axis and predicted using the model on the y-axis. Only assets matched with OPGEE fields are included. Both axis are transformed using the decimal logarithm. The size of each point reflects the log volume of oil reserves for each asset.

	Synthetic	Bitumen	Extra Heavy	Heavy 15–19	Heavy 20–22
0	157.1*** (6.51)	52.85*** (4.55)	48.12*** (4.17)	54.78*** (3.73)	58.86*** (3.56)
1	157.12*** (6.29)	53.14*** (4.40)	48.48*** (4.02)	55.42*** (3.60)	59.49*** (3.44)
10	133.06*** (6.51)	56.86*** (3.70)	51.35*** (3.37)	53.23*** (3.07)	50.40*** (2.99)
50	154.73*** (14.33)	51.56*** (2.88)	47.81*** (2.50)	49.14*** (2.40)	49.60*** (2.64)
	Regular	Light	Sour (>3%)	Condensate	NGL
0	46.52*** (1.11)	46.94*** (1.86)	43.31*** (5.14)	43.54*** (4.24)	48.05*** (6.64)
1	47.42*** (1.07)	46.87*** (1.80)	43.98*** (4.96)	44.40*** (4.09)	49.04*** (6.41)
10	47.62*** (0.90)	45.37*** (1.50)	44.17*** (4.12)	45.48*** (3.41)	53.37*** (5.37)
50	44.59*** (0.65)	41.86*** (1.07)	44.73*** (3.05)	43.83*** (2.51)	46.00*** (3.87)
	Offshore	SOR	GOR	FOR	Pred. R^2
0	-6.05*** (1.27)	26.32*** (0.98)	4.60*** (0.15)	83.91*** (0.53)	0.934
1	-6.85*** (1.23)	26.22*** (0.95)	4.62*** (0.15)	82.70*** (0.53)	0.931
10	-7.39*** (1.03)	24.97*** (0.81)	4.51*** (0.13)	82.67*** (0.54)	0.918
50	-5.21*** (0.74)	26.02*** (0.67)	4.54*** (0.09)	83.93*** (0.46)	0.910

Table A.5: Coefficient stability to influential observations (Cook's distance).

Notes: Each panel shows re-estimated coefficients using equation A.1 after excluding the 0, 1, 10, and 50 most influential observations. Cook's distances were computed to determine the influence of each observation. The last panel indicates predicted R-squared for each of the re-estimated models.

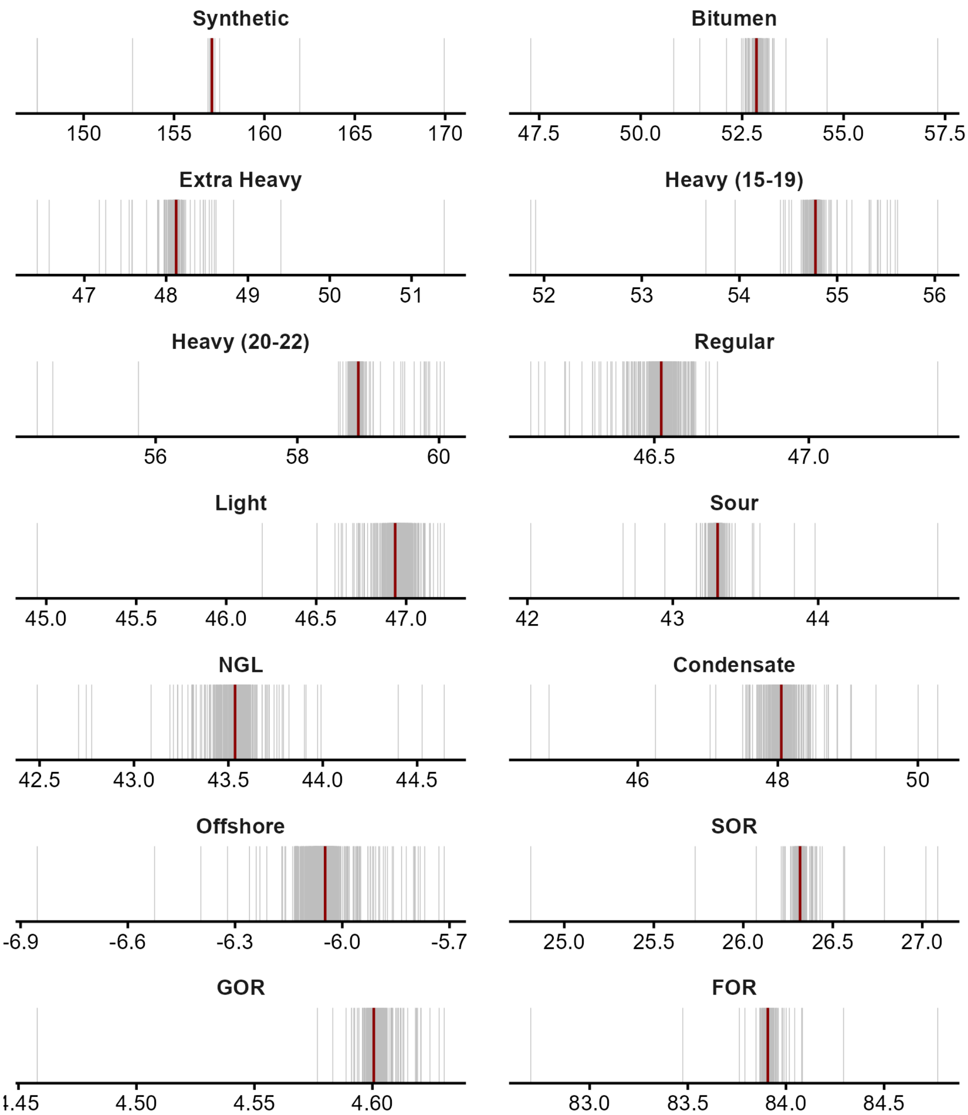

Figure A.9: Upstream CI estimation: Estimates of the coefficients on the explanatory variables removing the OPGEE fields one-by-one.

Notes: The panels show the estimated coefficients on each of the explanatory variables in equation (A.1) removing observations one-by-one. Each grey line corresponds to a different estimation with the same number of observations: $N(OPGEE_{sample}) - 1$. The thick red lines show the original estimates obtained with the full sample of OPGEE publicly-available fields (OPGEE sample).

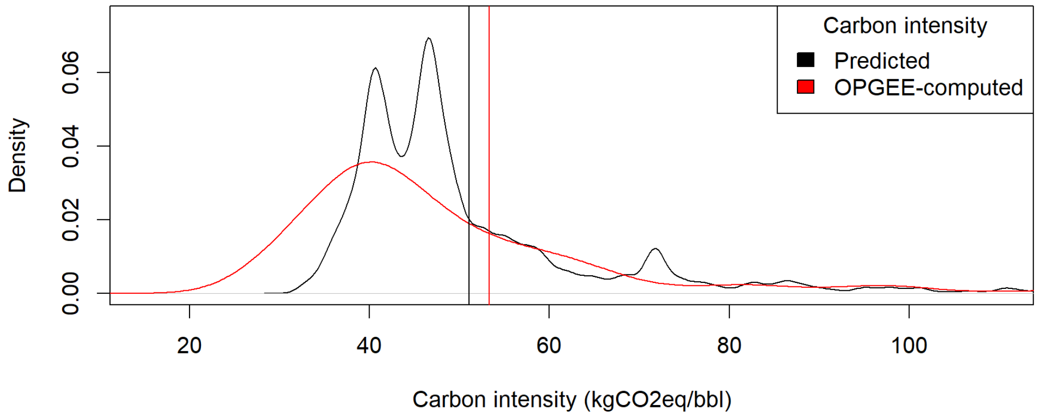

Figure A.10: Distribution of CI: Predicted vs OPGEE-computed.

Notes: The figures shows in red the distribution of baseline CI (100-years horizon, EBA) for the OPGEE sample, i.e. assets matched with OPGEE fields and for which the OPGEE value was kept, and in black predicted CI for all remaining assets. Densities are weighted by oil production volumes over the 2012-2024 period. Vertical lines indicate weighted averages for both samples.

OPGEE model and the average SOR value in OPGEE fields data from Masnadi et al. (2018). All assets containing "Steam", "Thermal" or "SAGD" in their name were identified as using steam injection. In addition, the list of Enhanced Oil Recovery projects made available by the EIA was used to manually identify remaining steam injecting assets (McGlade et al., 2018).

- ED_f: energy denominators for each asset were approximated by $ED_f = 1 + \frac{GOR_f - FOR_f}{6000}$, using a 6,000MJ average energy content per barrel. This approximation is remarkably close to energy denominators derived from the OPGEE model, with a correlation coefficient of 0.975 within the OPGEE matched assets sample.

When a unique Rystad asset was matched with a unique OPGEE field, the carbon intensity computed using the OPGEE model was kept. For all remaining assets, predicted values prevail. Figure A.10 shows the 2012-2024 production-weighted distribution of CI for both samples, with weighted means depicted by the vertical lines. Although averages are remarkably similar, the predictive model does not seem to fully replicate the variance across OPGEE-computed values. The general shape of the distribution remains coherent with OPGEE outputs.

A.2.2 Midstream-downstream CI

Midstream emissions correspond to emissions from refining crude oil, while downstream emissions include emissions from both transport to end-users and final use (often combustion) of the different oil products obtained from refining. Refining crude oil yields multiple oil products depending on its characteristics and refinery configurations. Midstream and downstream emissions are typically computed at the crude stream level, which corresponds to the actual crude stream arriving to the refinery inlet. In some instances, a crude stream originates in a single oil field, while the majority of crude streams consists of multiple fields outputs blended together. For instance, the Brent crude used as reference in oil pricing is a blend of crude oils produced in 13 separate assets in the North Sea.

A.2.2.1 PRELIM and OPEM models

Midstream and downstream CI can be computed using the Petroleum Refinery Life Cycle Inventory Model (PRELIM) and Oil Products Emissions Module (OPEM) respectively[9]. The versions used in this study to compute CI are PRELIM 1.6 and OPEM 1.1. The PRELIM model requires detailed information on crude assays, such as API gravity, sulphur content, and various boiling points. This data is typically not made available by producers, and publicly available data usually does not follow the same format and is therefore unusable in the models. Models authors thus included in the model data on 657 unique crude assays, that we match to Rystad assets they originate from.

Running the model. The PRELIM model allows for a large array of refinery configurations, designed to handle different types of oils and aimed at maximizing the production of specific oil products. Since I do not have sufficient information to match Rystad assets to the refineries treating their oil production, I follow the assumptions of the model on refinery configurations. Among the three main configurations used worldwide, the model selects the adapted configuration based on crude oil characteristics in terms of API gravity and sulphur content. All heavy crudes (API gravity below 22) are assigned the "deep conversion" configuration, entailing additional treatments aimed at reducing the length of carbon chains yielded. Light crudes with low sulphur levels (API gravity above 32 and sulphur content below 0.5%) are assigned the "hydroskimming" configuration, associated with less treatment stages and in turn less emissions. All other crudes are refined using the "medium conversion" configuration. The importance of refinery configuration is illustrated in figure A.11, depicting PRELIM-computed midstream CI. Intensive treatment of crude in the case of deep conversion is associated with higher emissions, while hydroskimming units yield much smaller emissions.

Once midstream emissions are computed, PRELIM output sheets are imported to the OPEM model in order to estimate downstream CI of each crude stream. Emissions from transport and final combustion (for oil products used as fuel) are computed, giving

[9]Information and most recent versions of the models can be found at: https://ociplus.rmi.org/methodology.

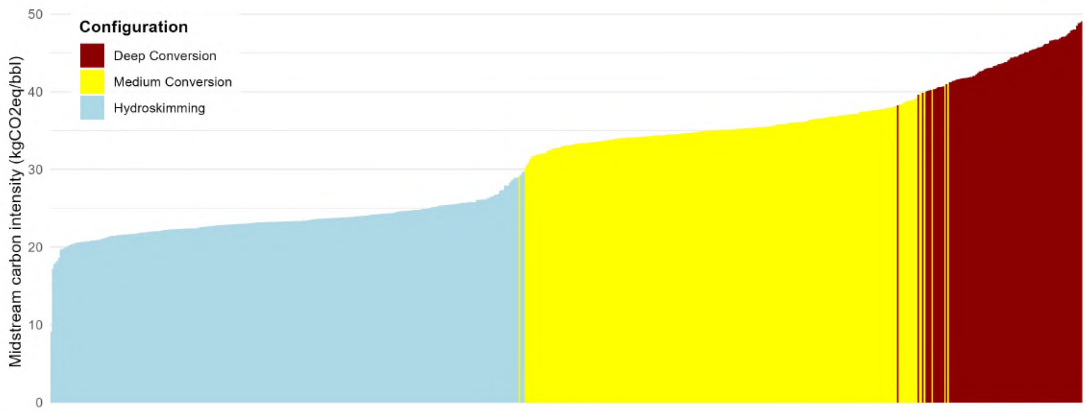

Figure A.11: PRELIM-computed midstream CI

me for a given barrel of crude oil its final products yield and associated emissions. Final combustion represents the larger share of emissions along oil lifecycle, and displays smaller variability across crudes.

A.2.2.2 Matching PRELIM crude assays to Rystad assets

Crude streams available in the PRELIM model are matched to Rystad assets using data made available by Rystad on crude streams and associated assets. The matching is performed using fuzzy matching methods on crudes names. After cleaning crudes names from both sources, Jaro-Winkler distances were calculated and capped at 0.25. Matches were then manually verified, leaving me with 493 crude assays from PRELIM (from the original 657) matched to 392 crude streams from Rystad[10]. Additionally, 25 unmatched PRELIM assets were manually matched to 41 Rystad assets with no specified crude streams. Overall, 518 PRELIM crude assays were linked to 8,386 Rystad assets, representing 64.4% of world production in 2024.

A unique crude stream often consists of a blend of crude oils from multiple assets, but in certain cases (mainly in Canada), production in a unique asset is split between different crudes. For these assets, midstream and downstream CI from PRELIM/OPEM were averaged weighting by the share of production being allocated to each crude stream.

A.2.2.3 Estimation models

Model and sample restrictions. Using PRELIM output, I fit a regression model to explain the variance in midstream and downstream CI separately. In both cases,

[10]The difference in number comes from unique crudes streams being split into different crude assays in PRELIM differing by the data source. For instance, the "Hibernia" crude in Canada is split into 3 assays, corresponding to information on this specific crude obtained from ExxonMobil, Statoil and Chevron.

I use the two most influential variables, API gravity and sulphur content, which are also available at the asset level in Rystad data. Refinery configuration plays the most important role in explaining variations. I follow the assumptions made in the PRELIM model and assign "deep conversion", "hydroskimming" and "medium conversion" based on the same rules on API gravity and sulphur content. Both variables (in log form for sulphur content) are subsequently introduced in quadratic form to account for as much of the residual variation as possible. Higher-order terms are iteratively dropped to avoid overfitting based on predicted R-squared. The final models used to explain midstream and downstream carbon emissions are as follow:

$$\begin{aligned} CI_f^{PRELIM} &= \textstyle\sum_1^3 \beta_i^C RefineryConfigf + \beta_4 API_f + \beta_5 API_f^2 + \beta_6 API_f^3 \\ &+\beta_7 log(Sulphur) + \beta_8 log(Sulphur)^2 + \epsilon_f \end{aligned} \tag{A.2}$$

$$\begin{aligned} CI_f^{OPEM} &= \textstyle\sum_1^3 \beta_i^C RefineryConfigf + \beta_4 API_f + \beta_5 API_f^2 + \beta_6 API_f^3 + \beta_7 API_f^4 \\ &+\beta_8 log(Sulphur) + \beta_9 log(Sulphur)^2 + \beta_{10} log(Sulphur)^3 + \epsilon_f \end{aligned} \tag{A.3}$$

Regression results. Table A.6 shows results from both estimations. All coefficients are significant and predicted R-squared are 0.95 and 0.88 respectively. As expected, refinery configuration explains a large share of variations across crudes, especially in the case of midstream emissions. Adding quadratically API gravity and log sulphur content improves the variance explained marginally, as the relatively low within R-squared indicates. On the contrary, API gravity and sulphur content play a much more important role in explaining the variance in downstream emissions.

Sensitivity analysis. The distribution of midstream and downstream CI are compared to predicted values using the model in figure A.12. Midstream values appear grouped according to each refinery configuration, illustrating the crucial role played by the dummies in the regression results. While the inclusion of API gravity and sulphur content in the model allows to explain a large share of within group variation for medium and deep conversion, it does not appear to be the case for crudes using the hydroskimming configuration (bottom-left of panel A), explaining the low within R-squared value. Downstream CI are more evenly distributed across the graph, and predicted values remain close to OPEM-computed ones.

Contrary to upstream CI, midstream and downstream CI computed using PRELIM and OPEM models do not contain visible outliers. The risk of overfitting particular values is thus low. Tables A.7 and A.8 reports re-estimated coefficients excluding most-influential observations for midstream and downstream CI respectively. In both cases,

Variables	Midstream	Downstream
Configuration: Deep conversion	56.552***	530.269***
	(1.374)	(17.341)
Configuration: Hydroskimming	40.093***	587.056***
	(1.608)	(17.764)
Configuration: Medium conversion	51.361***	515.626***
	(1.604)	(18.037)
API	-0.933***	-16.122***
	(0.124)	(2.159)
API^2	0.016***	0.781***
	(0.003)	(0.096)
API^3	-9.62e-05***	-0.017***
	(2.74e-05)	(0.002)
API^4		1.11e-04***
		(1.16e-05)
log(Sulphur)	-0.988***	-7.784***
	(0.103)	(0.832)
$\log(\text{Sulphur})^2$	-0.135***	-2.404***
	(0.025)	(0.474)
$\log(\text{Sulphur})^3$		-0.19***
		(0.072)
Corrected R^2	0.948	0.887
Predicted R^2	0.947	0.883
Within R^2	0.300	0.875

Table A.6: Regressions results from midstream and downstream CI estimations.

Notes: The table displays coefficients from the regressions using equations A.2 and A.3, with standard errors indicated between parenthesis. R^2 are corrected in order to account for the absence of intercept, as well as predicted R^2. Within R^2 are computed using fixed-effects regressions to capture the effect of configuration dummies.

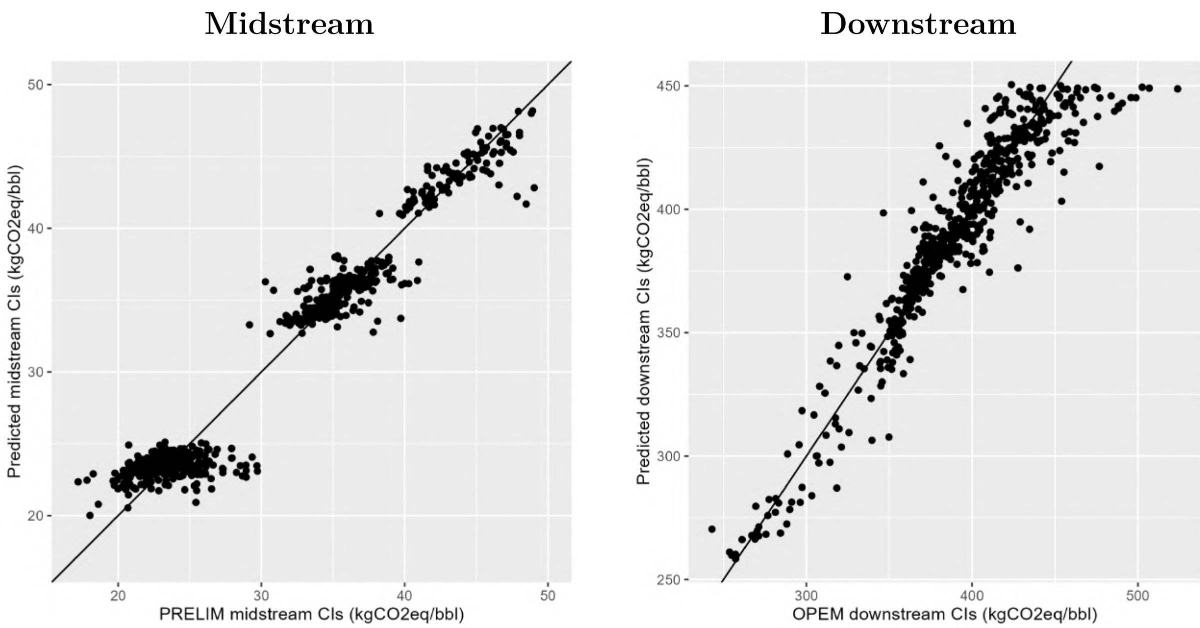

Figure A.12: Comparison between PRELIM/OPEM CI and predicted ones.

estimates remain significant and stable enough to exclude the risk of overfitting.

	Deep Conversion	Hydroskimming	Medium Conversion	API
0	56.55*** (1.37)	40.09*** (1.61)	51.36*** (1.60)	-0.93*** (0.12)
1	56.38*** (1.30)	40.05*** (1.52)	51.22*** (1.51)	-0.91*** (0.12)
10	56.69*** (1.23)	40.58*** (1.44)	51.73*** (1.44)	-0.93*** (0.11)
50	56.56*** (1.05)	40.80*** (1.22)	51.91*** (1.21)	-0.89*** (0.10)
	API^2	API^3	log(Sulphur)	$\log(\text{Sulphur})^2$
0	0.0159***	-9.62e−05***	-0.99*** (0.10)	-0.13*** (0.03)
1	0.0152***	-9.03e−05***	-1.00*** (0.10)	-0.13*** (0.02)
10	0.0153***	-8.77e−05***	-1.10*** (0.09)	-0.18*** (0.02)
50	0.0126***	-5.57e−05***	-1.18*** (0.08)	-0.22*** (0.02)

Table A.7: Midstream CI estimation: coefficient stability to influential observations (Cook's distance).

Notes: Each panel reports coefficients re-estimated after excluding the 0, 1, 10, and 50 most influential observations according to Cook's distance. Standard errors are reported in parentheses.

	Deep Conversion	Hydroskimming	Medium Conversion	API	API^2
0	530.27*** (17.34)	587.06*** (17.76)	515.63*** (18.04)	-16.12*** (2.16)	0.78*** (0.10)
1	543.82*** (18.09)	599.01*** (18.32)	527.93*** (18.62)	-18.39*** (2.33)	0.90*** (0.11)
10	544.71*** (16.65)	600.12*** (16.83)	528.63*** (17.12)	-18.46*** (2.15)	0.91*** (0.10)
50	555.93*** (14.05)	609.59*** (14.18)	540.02*** (14.41)	-19.59*** (1.83)	0.94*** (0.09)
	API^3	API^4	log(Sulphur)	$\log(\text{Sulphur})^2$	$\log(\text{Sulphur})^3$
0	-1.66e−02***	1.11e−04***	-7.78*** (0.83)	-2.40*** (0.47)	-0.19*** (0.07)
1	-1.92e−02***	1.30e−04***	-7.81*** (0.83)	-2.24*** (0.48)	-0.16** (0.07)
10	-1.93e−02***	1.31e−04***	-7.60*** (0.76)	-2.40*** (0.45)	-0.15** (0.07)
50	-1.95e−02***	1.30e−04***	-7.31*** (0.60)	-2.74*** (0.36)	-0.21*** (0.06)

Table A.8: Coefficient stability to influential observations (Cook's distance).

Notes: Each panel reports coefficients re-estimated after excluding the 0, 1, 10, and 50 most influential observations according to Cook's distance. Standard errors are reported in parentheses.

A.2.2.4 Predicting midstream/downstream CI

Rystad assets matched to PRELIM crudes are assigned midstream and downstream CIs computed using the PRELIM and OPEM models. These assets represent 59% of oil reserves in 2016. For remaining assets, I use the coefficients estimated in both regressions to predict their CI, at the crude level in the case of crude streams unmatched to PRELIM crude assays (17.3% of reserves) and at the asset level if the crude stream is unspecified (11.7%).

Several assets, representing 12% of world reserves, lack information on either API gravity or sulphur content. For these assets, reserve-weighted averages from similar oils were computed, according to the following order: reservoir, sub-basin, similar API within the same country, same main oil type within the same country, and finally same main oil type. The vast majority of assets are attributed values computed within the same reservoir or sub-basin. Geographically close assets having a greater probability to be blended together before refining, averages computed from assets nearby is preferable. Table A.9 sums up the methods and granularity levels of midstream and downstream CI predictions.

Method	Granularity level	Share of 2016 reserves
PRELIM/OPEM	Crude stream	58.9%
Predicted	Crude stream	17.3%
	Asset-specific	11.7%
Average	Reservoir	5.4%
	Sub-basin	5.7%
	API category $\times$ country	0.7%
	Oil type $\times$ country	0.2%
	Oil type	$< 0.1\%$

Table A.9: Data sources and aggregation levels for midstream and downstream carbon intensity values.

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Appendix B

Appendix to: The Climate-wise Values of Oil

B.1 Field-level data

B.1.1 Rystad upstream database

Our empirical investigation employs one of the most comprehensive datasets on oil and gas fields, the Rystad Upstream Database (UCube). This database covers World oil production since 1900, with over 90,000 oil and gas assets. It brings together precise field-level data on oil and gas production,[1] exploitable reserves, capital and operational expenditures from exploration to field decommission as well as production taxes, current governance (ownership and operating companies), field development dates (discovery, license, start-up, and production end), oil types, and reservoir characteristics (water depth, basin, and location), among others. For some parts of our analysis, we use geocoded boundaries of each field obtained from Rystad GIS team, as well as data on the annual number of drilled and active wells per asset from the Rystad WellCube. The Rystad data sources include governments and companies' operation reports.

B.1.2 Tax data

Revenue Allocation Analysis. For a given price and quantity vectors (p_t), $(q_d(t))$, we use our estimated tax rates (see 2.2.3), to then compute for field d the revenue distribution between firms and governments. We call $PaidTaxes_d$ the total taxes paid in field d for a given vector of future prices and quantities (p_t), $(q_d(t))$.

[1] All productions are expressed in energy-equivalent barrels, using Brent Crude as the benchmark.

$$\begin{aligned}
\text{Taxes Opex:} &\quad \sum_{2025}^{\infty} TO_i \cdot x_i(t)/(1+r)^{t-2025}, \\
\text{Royalties:} &\quad \sum_{2025}^{\infty} RO_i \cdot p(t) \cdot x_i(t)/(1+r)^{t-2025}, \\
\text{Gov't Profit:} &\quad \sum_{2025}^{\infty} GP_i \cdot \Big((1-RO_i)p(t)x_i(t) - (c_i + TO_i)x_i(t)\Big)/(1+r)^{t-2025}, \\
\text{Inc.Taxes:} &\quad \sum_{2025}^{\infty} IT_i \cdot (1-GP_i)((1-RO_i)p(t)x_i(t) - (c_i + TO_i)x_i(t))/(1+r)^{t-2025}.
\end{aligned}$$

B.2 Private extraction costs and extraction capacities

B.2.1 Data on private extraction costs

In our main specification, we measure annual costs as the sum of “well” and “facility” CAPEX, and the “selling, general and administrative”, “transportation”, “production” OPEX.

According to the Rystad definitions, “well CAPEX is capitalized costs related to well construction, including drilling costs, rig lease, well completion, well stimulation, steel costs and materials.” Facility CAPEX includes development CAPEX (“costs to develop, install, maintain and modify surface installations and infrastructure”), exploration CAPEX (“costs incurred to find and prove hydrocarbons”), and abandonment costs (“costs associated with shutting down and dismantling the surface and subsea facilities”). Production OPEX is “operational expenses directly related to the production activity. The category includes materials, tools, maintenance, equipment-lease costs and operation-related salaries. Depreciation and other non-cash items are not included.” Transportation OPEX includes “the costs of bringing the oil and gas from the production site/processing plant to the pricing point (only upstream transportation)”.

We use costs and productions recorded in Rystad for each field to compute field-level extraction costs per barrel: these costs and productions are observed for producing fields before 2025, and productions and costs are modeled for post-2024.[2] As we exclude fields discovered after 2024 from our analysis, most of the fields in our sample have been producing before 2022, thus our cost measure is partly computed ex post based on *realized* productions and costs. We assume that the marginal extraction cost is constant and equal to average cost as long as extraction is below capacity constraint.

[2]Following Rystad's guidelines, if a field is split into several assets for fiscal reasons, cost data are aggregated at the block level.

We calculate average costs as: $c_d = (\sum_{1970}^{2099} Exp_{dt})/(\sum_{1970}^{2099} x_{dt})$, where x_{dt} and Exp_{dt} denote the annual deposit production and *total* cost for all years post 2025.

Other approach: Only future costs. With this approach, we remove from the calculation of the average cost the production stream and the costs paid before the starting date of the optimization.

B.2.2 Extraction capacities

The literature usually describes field-level extraction as having two sequential phases: a plateau, and then a phase of decline in extraction (Fetkovich, 1980; Robelius, 2007; Höök and Aleklett, 2008; Höök et al., 2014; Jackson and Smith, 2014). In order to anchor the capacity constraint on observed extraction rates since 1970, the "plateau-decline" capacity constraint is included by adding *simultaneously* in the optimization programs two field-specific constraints over feasible extractions:

1. The "plateau" part reads for deposit d and year t, $x_{dt} \leq k_d$, where

$$k_d = \max\left(\max_{j\in[1970,2018]}(x_{d,j,observed}), 2.5\% \times R_{d,1970,observed}\right)$$

2. The "decline" part reads for deposit d and year t, $x_{dt} \leq \alpha_d R_{d,t}$, where

$$\alpha_d = \max\left(\max_{j\in[1970,2018]}\left(\frac{x_{d,j,observed}}{R_{d,j,observed}}\right), 5\%\right)$$

Where x_{dt} is production from deposit d at date t in any scenario, $x_{d,j,observed}$ is observed production from deposit d at date j, $R_{d,j,observed}$ is observed reserves of deposit d at date j, and $R_{d,t}$ remaining reserves from deposit d at date t in any scenario.

At each date, the production thus has to be less than the capacity:

$$x_{dt} \leq \min(k_d, \alpha_d R_{dt})$$

The "plateau" constraint forces a field's production to be below 2.5% of the field's reserves as of 1970 or to be below the maximal observed production of the field since 1970 if the latter is larger. For 95% of the fields that start producing after 1970, the maximal observed production of the field since 1970 is larger than 2.5% of the field's reserves as of 1970. For a typical field with no (or little) observed past production, forcing production to be below 2.5% of the field's 1970 reserves is really conservative.

The "decline" constraint forces a field's extraction-to-current-reserves to be below 5% or to its maximal historical extraction-to-current-reserves, if the latter is larger. The

5% decline rate is chosen because on average the maximal observed extraction-to-current-reserves rate during the decline phase (when production is less than 95% of the maximal production, as defined in Höök et al. (2014)) is twice the maximal observed extraction-to-1970-reserves rate.

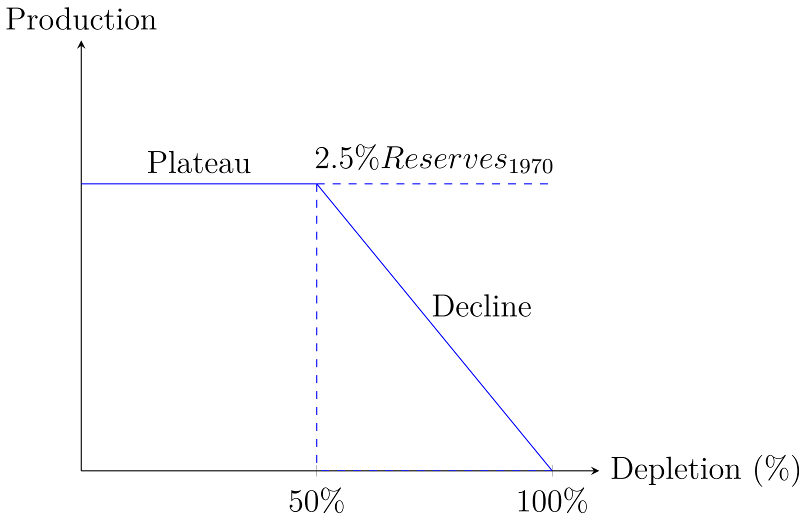


Figure B.1: The "Plateau-decline" constraint for a virtual field

This specification allows for field specific extraction. There is indeed significant variation in extraction rates across field sizes: numerous small fields, such as shale oil fields, have high extraction rates of about 15% to 20% and thus can be exhausted over the course of a few years, whereas larger fields have lower rates of extraction and longer plateau duration. Our chosen capacity constraint reflects this heterogeneity. Assume for instance a large field whose extraction is never more than 5% of 1970 reserves, but the maximal observed extraction-to-current-reserves is 10% as the field gets depleted, then the plateau lasts until there is only 50% of 1970 reserves left, a situation described in Figure B.1. For a small field whose extraction is never more than 6% of 1970 reserves, but the observed maximum rate of extraction is 10% as the field gets depleted, the plateau is shorter and the decline starts when 60% of 1970 reserves are left.

Even if median extraction rates are larger for small fields than for large fields, it remains that, for all usual field size categories, the 5^{th} percentile of maximal production over 1970 reserves is around 2.5% of the field's reserves as of 1970.

For a typical field with no (or little) observed past production, the depletion path (or equivalently the pattern of production at full capacity) looks like a plateau with production at 2.5% of 1970 reserves that lasts until 1970 reserves are half depleted, then a decline phase with annual productions at 5% of current remaining reserves, a situation described in Figure B.1.

We also provide a sensitivity analysis on capacity constraints. First, we consider two polar cases in two distinct exercises: i) "plateau-only": the extraction rate is below 2.5%

of 1970 reserves, or below the historical maximal extraction rate if the latter is larger; ii). "decline-only": a field capacity is strictly declining from the beginning of the field life, the extraction rate is below 5% of current reserves, or below the historical maximal extraction-to-current-reserves ratio if the latter is larger.

Finally, in Venables (2014), a faster rate of depletion can reduce the size of recoverable reserves in a field. This rate sensitivity is based on Nystad (1985, 1987) who categorize fields as "Hotelling" (recoverable reserves are not rate sensitive), "intermediate", and "geosensitive" (recoverable reserves are strongly rate sensitive) and consider the theoretical implications of reservoirs' rate-sensitivity on optimal extraction rate. Our conservative assumption regarding capacity constraint ensures that extraction rates in our counterfactuals are below what is observed in the data and likely below levels of depletion rates that would significantly reduce the size of recoverable reserves. We have included robustness checks on reserves' recoverability that indirectly address the issue of highly rate-sensitive reservoirs such as dropping 10% of each field reserves.

B.3 Other modeling and parameter choices

B.3.1 The selection of deposits

Our analysis is restricted to Rystad assets that have been discovered before 2025, with positive reserves at the beginning of 2022 of at least one of the following oil types: "Condensate", "NGL", "Light", "Regular", "Sour ($> 3\%$)", "Extra Heavy Oil", "Heavy Oil 15-19", "Heavy Oil 20-23", "Bitumen", and "Synthetic crude" (the numbers for heavy oils represent the API gravity range). The sample is composed of 21,031 oil fields.

B.3.2 Carbon intensities

Appendix A.2 describes OPGEE and PRELIM, the matching procedure and results, the estimation model and the robustness checks in detail.

B.3.3 Reserves

The reserve estimates are from the Rystad UCube dataset. A deposit reserve is defined as its economically-recoverable volume, assuming an oil price of US$120 per barrel (scenario "Resources High Case"). Current World reserves amount to 1,517 billion barrels in the Rystad dataset, as against 1,729 billion barrels for World proven reserves according to BP (2019). The difference in these numbers mostly reflects that the Rystad dataset does not record the proven reserves for some bitumen and extra heavy oils that are expected to be too expensive to be profitable, even in the long run. For instance, in Rystad, Canada and Venezuela have reserves estimated respectively at 101 and 34 billion barrels (Gb) at the end of 2018; the analogous figures in BP (2019) are 167 and 303 billion barrels. The resources not in the Rystad dataset are in general not only too expensive to be extracted but also very polluting. As a consequence, they would not be used in counterfactual scenarios in which social or private extraction costs are minimized. Overall, using a less-restrictive definition of reserves in our analysis would have little impact on our findings. There exist various definitions of oil reserves, as noted by McGlade (2012). Our reserve estimates are close to that of OPEC (2019) (1,498 Gb) and below those of BP 2019; EIA 2021; BGR 2020. The EIA estimates global reserves to amount to 1,659 Gb (EIA, 2021), the BGR to 1,790 Gb (BGR, 2020), the IEA to 1,700 Gb (IEA, 2019). In contrast, global reserves are lower at 1,276 Gb (2018 figure) in Welsby et al. (2021).

Rystad data include yet-to-find resources within already-discovered fields from unconventional assets. These yet-to-find resources that correspond to resource extension represent only a small part (about 8%) of total reserves available as of 2019 (fields discovered before 2019). In a robustness check, we reduce reserves by 10% to account for the possibility of overestimation in field reserve size.

B.3.4 Resource availability

All fields in our sample have been discovered before 2021. Yet, for fields that do not produce by 2022, we include development lags after a resource discovery before production can start as in Arezki et al. (2017); Bornstein et al. (2017). Bornstein et al. (2017), in particular, highlight that development lags vary with oil type, a piece of evidence we do verify with our data. We assume that for all resource types,—except shale oil and tight oil—the production can only start five years after the discovery date (always anterior to 2022), or after the first year of observed production if the latter is sooner. For tight and shale oil, we use a one-year lag.

B.3.5 The social cost of carbon

The value of the SCC in 2022. The development of Integrated Assessment Models (IAMs) in the 1990s—e.g., the Dynamic Integrated Climate and Economy (DICE), developed by William Nordhaus and used in Nordhaus (1994, 1992), the Policy Analysis of the Greenhouse Effect (PAGE), developed by Chris Hope and used in Stern (2008), and the Climate Framework for Uncertainty, Negotiation, and Distribution (FUND), developed by Richard Tol and used in Anthoff and Tol (2013)—constituted a turning point in the estimates of the social cost of carbon (SCC).[3] These models describe interactions between carbon-dioxide concentration, the climate, damage from climate change, and

[3]See Calel and Stainforth (2017) for a comparison of these IAMs.

human activities that produce carbon-dioxide emissions. Although criticisms about the limitations of IAM remain (e.g., Pindyck, 2013, 2019, which stress arbitrary modelling choices, the deterministic nature of most IAMs, and the exclusion of catastrophes), the IAM estimates have been widely discussed in the public debate.

The SCC estimates from IAMs vary greatly, in particular depending on the choice of the social discount rate (Nordhaus, 2007; Stern, 2008; Tol et al., 2013), the coverage of damage, and the modeling (if any) of uncertainty (Weitzman, 2009, 2011; Calel et al., 2015; Gillingham et al., 2018). A growing body of literature argues that social costs of carbon, such as those used by the U.S. Government Interagency Working Group, that are based on simulations from the DICE, PAGE and FUND models underestimate the true social costs, as they ignore key uncertainties, some climate-change related damage, irreversibilities and acceleration factors, such as the possibility of tipping points, and usually keep the valuation of ecosystems constant despite their rarefaction (for a summary of these criticisms, see e.g., Revesz et al., 2014; Van Den Bergh and Botzen, 2014).

? finds a SCC of \$37 in 2020 (2010 US dollars) along the optimized emission path, which jumps to \$87 with a discount rate of 3%. In the baseline parameterization of the DICE2016R, constraining the increase in temperature over the next 100 years as compared to 1900 to be under 2.5° C increases the SCC to \$229 per tCO_2 (2010 US dollars). Dietz and Stern (2015) bring three main changes to the 2010 version of the DICE model: (i) Climate change negatively affecting the accumulation of physical, technological and intellectual capital; (ii) Modifying the function translating temperature increases into GDP loss to account for possible tipping points (Weitzman, 2012; Lenton and Ciscar, 2013); (iii) The "climate-sensitivity" parameter, which links atmospheric concentrations of greenhouse gases to (expected) temperature increases, being updated to account for new climate-science knowledge, although Nordhaus' discount rate is left unchanged (at twice the level of that preferred by Stern). They find a carbon price in the \$32–103/tCO_2 range (2012 prices) in 2015. Lemoine (2021) estimates a 200-year social cost of carbon in 2014 of \$362 per tCO_2 in a calibration exercise accounting for uncertainty about both warming and the impact of warming on consumption, and including stochastic shocks to consumption growth. Rennert et al. (2022), using the new open-source Greenhouse Gas Impact Value Estimator (GIVE) model with up-to-date knowledge regarding socioeconomic projections, climate models, damage functions, and discounting methods, found a central value of \$185 per tCO_2 (2020 US dollars).

Overall, our choice of a SCC of US\$200 per tCO_2 in 2022 is consistent with the SCC in DICE2016R when the temperature increase is kept strictly below 2.5° C over the next 100 years.

The rate of increase of the SCC. The social cost of carbon in current value increases at the rate of the discount rate, thus the discounted cost of emissions is not time-dependent. The reason for this modeling choice is to be found in the scientific literature that shows that temperature increase is proportional to cumulative carbon emissions at a given time horizon, and independent of the carbon emission pathways (Allen et al., 2009; Matthews et al., 2009; Meinshausen et al., 2009; ?). This evidence serves as a rationale behind mitigation targets formulated in terms of carbon budgets (Rogelj et al., 2019).

B.3.6 Social welfare maximisation

The numerical analysis is carried out in R (version 3.6.0) using the following packages: data.table, Matrix, slam, gurobi, pryr, ggplot2, Hmisc, and rio. Optimization is carried out using the linear-programming solver Gurobi Optimizer available at https://www.gurobi.com/products/gurobi-optimizer/.

B.4 Simplified model: Theoretical Insights

B.4.1 The welfare-maximisation problem

We define the current-value Hamiltonian as:

$$
\begin{aligned}
H(t) = u\Big(\sum_{i\in\{a,b,c,d\}} x_i(t)\Big) - \sum_{i\in\{a,b,c,d\}} (c_i + \mu(t)\theta_i)x_i(t) \\
- \sum_{i\in\{a,b\}} \lambda_i(t)x_i(t)
\end{aligned}
$$

It represents the instantaneous social surplus net of the shadow values of the extracted resources and their associated pollution costs. This has the following slackness conditions:

$$
\begin{aligned}
\nu(t) \geq 0 \quad &\text{and} \quad \nu(t)\Big(\bar{Z} - Z(t)\Big) = 0 \\
\beta_i(t) \geq 0 \quad &\text{and} \quad \beta_i(t)X_i(t) = 0 \\
\gamma_i(t) \geq 0 \quad &\text{and} \quad \gamma_i(t)x_i(t) = 0
\end{aligned}
$$

For any control $x_i(t)$, $i = \{a, b, c, d\}$, there exist co-state variables $\lambda_i(t)$, and $\mu(t)$, that must satisfy the following conditions, as well as the transversality and slackness conditions:

$$\dot{\lambda}_i(t) = r\lambda_i(t) - \frac{\partial H(t)}{\partial X_i(t)} \iff \dot{\lambda}_i(t) = r\lambda_i(t) + \beta_i(t) \tag{B.1}$$

$$\dot{\mu}(t) = r\mu(t) - \frac{\partial H(t)}{\partial Z(t)} \iff \dot{\mu}(t) = r\mu(t) + \nu(t) \tag{B.2}$$

$$\frac{\partial H(t)}{\partial x_i(t)} = 0 \iff p_i(t) = c_i + \lambda_i(t) + \theta_i\mu(t) \tag{B.3}$$

$$\frac{\partial H(t)}{\partial x_d(t)} = 0 \iff p_d(t) = c_d + \theta_d\mu(t) \tag{B.4}$$

$$\frac{\partial H(t)}{\partial x_c(t)} = 0 \iff p_c(t) = c_c. \tag{B.5}$$

The transversality conditions are given by:

$$\lim_{t\to\infty} \lambda_i(t)e^{-rt}X_i(t) = 0 \tag{B.6}$$

$$\lim_{t\to\infty} \mu(t)e^{-rt}Z(t) = 0. \tag{B.7}$$

B.4.2 Expression of $\frac{d\lambda_a}{dZ}$ and of $\frac{d\lambda_b}{dZ}$

Expression of $\frac{d\lambda_a}{dZ}$

Differentiating Eq. 2.15, and remarking that $p_a(t) = c_a + (p^1 - c_a)e^{-r(t_1-t)}$, the change in the duration of the extraction of R_a is:

$$dt_1 = \frac{D(p^0) - D(p^1)}{D(p^0)} \frac{1}{r(p^1 - c_a)} dp^1. \tag{B.8}$$

Differentiating Eq. 2.16, and remarking that $p_b(t) = c_b + (p^2 - c_b)e^{-r(t_2-t)}$, the change in the duration of the extraction of R_b is given by:

$$d(t_2 - t_1) = \frac{D(p^1) - D(p^2)}{D(p^1)} \frac{1}{r(p^2 - c_b)} dp^2. \tag{B.9}$$

The switch price from R_a to R_b, p^1, and the switch price from R_b to R_d, p^2, satisfies $p^1 = c_b + (p^2 - c_b)e^{-r(t_2-t_1)}$. Differentiating this last equation, and using Eq. B.9, it comes that: $dp^1 = dp^2 \frac{D(p^2)}{D(p^1)} e^{-r(t_2-t_1)}$. Eq. B.8 can be rewritten into:

$$dt_1 = \frac{D(p^0) - D(p^1)}{D(p^0)D(p^1)} \frac{D(p^2)}{r(p^1 - c_a)} dp^2\, e^{-r(t_2-t_1)} \tag{B.10}$$

From Eq. 2.12, we have that $d\lambda_a^1 = dp^1 - \theta_a d\mu_1$. Since $dp^2 = d\mu^2\theta_d$ and $\mu^2 =$

$\mu_1 e^{-r(t_1-t_2)}$, thus $d\mu_1 = e^{-r(t_2-t_1)}\frac{dp^2}{\theta_d} - rd(t_2-t_1)\mu_1$, it comes that:

$$d\lambda_a^1 = dp^2\frac{D(p^2)}{D(p^1)}e^{-r(t_2-t_1)} - \theta_a(e^{-r(t_2-t_1)}\frac{dp^2}{\theta_d} - rd(t_2-t_1)\mu_1)$$

After simplifying,

$$d\lambda_a^1/dp^2 e^{r(t_2-t_1)} = \frac{D(p^2)}{D(p^1)} - \frac{\theta_a}{\theta_d} + \frac{D(p^1)-D(p^2)}{D(p^1)}\frac{\theta_a\mu}{\lambda_b+\theta_b\mu}.$$

The sign of $d\lambda_a^1$ is ambiguous as the ceiling falls. As $d\lambda_a = d\lambda_a^1 e^{-rt_1} - r\lambda_a dt_1$, it comes that

$$d\lambda_a = [\frac{D(p^2)}{D(p^0)} - \frac{\theta_a}{\theta_d} + (1-\frac{D(p^2)}{D(p^1)})\frac{\theta_a\mu}{\lambda_b+\theta_b\mu} + (\frac{D(p^2)}{D(p^1)} - \frac{D(p^2)}{D(p^0)})\frac{\theta_a\mu}{\lambda_a+\theta_a\mu}]dp^2 e^{-rt_2}$$

Expression of $\frac{d\lambda_b}{dZ}$

Let us now determine the impact of the fall of the ceiling on the scarcity rent of resource b. We have: $d\lambda_b = d\lambda_b^2 e^{-rt_2} - r\lambda_b dt_2$. Using Eq. 2.13, we get that $d\lambda_b^2 = (\theta_d-\theta_b)/\theta_d dp^2$. The increase of the switch price p^2 increases the scarcity rent of R_b at time t_2.

From Eq. B.9 and B.10,

$$dt_2 = \frac{D(p^1)-D(p^2)}{D(p^1)}\frac{1}{r(p^2-c_b)}dp^2 + \frac{D(p^0)-D(p^1)}{D(p^0)}\frac{1}{r(p^1-c_a)}dp^2\frac{D(p^2)}{D(p^1)}e^{-r(t_2-t_1)}$$

After simplification we get:

$$d\lambda_b/dp^2 e^{rt_2} = -\theta_b/\theta_d + \frac{D(p^2)}{D(p^1)} + (1-\frac{D(p^2)}{D(p^1)})\frac{\theta_b\mu}{\lambda_b+\theta_b\mu} - (\frac{D(p^2)}{D(p^1)} - \frac{D(p^2)}{D(p^0)})\frac{\lambda_b}{\lambda_a+\theta_a\mu}$$

B.4.3 Proof of Proposition 1

- Claim (i) writes $\forall c_a, c_b, c_d$ such that $c_a < c_b < c_d$, $\exists\epsilon_i$ such that:

$$\left\{\forall p, -\frac{D'(p)p}{D(p)} < \epsilon_i \text{ and } \bar{Z} > \theta_a X_a + \theta_b X_b\right\} \Longrightarrow \frac{d\lambda_i^0}{d\bar{Z}} < 0.$$

A sufficient condition to ensure that $\frac{d\lambda_a}{d\bar{Z}} < 0$ is that $1 - \frac{\theta_d D(p^2)}{\theta_a D(p^0)} < 0$. By the mean-value theorem, there exists a date t_i, satisfying $0 \leq t_i \leq t_2$ such that $\frac{D(p^2)}{D(p^0)} = 1 + \frac{D'(p(t_i))}{D(p^0)}(p(t_2)-p(0)) \geq 1 - (-\frac{D'(p(t_i))}{D(p(t_i))}p(t_i))\frac{c_c-c_a}{c_a}$. Indeed, since $p^2 \leq \frac{c_c}{c_a}p^0$, and $D(p(t_i)) \leq D^0$ and $p^0 \leq p(t_i)$, we have: $\frac{p^2-p^0}{D^0} \leq \frac{p(t_i)}{D^i}\frac{c_c-c_a}{c_a}$.

If for all p, $-\frac{D'(p)}{D(p)}p < \frac{c_a(1-\theta_a/\theta_d)}{c_c-c_a} \equiv \epsilon_a$, then tightening the carbon ceiling increases the profits of R_a owners.

Similarly, a sufficient condition to ensure that $\frac{d\lambda_b}{d\bar{Z}} < 0$ is that $1 - \frac{\theta_d D(p^2)}{\theta_b D(p^0)} < 0$. The same reasoning as above applies.

- Claim (ii) reads $\forall c_a, c_b, c_d$ such that $c_a < c_b < c_d$, for $i \in \{a, b\}$, $\exists \eta_i$, with $0 < \eta_i < 1$, such that:
$$\left\{\frac{\theta_i}{\theta_d} \leq 1 - \eta_i \text{ and } \bar{Z} > \theta_a X_a + \theta_b X_b\right\} \Longrightarrow \frac{d\lambda_i^0}{d\bar{Z}} < 0$$
Straightforward.

- Claim (iii) reads $\forall c_a, c_b, c_d$ such that $c_a < c_b < c_d$, $\forall \bar{Z} > \theta_a X_a + \theta_b X_b$, $\exists \epsilon_a > 0$:
$$c_d - c_b \leq \epsilon_a \implies \frac{d\lambda_b}{d\bar{Z}} < 0$$
$\forall c_a, c_b, c_d$ such that $c_a < c_b < c_d$, $\forall \bar{Z} > \theta_a X_a + \theta_b X_b$, $\exists \epsilon_a, \epsilon_b > 0$ such that:
$$c_d - c_b \leq \epsilon_a \text{ and } c_b - c_a \leq \epsilon_b \implies \frac{d\lambda_a}{d\bar{Z}} < 0$$

Using Eq. 2.13, it comes that $(\theta_d - \theta_b)\frac{\mu}{\lambda_b} = 1 - \frac{c_d-c_b}{\lambda_b}e^{-rt_2}$. $\frac{d\lambda_b}{d\bar{Z}}$ has the sign of: $\frac{c_d-c_b}{\lambda_b}e^{-rt_2} - \frac{D(p^2)}{D(p^1)}\frac{\theta_d}{\theta_b} + \frac{\lambda_b+\theta_b\mu}{\lambda_a+\theta_a\mu}\frac{\theta_d}{\theta_b}(\frac{D(p^0)}{D(p^1)} - 1)\frac{D(p^2)}{D(p^0)}$. But $\frac{c_d-c_b}{\lambda_b}e^{-rt_2} - \frac{D(p^2)}{D(p^1)}\frac{\theta_d}{\theta_b} + \frac{\lambda_b+\theta_b\mu}{\lambda_a+\theta_a\mu}\frac{\theta_d}{\theta_b}(\frac{D(p^0)}{D(p^1)} - 1)\frac{D(p^2)}{D(p^0)} < \frac{c_d-c_b}{\lambda_b}e^{-rt_2} - \frac{D(p^2)}{D(p^1)}\frac{\theta_d}{\theta_b} + \frac{\theta_d}{\theta_b}(\frac{D(p^0)}{D(p^1)} - 1)\frac{D(p^2)}{D(p^0)} \leq \frac{c_d-c_b}{\lambda_b}e^{-rt_2} - \frac{\theta_d}{\theta_b}\frac{D(p^2)}{D(p^0)}$. We know that $\frac{D(p^2)}{D(p^0)}\frac{\theta_d}{\theta_b} > \frac{\theta_d}{\theta_b}\frac{D(c_c)}{D(c_a)} > 0$ and $\lambda_b e^{rt_2} > (\theta_d - \theta_b)\mu e^{rt_2} > 0$. Thus, $\frac{d\lambda_b}{d\bar{Z}}$ is negative if $-\frac{D(c_c)}{D(c_a)}\frac{\theta_d}{\theta_a} + \frac{c_d-c_b}{(\theta_d-\theta_b)\mu e^{rt_2}}$ is negative. μe^{rt_2} does not depend on c_a and is strictly positive for any $\bar{Z}$. At μe^{rt_2} given, as $-\frac{D(c_c)}{D(c_a)}\frac{\theta_d}{\theta_a} + \frac{c_d-c_b}{(\theta_d-\theta_b)\mu e^{rt_2}}$ is continuous with c_b and decreases with c_b and is strictly negative for $c_b = c_d$, then there exists ϵ_a such that this claim holds.

For any μe^{rt_2} (which does not depend on c_a nor c_b). We know that $\frac{d\lambda_a}{dZ}$ has the sign of: $1 - \frac{\theta_d}{\theta_a}\frac{D(p^2)}{D(p^0)} - \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}(1 - \frac{D(p^2)}{D(p^1)}) - \frac{\theta_d\mu}{\lambda_a+\theta_a\mu}\frac{D(p^2)}{D(p^1)}(1 - \frac{D(p^1)}{D(p^0)})$. But $-\frac{\theta_d\mu}{\lambda_b+\theta_b\mu} = -(1 - \frac{c_d-c_b}{c_d-c_b+\theta_d\mu e^{rt_2}}) = -\frac{\theta_d\mu e^{rt_2}}{c_d-c_b+\theta_d\mu e^{rt_2}}$, which is decreasing with c_b, at c_d and μe^{rt_2} given, and equals -1 for $c_b = c_d$. Thus, $\frac{d\lambda_a}{dZ}$ has the sign of: $1 - \frac{\theta_d}{\theta_a}\frac{D(p^2)}{D(p^0)} - \frac{\theta_d\mu e^{rt_2}}{c_d-c_b+\theta_d\mu e^{rt_2}}(1 - \frac{D(p^2)}{D(p^1)}) - \frac{\theta_d\mu}{\lambda_a+\theta_a\mu}\frac{D(p^2)}{D(p^1)}(1 - \frac{D(p^1)}{D(p^0)})$. Moreover $\frac{\theta_d\mu}{\lambda_a+\theta_a\mu} = \frac{\theta_d\mu e^{rt_2}}{c_d-c_b+\theta_d\mu e^{rt_2}}\frac{(\lambda_b+\theta_b\mu)e^{rt_1}}{c_b-c_a+(\lambda_b+\theta_b\mu)e^{rt_1}}$ $\frac{d\lambda_a}{dZ}$ has the sign of: $1 - \frac{\theta_d}{\theta_a}\frac{D(p^2)}{D(p^0)} - \frac{\theta_d\mu e^{rt_2}}{c_d-c_b+\theta_d\mu e^{rt_2}}(1 - \frac{D(p^2)}{D(p^1)}) - \frac{\theta_d\mu e^{rt_2}}{c_d-c_b+\theta_d\mu e^{rt_2}}\frac{(\lambda_b+\theta_b\mu)e^{rt_1}}{c_b-c_a+(\lambda_b+\theta_b\mu)e^{rt_1}}\frac{D(p^2)}{D(p^1)}(1 - \frac{D(p^1)}{D(p^0)})$. But $\frac{(\lambda_b+\theta_b\mu)e^{rt_1}}{c_b-c_a+(\lambda_b+\theta_b\mu)e^{rt_1}} \geq \frac{\theta_b\mu e^{rt_2}e^{-rX_b/D(c_c)}}{c_b-c_a\theta_b\mu e^{rt_2}e^{-rX_b/D(c_c)}}$. As $\theta_b\mu e^{rt_2}e^{-rX_b/D(c_c)}$ does not depend on c_a, c_b or c_d, it is straightforward that for c_a close enough to c_b, the last term of the equation tends to one. We can choose ϵ_a and ϵ_b small enough to ensure that $\frac{d\lambda_a}{dZ}$ is less than $-\frac{\theta_d}{\theta_a}\frac{D(p^2)}{D(p^0)} + \frac{D(p^2)}{D(p^0)} + \alpha$ with α arbitrarily small. So that the result holds for ϵ_a and ϵ_b small enough.

B.4.4 Proof of Proposition 2

The result for $\theta_a > \theta_b$ is straightforward: as $\frac{\theta_d\mu}{\lambda_a+\theta_a\mu} \leq \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}$, then $1 - \frac{\theta_d\mu}{\lambda_a+\theta_a\mu}(\frac{D(p^2)}{D(p^1)} - \frac{D(p^2)}{D(p^0)}) - \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}(1 - \frac{D(p^2)}{D(p^1)}) \geq 1 - \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}(1 - \frac{D(p^2)}{D(p^0)}) \geq 0$.

Assume now that $\theta_a < \theta_b$, and $\bar{Z} > \theta_a X_a + \theta_b X_b$. We know that $0 < 1 - \frac{\theta_d\mu}{\lambda_a+\theta_a\mu}(\frac{D(p^2)}{D(p^1)} - \frac{D(p^2)}{D(p^0)}) - \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}(1 - \frac{D(p^2)}{D(p^1)}) < 1$, it follows that $\underset{\theta_b \to \theta_a}{lim}(\theta_a - \theta_b)(1 - \frac{\theta_d\mu}{\lambda_a+\theta_a\mu}(\frac{D(p^2)}{D(p^1)} - \frac{D(p^2)}{D(p^0)}) - \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}(1 - \frac{D(p^2)}{D(p^1)})) = 0$. We also know that $\frac{D(p^2)}{D(p^1)} > \frac{D(c_c)}{D(c_b)}$. Using Eq. 2.15, and the First Mean Value Theorem for Integrals there exists p^*, such that: $p(0) < p^* < p(t_1)$, and $(p(t_1) - p(0))D(p^*) = X_a$. Note that $D(c_c) < D(p^*) < D(c_a)$. It follows that $p(t_1) - p(0) > \frac{X_a}{D(c_a)} > 0$. Moreover $\exists p^{**}$ such that $D(p^1) - D(p^0) = D'(p^{**})(p^1 - p^0)$, with $-D'(p^{**}) \geq \min\limits_{p\in[c_a,c_c]}(-D'(p)) > 0$.

So that, we get that $\exists\beta$, $(\frac{D(p^2)}{D(p^1)} - \frac{D(p^2)}{D(p^0)}) > \beta > 0$. Using Eq. 2.12, we get that: $\lambda_a - \lambda_b + (\theta_a - \theta_b)\mu = (c_b - c_a)e^{rt_1}$. As $t_1 \leq \frac{X_a}{D(c_c)}$, it follows that $\lambda_a - \lambda_b + (\theta_a - \theta_b)\mu > (c_b - c_a)e^{r\frac{X_a}{D(c_c)}} > 0$, thus $\exists\gamma$, such that $(1 - \frac{\lambda_b+\theta_b\mu}{\lambda_a+\theta_a\mu}) > \gamma > 0$. Finally, $\theta_d(1 - \frac{\lambda_b+\theta_b\mu}{\lambda_a+\theta_a\mu})(1 - \frac{D(p^1)}{D(p^0)})\frac{D(p^2)}{D(p^1)} > \theta_d\beta\gamma\frac{D(c_c)}{D(c_b)} > 0$. It follows that there exists θ^* such that Prop. 2 holds.

B.4.5 Proof of Proposition 3

- Claim (i) reads:
  $\forall c_a, c_b, c_d$, $c_a < c_b < c_d$, $\forall\bar{Z} > \theta_a X_a + \theta_b X_b$, $\exists\epsilon^*$ such that:

$$\left\{\forall p, -\frac{D'(p)p}{D(p)} \leq \epsilon^*\right\} \implies (\theta_a - \theta_b)(\frac{d\lambda_a}{d\bar{Z}} - \frac{d\lambda_b}{d\bar{Z}}) > 0.$$

  Note that $\frac{d\lambda_a}{d\bar{Z}} - \frac{d\lambda_b}{d\bar{Z}} < 0$ iff. $(\theta_a - \theta_b) + \left(\theta_d(1 - \frac{\lambda_b+\theta_b\mu}{\lambda_a+\theta_a\mu}) - \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}\right)(1 - \frac{D(p^1)}{D(p^0)})\frac{D(p^2)}{D(p^1)} < 0$. Recall that $|1 - \frac{\lambda_b+\theta_b\mu}{\lambda_a+\theta_a\mu} - \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}| < 1$ and $\frac{D(p^2)}{D(p^1)} < \frac{D(c_a)}{D(c_c)}$, it comes that $|\left(\theta_d(1 - \frac{\lambda_b+\theta_b\mu}{\lambda_a+\theta_a\mu}) - \frac{\theta_d\mu}{\lambda_b+\theta_b\mu}\right)\frac{D(p^2)}{D(p^1)}| < \theta_d\frac{D(c_a)}{D(c_c)}$. As done above, using the Mean Value Theorem, there exists $t^i \in [0; t_1]$, such that: $\frac{D(p^1)}{D(p^0)} \geq 1 - (\frac{-D'(p(t^i))}{D(p(t^i))}p(t^i))\frac{p(t_1)-p(0)}{p(t^i)}$. We also have $\frac{p(t_1)-p(0)}{p(t^i)} \leq \frac{p(t_1)-p(0)}{p(0)} < \frac{c_b-c_a}{c_a}$. Thus, if $\frac{-D'(p(t^i))}{D(p(t^i))}p(t^i)\frac{c_b-c_a}{c_a} \leq \frac{|\theta_a-\theta_b|}{\theta_d\frac{D(c_a)}{D(c_c)}}$, then

$$|\theta_a - \theta_b| > |[\theta_d(1 - \frac{\lambda_b + \theta_b\mu}{\lambda_a + \theta_a\mu}) - \frac{\theta_d\mu}{\lambda_b + \theta_b\mu}](1 - \frac{D(p^1)}{D(p^0)})\frac{D(p^2)}{D(p^1)}|.$$

  Finally, if $\frac{-D'(p(t^i))}{D(p(t^i))}p(t^i) \leq \frac{|\theta_a-\theta_b|}{\theta_d\frac{D(c_a)}{D(c_c)}\frac{c_b-c_a}{c_a}} = \epsilon$, then $\frac{d\lambda_a - d\lambda_b}{d\bar{Z}}$ has the sign of $\theta_a - \theta_b$.

- Claim (ii) reads

$\forall c_a, c_d,\ c_a < c_d,\ \forall \bar{Z} > \theta_a X_a + \theta_b X_b,\ \exists c^*,\ c_a < c^* < c_d$:

$$\forall c_b,\ c_a < c_b \leq c^* \implies (\theta_a - \theta_b)(\frac{d\lambda_a}{d\bar{Z}} - \frac{d\lambda_b}{d\bar{Z}}) > 0$$

We replace $\theta_a - \theta_b$ by $\frac{c_b - c_a}{e^{rt_1}\mu} - \frac{\lambda_a - \lambda_b}{\mu}$, it comes that when c_b tends towards c_a, $\frac{d\lambda_a}{d\bar{Z}} - \frac{d\lambda_b}{d\bar{Z}}$ has the sign of $(\theta_b - \theta_a)(-\theta_d \frac{\mu}{\lambda_a + \theta_a \mu} \frac{D(p^2)}{D(p^0)} - 1 + \mu \frac{\theta_d}{\lambda_a + \theta_a \mu})$. Since $-\theta_d \frac{\mu}{\lambda_a + \theta_a \mu} \frac{D(p^2)}{D(p^0)} - 1 + \mu \frac{\theta_d}{\lambda_a + \theta_a \mu} < 0$, it follows that $\frac{d\lambda_a}{d\bar{Z}} - \frac{d\lambda_b}{d\bar{Z}}$ has the sign of $\theta_a - \theta_b$. Thus there exists c^* such that Claim (ii) holds.

B.4.6 Proof of Proposition 4

Call $X = (\lambda_a, \lambda_b, \mu, t_1, t_2, t_3)$ and F a function from $\mathbb{R}^6 \times \mathbb{R}^2$ to $\mathbb{R}^6$, defined as:

$$F : (X, \theta_a, \bar{Z}) \rightarrow \begin{cases} f_1(X, \theta_a, \bar{Z}) = c_a + \lambda_a e^{rt_1} + \theta_a \mu e^{rt_1} - (c_b + \lambda_b e^{rt_1} + \theta_b \mu e^{rt_1}), \\ f_2(X, \theta_a, \bar{Z}) = c_b + \lambda_b e^{rt_2} + \theta_b \mu e^{rt_2} - (c_d + \theta_d \mu e^{rt_2}), \\ f_3(X, \theta_a, \bar{Z}) = c_d + \theta_d \mu e^{rt_3} - c_c, \\ f_4(X, \theta_a, \bar{Z}) = \int_0^{t_1} u'^{-1}(c_a + \lambda_a e^{rt} + \theta_a \mu e^{rt})\, dt - X_a, \\ f_5(X, \theta_a, \bar{Z}) = \int_{t_1}^{t_2} u'^{-1}(c_b + \lambda_b e^{rt} + \theta_b \mu e^{rt})\, dt - X_b, \\ f_6(X, \theta_a, \bar{Z}) = \theta_a X_a + \theta_b X_b + \theta_d \int_{t2}^{t3} u'^{-1}(c_d + \theta_d \mu e^{rt})\, dt - \bar{Z}. \end{cases}$$

It is straightforward that F is a function of class $\mathcal{C}^1$. Let $(X_0, \theta_a^0, \bar{Z})$ be a point such that $F(X_0, \theta_a^0, \bar{Z}) = 0$.

Call J_F the Jacobian matrix of F evaluated at $(X_0, \theta_a^0, \bar{Z})$.

Calling

$$X = -\int_0^{t_1} u''^{-1}(c_a + \lambda_a e^{rt} + \theta_a^0 \mu e^{rt}) e^{rt}\, dt$$

,

$$X2 = \int_{t_1}^{t_2} u''^{-1}(c_b + \lambda_b e^{rt} + \theta_b \mu e^{rt}) e^{rt}$$

,

$$X3 = \theta_d \int_{t_2}^{t_3} u''^{-1}(c_d + \theta_d \mu e^{rt}) e^{rt}\, dt$$

,

$$D_1 = u'^{-1}(c_a + \lambda_a e^{rt_1} + \theta_a^0 \mu e^{rt_1})$$

,

$$D_2 = u'^{-1}(c_b + \lambda_b e^{rt_2} + \theta_b \mu e^{rt_2})$$

and

$$D_3 = u'^{-1}(c_d + \theta_d \mu e^{rt_3})$$

The Jacobian writes:

$$\begin{bmatrix} 1 & -1 & \theta_a^0 - \theta_b & r(\lambda_a + \theta_a^0\mu - (\lambda_b + \theta_b\mu)) & 0 & 0 \\ 0 & 1 & (\theta_b - \theta_d) & 0 & r\lambda_b + r\theta_b\mu - r\theta_d\mu & 0 \\ 0 & 0 & \theta_d & 0 & 0 & r\theta_d\mu \\ -X & 0 & -\theta_a^0 X & D1 & 0 & 0 \\ 0 & -X2 & -\theta_a^0 X2 & -D1 & D2 & 0 \\ 0 & 0 & -X3 & 0 & -D2 & D3 \end{bmatrix}.$$

Evaluated at $c_d = c_b > c_a$ and $\theta_d = \theta_b = \theta_a^0$, the matrix becomes

$$\begin{bmatrix} 1 & -1 & 0 & r(\lambda_a - \lambda_b) & 0 & 0 \\ 0 & 1 & 0 & 0 & r\lambda_b & 0 \\ 0 & 0 & \theta_b & 0 & 0 & r\theta_a\mu \\ -X & 0 & -\theta_b X & D1 & 0 & 0 \\ 0 & -X2 & -\theta_b X2 & -D1 & D2 & 0 \\ 0 & 0 & -X3 & 0 & -D2 & D3 \end{bmatrix}.$$

whose determinant has the sign of

$$\begin{aligned} &\Big(D_1D_2D_3 + D_1D_2X_3\mu r\Big) \\ &+ XD_1D_2\mu r\theta_d + XD_1D_3\lambda_b r + XD_1X_3\lambda_b\mu r \\ &\quad + XD_2D_3r(\lambda_a - \lambda_b) + XD_2X_2\mu r^2\theta_d(\lambda_a - \lambda_b) \\ &\quad + XD_2X_3\mu r^2(\lambda_a - \lambda_b) + XD_3X_2\lambda_b r^2(\lambda_a - \lambda_b) \\ &\quad + XX_2X_3\lambda_b\mu r^3(\lambda_a - \lambda_b) \\ &+ X_2\Big(D_1D_2\mu r\theta_d + D_1D_3\lambda_b r + D_1X_3\lambda_b\mu r\Big) \end{aligned}$$

As $c_a + \lambda_a e^{rt_1} = c_b + \lambda_b e^{rt_1}$, and $c_a < c_b$, it is straightforward that $\lambda_a - \lambda_b > 0$, so that $\det J_F > 0$.

If J_F is invertible ($\det J_F \neq 0$), by the implicit function theorem, then there exists a function ϕ of class $\mathcal{C}^1$ from $\mathbb{R}^2$ to $\mathbb{R}^6$, as well as neighborhoods V of $(\theta_a^0, \bar{Z})$ and W of X_0, such that for all $(\theta_a, \bar{Z}) \in V$ and $X \in W$, the following holds:

$$F(X, \theta_a, \bar{Z}) = 0 \iff X = \phi(\theta_a, \bar{Z}).$$

Since ϕ is of class $\mathcal{C}^1$, the partial derivatives $\frac{\partial\phi_1}{\partial\bar{Z}} \equiv \frac{\partial\lambda_a}{\partial\bar{Z}}$ and $\frac{\partial\phi_2}{\partial\bar{Z}} \equiv \frac{\partial\lambda_b}{\partial\bar{Z}}$ exist and are continuous. Therefore, $\frac{d\lambda_b}{d\bar{Z}}$ and $\frac{d\lambda_a}{d\bar{Z}}$ are continuous functions of θ_a.

As a result, for θ_a close enough to θ_b, we have:

$$\frac{d\lambda_b}{d\bar{Z}}(\theta_b, \bar{Z}) < 0 \implies \frac{d\lambda_b}{d\bar{Z}}(\theta_a, \bar{Z}) < 0.$$

and

$$\frac{d\lambda_a}{d\bar{Z}}(\theta_b, \bar{Z}) > 0 \implies \frac{d\lambda_a}{d\bar{Z}}(\theta_a, \bar{Z}) > 0.$$

B.4.7 Endogenous reserves

Following Gaudet and Lasserre (1988), we assume that reserves can be extended thanks to early exploration efforts. Reserves of resources a and b result from separate exploration activities. We write $E(.)$ the exploration cost function for each type of resource. Exploration costs are increasing and convex in the reserves. We assume that $E'() > 0$, and $E''() > 0$, and $E(0) = E'(0) = 0$.

The Hamiltonian becomes:

$$\begin{aligned} H(t) = u\Big(\sum_{i\in\{a,b,c,d\}} x_i(t)\Big) - \sum_{i\in\{a,b,c,d\}} (c_i + \mu(t)\theta_i)x_i(t) \\ - \textstyle\sum_{i\in\{a,b\}} E(X_i^0) - \displaystyle\sum_{i\in\{a,b\}} \lambda_i(t)x_i(t) \end{aligned}$$

And first-order conditions now include:
$\frac{\partial H(t)}{\partial X_i^0} = 0 \iff E'(X_i^0) = \lambda_i^0$.

We write $\mathbf{x}$ the vector of (x_{it}) for all (i,t) and $\mathbf{x}_i$ the vector of (x_{it}) at i given. Consider the function $V((\bar{X}_i), \bar{Z}) = \max_{\mathbf{x}} f(\mathbf{x})$ subject to $\forall i \; l_i(\mathbf{x}_i) \leq \bar{X}_i$ and $h(\mathbf{x}) \leq \bar{Z}$.

We first aim to prove that $V((\bar{X}_i), \bar{Z})$ is a concave function of $\bar{X}_j$. To do this, define the Lagrangian $\mathcal{L}$ associated with the optimization problem:

$$\mathcal{L}(\mathbf{x}, (\lambda_i), \mu, (\bar{X}_i), \bar{Z}) = f(\mathbf{x}) + \sum_i \lambda_i(\bar{X}_i - l(\mathbf{x}_i)) + \mu(\bar{Z} - h(\mathbf{x}))$$

Then:

$$\forall((\bar{X}_i), \bar{Z}), \quad V((\bar{X}_i), \bar{Z}) = \min_{(\lambda_i),\mu} \max_{\mathbf{x}} \mathcal{L}((\mathbf{x}), (\lambda_i), \mu, (\bar{X}_i), \bar{Z})$$

Let $\tilde{V}((\lambda_i), (\bar{X}_i), \bar{Z}) = \max_{\mathbf{x}} \mathcal{L}(\mathbf{x}, (\lambda_i), \mu, (\bar{X}_i), \bar{Z})$.

By the envelope theorem, for all $((\lambda_i), (\bar{X}_i), \bar{Z})$:

$$\frac{\partial \tilde{V}}{\partial \bar{X}_j} = \lambda_j$$

This implies that $\tilde{V}((\lambda_i), (\bar{X}_i), \bar{Z})$ is a linear function of $\bar{X}_j$. Therefore, $V((\bar{X}_i), \bar{Z})$

is a concave function of $\bar{X}_j$ as the minimum envelope of linear functions of $\bar{X}_j$.

Consider now $\bar{X}_j$ that maximizes

$$V((\bar{X}_i), \bar{Z}) - \sum_i g_i(\bar{X}_i)$$

for a given $\bar{Z}$.

By the implicit function theorem:

$$\frac{d\bar{X}_j}{d\bar{Z}} = \frac{\frac{\partial^2 V}{\partial \bar{X}_j \partial \bar{Z}}}{g_j'' - \frac{\partial^2 V}{\partial \bar{X}_j^2}}$$

Since V is concave in $\bar{X}_j$ and g_j is convex in $\bar{X}_j$, we have $g_j'' - \frac{\partial^2 V}{\partial \bar{X}_j^2} > 0$.

Thus, $\frac{d\bar{X}_j}{d\bar{Z}}$ has the same sign as

$$\frac{\partial^2 V}{\partial \bar{X}_j \partial \bar{Z}} = \left(\frac{\partial \lambda_j}{\partial \bar{Z}}\right)_{\bar{X}_j}$$

In other words, the sign of impact of the fall of the carbon ceiling on λ_j is the same as in the exogenous-reserves case (as $\lambda_j = g'(\bar{X}_j)$ and $g'' > 0$).

If c_b is close enough to c_d, resource b benefits from the increase in stringency.

B.5 GHG mitigation policies in the oil industry

There are three main messages considering GHG regulations in the oil industry.

First, most major oil-producing countries do not have internationally-binding mitigation targets (Panel A of Table B.1). Of the Top-10 oil producers in 2018, only the US and Russia belonged to the group of countries with emission-reduction targets in the Kyoto Protocol (Annex B). However, the US did not ratify the Kyoto Protocol, and Russia's target was to keep emissions over the 2008–2012 period at their 1990 level, a loose constraint in the context of the post-USSR collapse in industrial production.

Second, in countries with some oil-emission regulations, these usually cover emissions related to oil combustion, ignoring emissions related to extraction and refining. As oil heterogeneity in terms of CO_2 lays in these two sectors, current policies, such as taxing fuel, imposing fuel-efficiency standards or subsidizing oil alternatives, produce no incentives to extract oil from less-polluting deposits. Among the Top-10 oil producers in 2018 (Panel B of Table B.1), only the US, China, and Canada have implemented a carbon tax or an emissions-trading system (ETS). However, these schemes do not cover upstream emissions, with the exception of that in Alberta Province in Canada.[4][5][6]

Third, these instruments do not always ensure correct carbon pricing (for instance, as non market-based instruments are used, or carbon prices differ across jurisdictions, or are too low compared to the social cost of carbon[7]). Carbon mispricing leads to the current situation where polluting oil types are extracted, refined and combusted instead of less-polluting alternatives.

[4] China developed a regional pilot ETS (planned to be developed into a nation-wide ETS). The ETS systems in Hubei, Shanghai, Tianjin, Beijing, Guangdong partly regulate emissions from the petrochemical sector. However, there is no significant oil extraction in these regions. More information on the Chinese pilot ETS is available at https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6571708/.

[5] In the USA, some States have implemented carbon-pricing initiatives, including oil-producing States (e.g., California), but upstream and midstream emissions are not covered. None of the Top-5 US producing States (Texas, North Dakota, New Mexico, Oklahoma and Alaska, which jointly represent 68% of total US crude-oil production in 2018 according to U.S. Energy Information Administration 2020a) has set up a carbon-pricing scheme to price oil emissions as of 2019.

[6] In Alberta, Canada, under the Carbon Competitiveness Incentive Regulation (CCIR), since 2018 firms receive credits if their carbon intensity is below a product-specific threshold, and have to pay a levy per tonne of CO_2 above this threshold. The Alberta CCIR applies to GHG emissions from the industry, power generation and also to large oil-sands mines. The nominal carbon price was CAN\$30/tCO_2eq (US\$22/tCO_2eq) as of August 1^{st} 2019. In Ontario, a cap and trade system existed between 2017 and 2019, linked with California and Quebec within the Western Climate Initiative (WCI). The cap and trade system was canceled, with an effective end year of 2019. With a first compliance period starting in 2013, the Quebec ETS system initially covered electricity generation and industry, and after 2015 the distribution and import of fossil fuels (notably for transportation, building, and the small-business sectors). This includes industrial-process emissions. The Quebec ETS was linked in 2014 to that of California. Its price reached US\$18/tCO_2eq on August 1^{st} 2019. The Province of British Columbia implemented a carbon tax in 2008. All sectors are covered, with some exemptions for the industry, aviation, transport and agriculture sectors. Fugitive emissions, gas linked to extraction but not flared (most often methane), are not covered. The carbon price is US\$30/tCO_2eq (CAN\$40/tCO_2eq).

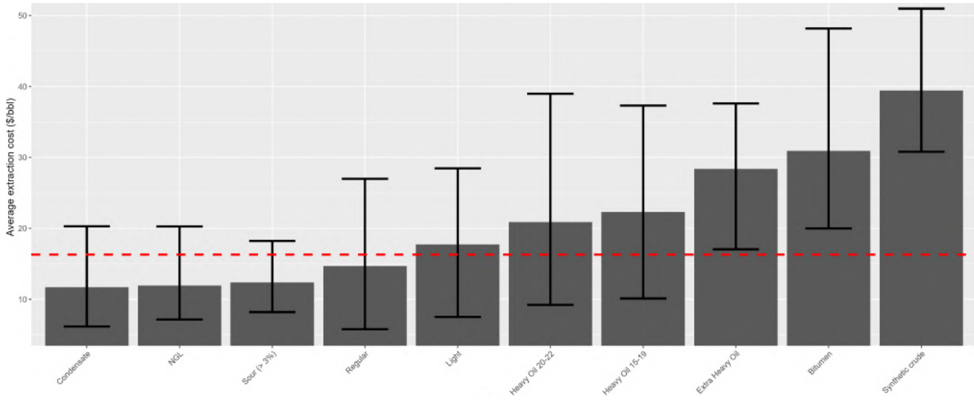
[7] For instance, the Chinese pilot ETS had carbon prices ranging from US\$6 to US\$13/tCO_2eq (from RMB39/tCO_2eq to RMB84/tCO_2eq) across the ETS as of August 1^{st} 2019.

Country	Share of world supply	Year (start-end)	Sectors
US ETS	18%		
RGGI †		2009-	power
Washington		2017-	industry, power, transport, waste, buildings
Massachusetts		2018-	power
California		2012-	power, road fuel distribution
Canada	5%		
Alberta		2007-17	industry, power
Alberta CCIR		2018-	industry, power, large oil-sands mines
Quebec ETS		2013-	power, industry, distribution, fossil-fuel imports
BC tax		2008-	all sectors except agriculture (from 2013)
Ontario CaT		2017-19	all sectors except agriculture, waste, aviation, marine transport
China ETS	5%		
Shanghai		2013-	power, petrochemicals, aviation, heavy industry
Shenzhen		2013-	power, manufacturing
Tianjin		2013-	petrochemicals, power, oil & gas, heavy industry
Guangdong		2013-	power, cement, steel, petrochemicals
Chongqing		2014-	power, heavy industry
Hubei		2014-	power, heavy industry, petrochemicals
Beijing		2013-	power, heavy industry, petrochemicals

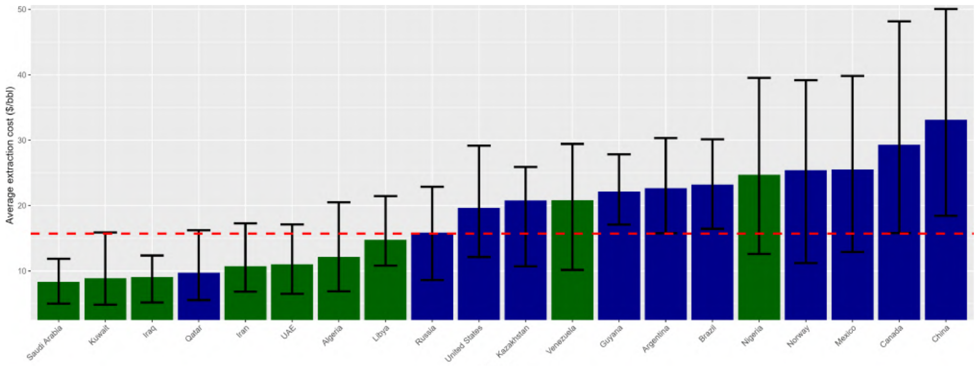
Table B.1: The global carbon-policy context: Carbon-pricing initiatives in the 10 largest oil producers as of 2018.

Notes: Production includes the domestic production of crude oil, all other petroleum liquids, biofuels, and refinery processing gain. Data from U.S. Energy Information Administration (2020b) and World Bank (2020). For the British Columbia (BC) carbon tax and Ontario CaT, data are from Murray and Rivers (2015) and Financial Accountability Office of Ontario (2018) respectively. †: The Regional Greenhouse Gas Initiative (RGGI) covers Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New York, Rhode Island, and Vermont.

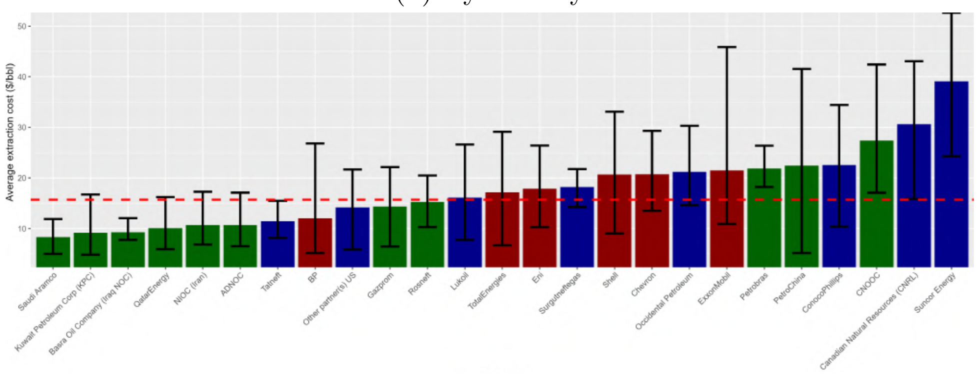
B.6 Additional figures and tables



(a) By oil type



(b) By country



(c) By company

Figure B.2: Heterogeneity in private extraction cost of 2025 proven oil reserves.

Notes: Panel (a), Panel (b) and Panel (c) represent the private extraction cost (in USD) per barrel of 2025 reserves by oil type, producing country, and company, respectively. The bar height represents the average (weighted by 2025 reserves), and the extremities of the lines the 10% and 90% deciles. The red dashed line corresponds to the World average figure. Panel (b): Only the top 20 oil producers over the period are represented, and OPEC country carbon intensity bars appear in blue. See the notes to Figure 2.1 for OPEC composition, as of 2025. Panel (c): NOCs', Majors', other firms' carbon intensity bars appear in green, red, and blue, respectively. Section 2.2.2 and Appendix B.3 provide detailed information on the estimation of private costs.

(a) Changes in deposit's per barrel social value (b) Changes in deposit's cumulative production

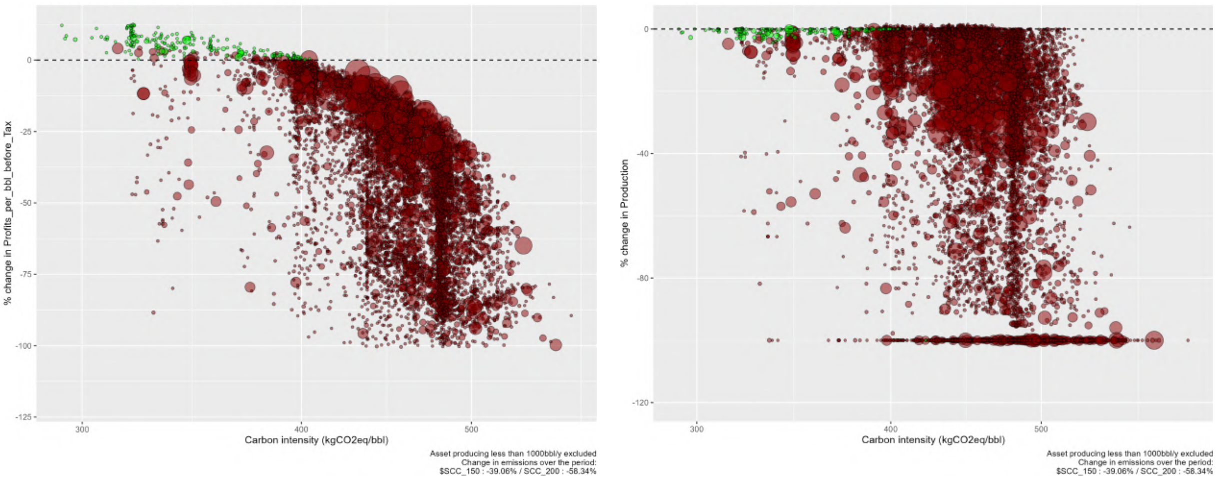
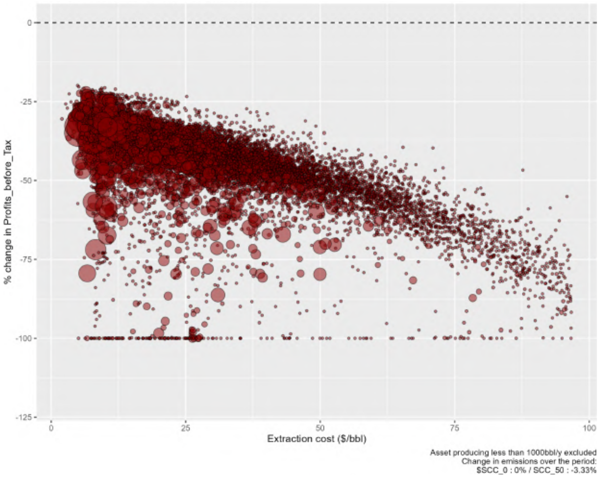


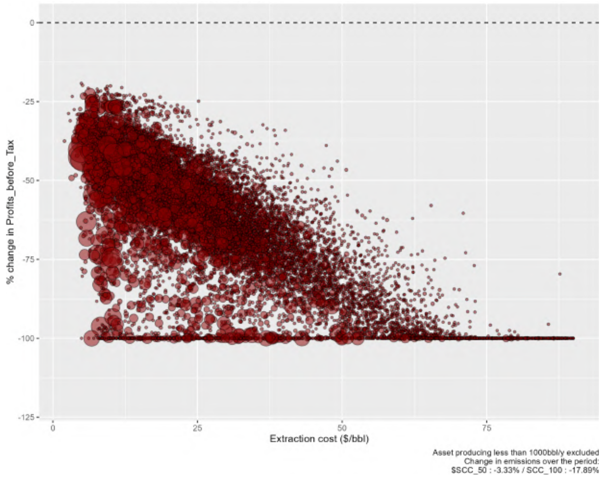
Figure B.3: Changes in deposits' per barrel social value of oil and in cumulative production: SCC increases from \$150 to \$200/tCO_2

Notes: Each circle represents an oil deposit, with the circle size an indication of the deposit volume. Deposits producing fewer than 1,000 barrels per year are omitted for visibility, as are outliers. For assessment of deposits' carbon intensity, measured in kgCO_2eq per barrel, refer to Section 2.2.4 and Appendix B.3.2. Panel (a): The vertical axis indicates the change in deposits' social value of oil after the SCC increases from \$150 to \$200/tCO_2eq. For each deposit i, the total discounted social value is computed as $\sum_0^\infty x_i(t)\,(\,p(t) - \theta_i\mu(t) - c_i)\,e^{-rt}/\sum_0^\infty x_i(t)$. Panel (b): The vertical axis indicates the change in deposits' cumulative production, that writes $\sum_0^\infty x_i(t)$ for deposit i, after the SCC increases from \$150 to \$200/tCO_2eq.

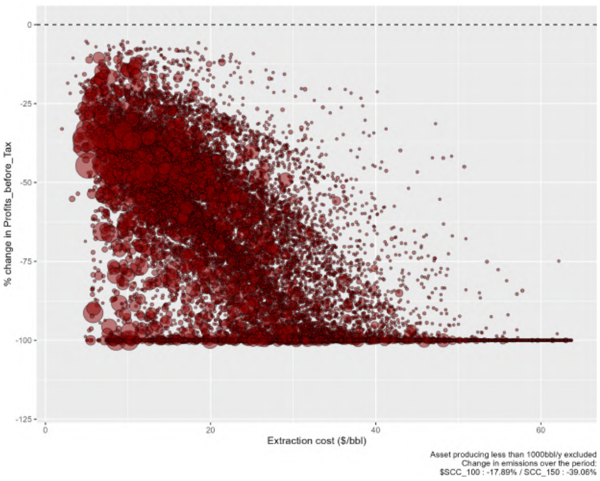
(a) Carbon tax increase: \$0 to \$50/tCO$_2$.



(b) Carbon tax increase: \$50 to \$100/tCO$_2$.



(c) Carbon tax increase: \$100 to \$150/tCO$_2$.



(d) Carbon tax increase: \$150 to \$200/tCO$_2$.

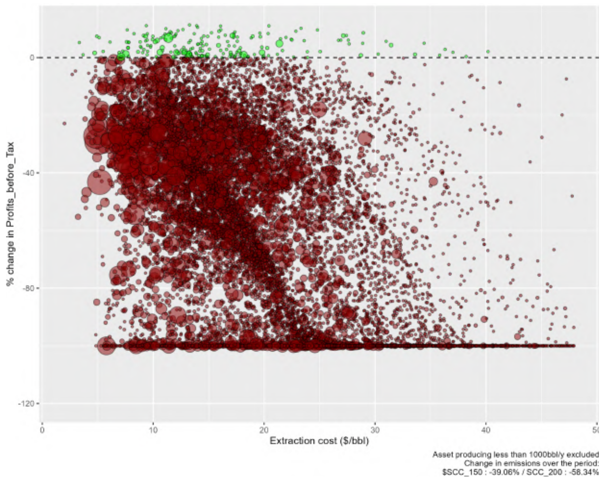


Figure B.4: Changes in deposits' per barrel social value as a function of deposits private costs.

Notes: Each circle represents an oil deposit, with the circle size an indication of the deposit volume. Deposits producing fewer than 1,000 barrels per year are omitted for visibility, as are outliers. The figure reproduces Figure 2.2 but plots deposits' social value changes as a function of deposits' private extraction costs.

(a) Carbon tax increase: \$0 to \$50/tCO_2.

(b) Carbon tax increase: \$50 to \$100/tCO_2.

(c) Carbon tax increase: \$100 to \$150/tCO_2.

(d) Carbon tax increase: \$150 to \$200/tCO_2.

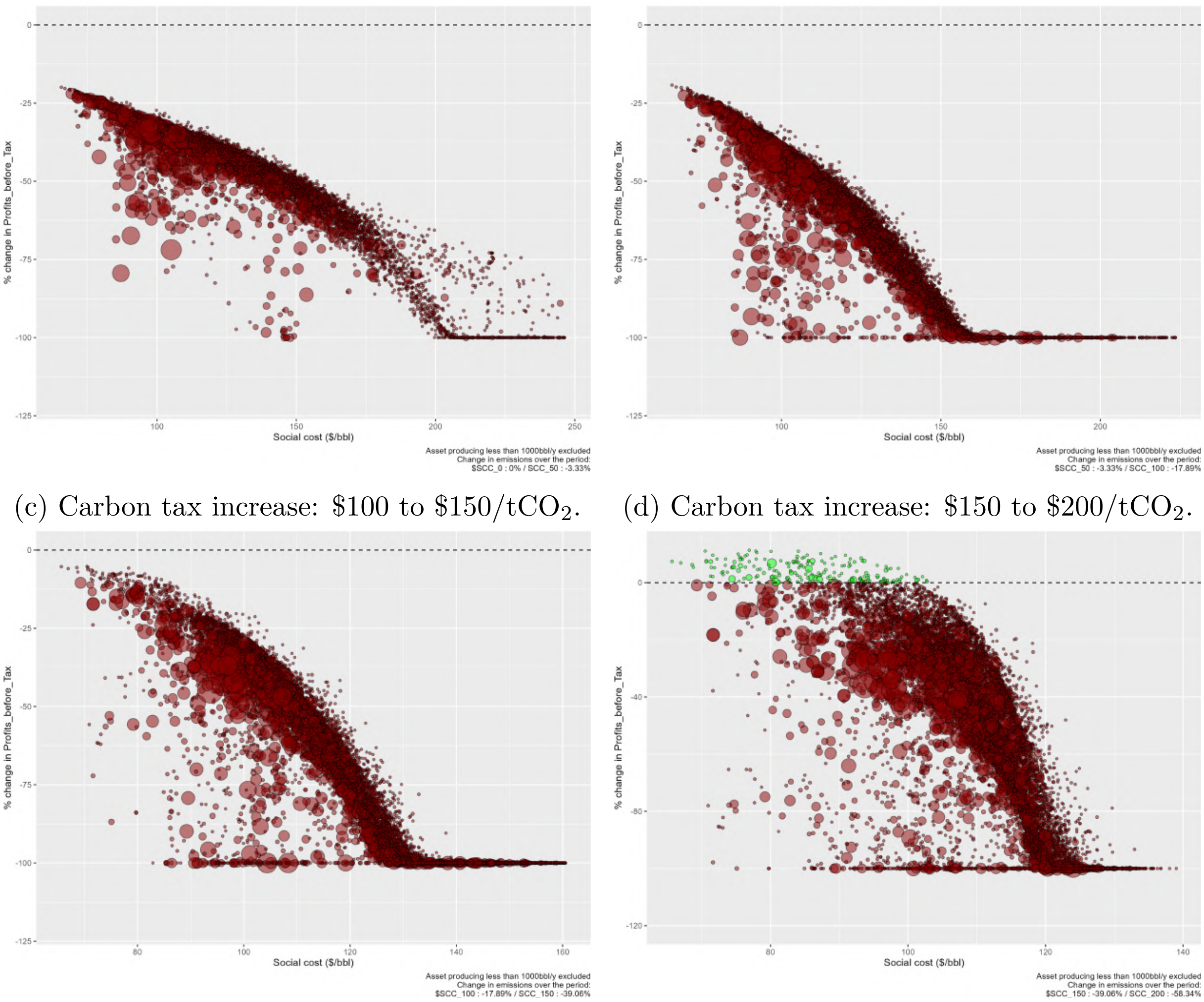


Figure B.5: Changes in deposits' per barrel social value as a function of deposits social cost.

Notes: Each circle represents an oil deposit, with the circle size an indication of the deposit volume. Deposits producing fewer than 1,000 barrels per year are omitted for visibility, as are outliers. The figure reproduces Figure 2.2 but plots deposits' social value changes as a function of deposits' social cost (computed as the sum of the deposits' private cost and its life-cycle emissions valued at \$200/tCO_2eq).

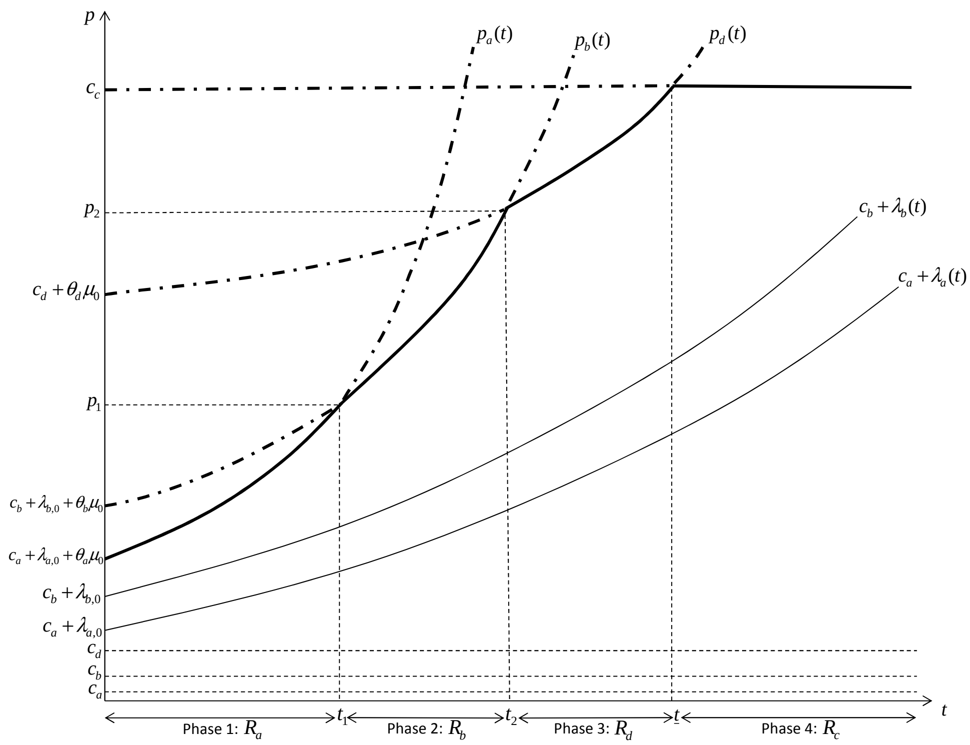


Figure B.6: Energy price path (bold solid line)

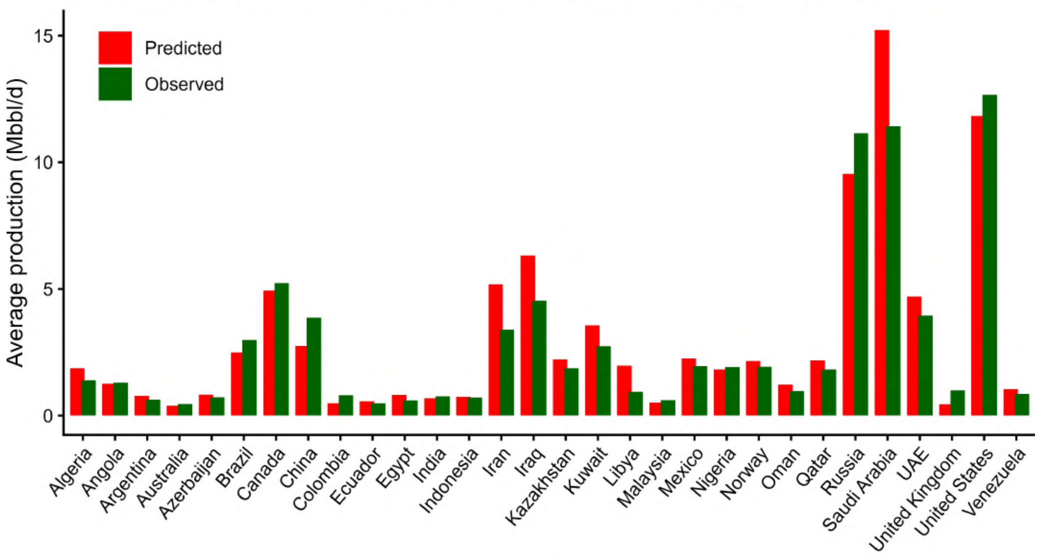


Figure B.7: Predicted and observed Country productions over the 2019–2021 period.

Notes: This figure represents predicted and observed country-level production in million barrels per day (Mbbl/d), summed over the 2019–2021 period for the 30 largest oil-producing countries. Predicted productions are the solution of our optimization program initialized in 2019, with no carbon taxes but incorporating distorting taxes such as royalties and OPEX taxes into the private extraction costs.

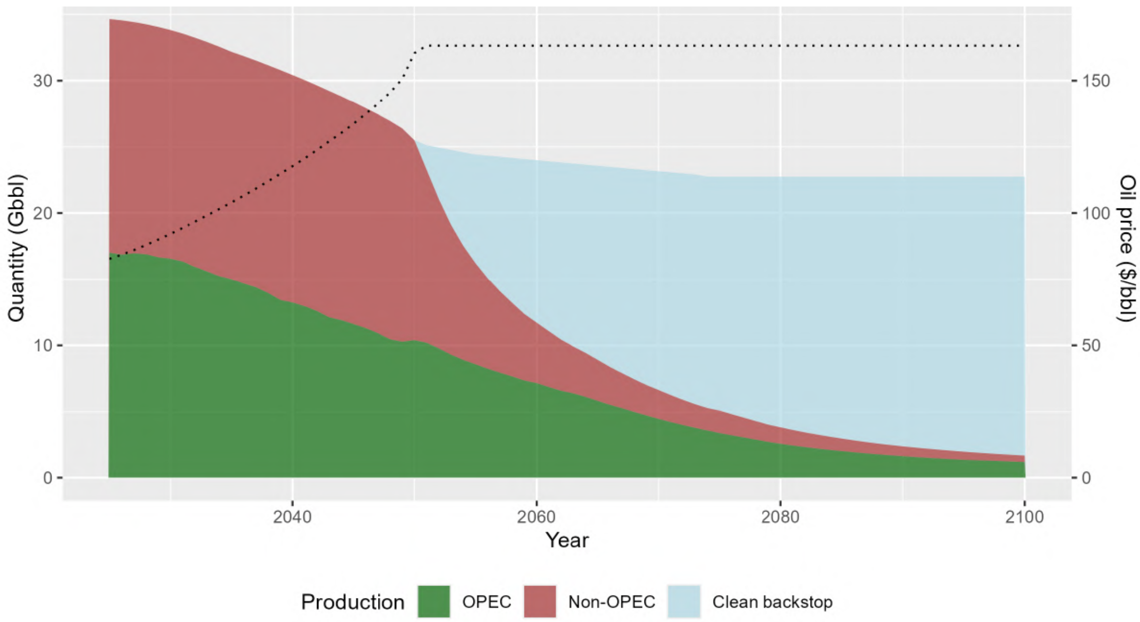
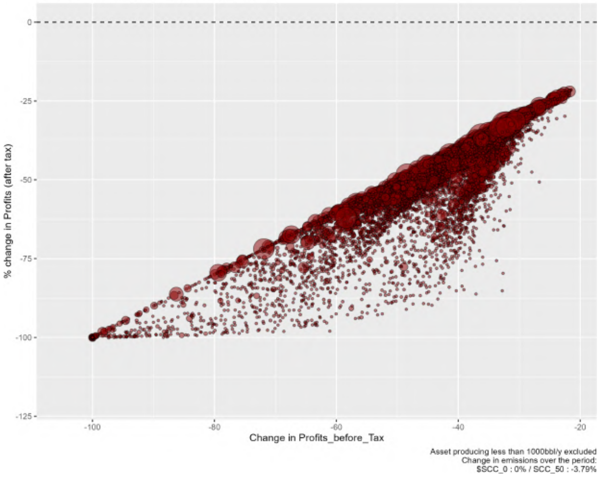


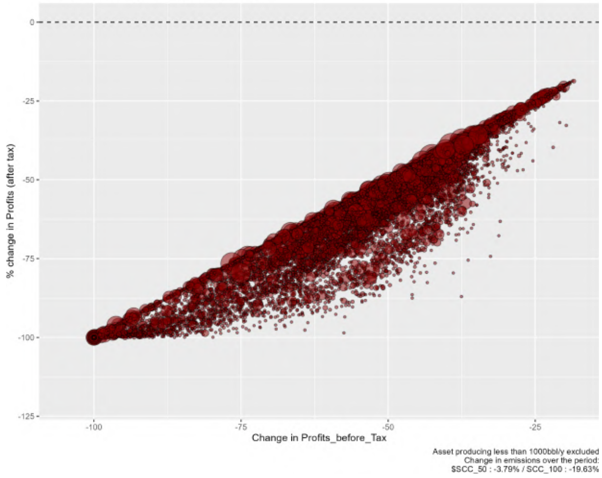
Figure B.8: Oil and energy demands in the future with no carbon tax.

Notes: The equilibrium is computed using $\mathcal{P}_1(\mu(t))$ with $\mu = 0$, and assuming private costs are augmented of production taxes (royalties and tax opex).

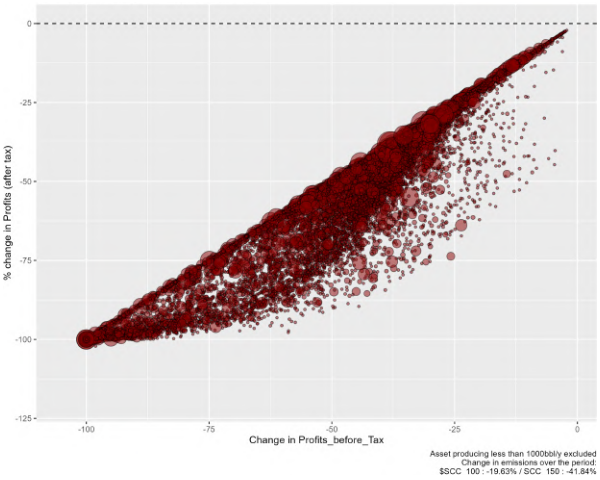
(a) Carbon tax increase: \$0 to \$50/tCO_2.



(b) Carbon tax increase: \$50 to \$100/tCO_2.



(c) Carbon tax increase: \$100 to \$150/tCO_2.



(d) Carbon tax increase: \$150 to \$200/tCO_2.

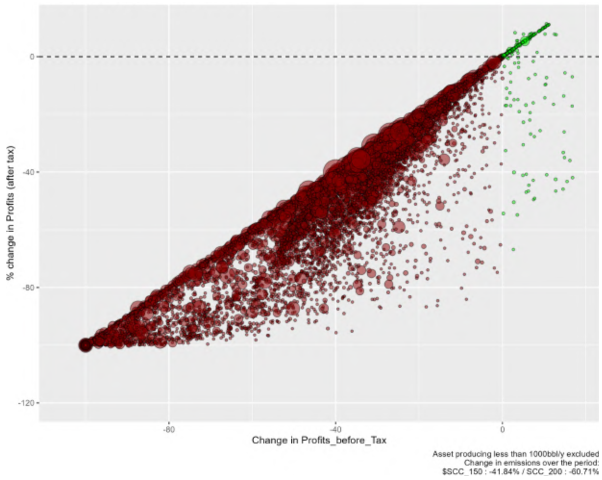


Figure B.9: Before- and after-tax profits at the deposit level: Various tax changes.

Notes: The figure plots the before-tax profits (horizontal axis) and after-tax profits (vertical axis) at the deposit level. Both profit metrics are net of carbon taxes: the terms "before-tax" and "after-taxes" refer to the inclusion or the exclusion of production taxes paid to government that are royalties, income tax, tax opex, and government profit oil. Each circle represents an oil deposit, with the circle size an indication of the deposit volume. Deposits producing fewer than 1,000 barrels per year are omitted for clarity, as are outliers for visibility.

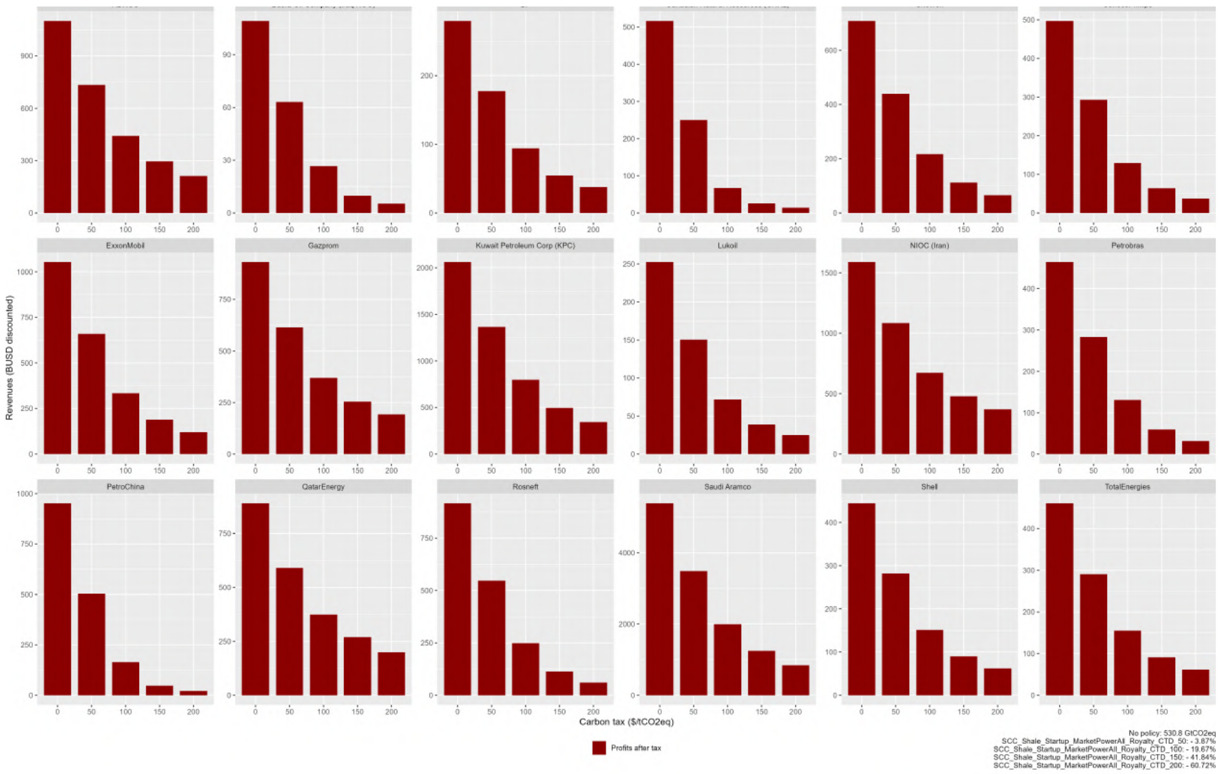


Figure B.10: Companies' profits and carbon taxes.

Notes: The figure displays companies' discounted profits in billions USD (2025 present value) for the top-18 companies (in terms of 2025 reserves) when the carbon taxes vary from 0 to \$200/tCO_2eq.

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Appendix C

Appendix to: Voice Until the Last Exit

C.1 Additional tables

Year	Companies
2015	BG, Eni, Kuwait Petroleum Corp (KPC), PetroAmazonas, Shell, SNH (Cameroon), SNPC (Congo), SOCAR, Statoil/Equinor, TotalEnergies, BP, ETAP (Tunisia), Niger Delta Petroleum Resources Ltd.
2016	Galp Energia SA, KazMunaiGaz, MOL Group, ONGC, Seven Energy Nigeria, Sonangol, Uzbekneftegaz, Wintershall / Wintershall Dea, Repsol, Nigerian National Petroleum Corporation (NNPC), Oil India Limited, Frontier Oil Limited, Pan Ocean Oil
2017	Oando Energy Resources, OMV, Petroleum Development Oman (PDO), Seplat Energy, DEA / Wintershall Dea, Lukoil, Woodside, Nile Petroleum Corporation
2018	Gazprom Neft, Sonatrach (Algeria), Petrobras
2019	KazPetrol Group, Saudi Aramco
2020	Ecopetrol, Occidental, Cairn Energy / Capricorn Energy, ConocoPhillips
2021	EOG Resources, QatarEnergy, Cepsa, Pioneer Natural Resources, Chevron, Harbour Energy, Neptune Energy, Range Resources, Petronas
2022	Hess Corporation, Savannah Energy, ExxonMobil, Vista, Civitas Resources
2024	ADNOC, Santos, Pertamina, Chord Energy, Ithaca Energy

Table C.1: Year of Endorsement to ZRF by Companies

	Nb Assets	N	Mean	SD	Min	Max
World - Oil						
Committed - Divested						
Years in data	672	8,300	12.7	1.3	1	13
Current age (years)	672	8,300	34.3	17.8	1	115
Oil production (kbbl/d)	672	8,300	3.8	13.0	0	226.8
Flared volumes (mmscf/d)	672	8,300	0.6	3.5	0	117.2
Committed - Non-divested						
Years in data	6,785	69,285	11.7	2.6	1	13
Current age (years)	6,785	69,285	28.1	21.9	1	182
Oil production (kbbl/d)	6,785	69,285	8.8	59.4	0	3,435.7
Flared volumes (mmscf/d)	6,785	69,285	0.9	5.8	0	355.2
Uncommitted - Non-divested						
Years in data	12,510	131,486	11.9	2.5	1	13
Current age (years)	12,510	131,486	32.7	24.3	1	166
Oil production (kbbl/d)	12,510	131,486	3.3	15.7	0	628.8
Flared volumes (mmscf/d)	12,510	131,486	0.6	5.4	0	301.4
World - Oil - No gas sold before commitment						
Committed - Divested						
Years in data	325	3,893	12.5	1.6	1	13
Current age (years)	325	3,893	35.1	18.0	1	114
Oil production (kbbl/d)	325	3,893	3.7	14.0	0	223.8
Flared volumes (mmscf/d)	325	3,893	0.7	4.5	0	117.2
Committed - Non-divested						
Years in data	3,318	30,861	11.5	3.0	1	13
Current age (years)	3,318	30,861	31.0	23.0	1	182
Oil production (kbbl/d)	3,318	30,861	7.3	38.1	0	1,272.9
Flared volumes (mmscf/d)	3,318	30,861	1.2	8.0	0	355.2
Uncommitted - Non-divested						
Years in data	4,596	48,891	12.0	2.4	1	13
Current age (years)	4,596	48,891	33.6	21.4	1	161
Oil production (kbbl/d)	4,596	48,891	3.6	16.6	0	628.8
Flared volumes (mmscf/d)	4,596	48,891	0.8	6.0	0	269.6

Table C.2: Summary statistics

Notes: This table reports summary statistics for the main samples used in the analysis. The unit of observation is the asset-year. "Nb Assets" denotes the number of distinct assets and "N" the total number of asset-year observations. "Committed – Divested" refers to assets with at least one ZRF-endorsed owner that subsequently divested to entirely uncommitted ownership. "Committed – Non-divested" refers to assets that gained at least one committed owner and retained committed ownership through the end of the sample period. "Uncommitted – Non-divested" refers to assets that never had a ZRF-endorsed owner during the sample period. The upper panel includes all oil-producing assets; the lower panel restricts the sample to assets with no recorded gas sales prior to commitment (indicating no connection to a gas market). Oil production is measured in thousands of barrels per day (kbbl/d). Flared volumes are measured in million standard cubic feet per day (mmscf/d). Current age is the number of years since the asset's first recorded production. The sample covers the period 2012–2024.

C.2 Additional figures

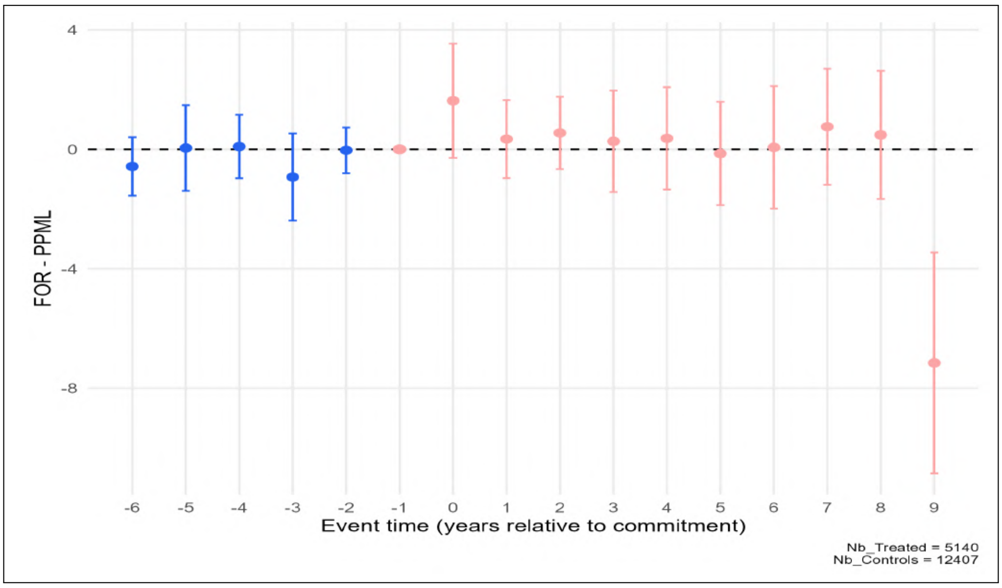


Figure C.1: ZRF commitment on flaring-to-oil ratio (FOR).

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is the flaring-to-oil ratio (scf of gas flared per barrel of oil). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is exp(β) - 1. Specification includes asset, year-by-continent, and operator fixed effects. Standard errors are clustered at the asset level. Point estimates are displayed if observations represent at least 10% of treated assets, and 95% error bars are reported.

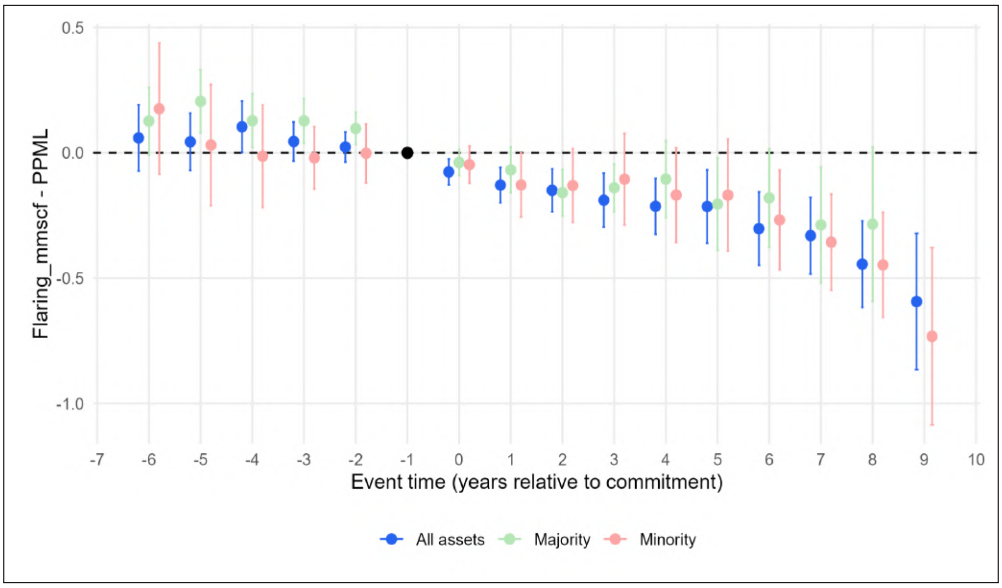
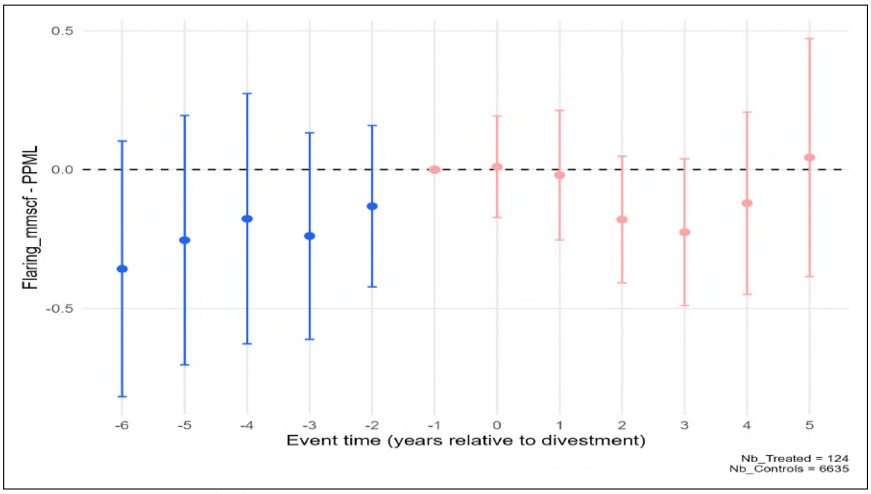
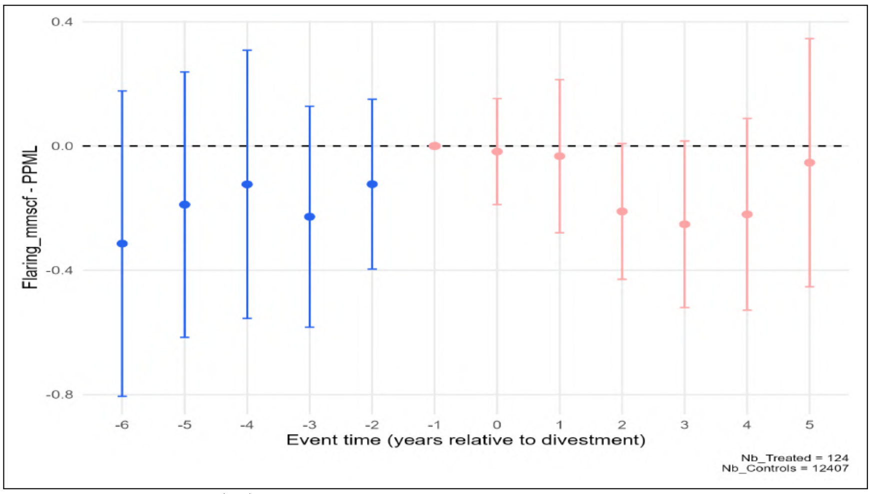


Figure C.2: ZRF commitment on flaring: majority vs minority ownership.

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is exp(β) - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.



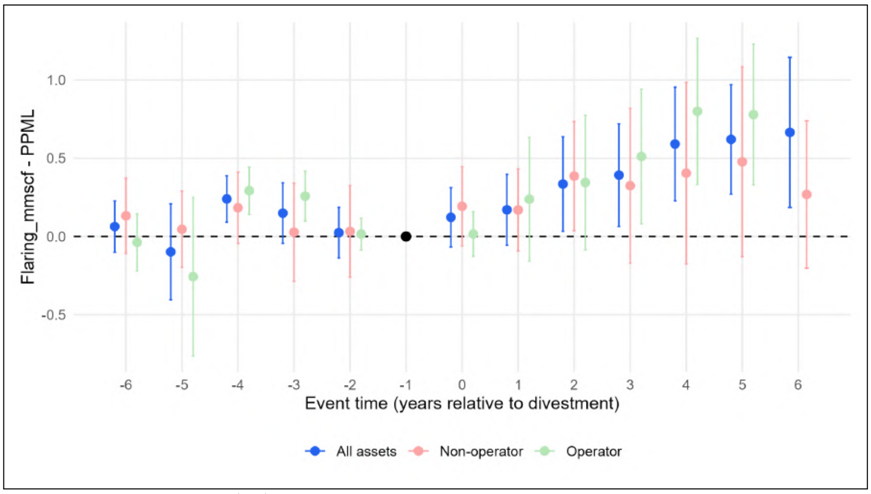
(A) Control group: always committed



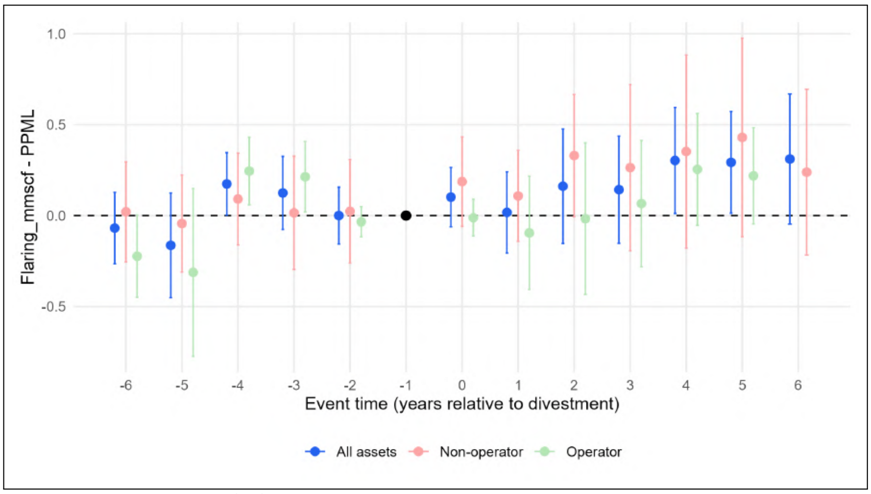
(B) Control group: never committed

Figure C.3: Incomplete divestment on flaring

Notes: This figure plots event study coefficients estimating the effect of "partial" divestment on flaring intensity. The treatment happens when one committed equity owner divests, while at least one other committed firm stays in the board. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before divestment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is exp(β) - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.



(A) Including operator fixed effect

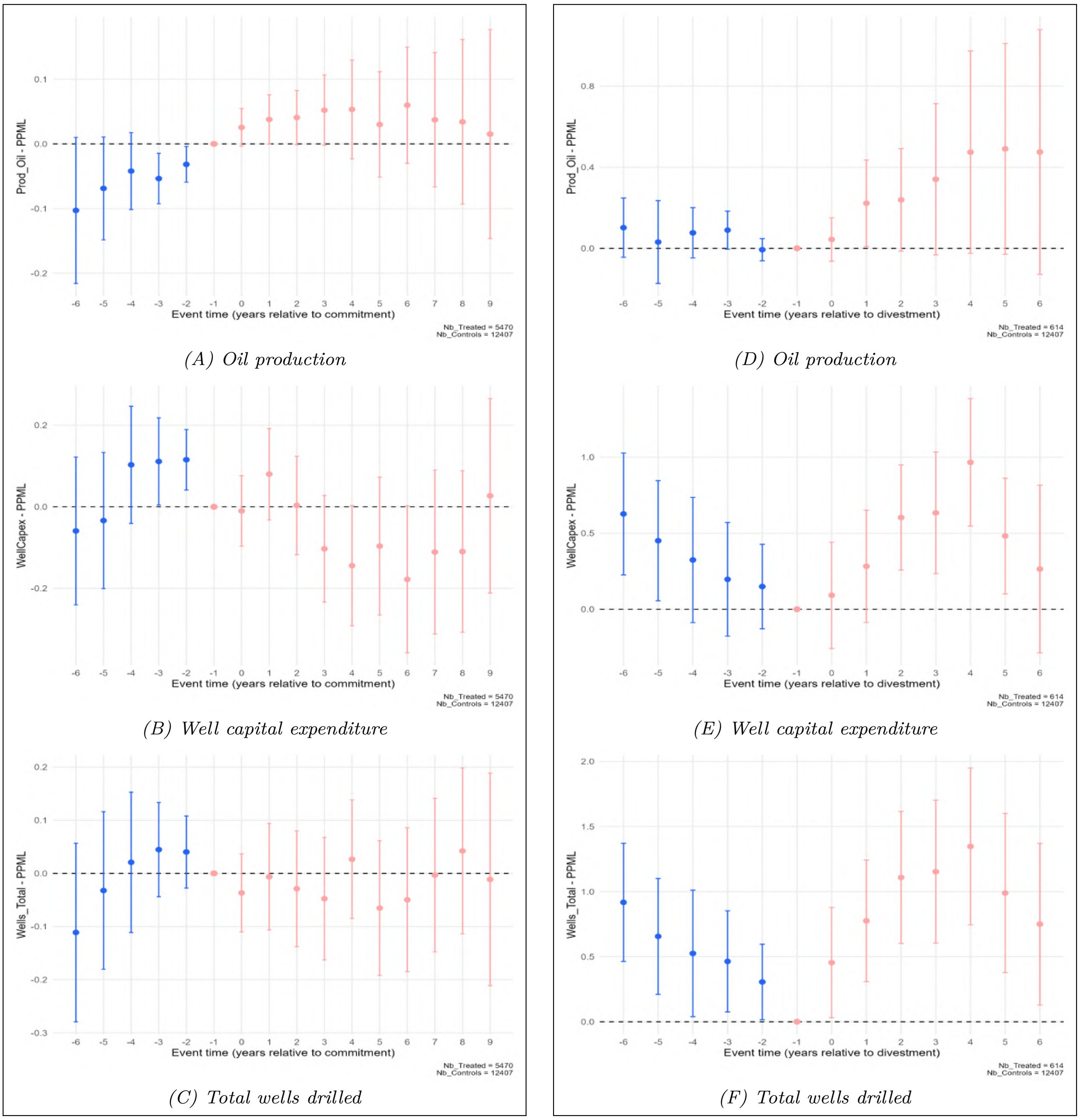


(B) Excluding operator fixed effect

Figure C.4: Divestment on flaring by operator commitment status.

Notes: This figure plots event study coefficients estimating the effect of complete divestment by committed companies on flaring intensity. The treatment happens when the last committed equity owner divests, leaving no committed companies on the management board. The control group are assets which remain committed until the end of the period. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before divestment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is exp(β) - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

Commitment **Divestment**



(A) Oil production *(D) Oil production*

(B) Well capital expenditure *(E) Well capital expenditure*

(C) Total wells drilled *(F) Total wells drilled*

Figure C.5: Mechanisms: commitment and divestment effects on production and investment

Notes: This figure plots event study coefficients estimating the effect of ZRF commitment (left) and complete divestment by the last committed owner (right) on oil production, well capital expenditure, and total wells drilled. The control group consists of assets never owned by committed companies. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

Commitment **Divestment**

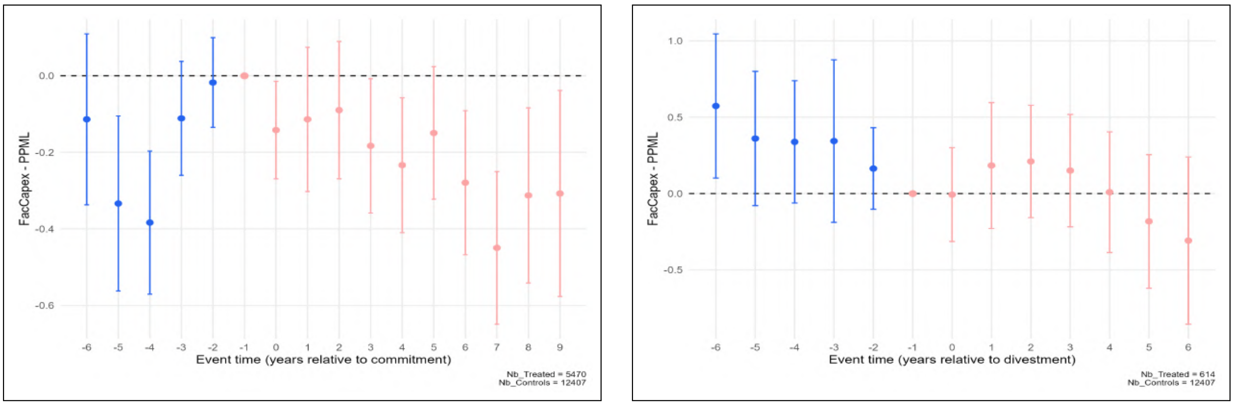
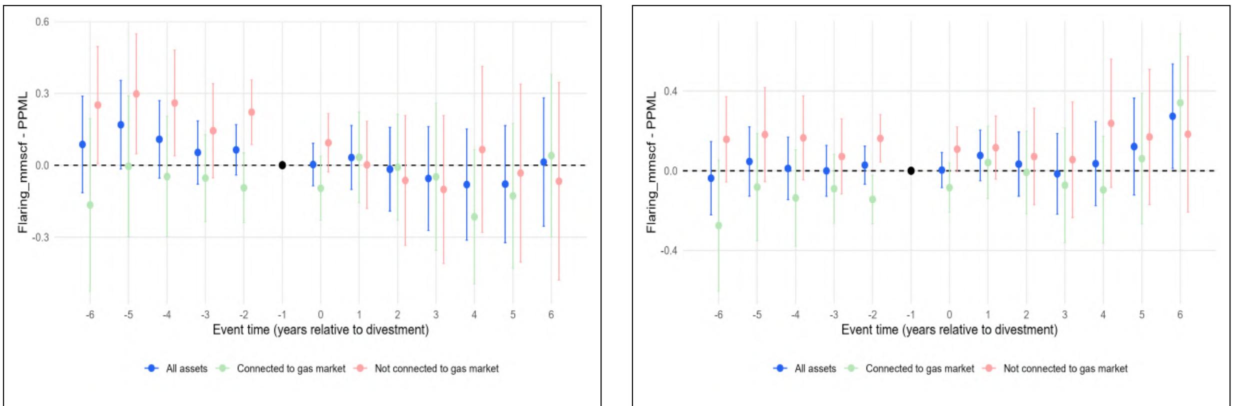


Figure C.6: Commitment and divestment effects on facility capital expenditure

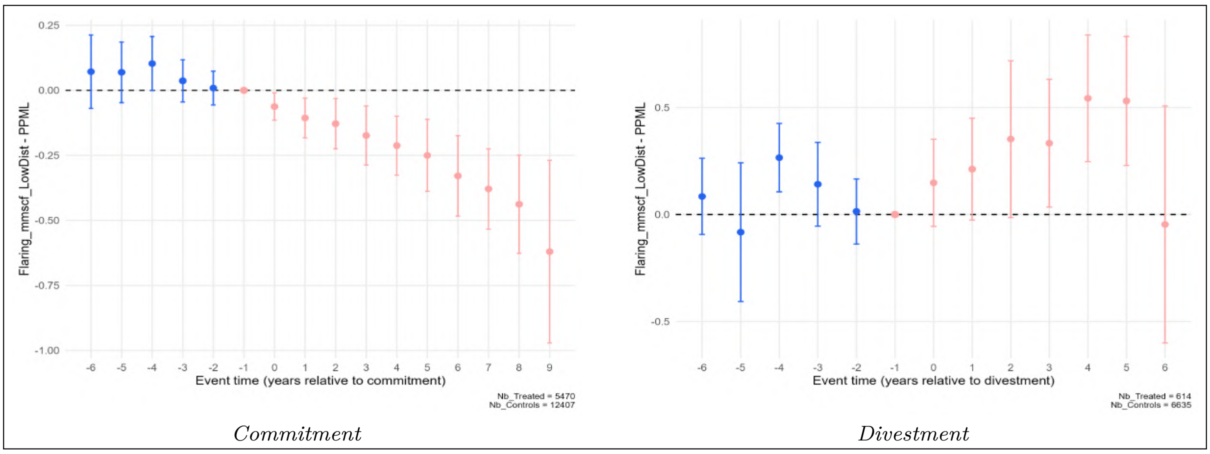
Notes: This figure plots event study coefficients estimating the effect of ZRF commitment (left) and complete divestment by the last committed owner (right) on facility capital expenditure. The control group consists of assets never owned by committed companies. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline; vertical bars show 95% confidence intervals.



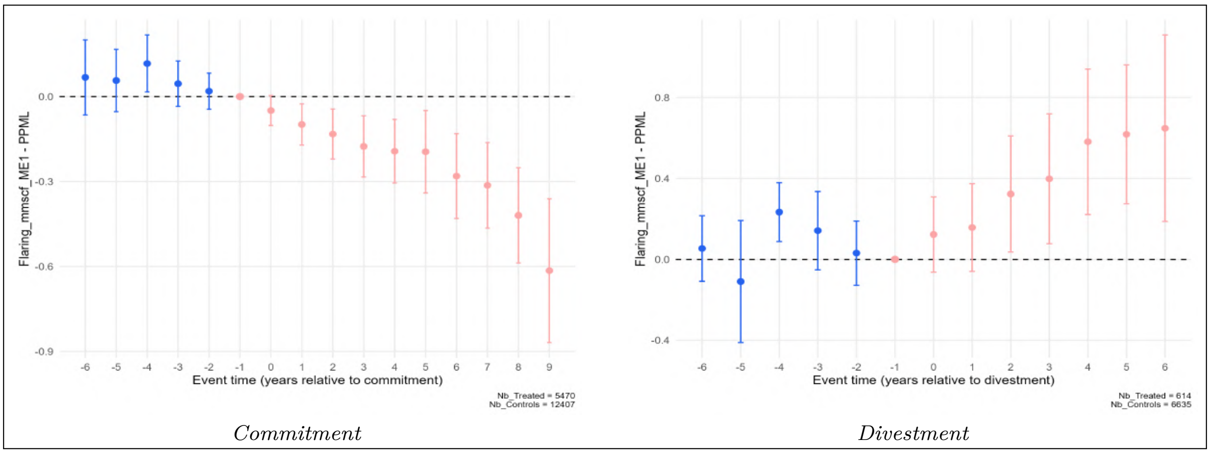
Control group: never committed *Control group: always committed*

Figure C.7: Divestment from uncommitted to uncommitted and flaring

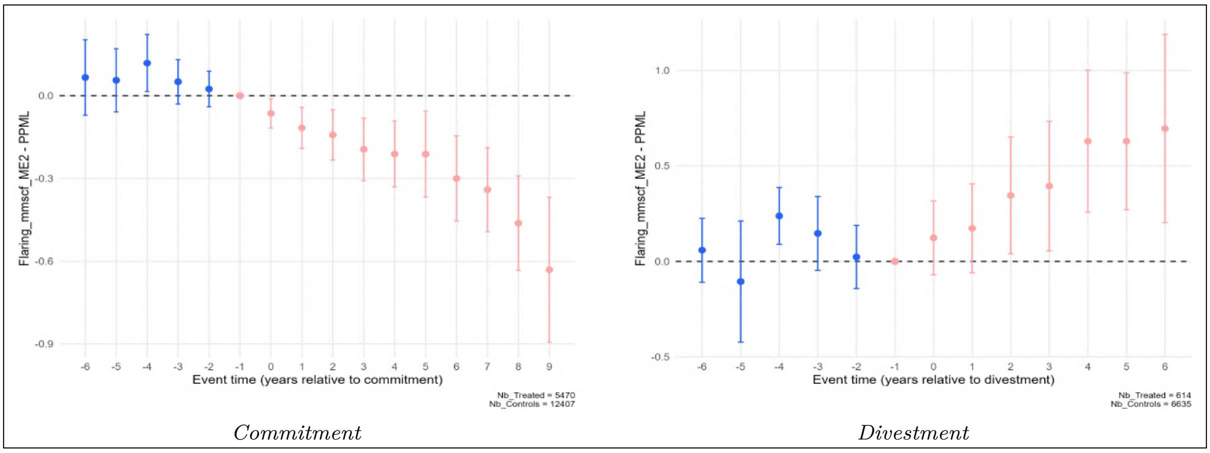
Notes: These figures plot event study coefficients estimating the effect of divestment by an uncommitted owner to another uncommitted owner on flaring intensity. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before divestment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is exp(β) - 1. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.



(A) Alternative flaring measure: lower distance thresholds



(B) Alternative flaring measure: λ = 1



(C) Alternative flaring measure: λ = 2

Figure C.8: Commitment and divestment effects under alternative flaring measures

Notes: This figure replicates the main event study results using three alternative measures of flaring. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline; vertical bars show 95% confidence intervals.

C.3 Matching satellite flares observations to oil assets.

Rystad UCube database does not contain information on gas flared volumes at the asset level. This is mainly due to the fact that most oil and gas operators do not publicly report these amounts of gas flared. Moreover, under-reporting by fields operators remains an important issue making the use of self-reported flaring less reliable.

To fill this gap, we use geocoded flaring volumes calculated by the Earth Observation Group (EOG) at the Payne Institute for Public Policy, Colorado School of Mines, in partnership with NOAA. The EOG analyzes satellite data from the Visible Infrared Imaging Radiometer Suite (VIIRS), launched in 2012 and 2017, which detects heat from gas flares at night. Flaring volumes are then estimated using the VIIRS Nightfire algorithm (VNF), as detailed by Elvidge et al. (2015) and Zhizhin et al. (2021). We use the 141,841 observations with positive flared volumes between 2012 and 2024 to assign them to specific assets from Rystad's database.

The matching relies on asset eligibility and geographic distance, using the two sources of information at our disposal described in the data section: geographic shapes of assets and geographic locations of wells drilled since 2000 within each asset.

Data on geographic shapes was available for 41,762 assets, among which 16,801 were generic. We added generic shapes for 1,916 missing assets present in our main Rystad dataset on production and ownership, based on their centroid, making the total number of generic shapes 18,785. The distribution of generic shapes is not random. For instance, onshore US (except Alaska) and Canada does not contain any defined shape. Luckily, these two countries (that represent a large share of generic shapes worldwide) are the ones in which data on well location was the most precise, making the use of this secondary database particularly useful. Moreover, some defined shapes overlap, and data on well location helped in many instances to assign flares observations to one specific asset instead of splitting flared volumes between all assets within which shapes the observation falls. The well location dataset contains 1,733,698 wells drilled since 2000, from 39,494 assets, with 857,900 unique locations (some wells being assigned the exact same location in some instances, usually the asset shape centroid).

From these two datasets, two restrictions were added. First, we tagged as ineligible assets defined as subsea tie backs. These oil and gas producing infrastructures being Under water, they cannot possess a flare themselves, and necessarily flare through another producing asset to which they are linked, usually a floating or onshore facility. This restriction removes 2,904 assets and 5,146 wells from our eligible sample. Additionally, for an asset to be eligible in a given year to be assigned a flaring observation, it needs to contain at least one well drilled and completed, whether this well is a producing well or an exploratory/wildcat one.

Once these restrictions are made, geographic distances between each flaring observations and wells or asset shapes were computed. The assignment to specific asset or the split between several follows a 3-phase procedure detailed below. For each of these procedures, a score is computed. For flaring observation o, the score of asset i is computed such as:

$$Score_{oi} = \sum(\frac{10{,}000}{dist_{oiw}})^{\lambda} * n_{iw}^{\kappa} + \sum(\frac{10{,}000}{dist_{oi}})^{\lambda}$$

Each w denotes unique wells locations of asset i, n_{iw} the number of wells in location w. κ is equal to 0.5 if w is generic (i.e. equal to asset's shape centroid) and 1 if non-generic. Finally, λ is an exponent equal to 4/3 in the main specification[1]. We cap the lowest distances to 100m since satellite pixels definition does not allow to be more precise. The first part of the score is thus based on well locations and the second part on proximity to assets boundaries. The rationale behind the first part of the score is that the probability that an asset is responsible for a flare detected by satellite depends on both the proximity and the number of wells of an asset i near the detection location. κ is reduced to 0.5 when multiple wells locations are reported to be the asset centroid since this probably indicate that Rystad does not know the precise locations of these wells.

Attribution follows this 3-phases algorithm:

- ***Phase 1***: for satellite observations with existing wells less than 2km away, only well location was used, this dataset being the most precise one. This means that only the first part of the score detailed above is used. An additional restriction is made: only wells located in a 4km radius are used to compute scores. For a given observation, once scores are computed, each asset scores are ranked from highest to lowest, and consecutive ratios are computed. If a ratio reaches 10, all following assets are eliminated. The rationale behind this selection is that if a ratio reaches such value, the probability that the observed flare actually correspond to an asset with a smaller score is extremely low. For approximately half of observations in this phase, this selection reduces the number of assets to which the observation is attributed to only one. In the remaining 50%, flared volumes are attributed to each

[1] Robustness checks include flared volumes computed with $\lambda = 1$ and $\lambda = 2$.

asset by splitting proportionally to its score and its oil production capacity.

- ***Phase 2***: for observations with at least one well within a 2km to 10km radius, assets shapes information is also used when possible. If at least one close-by asset has a non-generic shape, both assets shapes and wells locations are used (***phase 2A***). If all surrounding assets have generic shapes, only wells locations are used (***phase 2B***). The same process as in phase 1 is applied, except that the ratio threshold for assets with lower score to be dropped is now equal to 2. This reduction comes from the fact that scores are much lower than in phase 1 (due to the exponential nature of the score), and at these distances a ratio of 2 denotes a large difference in terms of number of wells close-by and distance to the assets' boundaries. Additionally, in phase 2A, if an observation falls within at least one asset boundaries, only these assets are kept (with a 100m tolerance), and the scores are computed among them.

- ***Phase 3***: this stage applies to observation with no well in a 10km radius, but existing wells or asset boundaries in a 20km radius. If non-generic shapes exist near the observation, asset shape only matching is performed, using only the second part of the score (***phase 3A***). If only generic shapes surround the observation (***phase 3B***), the procedure is the same as the one performed in phase 2B.

Table C.3 displays the number of observations matched in each phase, as well as the corresponding share of total flaring detected in terms of count and volume (2nd and 3rd columns). Phase 1 allows to match 58.3% of observations, representing 36.1% of the total volume flared. Along with phase 2A, the first phases allow to match over 70% of flaring observations worldwide. 7.261 observations remain unmatched, representing 4.7% of total flared volumes. For these observations, no completed well nor active asset boundaries was found in a 20km radius. The two last columns report, for each phase, the share of observations attributed to a unique Rystad asset. Overall, 56% of observations could be matched with a unique asset, representing 66% of total volumes.

Phase	Count	Share of total flaring		Attributed to unique asset	
		Count	Volume	Count	Volume
Phase 1	77,040	58.3%	36.1%	48.8%	64.0%
Phase 2A	24,240	18.3%	35.3%	76.6%	78.5%
Phase 2B	9,058	6.9%	5.0%	63.1%	70.9%
Phase 3A	12,088	9.1%	18.3%	49.4%	44.6%
Phase 3B	2,482	1.9%	0.6%	80.0%	80.7%
Unmatched	7,261	5.5%	4.7%	-	-

Table C.3: Summary statistics: matching of satellite flares observations to oil and gas assets.

References

Elvidge, Christopher D, Mikhail Zhizhin, Kimberly Baugh, Feng-Chi Hsu, and Tilottama Ghosh, "Methods for global survey of natural gas flaring from visible infrared imaging radiometer suite data," *Energies*, 2015, *9* (1), 14.

Zhizhin, Mikhail, Alexey Matveev, Tilottama Ghosh, Feng-Chi Hsu, Martyn Howells, and Christopher Elvidge, "Measuring gas flaring in Russia with multispectral VIIRS nightfire," *Remote Sensing*, 2021, *13* (16), 3078.